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4

The Inverse Kinematics

The direct kinematic model discussed in Chapter 3 determines the position and orientation of the end-effector (tool) for given values of joint-link displacements. In other words, it answers the question "Where is the origin of the end-effector?" The direct kinematic model, thus, specifies the end-effector frame, frame $\{n\}$ relative to the base frame $\{0\}$, for the n-DOF manipulator, which is expressed as

$$ {}^0T_n(q_1, q_2, \ldots, q_n) = \prod_{i=1}^{n} {}^{i-1}T_i = \begin{bmatrix} n_x & o_x & a_x & d_x \\ n_y & o_y & a_y & d_y \\ n_z & o_z & a_z & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \boldsymbol{T} \tag{4.1}$$

The joint displacements $(q_1, q_2, \ldots, q_n)$ that lead the end-effector to a certain position and orientation $\boldsymbol{T}$ can be found by solving the kinematic model equations for unknown joint displacements. Moving each joint by the respective joint displacement, the location (position and orientation) of the end-effector is achieved. This is the inverse kinematic problem already defined in the previous chapter, Section 3.3. It is possible that for the desired end-effector location, multiple or no solutions may exist. A rigorous definition for the inverse kinematic problem is:

"The determination of all possible and feasible sets of joint variables, which would achieve the specified position and orientation of the manipulator's end-effector with respect to the base frame."

In practice, a robot manipulator control requires knowledge of the end-effector position and orientation for the instantaneous location of each joint as well as knowledge of the joint displacements required to place the end-effector in a new location. Therefore, direct and inverse kinematics are the fundamental problems of utmost importance in the robot manipulator's position control. Many industrial applications such as welding and certain types of assembly operations require that a specific path should be negotiated by the end-effector. To achieve this, it is necessary to find corresponding motions of each joint, which will produce the desired tool-tip motion. This is a typical case of inverse kinematic application.

The manipulator transformation matrix T represents the orientation R and position D of the end-effector with respect to the base frame as:

$$T = \left[\begin{array}{ccc:c} & R & & D \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right] \tag{4.2}$$

The position and orientation of the end-effector is collectively referred as *configuration* of the end-effector. The configuration of the end-effector is represented by three position components as displacements along three orthogonal axes of base frame and three rotations about the base frame axes. These six components can be represented by a six dimensional space called *configuration space* or *Cartesian space*. The kinematic description of orientation of the end-effector with respect to the base frame can be according to any of the conventions outlined in Chapter 2. The configuration, or position and orientation, of the end-effector is a function of joint displacement variables $q_1, q_2, \ldots q_n$ as shown in Fig. 4.1.

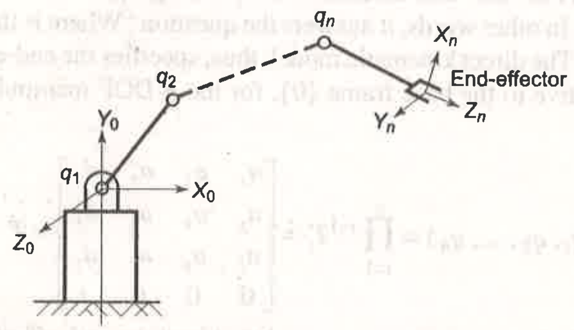


Fig. 4.1 *Configuration of end-effector as a function of joint displacements*

For an n-DOF manipulator the set of n joint displacement variables is represented by a $n \times 1$ vector. The set of all $n \times 1$ joint displacement vectors generates the *joint vector space* or *joint space*. The Cartesian space and joint space representations of a manipulator's end-effector position and orientation are related to each other by mappings shown in Fig. 4.2. The direct kinematic model is the mapping of joint space to Cartesian space, and the mapping from the Cartesian space to the joint space is the inverse problem.

In other words, the inverse kinematics is the determination of the set of positions and orientations in Cartesian space that are reachable by the origin of the end-effector frame as the joint displacements vector q ranges over the joint-space.

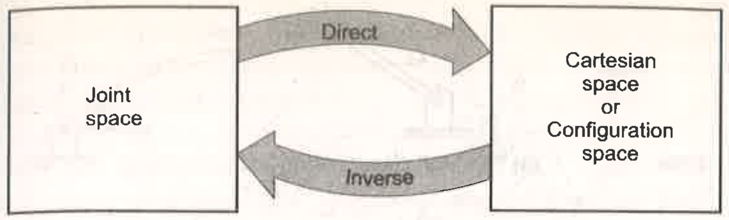


Fig. 4.2 *Mappings between kinematic descriptions*

The inverse kinematic problem is more difficult than the direct problem because no systematic procedures exist for its solution. Inverse problem of every manipulator has to be worked out separately. In this chapter, techniques to solve the inverse problem are presented. First, the concepts of manipulator workspace, solvability of kinematic equations, and existence of multiple solutions are discussed.

4.1 MANIPULATOR WORKSPACE

The workspace of a manipulator is defined as the volume of space in which the manipulator is able to locate its end-effector. The workspace gets specified by the existence or nonexistence of solutions to the inverse problem. The region that can be reached by the origin of the end-effector frame with at least one orientation is called the *reachable workspace* (RWS). If a point in workspace can be reached only in one orientation, the manipulatability of the end-effector is very poor and it is not possible to do any practical work satisfactorily with just one fixed orientation. It is, therefore, necessary to look for the points in workspace, which can be reached in more than one orientation. Some points within the RWS can be reached in more than one orientation. The space where the end-effector can reach every point from all orientations is called *dexterous workspace* (DWS). It is obvious that the dexterous workspace is either smaller (subset) or same as the reachable workspace.

As an example, consider a two-link nontrivial (2-DOF)-planar manipulator having link lengths L_1 and L_2, as shown in Fig. 4.3(a). The RWS for this manipulator is plane annular space with radii $r_1 = L_1 + L_2$ and $r_2 = |L_1 - L_2|$, as shown in Fig. 4.3(b). The DWS for this case is null. Inside the RWS there are two possible orientations of the end-effector for a given position, while on the boundaries of RWS, end-effector has only one possible orientation. For the special case of $L_1 = L_2$, the RWS is a circular area and DWS is a point at the center, as shown in Fig. 4.3(c). It can be shown that for a 3-DOF redundant planar manipulator having link lengths L_1, L_2, and L_3 with $(L_1 + L_2) > L_3$, the RWS is a circle of radius $(L_1 + L_2 + L_3)$, while the DWS is a circle of radius $(L_1 + L_2 - L_3)$.

The reachable workspace of an n-DOF manipulator is the geometric locus of the points that can be achieved by the origin of the end-effector frame as determined by the position vector of direct kinematic model. To locate the tool point or end-effector at an arbitrary position with an arbitrary orientation in 3-D space, a minimum of 6-DOF are required. Thus, for a 6-DOF-manipulator arm,

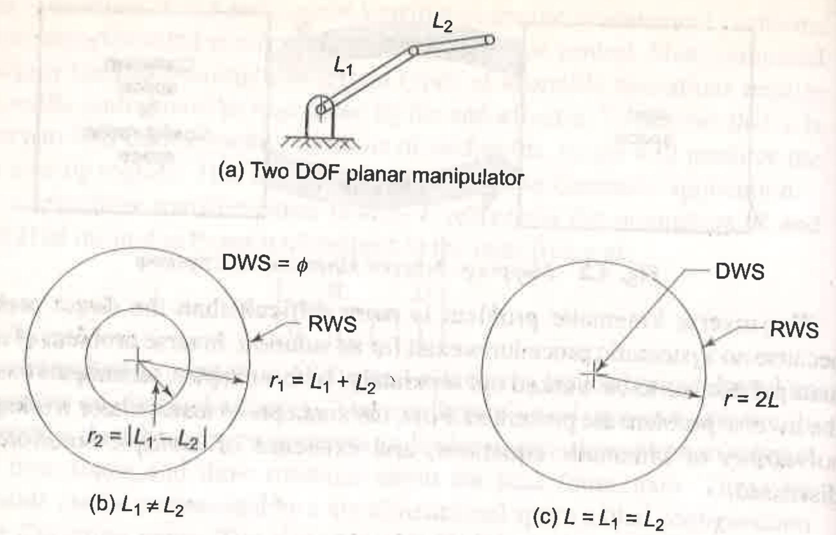


Fig. 4.3 *Workspace a of a two-link planar manipulator*

the dexterous workspace may almost approach the reachable workspace. The reachable work envelops of standard manipulator configurations have been discussed in Chapter 1.

The manipulator workspace is characterized by the mechanical joint limits in addition to the configuration and the number of degrees of freedom of the manipulator. It is important to note that in Fig. 4.3, the workspaces are specified assuming a full 360° rotational joint range for each revolute joint. In practice, the joint range of revolute motion is much less than 360° for the revolute joints and is severely limited for prismatic joints, due to mechanical constraints. This limitation greatly reduces the workspace of the manipulator and the shape of workspace may not be similar to the ideal case.

To understand the effect of mechanical joint limits on the workspace, consider the 2-DOF planar manipulator with $L_1 > L_2$ and joint limits on θ_1 and θ_2 as:

$$\begin{aligned} -60° \le \theta_1 \le 60° \\ -100° \le \theta_2 \le 100° \end{aligned} \tag{4.3}$$

For these joint limits, considering $\theta_1 = \theta_2 = 0$ as home position, the annular workspace in Fig. 4.3(b) gets severely limited. The workspace, obtained geometrically, is not annular any more, rather it has a complex shape and is defined by contour ABCDEFA in Fig. 4.4.

Thus, the factors that decide the workspace of a manipulator apart from the number of degrees of freedom are the manipulator's configuration, link lengths, and the allowed range of joint motions.

4.2 SOLVABILITY OF INVERSE KINEMATIC MODEL

Inverse kinematics is complex because the solution is to be found for nonlinear simultaneous equations, involving transcendental (harmonic sine and cosine)

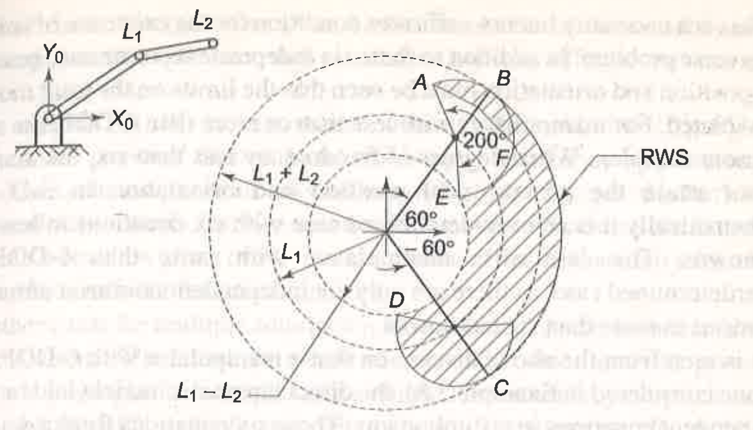


Fig 4.4 *Reachable workspace of a two-link planar manipulator with joint limits*

functions. The number of simultaneous equations is also generally more than the number of unknowns, making some of the equations mutually dependent. These conditions lead to the possibility of multiple solutions or nonexistence of any solution for the given end-effector position and orientation. The existence of solutions, multiple solutions, and methods of solutions are discussed in the following sections.

4.2.1 Existence of Solutions

The conditions for existence of solutions to the inverse kinematic problem are examined first. It is obvious that if the desired point P lies outside the reachable workspace then no solution exists. Even when P is within reachable workspace, not all orientations are realizable, unless P lies within dexterous workspace. If the wrist has fewer than 3-DOF to orient the end-effector, then certain classes of orientations are not realizable. To examine the reasons for this consider Eq. (4.1), the direct kinematic model.

As the last row of the matrix ${}^0T_n$ in Eq. (4.1) is always constant, it can yield a maximum of 12 simultaneous equations, which are nonlinear algebraic equations involving transcendental functions in n unknowns (the joint variables). Out of these, nine equations arise from the rotation matrix (3 × 3) and three from the displacement vector. The nine equations of the rotation matrix involve only three unknowns corresponding to the orientation of the end-effector expressed by Euler angles or roll-pitch-yaw angles. This means that there are only six (three from orientation and three from displacement) independent constraints in n unknowns. This leads to a very important conclusion. For a manipulator to have all position and orientation solutions, the number of DOF n (equal to the number of unknowns) must at least match the number of independent constraints. That is, for general dexterous manipulation

$$n \ge 6 \tag{4.4}$$

This is a necessary but not sufficient condition for the existence of solutions to the inverse problem. In addition to these six independent constraint equations, the tool position and orientation must be such that the limits on the joint motions are not violated. For manipulators with less than or more than 6-DOF, the solutions are more complex. When degrees of freedom are less than six, the manipulator cannot attain the general goal position and orientation in 3-D space—mathematically it is an over-determined case with six equations in less than six unknowns. The case of a manipulator, with more than 6-DOF, is an underdetermined case, as there are only six independent nonlinear simultaneous equations in more than six unknowns.

It is seen from the above discussion that a manipulator with 6-DOF (such as the one considered in Example 3.8), the direct kinematic model yields a set of six independent equations in six unknowns. These six equations form a determinate set of simultaneous equations, which can be quite difficult to solve. It may be recalled that in direct kinematic model it was emphasized to choose the frames, which make as many of the joint-link parameters as possible, to zero. This leads to less complex kinematic equations and, hence, relatively simpler inverse solutions. For the case of a general mechanism with all nonzero-link parameters, the direct kinematic equations are much more complex and so will be the inverse solutions.

4.2.2 Multiple Solutions

The existence of multiple solutions is a common situation encountered in solving inverse kinematic problem. Multiple solutions pose further problem because the robot system has to have a capability to choose one, probably the best one. Multiple solutions can arise because of different factors. Some common situations, which lead to multiple solutions, are discussed as follows.

Consider the 2-DOF planar arm of Fig. 4.3(a) with a wrist having just one DOF. There are two sets of values of joint displacements (θ_1, θ_2) and (θ'_1, θ'_2), as illustrated in Fig. 4.5, which lead to the same end-effector position and orientation

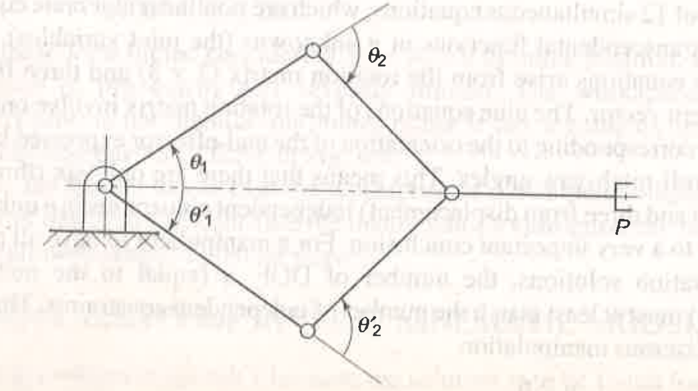


Fig. 4.5 *Multiple solutions due to parallel axes of revolute joints*

for point P. In the configuration space, both solutions are identical as they produce same configuration (position and orientation) of end-effector but are clearly distinct in joint space.

The solution (θ_1, θ_2) is 'elbow-up' position, while solution (θ'_1, θ'_2) is the 'elbow-down' position. Out of these, elbow-up solution may be preferred as in elbow-down solution joint-link may collide with objects lying on the work surface or the work table itself. The two solutions are obtained because the axes of two consecutive revolute joints of the arm are parallel. If more than two joint axes are parallel, the numbers of solutions multiply.

Another cause for multiple solutions is the existence of trigonometric functions in the equations. The harmonic nature of sine and cosine functions gives same magnitude for angles in multiples of π radians. For example, yaw, pitch, and roll motions of the RPY wrist for two sets of joint displacements (θ_1, θ_2, θ_3) and (θ'_1, θ'_2, θ'_3) with $\theta'_1 = 180° + \theta_1$, $\theta'_2 = 180° - \theta_2$ and $\theta'_3 = 180 + \theta_3$ will lead to the same orientation of the wrist. This can be easily verified.

Similarly, if the three motions of the wrist are roll, pitch, and roll; two sets of joint displacements (θ_1, θ_2, θ_3) and (θ'_1, θ'_2, θ'_3) with $\theta'_1 = 180° + \theta_1$, $\theta'_2 = -\theta_2$ and $\theta'_3 = 180° + \theta_3$ will lead to the same orientation of the wrist. With multiple solutions for positioning and orienting the end-effector, the number of solutions may multiply factorially.

The number of solutions also depends on the number of nonzero joint-link parameters and the range of joint motions allowed. In general, the number of ways to reach a certain goal is directly related to the number of nonzero link parameters. For example, for a completely general rotary-jointed, 6-DOF manipulator with all six $a_i \neq 0$, up to sixteen solutions are possible. A manipulator is said to be solvable, if it is possible to find all the solutions to its inverse kinematics problem for a given position and orientation.

Multiple solutions also arise from number of degree of freedom. For example, a manipulator with more than 6-DOF may have infinitely many solutions to the inverse kinematic problem. A manipulator with more degrees of freedom than are necessary is called *kinematically redundant* manipulator. The SCARA configuration is an example of redundant manipulator. It has one redundant degree of freedom in horizontal plane because only two joints (2-DOF) are needed to establish any horizontal position. Redundant manipulators have added flexibility, which can be useful in avoiding obstacles or reaching inaccessible locations, as illustrated in Fig. 4.6.

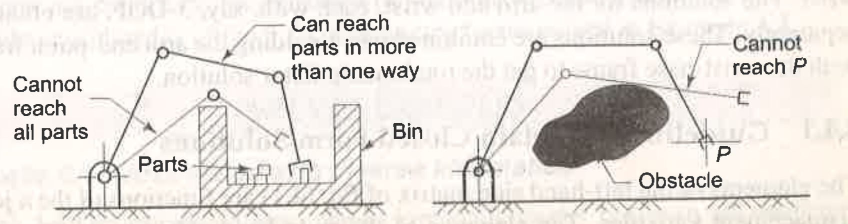


Fig. 4.6 *Use of redundant manipulator to avoid obstacles or reach around them*

4.3 SOLUTION TECHNIQUES

There are two approaches to the solutions to the inverse problem: *closed form solutions* and *numerical solutions*. In the closed form solution, joint displacements are determined as explicit functions of the position and orientation of the end-effector. In numerical methods, iterative algorithms such as the Newton-Raphson method are used. The numerical methods are computationally intensive and by nature slower compared to closed-form methods. Iterative solutions do not guarantee convergence to the correct solution in singular and degenerate cases. Iterative numerical techniques are not discussed in this text.

The "*closed form*" in the present context means a solution method based on analytical algebraic or kinematic approach, giving expressions for solving unknown joint displacements. The closed form solutions may not be possible for all kinds of structures. *A sufficient (but not necessary) condition for a 6-DOF manipulator to possess closed form solutions is that either its three consecutive joint axes intersect or its three consecutive joint axes are parallel.* The kinematic equations under either of these conditions can be reduced to algebraic equations of degree less than or equal to four for which closed form solutions exist. Almost every industrial manipulator manufactured today satisfies one of these conditions so that closed form solutions may be obtained. Manipulator arms with other kinematic structures may be solvable by analytical methods.

4.4 CLOSED FORM SOLUTIONS

Twelve equations, out of which only six are independent, are obtained by equating the elements of the manipulator transformation matrix with end-effector configuration matrix $\boldsymbol{T}$. At the same time, only six of the twelve elements of $\boldsymbol{T}$ specified by the end-effector position and orientation are independent. For a manipulator with less than 6-DOF, the number of independent equations may also be fewer than six. Several approaches such as, inverse transform, screw algebra, and kinematic approach and so on, can be used for solving these equations but none of them is general so as to solve the equations for every manipulator. A composite approach based on direct inspection, algebra, and inverse transform is presented here, which can be used to solve the inverse equations for a class of simple manipulators.

Another useful technique to reduce the complexity is dividing the problem into two smaller parts — the inverse kinematics of arm and the inverse kinematics of wrist. The solutions for the arm and wrist, each with, say, 3-DOF, are obtained separately. These solutions are combined by coinciding the arm end-point frame with the wrist-base frame to get the total manipulator solution.

4.4.1 Guidelines to Obtain Closed Form Solutions

The elements of the left-hand side matrix of Eq. (4.1) are functions of the n joint displacement variables. The elements of the right-hand side matrix $\boldsymbol{T}$ are the desired position and orientation of the end-effector and are either zero or constant. As the matrix equality implies element-by-element equality, 12 equations are obtained. To find the solution for n joint displacement variables from these 12 equations, the following guidelines are helpful.

(a) Look for equations involving only one joint variable. Solve these equations first to get the corresponding joint variable solutions.

(b) Next, look for pairs or set of equations, which could be reduced to one equation in one joint variable by application of algebraic and trigonometric identities.

(c) Use *arc tangent* (Atan2) function instead of *arc cosine* or *arc sine* functions. The two argument Atan2(y, x) function returns the accurate angle in the range of $-\pi \leq \theta \leq \pi$ by examining the sign of both y and x and detecting whenever either x or y is zero.

(d) Solutions in terms of the elements of the position vector components of ${}^0\boldsymbol{T}_n$ are more efficient than those in terms of elements of the rotation matrix, as latter may involve solving more complex equations.

(e) In the inverse kinematic model, the right-hand side of Eq. (4.1) is known, while the left-hand side has n unknowns ($q_1, q_2, \ldots, q_n$). The left-hand side consists of product of n link transformation matrices, that is

$$ {}^0\boldsymbol{T}_n = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2\,{}^2\boldsymbol{T}_3 \ldots {}^{n-1}\boldsymbol{T}_n = \boldsymbol{T} \tag{4.5}$$

Recall that each ${}^{i-1}\boldsymbol{T}_i$ is a function of only one unknown q_i. Premultiplying both sides by the inverse of ${}^0\boldsymbol{T}_1$ yields

$$ {}^1\boldsymbol{T}_n = {}^1\boldsymbol{T}_2\,{}^2\boldsymbol{T}_3 \ldots {}^{n-1}\boldsymbol{T}_n = [{}^0\boldsymbol{T}_1]^{-1}\boldsymbol{T} \tag{4.6}$$

The left-hand side of Eq. (4.6) has now (n–1) unknowns ($q_2, q_3, \ldots, q_n$) and the right-hand side matrix has only one unknown, the q_1. The matrix elements on the right-hand side are zero, constant, or function of the joint variable q_1. A new set of 12 equations is obtained and it may now be possible to determine q_1 from the elements of resulting equations using guideline (a) or (b) above. Similarly, by postmultiplying both sides of Eq. (4.5) by inverse of ${}^{n-1}\boldsymbol{T}_n$, unknown q_n can be determined. This process can be repeated by solving for one unknown at a time, sequentially from q_1 to q_n or q_n to q_1, until all unknowns are found. This is known as *inverse transform* approach.

The closed form solutions for the inverse kinematic model has been using the above guidelines are now illustrated with examples. For some of these examples the direct kinematic models have been obtained in Chapter 3. The multiple solutions and conditions for existence of solutions are also discussed. The first example considered is of the 3-DOF articulated arms solved in Example 3.3.

SOLVED EXAMPLES

Example 4.1 ***Articulated arm inverse kinematics***

For the 3-DOF articulated arm, whose kinematic model has been obtained in Example 3.3, determine the joint displacements for known position and orientation of the end of the arm point.

Solution Let the known position and orientation of the endpoint of arm be given by

$$T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ r_{31} & r_{32} & r_{33} & r_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.7}$$

where each r_{ij} has a numeric value.

To obtain the solutions for joint variables (θ_1, θ_2, θ_3), in Eq. (4.7) T is equated to overall transformation matrix for the 3-DOF articulated arm ${}^0T_3$ derived in Example 3.3, that is

$$\begin{bmatrix} C_1C_{23} & -C_1S_{23} & -S_1 & C_1(L_3C_{23} + L_2C_2) \\ S_1C_{23} & -S_1S_{23} & C_1 & S_1(L_3C_{23} + L_2C_2) \\ S_{23} & C_{23} & 0 & L_3S_{23} + L_2S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ r_{31} & r_{32} & r_{33} & r_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.8}$$

Equation (4.8) gives 11 nontrivial equations for the three unknown joint variables, θ_1, θ_2, and θ_3 appearing on the left-hand side. The determination of solution for these three joint variables for known r_{ij} is the inverse kinematic problem and is worked out as follows.

Step 1 Applying guideline (a), an inspection of elements of the matrices on both the sides of Eq. (4.8) gives that θ_1 can be obtained from element 3 of row 1. The element (1,3) of left-hand side matrix has a term ($-S_1$) in only one variable θ_1 and a constant r_{13} on right-hand side and, hence, it can give angle θ_1 from $-\sin\theta_1 = r_{13}$. However, according to guideline (c) this is not preferred as correct quadrant of the angle can not be found. Alternatively, applying guideline (b), θ_1 can be isolated by dividing element (2, 1) by (1, 1) or (2, 2) by (1, 2) or (1, 3) by (2, 3) or (2, 4) by (1, 4). Out of these, the last one is preferred as per guideline (d). Thus, equating element (1, 4) and (2, 4), on both sides of the matrix two equations are obtained as

$$C_1(L_3C_{23} + L_2C_2) = r_{14} \tag{4.9}$$

$$S_1(L_3C_{23} + L_2C_2) = r_{24} \tag{4.10}$$

Dividing Eq. (4.10) by Eq. (4.9) gives

$$\frac{S_1}{C_1} = \frac{r_{24}}{r_{14}} \tag{4.11}$$

Therefore, according to guidline (c),

$$\theta_1 = \text{Atan2}(r_{24}, r_{14}) \tag{4.12}$$

Step 2 The other two unknowns, θ_2 and θ_3 cannot be obtained directly. To get a solution for θ_2 and θ_3, inverse transform approach, guideline (e), is used. To isolate θ_3, both sides of Eq. (4.8) are postmultiplied by $({}^2T_3)^{-1}$. This will give

$${}^0T_1\,{}^1T_2 = T[{}^2T_3]^{-1} \tag{4.13}$$

From Eq. (3.21), ${}^2T_3$ is

$${}^2T_3 = \begin{bmatrix} C_3 & -S_3 & 0 & L_3C_3 \\ S_3 & C_3 & 0 & L_3S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.14}$$

The inverse of ${}^2T_3$ is obtained using Eq. (2.53) as

$$[{}^2T_3]^{-1} = \left[\begin{array}{c:c} {}^2R_3^T & -{}^2R_3^{T\,2}D_3 \\ \hdashline 0\ 0\ 0 & 1 \end{array}\right] = \begin{bmatrix} C_3 & S_3 & 0 & -L_3 \\ -S_3 & C_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.15}$$

Substituting ${}^0T_1$ and ${}^1T_2$ from Eqs. (3.19) and (3.20), Example 3.3, T from Eq. (4.7) and $[{}^2T_3]^{-1}$ from Eq. (4.15), in Eq. (4.13) gives

$$\begin{bmatrix} C_1C_2 & -C_1S_2 & S_1 & L_2C_1C_2 \\ S_1C_2 & -S_1S_2 & -C_1 & L_2S_1C_2 \\ S_2 & C_2 & 0 & L_2S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} C_3r_{11} - S_3r_{12} & S_3r_{11} + C_3r_{12} & r_{13} & -L_3r_{11} + r_{14} \\ C_3r_{21} - S_3r_{22} & S_3r_{21} - S_3r_{22} & r_{23} & -L_3r_{21} + r_{24} \\ C_3r_{31} - S_3r_{32} & S_3r_{31} - S_3r_{32} & r_{32} & -L_3r_{31} + r_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.16}$$

Note that the left-hand side of Eq. (4.16) has only θ_1 and θ_2 terms and the right-hand side has only θ_3 terms. A close examination of both sides reveals that equations obtained from elements (1, 4), (2, 4), and (3, 4) are only function of θ_1 and θ_2. Thus, with use of some algebra and trigonometric identities, θ_1 can be eliminated and solution for θ_2 is obtained. Equating the elements (1, 4), (2, 4), and (3, 4) of the two matrices, three equations obtained are

$$L_2C_1C_2 = -L_3r_{11} + r_{14} \tag{4.17}$$

$$L_2S_1C_2 = -L_3r_{21} + r_{24} \tag{4.18}$$

$$L_2S_2 = -L_3r_{31} + r_{34} \tag{4.19}$$

By squaring Eqs. (4.17) and (4.18), and adding gives,

$$L_2^2C_2^2(C_1^2 + S_1^2) = (-L_3r_{11} + r_{14})^2 + (-L_3r_{21} + r_{24})^2$$

From this θ_1 is eliminated because $C_1^2 + S_1^2 = 1$, thus

$$L_2C_2 = \pm\sqrt{(-L_3r_{11} + r_{14})^2 + (-L_3r_{21} + r_{24})^2} \tag{4.20}$$

Dividing Eq. (4.19) by Eq. (4.20), gives

$$\frac{S_2}{C_2} = \frac{-L_3r_{31} + r_{34}}{\pm\sqrt{(-L_3r_{11} + r_{14})^2 + (-L_3r_{21} + r_{24})^2}} \tag{4.21}$$

Hence,

$$\theta_2 = \text{Atan2}\left((-L_3r_{31} + r_{34}), \pm\sqrt{(-L_3r_{11} + r_{14})^2 + (-L_3r_{21} + r_{24})^2}\right) \tag{4.22}$$

Step 3 The solution for θ_3 is obtained by first solving for $(\theta_2 + \theta_3)$. Dividing element (3,1) of Eq. (4.8) by element (3,2) gives

$$\frac{S_{23}}{C_{23}} = \frac{r_{31}}{r_{32}} \tag{4.23}$$

or

$$\theta_2 + \theta_3 = \text{Atan2}(r_{31}, r_{32}) \tag{4.24}$$

Thus,

$$\theta_3 = \text{Atan2}\,(r_{31}, r_{32}) - \theta_2 \tag{4.25}$$

Equations (4.12), (4.22), and (4.25) give the complete solution for the 3-DOF articulated arm as expressions for the joint displacements θ_1, θ_2, and θ_3 in terms of known arm end-point position and orientation. Note that the above solution is one of the possible sets of expressions. Alternate expressions for θ_1, θ_2, and θ_3 would be obtained if instead of equating the chosen elements of the matrices, other elements are used, or instead of isolating θ_3, θ_1 is isolated by premultiplying both sides with $({}^0T_1)^{-1}$. It is also possible to find solution without use of the inverse matrix approach and instead of using algebra and trigonometry. For instance, after solving for θ_1, $(\theta_2 + \theta_3)$ can be obtained from elements (3, 1) and (3, 2) and through trigonometric manipulation, θ_2 and θ_3 are obtained.

Example 4.2 *Inverse kinematics of RPY wrist*

For the 3-DOF RPY wrist kinematic model was obtained in Example 3.4, Eq. (3.27), as

$${}^0T_3 = \begin{bmatrix} -C_1S_2C_3 + S_1S_3 & C_1S_2S_3 + S_1C_3 & C_1C_2 & 0 \\ -S_1S_2C_3 - C_1S_3 & S_1S_2S_3 - C_1C_3 & S_1C_2 & 0 \\ C_2C_3 & -C_2S_3 & S_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.26}$$

Determine the solution for the three joint variables for a given end-effector orientation matrix T_E.

$$T_E = \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.27}$$

Solution The overall transformation matrix ${}^0T_3$ and end-effector matrix T_E represent the same transformations. Thus, equating Eqs. (4.26) and (4.27) gives

$$\begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -C_1S_2C_3 + S_1S_3 & C_1S_2S_3 + S_1C_3 & C_1C_2 & 0 \\ -S_1S_2C_3 - C_1S_3 & S_1S_2S_3 - C_1C_3 & S_1C_2 & 0 \\ C_2C_3 & -C_2S_3 & S_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.28}$$

The elements of the matrix on left-hand side of matrix equation are known (given), while, the elements of matrix on right-hand side have three unknown joint variables θ_1, θ_2 and θ_3. To get the solution for these joint variables, the more consistent analytical approach (guideline (e)) is used here.

Guideline (e) suggests premultiplying the matrix equation, Eq. (4.28) by inverse of transformation matrix ${}^0T_1$ involving the unknown θ_1 and from the elements of the resultant matrix equation determine the unknown. Recall that the right-hand side of Eq. (4.28) is the product of three transformation matrices ${}^0T_1$, ${}^1T_2$ and ${}^2T_3$ each involving one unknown θ_1, θ_2 and θ_3, respectively.

This process is continued successively, that is, moving one unknown (by its inverse transform) from right-hand side of the matrix equation to the left-hand side of the matrix equation and solving it, then moving the next unknown to the left-hand side, until all unknown are solved.

To solve for θ_1, both sides of Eq. (4.28) are premultiplied by ${}^0T_1^{-1}$. From Eqs. (3.24) – (3.26)

$$\begin{bmatrix} C_1 & S_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ S_1 & -C_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -S_2 & 0 & C_2 & 0 \\ C_2 & 0 & S_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_3 & -S_3 & 0 & 0 \\ S_3 & C_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

or

$$\begin{bmatrix} C_1n_x + S_1n_y & C_1o_x + S_1o_y & C_1a_x + S_1a_y & 0 \\ n_z & o_z & a_z & 0 \\ S_1n_x - C_1n_y & S_1o_x - C_1o_y & S_1a_x - C_1a_y & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -S_2C_3 & S_2S_3 & C_2 & 0 \\ C_2C_3 & -C_2S_3 & S_2 & 0 \\ S_3 & C_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.29}$$

The left-hand side of Eq. (4.29) has one unknown (θ_1) and the right-hand side has two unknown (θ_2 and θ_3). Scanning the elements of both the matrices in Eq. (4.29), the equation in one unknown (θ_1) is obtained by equating elements (3, 3). That is,

$$S_1a_x - C_1a_y = 0 \tag{4.30}$$

or

$$\frac{S_1}{C_1} = \tan\theta_1 = \frac{a_y}{a_x}$$

which gives

$$\theta_1 = \text{Atan2}(a_y, a_x) \tag{4.31}$$

The process of further premultiplication is not necessary because the solutions for the remaining two unknowns (θ_2 and θ_3) can be obtained from Eq. (4.29). Equating (1, 3) and (2, 3) elements on both sides in Eq. (4.29) gives

$$C_2 = C_1a_x + S_1a_y$$
$$S_2 = a_z \tag{4.32}$$

From these two equations the solution for θ_2 is obtained as

$$\theta_2 = \text{Atan2}(a_z,\ C_1 a_x + S_1 a_y) \tag{4.33}$$

Equating elements (3,1) and (3, 2) of Eq. (4.29) gives

$$S_3 = S_1 n_x - C_1 n_y$$
$$C_3 = S_1 o_x - C_1 o_y \tag{4.34}$$

Which lead to the solution for θ_3 as

$$\theta_3 = \text{Atan2}(S_1 n_x - C_1 n_y,\ S_1 o_x - C_1 o_y) \tag{4.35}$$

The inverse transform technique is to move one unknown to the left-hand side at a time and solve it. Therefore, it is also possible to achieve this by postmultiplying instead of premultiplying by the inverse transform matrix involving an unknown. This is illustrated here.

To solve for θ_3, that is, to move it to the left-hand side, postmultiplying both sides of matrix equation Eq. (4.28) by ${}^2T_3^{-1}$ gives

$$\begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} C_3 & S_3 & 0 & 0 \\ -S_3 & C_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} -S_2 & 0 & C_2 & 0 \\ C_2 & 0 & S_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

or

$$\begin{bmatrix} C_3 n_x - S_3 o_x & S_3 n_x + C_3 o_x & a_x & 0 \\ C_3 n_y - S_3 o_y & S_3 n_y + C_3 o_y & a_y & 0 \\ C_3 n_z - S_3 o_z & S_3 n_z + C_3 o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -C_1 S_2 & S_1 & C_1 C_2 & 0 \\ -S_1 S_2 & -C_1 & S_1 C_2 & 0 \\ C_2 & 0 & S_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.36}$$

Comparing elements of the matrices on both sides, the elements (3, 2) gives

$$S_3 n_z + C_3 o_z = 0$$

and, thus, the solution for θ_3 is

$$\theta_3 = \text{Atan2}\ (-o_z,\ n_y) \tag{4.37}$$

Similarly, from the elements (3,1) and (3, 3), θ_2 is obtained as

$$\theta_2 = \text{Atan2}\ (a_z,\ C_3 n_z - S_3 o_z) \tag{4.38}$$

and from the elements (1, 2) and (2, 2), θ_1 is obtained as

$$\theta_1 = \text{Atan2}(S_3 n_x + C_3 o_x,\ -S_3 n_y - C_3 o_y) \tag{4.39}$$

In this example, premulitplying or postmultiplying gives solution of similar complexity but this may not be always the case. The decision to premultiply or postmultiply is left to the discretion of the reader.

Example 4.3 ***SCARA manipulator inverse kinematics***

Analytically solve the inverse kinematic problem for the 4-DOF SCARA configuration manipulator given in Fig. 3.22, Example 3.6. Discuss the conditions for existence and multiplicity of solutions.

Solution For the SCARA manipulator of Example 3.6, equating ${}^0T_4$ from Eq. (3.40) with T in Eq. (4.7) gives

$$\begin{bmatrix} C_{124} & S_{124} & 0 & L_2 C_{12} + L_{11} C_1 \\ S_{124} & -C_{124} & 0 & L_2 S_{12} + L_{11} S_1 \\ 0 & 0 & 1 & L_{12} + d_3 - L_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ r_{31} & r_{32} & r_{33} & r_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.40}$$

The solution for joint displacement d_3 is directly obtained by equating the elements (3, 4) on both sides of Eq. (4.40),

$$L_{12} + d_3 - L_4 = r_{34}$$

or

$$d_3 = r_{34} + L_4 - L_{12} \tag{4.41}$$

Next, to solve for θ_1 elements (1, 4) and (2, 4) are compared. This gives

$$L_2 C_{12} + L_{11} C_1 = r_{14} \tag{4.42}$$
$$L_2 S_{12} + L_{11} S_1 = r_{24} \tag{4.43}$$

Squaring Eqs. (4.42) and (4.43), adding and simplifying using the trigonometric identity cos $(\alpha + \beta)$ = cos α cos β – sin α sin β, gives

$$L_{11}^2 + L_2^2 + 2L_{11}L_2C_2 = r_{14}^2 + r_{24}^2 \tag{4.44}$$

$$C_2 = \frac{r_{14}^2 + r_{24}^2 - L_{11}^2 - L_2^2}{2L_{11}L_2} \tag{4.45}$$

Since

$$S_2 = \pm\sqrt{1 - C_2^2} \tag{4.46}$$

the solution for θ_2 is obtained from Eqs. (4.45) and (4.46) as

$$\theta_2 = \text{Atan2}(S_2, C_2) \tag{4.47}$$

Now that θ_2 being known, Eqs. (4.42) and (4.43) can be used to compute θ_1. These equations are written as

$$L_2(C_1C_2 - S_1S_2) + L_{11}C_1 = r_{14} \tag{4.48}$$
$$L_2(S_1C_2 + C_1S_2) + L_{11}S_1 = r_{24} \tag{4.49}$$

or

$$(L_{11} + L_2C_2)C_1 - (L_2S_2)S_1 = r_{14} \tag{4.50}$$
$$(L_{11} + L_2C_2)S_1 + (L_2S_2)C_1 = r_{24} \tag{4.51}$$

Let $(L_{11} + L_2C_2) = r\cos\phi$ and $(L_2S_2) = r\sin\phi$ (4.52)

with

$$r = \sqrt{(L_{11} + L_2C_2)^2 + (L_2S_2)^2} \tag{4.53}$$

and

$$\phi = \text{Atan2}\left(\frac{L_2S_2}{r},\ \frac{L_{11} + L_2C_2}{r}\right) \tag{4.54}$$

Eqs. (4.50) and (4.51) reduce to

$$r\cos(\theta_1 + \phi) = r_{14} \tag{4.55}$$
$$r\sin(\theta_1 + \phi) = r_{24} \tag{4.56}$$

From Eqs. (4.55) and (4.56), θ_1 is obtained as

$$\theta_1 = \text{Atan2}\left(\frac{r_{24}}{r}, \frac{r_{14}}{r}\right) - \phi \tag{4.57}$$

$$\theta_1 = \text{Atan2}\left(\frac{r_{24}}{r}, \frac{r_{14}}{r}\right) - \text{Atan2}\left(\frac{L_2 S_2}{r}, \frac{L_{11} + L_2 C_2}{r}\right) \tag{4.58}$$

With θ_1, θ_2 and d_3 determined, only one variable, θ_4 is unknown. From elements (1,1) and (2,1), the equations are

$$C_{124} = r_{11} \tag{4.59}$$

$$S_{124} = r_{21} \tag{4.60}$$

Equations (4.59) and (4.60) give θ_4 as

$$\theta_1 + \theta_2 - \theta_4 = \text{Atan2}(r_{21}, r_{11})$$

or

$$\theta_4 = \theta_2 + \theta_1 - \text{Atan2}\,(r_{21}, r_{11}) \tag{4.61}$$

The complete closed form solution for the joint displacements θ_1, θ_2, d_3 and θ_4, of SCARA manipulator, is given by Eqs. (4.58), (4.47), (4.41) and (4.61), respectively, as explicit functions of the manipulator's tool position and orientation.

Existence of Solutions

Solutions to the inverse kinematic problem of a given arm exist, that is, the given Cartesian position and orientation of the tool is within the manipulator's workspace if the following condition is satisfied:

"Since *sin* and *cos* functions take values in the range [–1,1], right-hand side of the Eq. (4.45) must lie in the range [–1,1]."

Again, these solutions are for full 360 degrees of rotation for the revolute joints and limitless translation for the prismatic joint. The mechanical constraints, however, will permit only such solutions for which the joint variables take a value that lies in the range of motions allowed.

Multiplicity of Solutions

Due to the presence of the square root in Eq. (4.46), there are two solutions for θ_2 for a given position and orientation of the tool with respect to the base. From Eqs. (4.57) and (4.61), observe that there is one set of solution for θ_1 and θ_4 corresponding to each value of θ_2. Thus, the number of solutions to the inverse kinematics problem of the given SCARA arm is two. Note that multiple solutions exist due to the fact that revolute joint axes 1 and 2 are parallel.

Example 4.4 ***Numerical solutions for a 3-DOF manipulator***

For the 3-DOF (RRP) configuration manipulator, shown in Fig. 4.7, the position and orientation of point P in Cartesian space is given by

$$T = \begin{bmatrix} 0.354 & 0.866 & 0.354 & 0.106 \\ -0.612 & 0.500 & -0.612 & -0.184 \\ 0.707 & 0 & 0.707 & 0.212 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.62}$$

Determine all values of all joint variables, that is, all solutions to the inverse kinematic problem. The joint displacements allowed (joint limits) for three joints are: $-100° < \theta_1 < 100°$, $-30° < \theta_2 < 70°$ and $0.05\text{m} < d_3 < 0.5$ m. Identify the feasible solutions.

Solution A manipulator with this configuration is another common structure widely used in industrial robots, as it is very effective in material handling and other applications, and gives a spherical workspace. The first two joints are revolute joints and provide motion in two perpendicular planes. Their sweep generates a constant radius sphere. The third prismatic joint provides the reach to the arm point where the wrist is attached. The three joint axes intersect at a point.

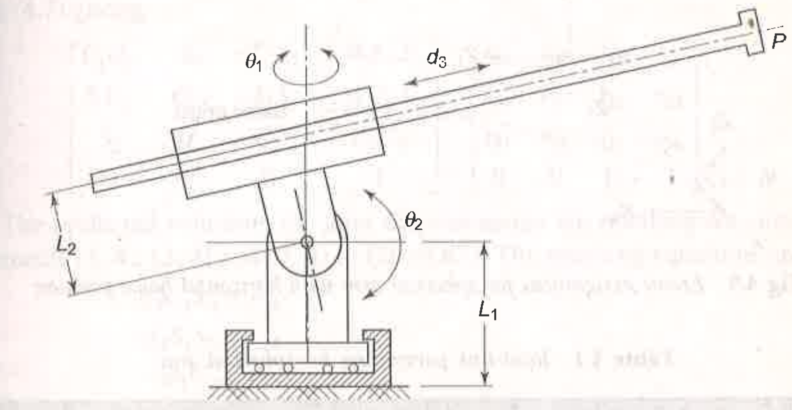


Fig. 4.7 *A 3-DOF spherical configuration arm*

The forward kinematic model is obtained first. For the forward kinematic model, the frame assignment for the home position is carried out first. While assigning frames it is observed that the link dimension L_1 can be eliminated from the kinematic model by choosing the origin of frame {0} to coincide with origin of frame {1} at joint 2 (see Example 3.3). The link dimension L_2 can be made zero by modifying the design slightly as shown in Fig. 4.8 such that the axis of prismatic link passes through the origin of frame {1}.

The final frame assignment with the origin of three frames, frame {0}, frame {1} and frame {2} at the same point is shown in Fig. 4.9. This minimizes the number of non-zero parameters as well as satisfies the necessary condition for existence of closed form solutions.

The joint-link parameters are tabulated in Table 4.1

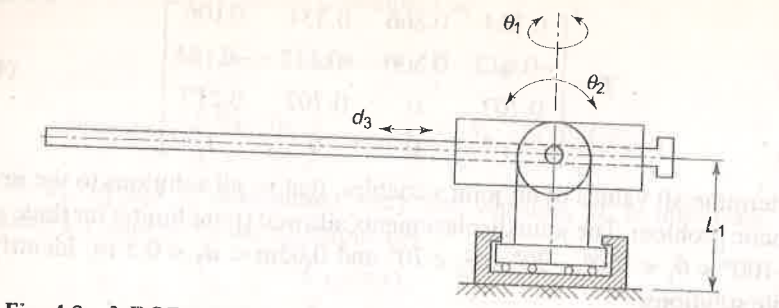


Fig. 4.8 *3-DOF spherical arm in home position:* $\theta_1 = \theta_2 = 0$ and $d_3 = 0.05$

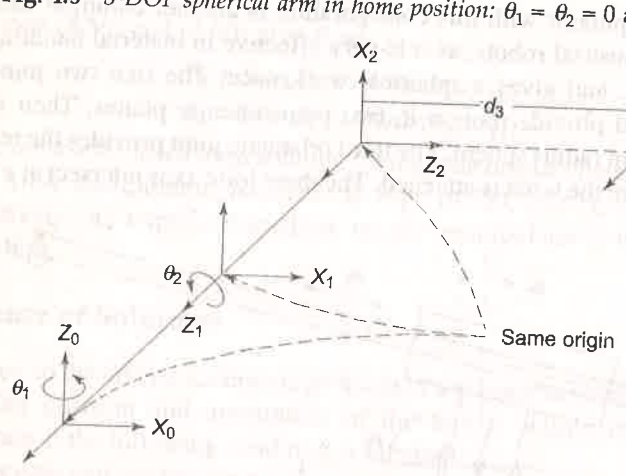


Fig 4.9 *Frame assignment for spherical arm with horizontal home position*

Table 4.1 *Joint-link parameters for spherical arm*

Link i	a_i	α_i	d_i	θ_i	q_i	$C\theta_i$	$S\theta_i$	$C\alpha_i$	$S\alpha_i$
1	0	90°	0	θ_1	θ_1	C_1	S_1	0	1
2	0	−90°	0	θ_2	θ_2	C_2	S_2	0	1
3	0	0	d_3	0	d_3	1	0	1	0

The three link transformation matrices ${}^0\boldsymbol{T}_1$, ${}^1\boldsymbol{T}_2$, ${}^2\boldsymbol{T}_3$ and the overall arm transformation matrix ${}^0\boldsymbol{T}_3$ are obtained as

$$
{}^0\boldsymbol{T}_1(\theta_1) = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.63}
$$

$$
{}^1\boldsymbol{T}_2(\theta_2) = \begin{bmatrix} C_2 & 0 & -S_2 & 0 \\ S_2 & 0 & C_2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.64}
$$

$$
{}^2\boldsymbol{T}_3(d_3) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.65}
$$

and, thus,

$$
{}^0\boldsymbol{T}_3 = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2\,{}^2\boldsymbol{T}_3 = \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 & -d_3C_1S_2 \\ S_1C_2 & C_1 & -S_1S_2 & -d_3S_1S_2 \\ S_2 & 0 & C_2 & d_3C_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.66}
$$

First, a generalized solution is worked out, as in previous examples and then the values from Eq. (4.62) will be substituted to get specific solutions. Let the arm point position and orientation be specified as in Eq. (4.7).

The kinematic model equations are thus, obtained by equating Eq. (4.66) and Eq. (4.7) giving

$$
\begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 & -d_3C_1S_2 \\ S_1C_2 & C_1 & -S_1S_2 & -d_3S_1S_2 \\ S_2 & 0 & C_2 & d_3C_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ r_{31} & r_{32} & r_{33} & r_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.67}
$$

The preferred solutions for joint displacements are obtained by comparing elements (1, 4), (2, 4) and (3, 4) in Eq. (4.67). The resulting equations are

$$-d_3C_1S_2 = r_{14} \tag{4.68}$$

$$-d_3S_1S_2 = r_{24} \tag{4.69}$$

$$d_3C_2 = r_{34} \tag{4.70}$$

Dividing Eq. (4.69) by Eq. (4.68) the solution for θ_1 is

$$\theta_1 = \text{Atan2}(-r_{24}, -r_{14}) \tag{4.71}$$

Squaring and adding Eq. (4.68) and Eq. (4.69) gives

$$d_3^2\, S_2^2\,(C_1^2 + S_1^2) = r_{14}^2 + r_{24}^2$$

or

$$d_3S_2 = \pm\sqrt{r_{14}^2 + r_{24}^2} \tag{4.72}$$

Dividing Eq. (4.72) by Eq. (4.70) gives solution of θ_2 as

$$\theta_2 = \text{Atan2}\left(\pm\sqrt{r_{14}^2 + r_{24}^2},\ r_{34}\right) \tag{4.73}$$

The joint displacement d_3 for joint 3 is obtained by squaring and adding Eqs. (4.68), (4.69) and (4.70). Since the displacement d_3 cannot be negative, only positive sign is used. Thus,

$$d_3 = +\sqrt{r_{14}^2 + r_{24}^2 + r_{34}^2} \tag{4.74}$$

The numerical values for joint displacements are obtained by substituting the values from given arm point position and orientation matrix, Eq. (4.62), in to Eqs. (4.71), (4.73) and (4.74). The specific solutions are:

$$\theta_1 = \text{Atan2}\,(0.184, -0.106) = -60°$$

$$\theta_2 = \text{Atan2}\left(\pm\sqrt{(0.106)^2 + (-0.184)^2},\ 0.212\right) = \pm 45° \tag{4.75}$$

$$\theta_3 = \text{Atan2}\ \sqrt{(0.106)^2 + (-0.184)^2 + (0.212)^2} = 0.30$$

The two possible solutions are tabulated in Table 4.2.

Table 4.2 *Two possible solutions for the specified arm point*

Solution No.	θ_1	θ_2	d_3
1	-60°	-45°	0.30
2	-60°	45°	0.30

The joint range specified for joint 2 is: $-30° < \theta_2 < 70°$. The solution 1 specifies angle θ_2 as $\theta_2 = -45°$. This violates the joint range constraint and, hence, solution 1 is not feasible.

Example 4.5 *Inverse Kinematics of 5-DOF Manipulator*

For the 5-DOF industrial manipulator discussed in Example 3.7, obtain the analytical solutions of joint variables.

Solution Let the end-effector tool point transformation matrix be given by

$$T_E = \begin{bmatrix} n_x & o_x & a_x & d_x \\ n_y & o_y & a_y & d_y \\ n_z & o_z & a_z & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.76}$$

From the kinematic model obtained in Example 3.7, the overall transformation matrix for the end-effector tool point is (Eq. (3.48))

$${}^0T_5 = \begin{bmatrix} C_1S_{234}C_5 + S_1S_5 & -C_1S_{234}S_5 + S_1C_5 & C_1C_{234} & C_1(L_2C_2 + L_3C_{23} + L_5C_{234}) \\ S_1C_{234}C_5 - C_1S_5 & -S_1S_{234}S_5 - C_1C_5 & S_1C_{234} & S_1(L_2C_2 + L_3C_{23} + L_5C_{234}) \\ -C_{234}C_5 & C_{234}S_5 & -S_{234} & L_1 - L_2S_2 - L_3S_{23} - L_5S_{234} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.77}$$

A close examination of Eqs.(4.76) and (4.77) clearly shows that no direct solution can be found for any of the joint-variables. Hence, the inverse matrix approach of guideline (e) is used here.

In order to solve for first joint variable θ_1, both matrices, Eq. (4.76) and Eq. (4.77) are premultiplied by inverse of ${}^0T_1$ (that is ${}^0T_1^{-1}$. This given left-hand side matrix of equation as

$${}^0T_1^{-1}T_E = \begin{bmatrix} C_1n_x + S_1n_y & C_1o_x + S_1o_y & C_1a_x + S_1a_y & C_1d_x + S_1d_y \\ -n_z & -o_z & -a_z & -d_z + L_1 \\ -S_1n_x + C_1n_y & -S_1o_x + C_1o_y & -S_1a_x + C_1a_y & -S_1d_x + C_1d_y \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.78}$$

and the right-hand side of matrix equation is

$${}^1T_5 = {}^1T_2\,{}^2T_3\,{}^3T_4\,{}^4T_5 = \begin{bmatrix} S_{234}C_5 & S_{234}S_5 & C_{234} & L_5C_{234} + L_3C_{23} + L_2C_2 \\ -C_{234}C_5 & C_{234}S_5 & S_{234} & L_5S_{234} + L_3S_{23} + L_2S_2 \\ -S_5 & -C_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.79}$$

or

$$\begin{bmatrix} C_1n_x + S_1n_y & C_1o_x + S_1o_y & C_1a_x + S_1a_y & C_1d_x + S_1d_y \\ -n_z & -o_z & -a_z & -d_z + L_1 \\ -S_1n_x + C_1n_y & -S_1o_x + C_1o_y & -S_1a_x + C_1a_y & -S_1d_x + C_1d_y \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} S_{234}C_5 & S_{234}S_5 & C_{234} & L_5C_{234} + L_3C_{23} + L_2C_2 \\ -C_{234}C_5 & C_{234}S_5 & S_{234} & L_5S_{234} + L_3S_{23} + L_2S_2 \\ -S_5 & -C_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.80}$$

Comparing elements of Eq. (4.80), the single variable equations in θ_1 are obtained from elements (3, 3) and (3, 4) as:

$$\begin{aligned} -S_1a_x + C_1a_y &= 0 \\ -S_1d_x + C_1d_y &= 0 \end{aligned} \tag{4.81}$$

Using the later, the solution for θ_1 is

$$\theta_1 = \text{Atan2}(d_y,\ d_x) \tag{4.82}$$

From ratio of elements (3, 1) and (3, 2), solution for θ_5 is obtained as

$$\begin{aligned} -S_5 &= -S_1n_x + C_1n_y \\ -C_5 &= -S_1o_x + C_1o_y \end{aligned} \tag{4.83}$$

or
$$\theta_5 = \text{Atan2}(-S_1n_x + C_1n_y, -S_1o_x + C_1o_y) \tag{4.84}$$

The solution for remaining three variables θ_2, θ_3 and θ_4 is obtained by first solving for $\theta_2 + \theta_3 + \theta_4$. Equating elements (1, 3) and (2, 3) gives

$$\begin{aligned} C_{234} &= C_1a_x + S_1a_y \\ S_{234} &= -a_z \end{aligned} \tag{4.85}$$

which leads to

$$\theta_{234} = \theta_2 + \theta_3 + \theta_4 = \text{Atan2}(-a_z,\ C_1 a_x + S_1 a_y) \tag{4.86}$$

Premultiplying both sides of Eq. (4.80) by $({}^1\boldsymbol{T}_2)^{-1}$, gives

$$({}^1\boldsymbol{T}_2)^{-1}({}^0\boldsymbol{T}_1)^{-1}\boldsymbol{T}_E = {}^2\boldsymbol{T}_3\,{}^3\boldsymbol{T}_4\,{}^4\boldsymbol{T}_5 \tag{4.87}$$

$$\begin{bmatrix} C_2(C_1n_x + S_1n_y) - S_2n_z & C_2(C_1o_x + S_1o_y) - S_2o_z & C_2(C_1a_x + S_1a_y) - S_2a_z & C_2(C_1d_x + S_1d_y) - S_2(d_z - L_1) - L_2 \\ -S_2(C_1n_x + S_1n_y) - C_2n_z & -S_2(C_1o_x + S_1o_y) - C_2o_z & -S_2(C_1a_x + S_1a_y) - C_2a_z & -S_2(C_1d_x + S_1d_y) - C_2(d_z + L_1) \\ -S_1n_x + C_1n_y & -S_1o_x + C_1o_y & -S_1a_x + C_1a_y & -S_1d_x + C_1d_y \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} S_{34}C_5 & -S_{34}S_5 & C_{34} & L_5C_{34} + L_3C_3 \\ -C_{34}C_5 & C_{34}S_5 & S_{34} & L_5S_{34} + L_3C_3 \\ -S_5 & -C_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.88}$$

From elements (1, 1) and (1, 2), two equations are obtained:

$$C_2\left(C_1n_x + S_1n_y\right) - S_2n_z = S_{34}C_5 \tag{4.89}$$

$$C_2\left(C_1o_x + S_1o_y\right) - S_2o_z = -S_{34}S_5 \tag{4.90}$$

Dividing Eq. (4.90) by Eq. (4.89) and rearranging gives

$$-(C_1o_x + S_1o_y)C_2 + S_2o_z = \tan\theta_5\left[(C_1n_x + S_1n_y)C_2 - S_2n_z\right] \tag{4.91}$$

or

$$\frac{S_2}{C_2} = \frac{\tan\theta_5\left(C_1n_x + S_1n_y\right) + \left(C_1o_x + S_1o_y\right)}{n_z\tan\theta_5 + o_z}$$

or

$$\theta_2 = \text{Atan2}\left[\tan\theta_5\left(C_1n_x + S_1n_y\right) + C_1o_x + S_1o_y,\ n_z\tan\theta_5 + o_z\right] \tag{4.92}$$

Similarly from elements (1, 4) and (2, 4) of Eq. (4.88), two equations are obtained after rearrangements as

$$\begin{aligned} L_5C_{34} + L_3C_3 &= C_1C_2d_x + S_1C_2d_y - S_2(d_z - L_1) - L_2 \\ L_5S_{34} + L_3S_3 &= -C_1S_2d_x + S_1S_2d_y - C_2(d_z + L_1) \end{aligned} \tag{4.93}$$

Substituting for C_{34} and S_{34} from elements (1, 3) and (2, 3), respectively form the left-hand matrix of Eq. (4.88) and simplifying gives

$$\begin{aligned} L_3C_3 &= C_1C_2d_x + S_1C_2d_y - S_2(d_z - L_1) - L_2 - L_5(C_1C_2a_x + S_1C_2a_y - S_2a_z) \\ L_3S_3 &= -C_1S_2d_x - S_1S_2d_y - C_2(d_z + L_1) + L_5(C_1S_2a_x + S_1S_2a_y + C_2a_z) \end{aligned} \tag{4.94}$$

Form Eq. (4.94) θ_3 is obtained as

$$\theta_3 = \text{Atan2}\begin{pmatrix} L_3S_3 = -C_1S_2d_x - S_1S_2d_y - C_2(d_z + L_1) + L_5(C_1S_2a_x + S_1S_2a_y + C_2a_z), \\ L_3C_3 = C_1C_2d_x + S_1C_2d_y - S_2(d_z - L_1) - L_2 - L_5(C_1C_2a_x + S_1C_2a_y - S_2a_z) \end{pmatrix} \tag{4.95}$$

The last unknown joint variable θ_4 is computed from Eq. (4.86).

$$\theta_4 = \theta_{234} - \theta_2 - \theta_3 \tag{4.96}$$

Example 4.6 *Inverse kinematics for 6-DOF Stanford manipulator*

Obtain all the joint displacements as explicit functions of the position and orientation of the end-effector for the 6-DOF manipulator of Example 3.8 by analytical method. Also, discuss the existence and multiplicity of solutions.

Solution The given 6-DOF-manipulator arm has a wrist whose joint axes 4, 5 and 6 intersect at a point. This satisfies the necessary condition for existence of closed form solutions. Hence, it is possible to solve the inverse problem in closed form.

Let the given position and orientation of the end-effector with respect to the base in Cartesian space be

$$\boldsymbol{T} = \begin{bmatrix} n_x & o_x & a_x & d_x \\ n_y & o_y & a_y & d_y \\ n_z & o_z & a_z & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.97}$$

From Example 3.8, the manipulator transformation matrix (the forward kinematic model) for the given manipulator is given by Eq. (3.58) that is

$${}^0\boldsymbol{T}_6 = \begin{bmatrix} \begin{matrix} C_1C_2C_4C_5C_6 \\ -S_1S_4C_5C_6 \\ -C_1S_2S_5C_6 \\ -C_1C_2S_4S_6 \\ -S_1C_4S_6 \end{matrix} & \begin{matrix} C_1C_2C_4C_5S_6 \\ +S_1S_4C_5S_6 \\ +C_1S_2S_5S_6 \\ -C_1C_2S_4C_6 \\ -S_1C_4C_6 \end{matrix} & \begin{matrix} C_1C_2C_4S_5 \\ -S_1S_4S_5 \\ +C_1S_2C_5 \end{matrix} & \begin{matrix} C_1C_2C_4S_5L_6 \\ -S_1S_4S_5L_6 \\ +C_1S_2C_5L_6 \\ +C_1S_2d_3 - S_1L_2 \end{matrix} \\ \begin{matrix} S_1C_2C_4C_5C_6 \\ +C_1S_4C_5C_6 \\ -S_1S_2S_5C_6 \\ -S_1C_2S_4S_6 \\ +C_1C_4S_6 \end{matrix} & \begin{matrix} -S_1C_2C_4C_5S_6 \\ -C_1S_4C_5S_6 \\ +S_1S_2S_5S_6 \\ -S_1C_2S_4C_6 \\ +C_1C_4C_6 \end{matrix} & \begin{matrix} S_1C_2C_4S_5 \\ +C_1S_4S_5 \\ +S_1S_2C_5 \end{matrix} & \begin{matrix} S_1C_2C_4S_5L_6 \\ +C_1S_4S_5L_6 \\ +S_1S_2C_5L_6 \\ +S_1S_2d_3 + C_1L_2 \end{matrix} \\ \begin{matrix} S_2C_4C_5C_6 \\ -C_2S_5C_6 \\ +S_2S_4S_6 \end{matrix} & \begin{matrix} S_2C_4C_5S_6 \\ +C_2S_5S_6 \\ +S_2S_4C_6 \end{matrix} & \begin{matrix} -S_2C_4S_5 \\ +C_2C_5 \end{matrix} & \begin{matrix} -S_2C_4S_5L_6 \\ +C_2C_5L_6 + C_2d_3 \end{matrix} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.98}$$

Observe that in Eq. (4.98) neither single variable terms are present nor simple algebra (like division of two elements) will give single variable isolation. Hence, to find a solution two alternatives are: (i) matrix premultiplication or postmultiplication to isolate one variable at a time, as was done in Example 4.1, and (ii) use more involved algebra and trigonometry. In this example latter is used.

Equating elements (1, 3), (1, 4), (2, 3), (2, 4), (3, 3) and (3, 4) in Eqs. (4.97) and (4.98), six equations are obtained.

$$C_1C_2C_4S_5 - S_1S_4S_5 + C_1S_2C_5 = a_x \tag{4.99}$$

$$L_6(C_1C_2C_4S_5 - S_1S_4S_5 + C_1S_2C_5) + C_1S_2d_3 - S_1L_2 = d_x \tag{4.100}$$

$$S_1C_2C_4S_5 + C_1S_4S_5 + S_1S_2C_5 = a_y \tag{4.101}$$

$$L_6(S_1C_2C_4S_5 + C_1S_4S_5 + S_1S_2C_5) + S_1S_2d_3 + C_1L_2 = d_y \tag{4.102}$$

$$-S_2C_4S_5 + C_2C_5 = a_z \tag{4.103}$$

$$L_6(-S_2C_4S_5 + C_2C_5) + C_2d_3 = d_z \tag{4.104}$$

Substituting Eq. (4.99) into Eq. (4.100) gives

$$L_6a_x + C_1S_2d_3 - S_1L_2 = d_x$$

or

$$C_1S_2d_3 - S_1L_2 = d_x - L_6a_x \tag{4.105}$$

Similarly, substituting Eq. (4.101) into Eq. (4.102) gives

$$L_6a_y + S_1S_2d_3 + C_1L_2 = d_y$$

or

$$S_1S_2d_3 + C_1L_2 = d_y - L_6a_y \tag{4.106}$$

Squaring Eqs. (4.105) and (4.106), adding and simplifying gives

$$S_2^2d_3^2 + L_2^2 = (d_x - L_6a_x)^2 + (d_y - L_6a_y)^2$$

or

$$S_2^2d_3^2 = (d_x - L_6a_x)^2 + (d_y - L_6a_y)^2 - L_2^2 \tag{4.107}$$

Also, combining Eqs. (4.103) and (4.104), rearranging and squaring gives

$$C_2d_3 = d_z - L_6a_z \tag{4.108}$$

or

$$C_2^2d_3^2 = (d_z - L_6a_z)^2 \tag{4.109}$$

Equations (4.107) and (4.109) are solved for d_3

$$d_3 = \pm\sqrt{(d_x - L_6a_x)^2 + (d_y - L_6a_y)^2 + (d_z - L_6a_z)^2 - L_2^2}$$

The displacement of prismatic joint d_3 is always positive; therefore, the negative solution is not valid. Thus

$$d_3 = \sqrt{(d_x - L_6a_x)^2 + (d_y - L_6a_y)^2 + (d_z - L_6a_z)^2 - L_2^2} \tag{4.110}$$

Thus, solution for the joint variable d_3, which gives the joint displacement for the prismatic joint, is found. Next, solve for joint displacement θ_2. From Eq. (4.107),

$$S_2d_3 = \pm\sqrt{(d_x - L_6a_x)^2 + (d_y - L_6a_y)^2 - L_2^2} \tag{4.111}$$

Dividing Eq. (4.111) by Eq. (4.108), solution of joint variable θ_2 is obtained as

$$\theta_2 = A\tan2\left(\pm\sqrt{(d_x - L_6a_x)^2 + (d_y - L_6d_y)^2 - L_2^2}\,,\,(d_z - L_6a_z)\right) \tag{4.112}$$

Next, to solve for θ_1, Eqs. (4.105) and (4.106) can be used as they have only θ_1 as unknown. First, let the constants K_1 and K_2 be defined as

$$K_1 = S_2d_3 \quad \text{and} \quad K_2 = L_2$$

Substituting these constants in Eqs. (4.105) and (4.106):

$$K_1C_1 - K_2S_1 = d_x - L_6a_x \tag{4.113}$$

$$K_1S_1 + K_2C_1 = d_y - L_6a_y \tag{4.114}$$

The equations of this form can be solved by making trigonometric substitutions

$$K_1 = r\sin\phi \quad \text{and} \quad K_2 = r\cos\phi \tag{4.115}$$

where

$$r = +\sqrt{K_1^2 + K_2^2}$$

$$\phi = A\tan2\left(\frac{K_1}{r}, \frac{K_2}{r}\right) \tag{4.116}$$

Substituting K_1 and K_2 from Eq. (4.115) in Eqs. (4.113) and (4.114)

$$\sin(\phi - \theta_1) = \frac{d_x - L_6a_x}{r} \tag{4.117}$$

$$\cos(\phi - \theta_1) = \frac{d_y - L_6a_y}{r} \tag{4.118}$$

From Eqs. (4.117) and (4.118),

$$\phi - \theta_1 = A\tan2\left(\frac{d_x - L_6a_x}{r}, \frac{d_y - L_6a_y}{r}\right) \tag{4.119}$$

Substituting for ϕ, r, K_1, and K_2 and solving for θ_1 gives

$$\theta_1 = A\tan2\left(\frac{S_2d_3}{\sqrt{S_2^2d_3^2 + L_2^2}}, \frac{L_2}{\sqrt{S_2^2d_3^2 + L_2^2}}\right) - A\tan2\left(\frac{d_x - L_6d_x}{\sqrt{S_2^2d_3^2 + L_2^2}}, \frac{d_y - L_6a_y}{\sqrt{S_2^2d_3^2 + L_2^2}}\right) \tag{4.120}$$

Thus, Eqs. (4.110), (4.112) and (4.120) give solutions for the first three joint displacements θ_1, θ_2, and d_3, respectively.

To obtain the solutions for the remaining three joint displacements θ_4, θ_5, and θ_6, another approach will be used. It is possible to view the description of tool frame, frame {6}, with respect to frame {3}, the arm endpoint frame, along two different paths.

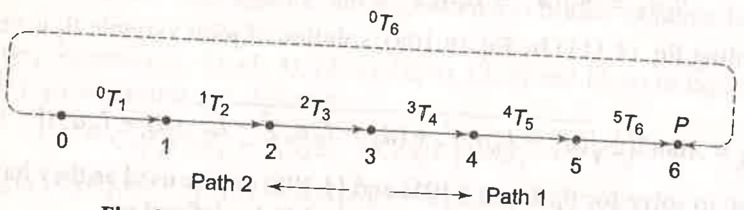


Fig. 4.10 *The transform graph for the manipulator*

The manipulator transformation matrix ${}^0T_6$ is equal to the product of six link transformation matrices

$$ {}^0T_6 = {}^0T_1\,{}^1T_2\,{}^2T_3\,{}^3T_4\,{}^4T_5\,{}^5T_6 \tag{4.121} $$

This can be represented graphically as shown in Fig. 4.10 with nodes representing frames and edges representing the transform. From the graph in Fig. 4.10 observe that there are two paths to traverse from frame {3} to frame {6}, one is via frame {4} and {5} in the forward direction and other is via frame {0}, the base, in the reverse direction. The use of these two paths to get the solutions is discussed below.

Path 1 frame {3} → frame {4} → frame {5} → frame {6}

Along this path the transformation ${}^3T_6$ can be obtained as

$$ {}^3T_6 = {}^3T_4\,{}^4T_5\,{}^5T_6 \tag{4.122} $$

In Example 3.8, the transformation matrices ${}^3T_4$, ${}^4T_5$, and ${}^5T_6$ were obtained in terms of θ_4, θ_5, and θ_6 respectively. On multiplying these matrices,

$$ {}^3T_6 = \begin{bmatrix} C_4C_5C_6 - S_4S_6 & -C_4C_5S_6 - S_4C_6 & -C_4S_5 & -L_6C_4S_5 \\ S_4C_5C_6 + C_4S_6 & -S_4C_5S_6 - C_4C_6 & -S_4S_5 & -L_6S_4S_5 \\ S_5C_6 & -S_5S_6 & C_5 & L_6C_5 + L_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.123} $$

Path 2 frame {3} → frame {2} → frame {1} → frame {0} → frame {6}

This is a path via the base. The tool frame is defined with respect to the base and hence, it can be reached from frame {3} traversing the links of the arm. Thus,

$$ {}^3T_6 = {}^3T_2\,{}^2T_1\,{}^1T_0\,{}^0T_6 $$

It is known that ${}^{i-1}T_i = ({}^iT_{i-1})^{-1}$ and ${}^0T_6 = T$ hence

$$ {}^3T_6 = \left({}^2T_3\right)^{-1}\left({}^1T_2\right)^{-1}\left({}^0T_1\right)^{-1} T \tag{4.124} $$

Since θ_1, θ_2, and d_3 are known, the matrices ${}^2T_3$, ${}^1T_2$ and ${}^0T_1$ and their inverse can be computed. Substituting these values and multiplying the matrices, ${}^3T_6$ can be computed. Let it be denoted as

$$ {}^3T_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ r_{31} & r_{32} & r_{33} & r_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.125} $$

Since matrices in Eq. (4.123) and (4.125) represent the same point, the arm point, θ_4, θ_5, θ_6 can be found by equating the corresponding elements of matrices. First, for θ_5, the elements in row 3 are equated to give three equations as:

$$ S_5C_6 = r_{31} \tag{4.126} $$

$$ -S_5S_6 = r_{32} \tag{4.127} $$

$$ C_5 = r_{33} \tag{4.128} $$

Squaring Eqs. (4.126) and (4.127), adding and dividing the result by Eq. (4.128) gives solution for θ_5 as

$$ \theta_5 = \text{Atan2}(\pm\sqrt{(r_{31}^2 + r_{32}^2)},\, r_{33}) \tag{4.129} $$

Next, joint angle θ_4 is obtained by equating the elements (2, 4) and (1, 4) as

$$ -L_6S_4S_5 = r_{24} \tag{4.130} $$

$$ -L_6S_4S_5 = r_{14} \tag{4.131} $$

Solving for θ_4 from Eqs. (4.130) and (4.131) gives

$$ \theta_4 = \text{Atan2}\left(\frac{r_{24}}{S_5}, \frac{r_{14}}{S_5}\right) \tag{4.132} $$

Note that since S_5 is a variable and may have more than one value, it is going to influence solution of θ_4. Finally, to solve for θ_6, Eqs. (4.127) is divided by Eq. (4.126) to give

$$ \theta_6 = \text{Atan2}\left(-\frac{r_{32}}{S_5}, \frac{r_{31}}{S_5}\right) \tag{4.133} $$

Thus, expressions for all the six joint displacements of 6-DOF manipulator are obtained as explicit functions of the desired position and orientation of end-effector. The values of joint displacements can be determined from these functions for the desired end-effector location data. Note that the inverse kinematics solutions could have been obtained by following alternate approaches. The issues of existence and multiplicity of solutions are discussed next.

Existence of Solutions

A close examination of the expressions of the joint variables reveals that the solutions to the inverse kinematics problem of the given 6-DOF arm exist only if the following conditions are satisfied:

(i) The term under the square root in Eq. (4.110) is non-negative, that is,

$$ [(d_x - L_6a_x)^2 + (d_y - L_6a_y)^2 + (d_z - L_6a_z)^2 - L_2^2] \ge 0 \tag{4.134} $$

(ii) Similarly, the term under the square root in Eq. (4.111), is nonnegative, that is,

$$[(d_x - L_6 a_x)^2 + (d_y - L_6 a_y)^2 - L_2^2] \geq 0 \tag{4.135}$$

The above conditions are clearly geometric constraints on the dimensions of the links of the manipulator.

There are two ways to look at these. One, for given link dimensions (L_2 and L_6) the end-effector will not be able to attain the desired position (d_x, d_y, d_z) and orientation (a_x, a_y, a_z) if the above conditions are not satisfied, and second, the conditions can be used to design link dimensions L_2 and L_6 for the desired workspace.

Note that, if condition (ii) is satisfied, condition (i) is automatically satisfied. Also, note that here it is assumed that full 360° of rotation for all revolute joints and limitless translations of the prismatic joint are possible. But, due to mechanical constraints, joint motions are restricted and solutions exist only if, in addition to satisfying the conditions given above, each of the joint variables takes a value that lies within the range of motion allowed for that joint.

Multiplicity of Solutions

Equation (4.112) gives two solutions of θ_2 due to the presence of the square root. Since θ_1 depends on θ_2 (Eq. (4.120)), there is one value of θ_1 for each value of θ_2. Similarly, from Eq. (4.129), there are two solutions for θ_5 and hence, it gives one set of solutions for θ_4 and θ_6 for each value of θ_5.

Thus, the number of solutions to the inverse kinematics problem of the given 6-DOF-manipulator arm is four.

Example 4.7 ***Determination of joint variables for a 4-DOF RPPR manipulator***

For a 4-DOF, RPPR manipulator, the joint-link transformation matrices, with joint variables θ_1, d_2, d_3 and θ_4 are

$${}^0T_1(\theta_1) = \begin{bmatrix} C_1 & -S_1 & 0 & 0 \\ S_1 & C_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.136}$$

$${}^1T_2(\theta_2) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.137}$$

$${}^2T_3(d_3) = \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.138}$$

$${}^3T_4(\theta_4) = \begin{bmatrix} C_4 & -S_4 & 0 & 0 \\ S_4 & C_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.139}$$

If the tool configuration matrix at a given instant is as given below, obtain the magnitude of each joint variable.

$$T_E = \begin{bmatrix} -0.250 & 0.433 & -0.866 & -89.10 \\ 0.433 & -0.750 & -0.500 & -45.67 \\ -0.866 & -0.500 & 0.000 & 50.00 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.140}$$

Solution The kinematic model of the manipulator will be

$${}^0T_1\,{}^1T_2\,{}^2T_3\,{}^3T_4 = T_E \tag{4.141}$$

Substituting from Eqs. (4.136) – (4.140) gives

$$\begin{bmatrix} C_1 & -S_1 & 0 & 0 \\ S_1 & C_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_4 & -S_4 & 0 & 0 \\ S_4 & C_4 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = T_E \tag{4.142}$$

or

$$\begin{bmatrix} C_1C_4 & -C_1S_4 & -S_1 & -d_3S_1 + 5C_1 \\ S_1C_4 & -S_1S_4 & C_1 & d_3C_1 + 5S_1 \\ -S_4 & -C_4 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.250 & 0.433 & -0.866 & -89.10 \\ 0.433 & -0.750 & -0.500 & -45.67 \\ -0.866 & -0.500 & 0.000 & 50.00 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.143}$$

The solutions for joint-variables are found by using the direct approach, guideline (a)–(d). The solution for first joint variable θ_1 is obtained by comparing elements (1, 3) and (2, 3)

$$-S_1 = -\sin\theta_1 = -0.866 \tag{4.144}$$

$$C_1 = \cos\theta_1 = -0.5$$

or $\theta_1 = Atan2(0.866, -0.5) = -60°$ (that is: $\theta_1 = Atan2(-a_x, a_y)$) (4.145)

The solution for second variable d_2 is obtained from element (3, 4) as:

$$d_2 = 50 \text{ (that is : } d_2 = d_z) \tag{4.146}$$

Similarly, the third joint variable d_3 is obtained from elements (1, 4) and elements (2, 4) by squaring, adding and simplifying

$$d_3 = \pm\sqrt{d_x^2 + d_y^2 - 25} \tag{4.147}$$

or $$d_3 = 100 \tag{4.148}$$

Note that d_3 can not be negative. The fourth joint variable θ_4 is computed from elements (3, 1) and (3, 2) as

$$\theta_4 = \text{Atan2}(0.866, 0.5) = 60° \text{ (that is: } \theta_4 = \text{Atan2}(-n_z, -o_z)\text{)} \tag{4.149}$$

The given end-effector position and orientation, T_E will be achieved by setting the joint variable vector to

$$q = [-60° \quad 50 \quad 100 \quad 60°] \tag{4.150}$$

EXERCISES

4.1 For the two link planar manipulator in Fig. 4.3(a) the first link as is twice as long as the second link ($L_1 = 2L_2$). Sketch the reachable workspace of the manipulator if the joint range limits are

$$0 < \theta_1 < 170°,$$
$$-90° < \theta_2 < 110°. \tag{4.151}$$

4.2 Sketch the approximate reachable workspace of the tip of a two-link planar arm with revolute joints. For this arm the first link is thrice as long as the second link, that is, $L_1 = 3L_2$ and the joint limits are $30° < \theta_1 < 180°$ and $-100° < \theta_2 < 160°$.

4.3 Sketch the approximate reachable workspace and the dexterous workspace of the 3-DOF planar manipulator shown in Fig. 4.5.

4.4 Show that for a 3R planar manipulator having link lengths as L_1, L_2 and L_3 with $(L_1 + L_2) > L_3$, the RWS is a circle with radius $r_{RWS} = (L_1 + L_2 + L_3)$ and DWS is a circle with radius $r_{DWS} = (L_1 + L_2 - L_3)$.

4.5 Explain why closed form analytical solutions are preferred over numerical iterative solutions.

4.6 Discuss the existence of multiple solutions for Example 4.1.

4.7 For a three-link planar manipulator (2-DOF for position and 1-DOF for orientation) two solutions are possible for a given position and orientation, as discussed in Section 4.3.2. If one more degree of freedom is added such that the manipulator is still planar, how many solutions will be possible for a given position and orientation when the added joint is
(a) revolute.
(b) prismatic.

4.8 How many solutions are possible, assuming no joint range limits for the following 3-DOF arm configuration?
(a) PPP manipulator shown in Fig. E4.8.
(b) RPP manipulator shown in Fig. 3.13.
(c) RRP manipulator shown in Fig. 4.7.
(d) RRR manipulator shown in Fig. 3.15.

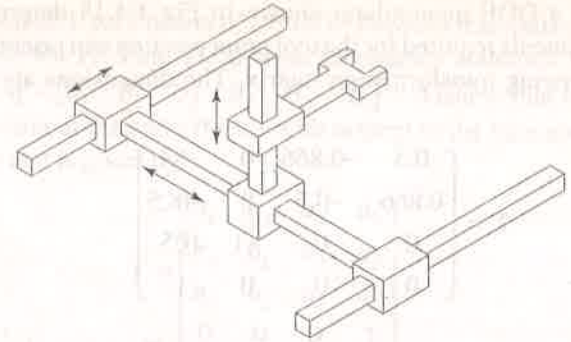


Fig. E4.8 *A three degree of freedom PPP configuration*

4.9 For the two degree of freedom planar RP configuration arm discussed in Example 3.1, how many solutions can be found for a given position and orientation. What will be the number of solutions if following alterations are made in the manipulator configuration?
(a) Add one revolute joint after the prismatic joint.
(b) Add one revolute joint before the prismatic joint.
(c) Add one degree of freedom (1R) wrist.
In each case, the manipulator remains planar.

4.10 For the 2-DOF manipulator shown in Fig. 3.11 determine the solution for all joint displacements q for a given tool point position and orientation.

4.11 Obtain the inverse kinematics solution for the 3-DOF planar manipulator shown in Fig. 4.5.

4.12 Workout Example 4.1 without using inverse transform approach.

4.13 Obtain the closed form solutions for the joint displacements of the cylindrical configuration arm described in Exercise 3.2.

4.14 For the manipulator arm consisting of 3-DOF described in Exercise 3.6, obtain the inverse kinematics solutions.

4.15 Obtain the inverse kinematics model for the 3-DOF articulated arm discussed in Example 3.3.

4.16 For the 3-DOF-RPY wrist shown in Fig. 3.18, obtain the conditions of singularities.

4.17 For the 3-DOF arm shown in Fig. E4.17, determine the forward kinematic model and, there from, obtain the general solution for inverse dinematics.

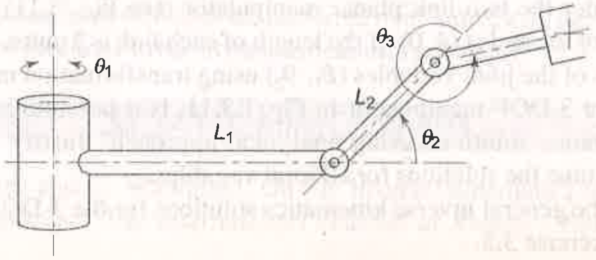
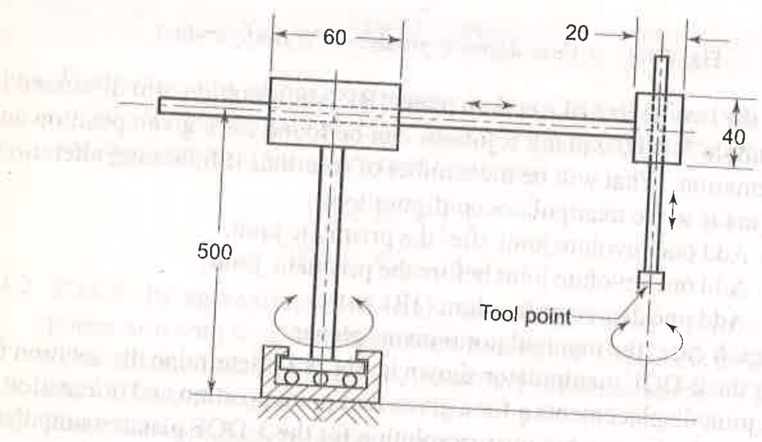


Fig. E4.17 *A three degree of freedom manipulator arm*

4.18 For the 4-DOF manipulator shown in Fig. E4.18 determine the joint displacements required for the tool point position and orientation given by the following transformation matrix. The dimensions are shown in the figure.

$$T = \begin{bmatrix} 0.5 & -0.866 & 0 & -84 \\ 0.866 & -0.5 & 0 & -48.5 \\ 0 & 0 & -1 & 105 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.152}$$



Note: Not to scale and all dimension in mm

Fig. E4.18 *A 4-DOF manipulator*

4.19 For a 5-DOF, RRR-RR articulated configuration manipulator shown in Fig. E3.14 obtain the inverse kinematics model.

4.20 Consider the 6-DOF manipulator in Exercise 3.18; find solutions for all the joint variables in terms of end-effector position and orientation by analytical method and inverse transform method.

4.21 For the SCARA robot discussed in Example 3.6, end-effector configuration (orientation and position) has been calculated in Exercise 3.13. Taking this as the desired end-effector configuration compute the joint displacement vector.

4.22 Consider the two-link planar manipulator (see Fig. 3.11) with its end-effector located at (4, 0). If the length of each link is 2 units, determine the values of the joint variables (θ_1, θ_2) using transformation matrices.

4.23 For the 3-DOF manipulator in Fig. E3.11, is it possible to find inverse kinematics solutions using analytical approach? Justify your answer. Determine the solutions for all joint variables.

4.24 Find the general inverse kinematics solutions for the 3-DOF Euler wrist, see Exercise 3.5.

4.25 The joint-link parameters of a 6-DOF freedom manipulator are described in Table E4.17. Find the inverse kinematics solutions for all the joint angles $q = [\theta_1 \;\; \theta_2 \;\; \theta_3 \;\; \theta_4 \;\; \theta_5 \;\; \theta_6]^T$. Assume that the position and orientation of the end-effector with respect to the base coordinates ${}^0T_6$ is known and is given by

$$T = \begin{bmatrix} n_x & o_x & a_x & d_x \\ n_y & o_y & a_y & d_y \\ n_z & o_z & a_z & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.153}$$

Table E4.17 *Joint-link parameters for a 6-DOF manipulator*

Link i	a_i	α_i	d_i	θ_i	q_i
1	0	−90°	d_1	θ_1	θ_1
2	a_2	0	0	θ_2	θ_2
3	a_3	0	0	θ_3	θ_3
4	a_4	−90°	0	θ_4	θ_4
5	0	90°	d_5	θ_5	θ_5
6	0	0	d_6	θ_6	θ_6

4.26 Describe the workspace of a manipulator. Make a list of factors on which the workspace, the dexterous and reachable workspace, of a given manipulator depends.

4.27 Solutions to inverse kinematics problem are generally difficult. Explain why.

4.28 Explain the factors on which the number of solutions to given inverse kinematics model depend.

4.29 How are the feasible solutions determined? What parameters have control on the number of feasible solutions to the given inverse kinematics problem?

4.30 Is it always possible to find analytical solutions to the inverse kinematics problem? Give a situation when the analytical solution to inverse kinematics problem cannot be found.

4.31 What are closed form solutions to inverse kinematics problem? Explain the methods for obtaining closed form solutions.

4.32 Why closed form solutions are preferred over numerical, iterative or other forms of solutions to the inverse kinematics problem?

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5

Manipulator Differential Motion and Statics

In the preceding chapters, the direct and inverse kinematic models were presented, which establish the relationship between the manipulator's joint displacements and position and orientation of its end-effector. These relationships permit the static control of the manipulator to place the end-effector at a specified location and make it traverse a specific path in space. However, for the manipulator, not only the final location of the end-effector is of concern, but also the velocities at which the end-effector would move to reach the final location is an equally important concern. This requires the coordination of the instantaneous end-effector velocity and joint velocities. One way to achieve this is to take the time derivative of kinematic equations of the manipulator.

The transformation from joint velocities to the end-effector velocity is described by a matrix, called the *Jacobian*. The Jacobian matrix, which is dependent on manipulator configuration is a linear mapping from velocities in joint space to velocities in Cartesian space. The inverse problem, where the joint velocities are to be determined for a given end-effector velocity is of practical importance and requires the inverse of the Jacobian.

The Jacobian is one of the most important tools for characterization of differential motions of the manipulator. At certain locations in joint space, the Jacobian matrix may lose rank and it may not be possible to find its inverse. These locations are referred to as *singular* configurations. In this chapter manipulator singularities are also discussed. These singularities are used to establish some indices of merit to judge a manipulator design, while it is still on the drawing board.

In addition, the Jacobian is also useful for describing the mapping between forces applied to the end-effector and resulting torques at joints. This is *statics* and is discussed in the later part of this chapter.

Before discussing the Jacobian, the requisite background will be prepared in the sections that follow.

5.1 LINEAR AND ANGULAR VELOCITY OF A RIGID BODY

A brief overview of the concepts of linear and angular velocity of a rigid body, as applicable to robot analysis, is carried out first in this section. Consider a link i of a manipulator in isolation, as shown in Fig. 5.1. The link joins with two other links at points O_{i-1} and O_i. Time-variation of position and orientation of the link in space produces linear and angular velocities $\boldsymbol{v}$ and ω, respectively. The concepts of translation and rotation, discussed in Chapter 2, are extended to the time-variation of location, that is, velocities. The motion of rigid body (the link), therefore, can be equivalently studied by attaching coordinate frames to the link and studying the motion of frames relative to one another.

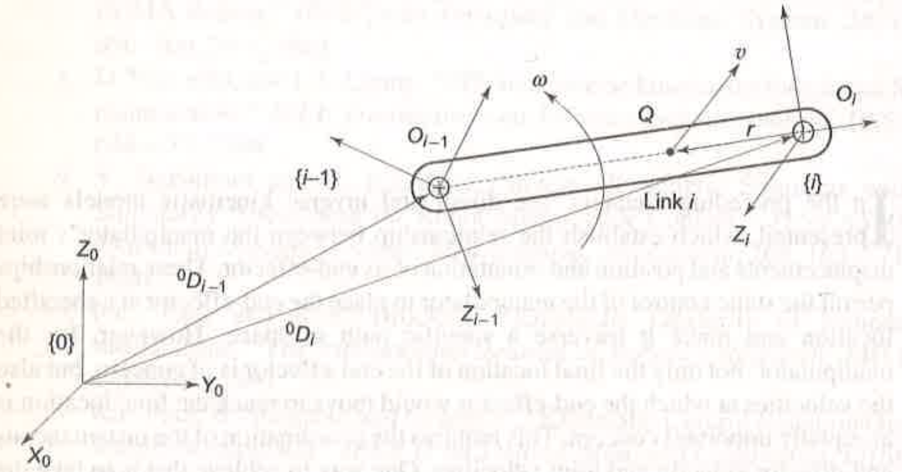


Fig. 5.1 *One link of a manipulator moving with linear velocity v and angular velocity ω*

It is seen from Fig. 5.1 that frame $\{i\}$ is attached to the link i and it moves with the link. The motion of link is, described relative to the fixed frame $\{0\}$. The position vector ^0D_i describes the location of frame $\{i\}$ relative to frame $\{0\}$, the rotation matrix ^0R_i describes the orientation of frame $\{i\}$ relative to frame $\{0\}$, and the homogeneous transformation ^0T_i combines both, the translation and rotation of frame $\{i\}$ relative to frame $\{0\}$.

5.1.1 Linear Velocity

Suppose the link i is only translating relative to frame $\{0\}$. Every point on the link will have the same linear velocity. The linear velocity of link i or that of any point Q on it, relative to frame $\{0\}$ is given by

$$^0v_i = {^0v_Q} = \frac{d(^0D_i)}{dt} = \lim_{\Delta t \to 0} \frac{^0D_i(t+\Delta t) - {^0D_i(t)}}{\Delta t} \tag{5.1}$$

For pure translation, linear link velocity can be obtained by taking the time derivative of the homogeneous transformation matrix because rotation matrix $^0\boldsymbol{R}_i$ is constant

$$^0T_i = \left[\begin{array}{ccc|c} & {^0\boldsymbol{R}_i} & & {^0\boldsymbol{D}_i} \\ & \text{(Constant)} & & \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \tag{5.2}$$

The link linearity velocity is thus given as

$$^0\boldsymbol{v}_i = \lim_{\Delta t \to 0} \frac{^0\boldsymbol{T}_i(t+\Delta t) - {^0\boldsymbol{T}_i(t)}}{\Delta t} = {^0\dot{\boldsymbol{T}}_i} \tag{5.3}$$

Note that in Eq. (5.3), $^0\boldsymbol{v}_i$ is a 3×1 vector with the components $[^0v_{ix}\ {^0v_{iy}}\ {^0v_{iz}}]^T$ each of which can be obtained by differentiating the corresponding components of the position vector $^0\boldsymbol{D}_i = [d_{ix}\ d_{iy}\ d_{iz}]^T$ with respect to time.

As with any other vector, a velocity vector can be mapped from one frame to another. But unlike position vectors, the velocity vectors are free vectors because they do not depend on their line of action and, hence, the mathematics of their mapping is different from that discussed in Chapter 2 as explained in the following subsection.

5.1.2 Angular Velocity

Consider now the rotation of the link in Fig. 5.1 with an angular velocity ω about some axis. In general, the axis of rotation itself may be changing with time. Therefore, there is an instantaneous axis of rotation and every point on the body as it rotates also has an instantaneous linear velocity, which is tangential to the arc of rotation at that instant. The angular velocity is discussed first.

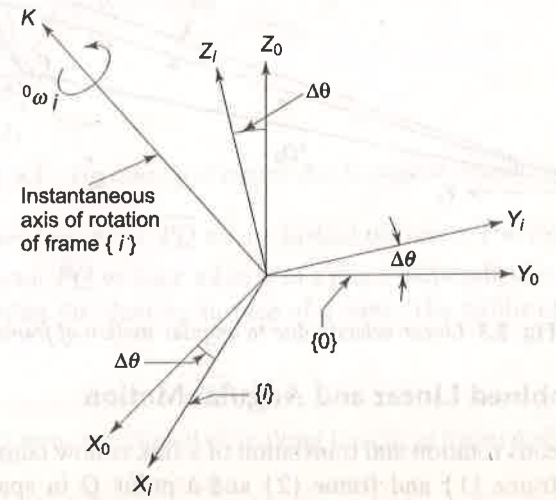


Fig. 5.2 *Angular velocity of frame {i} relative to frame {0} about axis k*

Let frame $\{i\}$ attached to the body be rotating with respect to a known stationary reference frame $\{0\}$, as shown in Fig. 5.2. The origins of the two frames are considered to be coincident such that only relative rotation is possible. The angular velocity of frame $\{i\}$, relative to frame $\{0\}$, is represented by angular velocity vector ${}^0\omega_i$ directed along the instantaneous axis of rotation of frame $\{i\}$, the axis k. The magnitude of angular velocity is equal to the speed of rotation. The instantaneous angular velocity ${}^0\omega_i$ is defined as

$$ {}^0\omega_i = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} = \dot{\theta} \tag{5.4} $$

5.1.3 Linear Velocity due to Angular Motion

To find the linear velocity arising due to angular motion consider frames $\{1\}$ and $\{2\}$ and a point Q on the link as shown in Fig. 5.3 (the link is not shown in figure for clarity). With respect to Fig. 5.1 for link i, frame $\{1\}$ corresponds to frame $\{i\text{–}1\}$ and frame $\{2\}$ to frame $\{i\}$. The frame $\{2\}$ along with the point Q is rotating about axis $\boldsymbol{k}$ with an instantaneous angular velocity ω. In small time interval Δt, the point Q describes an arc with radius $r = O_1Q$ and moves through an angle $\Delta\theta$, with respect to frame $\{1\}$. The linear distance translated in this time is $\Delta Q = r\Delta\theta$. Therefore, the linear velocity of Q is

$$ v = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = \lim_{\Delta t \to 0} \frac{r\Delta\theta}{\Delta t} = r\dot{\theta} = r\omega \tag{5.5} $$

The direction of this velocity is tangential to the arc described by Q.

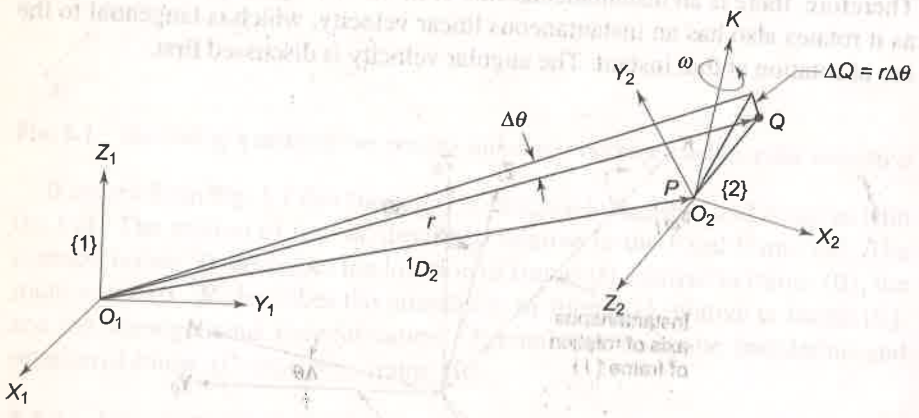


Fig. 5.3 *Linear velocity due to angular motion of frame {2}*

5.1.4 Combined Linear and Angular Motion

The simultaneous rotation and translation of a link is now considered. Consider two frames, frame $\{1\}$ and frame $\{2\}$ and a point Q in space, as shown in Fig. 5.3. The frame $\{2\}$ is described with reference to frame $\{1\}$ by the transformation matrix ${}^1\boldsymbol{T}_2$, having components as the translation vector ${}^1\boldsymbol{D}_2$ and

rotation matrix ${}^1\boldsymbol{R}_2$. Small motion of frame $\{2\}$ along with point Q, relative to frame $\{1\}$, results in a small displacement ΔQ of point Q along a circular arc of radius r in time Δt, subtending small angle $\Delta\theta$. Initially assume that the orientation of frame $\{2\}$ is not changing with time, that is, ${}^1\boldsymbol{R}_2$ is constant. Then, the motion of point Q relative to frame $\{1\}$ can be due to translation of origin of frame $\{2\}$ (point P or origin O_2) and/or translation of point Q relative to frame $\{2\}$. The linear velocity of point Q with respect to frame $\{1\}$ is, therefore,

$$ {}^1v_Q = {}^1v_P + {}^1T_2\, {}^2v_Q \tag{5.6} $$

where ${}^2v_Q$ is the velocity of point Q with respect to frame $\{2\}$ and ${}^1v_p$ is the velocity of frame $\{2\}$, with origin P, with respect to frame $\{1\}$.

Consider now the rotation of frame $\{2\}$ about k-axis with an angular velocity ${}^1\omega_2$. Let Q be a fixed point in frame $\{2\}$, as shown in Fig. 5.4. Assume that frame $\{1\}$ is translated to frame $\{1'\}$ such that its origin coincides with origin of frame $\{2\}$ and the axis of rotation (k-axis) is fixed with respect to frame $\{1'\}$. It means that vector $\overrightarrow{PQ}$ rotates about k-axis at a fixed angular velocity ${}^1\omega_2$.

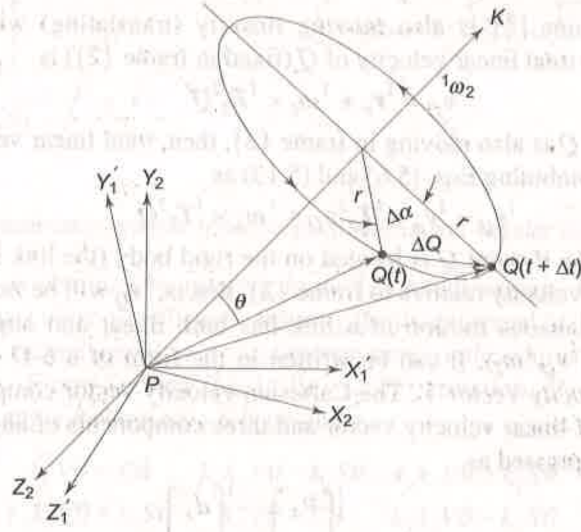


Fig. 5.4 *The velocity of a point due to angular velocity*

The figure shows the vector $\overrightarrow{PQ}$ at any instant of time t. The rotation causes the tip Q of the vector $\overrightarrow{PQ}$ to trace a circle in a plane perpendicular to the k-axis and line PQ describes the slanting surface of a cone. The radius of the circular path of Q is

$$ r = |\overrightarrow{PQ}| \sin\theta \tag{5.7} $$

In time Δt, point Q turns by an angle $\Delta\alpha$ along the circle traversing a distance

$$ \Delta Q = r\Delta\alpha $$

or

$$ \Delta Q = |\overrightarrow{PQ}| \sin\theta\, \Delta\alpha \tag{5.8} $$

Thus, the magnitude of linear velocity of Q is (in frame {1})

$$\lim_{\Delta t\to 0}\left|\frac{\Delta Q}{\Delta t}\right| = \left|\overrightarrow{PQ}\right|\sin\theta \lim_{t\to 0}\frac{\Delta\alpha}{\Delta t} \tag{5.9}$$

as

$$\lim_{\Delta t\to 0}\frac{\Delta\alpha}{\Delta t} = \left|{}^{1}\omega_2\right|$$

Therefore,

$$\left|{}^{1}v_Q\right| = \left|{}^{1}\omega_2\right|\left|\overrightarrow{PQ}\right|\sin\theta \text{ or } {}^{1}v_Q = {}^{1}\omega_2 \times {}^{1}Q \tag{5.10}$$

Consider now the direction of the linear velocity of Q. It is tangential to the circle in the instantaneous direction of motion. The linear velocity direction at Q is, therefore, at 90° to both k-axis and $\overrightarrow{PQ}$ in right-handed direction. By virtue of these two observations, the linear velocity vector of point Q about the k-axis caused by rotation of $\overrightarrow{PQ}$ is the cross product

$${}^{1}v_Q = {}^{I}\omega_2 \times {}^{1}Q \tag{5.11}$$

Since the point Q is known in frame {2}, it can also be written as

$${}^{1}v_Q = {}^{1}\omega_2 \times {}^{1}T_2{}^{2}Q \tag{5.12}$$

Now, if frame {2} is also moving linearly (translating) with respect to frame {1}, the total linear velocity of Q (fixed in frame {2}) is

$${}^{1}v_Q = {}^{1}v_p + {}^{1}\omega_2 \times {}^{1}T_2{}^{2}Q \tag{5.13}$$

If the point Q is also moving in frame {2}, then, total linear velocity of Q is obtained by combining Eqs. (5.6) and (5.13) as

$${}^{1}v_Q = {}^{1}v_p + {}^{1}T_2{}^{2}v_Q + {}^{1}\omega_2 \times {}^{1}T_2{}^{2}Q \tag{5.14}$$

In Eq. (5.14), if point Q is located on the rigid body (the link in Fig. 5.1), it cannot have a velocity relative to frame {2}, that is, ${}^{2}v_Q$ will be zero.

The instantaneous motion of a link has both linear and angular velocity components, $({}^{1}v_Q, {}^{1}\omega_2)$. It can be written in the form of a 6-D vector, called *Cartesian Velocity Vector* **V**. The Cartesian velocity vector comprises of three components of linear velocity vector and three components of angular velocity vector. It is expressed as

$${}^{1}V_Q = {}^{1}\begin{bmatrix} v \\ \omega \end{bmatrix}_Q = {}^{1}\begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}_Q = {}^{1}\begin{bmatrix} \dot{d}_x \\ \dot{d}_y \\ \dot{d}_z \\ \dot{\theta}_x \\ \dot{\theta}_y \\ \dot{\theta}_z \end{bmatrix}_Q \tag{5.15}$$

5.2 RELATIONSHIP BETWEEN TRANSFORMATION MATRIX AND ANGULAR VELOCITY

In Chapter 2, it was seen that finite rotations between two coordinate frames could be represented using a rotation matrix. The angular velocity vector also

represents the rotational motion between frames. Hence, the homogeneous transformation matrix T and the angular velocity vector ω, at any instant of time, must be related. To analyze this relationship consider frame {2} attached to rigid body as it rotates with respect to frame {2′}, as shown in Fig. 5.5, in time t to $(t + \Delta t)$.

Let ${}^{1}T_2$ and ${}^{1}T_{2'}$ represent the homogeneous transformation matrix describing frame {2} and frame {2′} with respect to frame {1} at time instant t and $(t + \Delta t)$, respectively. The derivative of the transformation matrix ${}^{1}T_2(t)$ with respect to time is

$${}^{1}\dot{T}_2 = \lim_{\Delta t\to 0}\frac{{}^{1}T_{2'} - {}^{1}T_2}{\Delta t} \tag{5.16}$$

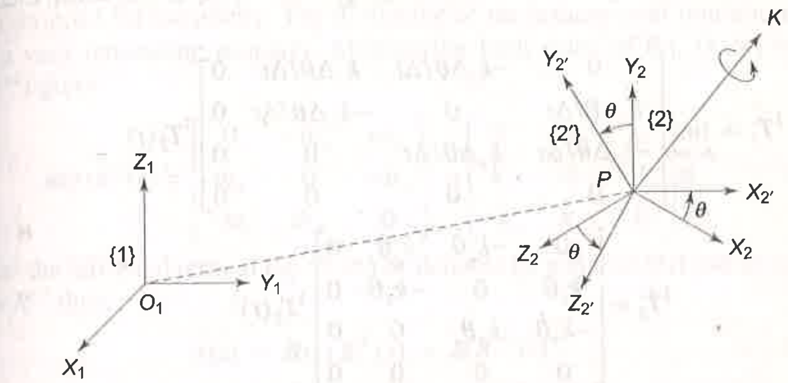


Fig. 5.5 *Relationship between transformation matrix and angular velocity vector*

Frame {2′} is obtained by rotating frame {2} about an axis K in the fixed reference frame {1} by a small angle θ during the time interval Δt. Here, axis K is nothing but the instantaneous axis of rotation at time t or the equivalent angle axis representation. From Section 2.6.4, Eq. (2.71), rotation of frame {2} to frame {2′} by angle θ about axis K is given by

$$T_k(\theta) = \begin{bmatrix} k_x^2 V\theta + C\theta & k_x k_y V\theta - k_z S\theta & k_x k_z V\theta + k_y S\theta & 0 \\ k_x k_y V\theta + k_z S\theta & k_y^2 V\theta + C\theta & k_y k_z V\theta - k_x S\theta & 0 \\ k_x k_z V\theta - k_y S\theta & k_y k_z V\theta + k_x S\theta & k_z^2 V\theta + C\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.17}$$

where $\hat{k} = [k_x\ k_y\ k_z]^T$, $C\theta = \cos\theta$, $S\theta = \sin\theta$ and $V\theta = 1 - \cos\theta$.

When θ is very small, $C\theta \approx 1$, $S\theta \approx \Delta\theta$, and $V\theta \approx 0$, such that, Eq. (5.17) reduces to

$$T_k(\Delta\theta) = \begin{bmatrix} 1 & -k_z\Delta\theta & k_y\Delta\theta & 0 \\ k_z\Delta\theta & 1 & -k_x\Delta\theta & 0 \\ -k_y\Delta\theta & k_x\Delta\theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.18}$$

The description of frame $\{2'\}$ can be obtained by two successive transformations relative to the fixed frame $\{1\}$ as

$$\text{frame}\{1\} \xrightarrow{{}^1T_2(t)} \text{frame}\{2\} \xrightarrow{T_k(\Delta\theta)} \text{frame}\{2'\}$$

That is

$${}^1T_{2'} = T_k(\Delta\theta)\ {}^1T_2(t) \tag{5.19}$$

Equation (5.16) is simplified using Eq. (5.19) giving

$${}^1\dot{T}_2 = \lim_{\Delta t\to 0}\left(\frac{T_k(\Delta\theta)-\boldsymbol{I}}{\Delta t}\right){}^1T_2(t) \tag{5.20}$$

where $\boldsymbol{I}$ is 4×4 identity matrix. Substituting $T_k(\Delta\theta)$ from Eq. (5.18) in Eq. (5.20) gives

$${}^1\dot{T}_2 = \lim_{\Delta t\to 0}\begin{bmatrix} 0 & -k_z\Delta\theta/\Delta t & k_y\Delta\theta/\Delta t & 0 \\ k_z\Delta\theta/\Delta t & 0 & -k_x\Delta\theta/\Delta t & 0 \\ -k_y\Delta\theta/\Delta t & k_x\Delta\theta/\Delta t & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}{}^1T_2(t)$$

or

$${}^1\dot{T}_2 = \begin{bmatrix} 0 & -k_z\dot\theta & k_y\dot\theta & 0 \\ k_z\dot\theta & 0 & -k_x\dot\theta & 0 \\ -k_y\dot\theta & k_x\dot\theta & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}{}^1T_2(t)$$

or

$${}^1\dot{T}_2 = \begin{bmatrix} 0 & -k_z & k_y & 0 \\ k_z & 0 & -k_x & 0 \\ -k_y & k_x & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\dot\theta\ {}^1T_2(t) \tag{5.21}$$

where $\dot\theta = \lim_{\Delta t\to 0}\left(\frac{\Delta\theta}{\Delta t}\right)$ is the speed of angular rotation of frame $\{2\}$ about axis K.

The angular rotation at speed $\dot\theta$ about axis K gives angular velocity of frame $\{2\}$ relative to frame $\{1\}$ as a vector ${}^1\omega_2$ directed along axis K. This is equivalent to three rotations of ω_x, ω_y, and ω_z made in that order about the x-, y-, and z-axes, respectively, where

$${}^1\omega_2 = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix}\dot\theta \tag{5.22}$$

Thus, Eq. (5.21) is rewritten as

$${}^1\dot{T}_2 = \begin{bmatrix} 0 & -\omega_z & \omega_y & 0 \\ \omega_z & 0 & -\omega_x & 0 \\ -\omega_y & \omega_x & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}{}^1T_2(t) \tag{5.23}$$

Equations (5.22) and (5.23) give the relationship between the angular velocity of the link and the transformation matrix. If a link undergoes only angular motion with no translation of the origin of frame $\{2\}$, in place of transformation matrix $\boldsymbol{T}$ the rotation matrix $\boldsymbol{R}$ may be used in the above derivation. For pure rotation, the relation between angular velocity of the link and rotation matrix follows from Eqs. (5.22) and (5.23) as

$$\dot{\boldsymbol{R}}(t) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}\boldsymbol{R}(t) = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}\dot\theta\,\boldsymbol{R}(t) \tag{5.24}$$

The detailed derivation is left to the reader. The superscript and subscript have been dropped for simplicity. The derivative of the orthonormal rotation matrix has a very interesting property. Multiplying both sides of Eq. (5.24) by $\boldsymbol{R}^T$ $(= \boldsymbol{R}^{-1})$ gives

$$\dot{\boldsymbol{R}}(t)\boldsymbol{R}^T(t) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}\dot\theta \tag{5.25}$$

Let the left-hand term in Eq. (5.25) be denoted by a matrix $\boldsymbol{S}(t)$ and recall that $\boldsymbol{R}^T = \boldsymbol{R}^{-1}$ thus,

$$\boldsymbol{S}(t) = \dot{\boldsymbol{R}}(t)\,\boldsymbol{R}^T(t) = \dot{\boldsymbol{R}}\,\boldsymbol{R}^{-1}(t) \tag{5.26}$$

where

$$\boldsymbol{S}(t) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \tag{5.27}$$

It is observed that $\boldsymbol{S}(t)$ is a skew-symmetric matrix. It immediately follows from the property of skew symmetric matrix that

$$\boldsymbol{S}(t) + \boldsymbol{S}^T(t) = \boldsymbol{0} \tag{5.28}$$

where $\boldsymbol{0}$ is a 3×3 zero matrix.

Equation (5.26) reveals an important property of the rotation matrix: the product of derivative of the orthonormal rotation matrix with its transpose or inverse yields a skew-symmetric matrix.

It is concluded from Eqs. (5.22) and (5.28), that at any instant the elements of the skew-symmetric matrix $\boldsymbol{S}(t)$ are indeed the components of the angular velocity vector ${}^1\omega_2$. Therefore, a change in orientation of a rotating frame can be viewed as a rotation about some axis K.

It follows from Eq. (5.26), that

$$\dot{\boldsymbol{R}}(t) = \boldsymbol{S}(t)\,\boldsymbol{R}(t) \tag{5.29}$$

Consider a constant vector ${}^2\boldsymbol{P}$ undergoing rotation. Then

$${}^1\boldsymbol{P} = \boldsymbol{R}(t)\,{}^2\boldsymbol{P} \tag{5.30}$$

The time derivative of ${}^1P(t)$ is

$${}^1\dot{P} = \dot{R}(t)\ {}^2P \tag{5.31}$$

which, from Eq. (5.29) can be written as

$${}^1\dot{P} = S(t)\,R(t)\ {}^2P \tag{5.32}$$

The vector $\omega(t)$ of Eq. (5.22) denotes the angular velocity of frame {2} with respect to frame {1} at time t. Using $\omega(t)$ and $S(t)$ of Eq. (5.27); it is easily verified that

$$S(t)P = \omega(t) \times P \tag{5.33}$$

Comparison of Eq. (5.32) and Eq. (5.33) gives

$$\dot{P}(t) = \omega(t) \times R(t)\ {}^2P \tag{5.34}$$

where × is the vector cross product and superscripts have been dropped for generalization discussed in next section.

5.3 MAPPING VELOCITY VECTORS

Velocity vectors belong to a class of vectors called *free vectors*. In the case of free vectors, all that needs to be preserved is the magnitude and direction. Thus, free vectors could be displaced anywhere in space without changing their meaning. A moment vector is another example of a free vector.

The mapping of the linear angular velocity vector, from frame {1} to frame {2}, can have reference to two frames, the frame of description and the frame of differentiation. In the study of motion of links of manipulators, the frame of differentiation for velocity vectors is always the base frame, that is, frame {0}. Hence, the leading superscript for the frame of differentiation can be dropped and a single leading superscript in velocity vector will denote the frame of description. Further, the absence of a superscript will indicate that the base frame serves as both the frame of description and the frame of differentiation. At any instant, each link of the manipulator in motion has both linear and angular velocities. The subscript describes the link or the link's frame. For example, for link i, v_i is linear velocity (of origin) of frame $\{i\}$, and ω_i is the angular velocity of frame $\{i\}$, with base frame {0} as the frame of description as well as the frame of differentiation.

5.4 VELOCITY PROPAGATION ALONG LINKS

Having discussed the concepts of linear and angular velocity of a rigid body, we now proceed to determine the linear and angular velocity of any link of a given manipulator.

The n-DOF manipulator is an open kinematic chain with n-links, each one capable of motion relative to its neighbour. The base, link 0, is the stationary reference and velocity of all other links is defined with respect to this. Therefore, the velocity of each link can be computed starting from the base. The velocity of link (i) will be the velocity of link (i – 1) plus velocity component added by joint (i).

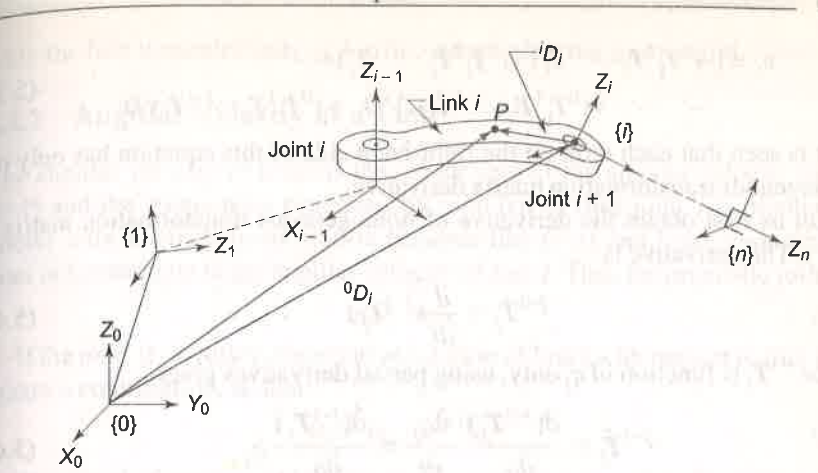


Fig. 5.6 *Velocity propagation along links of a n-DOF manipulator*

To determine the velocity of link i of a given n-DOF manipulator, consider link i with frame $\{i-1\}$ and frame $\{i\}$ attached to two ends of link i, as shown in Fig. 5.6. The origin of frame $\{i\}$, which is attached to the distal end of link will specify the velocity of link i.

5.4.1 Linear Velocity of a Link

To find the linear velocity of link i with respect to base, consider any point P on the link. The kinematic relationship between link i and link 0 (base) as given by Eq. (2.46) is

$${}^0T_i = {}^0T_1\ {}^1T_2 \cdots {}^{i-1}T_i \tag{5.35}$$

The position vector ${}^iD_i$ of this point P referred to frame $\{i\}$, can be referred to the base frame {0} using Eq. (2.26). Thus,

$${}^0D_i = {}^0T_i\ {}^iD_i \tag{5.36}$$

The linear velocity of P and so of the link i, using Eq. (5.1) is given as

$${}^0v_i = \frac{d}{dt}({}^0D_i) = \frac{d}{dt}({}^0T_i\,{}^iD_i) \tag{5.37}$$

In Eq. (5.37), as frame {0} is the frame of differentiation as well as description, we can write ${}^0v_i$ *as* v_i. In each ${}^{j-1}T_j$, only one variable exists, the j^{th} joint displacement variable, q_j. It is useful to recall that

$$q_j = \begin{cases} \theta_j & \text{if joint } j \text{ is revolute.} \\ d_j & \text{if joint } j \text{ is prismatic.} \end{cases}$$

Substituting Eq. (5.35) in Eq. (5.37)

$$v_i = \frac{d}{dt}\left[({}^0T_1\ {}^1T_2 \ldots {}^{i-1}T_i)\,{}^iD_i\right] \tag{5.38}$$

or

$$\boldsymbol{v}_i = [\ ({}^0\dot{\boldsymbol{T}}_1\,{}^1\boldsymbol{T}_2 \cdots {}^{i-1}\boldsymbol{T}_i) + ({}^0\boldsymbol{T}_1\,{}^1\dot{\boldsymbol{T}}_2 \cdots {}^{i-1}\boldsymbol{T}_i) + \cdots \quad \cdots$$
$$\cdots \quad \cdots \quad + ({}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2 \cdots {}^{i-1}\dot{\boldsymbol{T}}_i)\ ]\ {}^i\boldsymbol{D}_i + ({}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2 \cdots {}^{i-1}\boldsymbol{T}_i)\,{}^i\dot{\boldsymbol{D}}_i \tag{5.39}$$

It is seen that each term on the right-hand side of this equation has only one homogenous transformation matrix derivative.

Let us first obtain the derivative of homogeneous transformation matrix of link j. The derivative is

$${}^{j-1}\dot{\boldsymbol{T}}_j = \frac{d}{dt}({}^{j-1}\boldsymbol{T}_j) \tag{5.40}$$

Since ${}^{j-1}\boldsymbol{T}_j$ is function of q_j only, using partial derivatives gives

$${}^{j-1}\dot{\boldsymbol{T}}_j = \frac{\partial({}^{j-1}\boldsymbol{T}_j)}{\partial q_j}\frac{\partial q_j}{\partial t} = \frac{\partial({}^{j-1}\boldsymbol{T}_j)}{\partial q_j}\dot{q}_j \tag{5.41}$$

Now consider the j^{th} term in Eq. (5.39) containing term ${}^{j-1}\dot{\boldsymbol{T}}_j$

$$j^{\text{th}} \text{ term} = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2 \cdots {}^{j-1}\dot{\boldsymbol{T}}_j \cdots {}^{i-1}\boldsymbol{T}_i$$

and substitute result of Eq. (5.41) in it. This gives

$$j^{\text{th}} \text{ term} = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2 \cdots \frac{\partial({}^{j-1}\boldsymbol{T}_j)}{\partial q_j}\dot{q}_j \cdots {}^{i-1}\boldsymbol{T}_i \tag{5.42}$$

In Eq. (5.42), the partial derivative can be taken out because no other transformation matrix except ${}^{j-1}\boldsymbol{T}_j$ is a function of q_j. Thus,

$$j^{\text{th}} \text{ term} = \frac{\partial}{\partial q_j}({}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2 \cdots {}^{j-1}\boldsymbol{T}_j \cdots {}^{i-1}\boldsymbol{T}_i)\dot{q}_j \tag{5.43}$$

Combining transformation matrices with in brackets, Eq. (5.43) reduces to

$$j^{\text{th}} \text{ term} = \frac{\partial({}^0\boldsymbol{T}_i)}{\partial q_j}\dot{q}_j \tag{5.44}$$

where $\dot{q}_j$ is the time-rate of change of joint displacement q_j. The last term of Eq. (5.39) involves a derivative of ${}^i\boldsymbol{D}_i$. Because the link is a rigid body, ${}^i\boldsymbol{D}_i$ will be constant and, hence, its derivative will be zero. That is

$${}^i\dot{\boldsymbol{D}}_i = 0 \tag{5.45}$$

Substituting Eq. (5.44) and Eq. (5.45) in Eq. (5.39) and simplifying gives

$$\boldsymbol{v}_i = \sum_{j=1}^{i}\frac{\partial({}^0\boldsymbol{T}_i)}{\partial q_j}\dot{q}_j\,{}^i\boldsymbol{D}_i \tag{5.46}$$

The result of Eq. (5.46) may be extended to obtain the linear velocity of the end-effector of an n-DOF manipulator as

$$\boldsymbol{v}_{\text{end-effector}} = \boldsymbol{v}_n = \sum_{j=1}^{n}\frac{\partial({}^0\boldsymbol{T}_n)}{\partial q_j}\dot{q}_j\,{}^n\boldsymbol{D}_n \tag{5.47}$$

Only the first three elements of 4×1 column vector are meaningful.

5.4.2 Angular Velocity of a Link

The angular velocity of link i is the vector sum of the angular velocity of link $(i-1)$ and the component generated by joint (i). In case joint i is prismatic, it allows only a translational motion between link $(i-1)$ and link i and, hence, it does not contribute to the angular velocity of link i. Thus for prismatic joint i

$$\omega_i = \omega_{i-1} \tag{5.48}$$

If the joint (i) is rotary, the relative rotation of link i with respect to link $(i-1)$ makes a contribution, so that

$$\omega_i = \omega_{i-1} + {}^{i-1}\omega_i \tag{5.49}$$

The speed of rotation of link i with respect to link $(i-1)$, is $\dot{\theta}_i$ about the z_{i-1}-axis. When referred to frame $\{0\}$

$${}^{i-1}\omega_i = {}^0\boldsymbol{R}_{i-1}\,z_{i-1}\,\dot{\theta}_i \tag{5.50}$$

Substituting in Eq. (5.49)

$$\omega_i = \omega_{i-1} + {}^0\boldsymbol{R}_{i-1}\,z_{i-1}\,\dot{\theta}_i = \omega_{i-1} + {}^0\boldsymbol{R}_{i-1}\,z_{i-1}\,\dot{q}_i \tag{5.51}$$

It is easily concluded from Eq. (5.51) that the rotary joint i contributes to the angular velocities of all links from link i to n. Further, it is to be noted that, both, the angular velocities of the manipulator links and the joint velocity are relative to the base frame $\{0\}$.

It is thus found from the above account that (i) linear velocities of all manipulator links are obtained using Eq. (5.46), and (ii) angular velocities of all the links are obtained using Eqs. (5.48) and (5.51).

5.5 MANIPULATOR JACOBIAN

From the results of the previous section, it is observed that the Cartesian velocities (linear as well as angular) of the end-effector are linearly related to the joint velocities. The relationship between infinitesimal (differential) joint motions with infinitesimal (differential) changes in end-effector position (and orientation) is investigated now. As these changes take place in infinitesimal (differential) time, we are really looking for a mapping between instantaneous end-effector velocity, $\boldsymbol{V}_e$ (in Cartesian space) to instantaneous joint velocities (in joint space). This mapping between differential changes is linear and can be expressed as

$$\boldsymbol{V}_e(t) = \boldsymbol{J}(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{5.52}$$

where $\boldsymbol{V}_e(t) = 6 \times 1$ Cartesian velocity vector, [Eq. (5.15)],
$\boldsymbol{J}(\boldsymbol{q}) = 6 \times n$ *Manipulator Jacobian* or *Jacobian* matrix,
$\dot{\boldsymbol{q}} = n \times 1$ vector of n joint velocities.

Equation (5.52) can be written in column vectors of the Jacobian, that is,

$$\boldsymbol{V}_e(t) = [\boldsymbol{J}_1(\boldsymbol{q})\,\boldsymbol{J}_2(\boldsymbol{q}) \cdots\cdots \boldsymbol{J}_n(\boldsymbol{q})]\,\dot{\boldsymbol{q}}(t) \tag{5.53}$$

In the above equation, $J_i(q)$ is the i^{th} column of the Jacobian matrix. From Eq. (5.15) and Eq. (5.53)

$$V_e(t) = \begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} \dot{d} \\ \dot{\theta} \end{bmatrix} = J(q)\, \dot{q}(t) \tag{5.54}$$

Equation (5.54) represents the ***forward differential motion model*** or ***differential kinematics model*** presented schematically in Fig. 5.7, which is similar to the forward kinematic model. Note that the Jacobian $J(q)$ is a function of the joint variables.

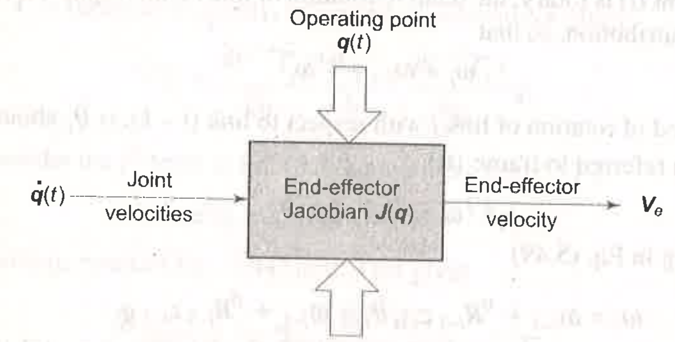


Fig. 5.7 *The forward differential motion model*

The first three rows of Jacobian $J(q)$ are associated with the linear velocity of end-effector $\boldsymbol{v}$, while the last three rows correspond to the angular velocity $\boldsymbol{\omega}$ of the end-effector. Each joint of the manipulator generates some linear and/or some angular velocity at the end-effector. Column i, of Jacobian matrix $J_i(q)$ is, thus, made up from three linear velocity components j_{vi} and three angular velocity components $j_{\omega i}$ and can be expressed as

$$J_i(q) = \begin{bmatrix} j_{vi} \\ j_{\omega i} \end{bmatrix} = \begin{bmatrix} j_{vxi} \\ j_{vyi} \\ j_{vzi} \\ j_{\omega xi} \\ j_{\omega yi} \\ j_{\omega zi} \end{bmatrix} \tag{5.55}$$

where j_{vki} and $j_{\omega ki}$ represents the component k of linear velocity and angular velocity, respectively, contributed by joint i with $k = x, y,$ or z and $i = 1, 2, 3, ..., n$. The contribution of joint i to the column J_i is computed depending on whether joint i is prismatic or rotary.

5.5.1 Jacobian Computation

The contribution of joint i to linear velocity of end-effector is $J_{vi}\,\dot{q}_i$ and to angular velocity of end-effector is $J_{\omega i}\,\dot{q}_i$. One simple method of computation of Jacobian

is to determine J_{vi} and $J_{\omega i}$ for joint i. This is carried out in the following sections for both prismatic and revolute joints.

5.5.2 The Prismatic Joint Jacobian

Consider the prismatic joint i, as shown in Fig. 5.8. The link i translates along z_{i-1}-axis with a joint (scalar) velocity $\dot{d}_i$. To find the i^{th} column $J_i(q)$ of the Jacobian matrix $J(q)$, assume that all joints except joint i are locked in position.

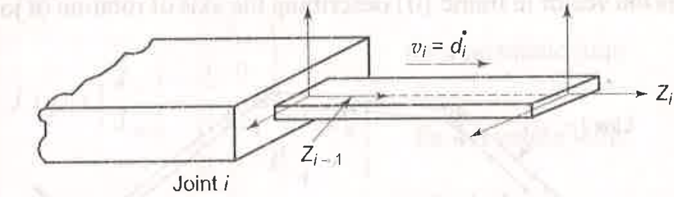


Fig. 5.8 *Prismatic joint contributing to end-effecter velocity*

In this situation, the translating prismatic joint i will produce a linear velocity at the end-effector in the same direction as the joint i axis, z_{i-1}. If a vector pointing along the direction z_{i-1}, with respect to base frame is denoted as P_{i-1}, the linear velocity produced at the end-effector by translating link i is

$$v_i = P_{i-1}\, \dot{d}_i \tag{5.56}$$

Note that for convenience P_{i-1} is written in place of ${}^0P_{i-1}$, that is, the leading superscript 0 is dropped. The prismatic joint cannot contribute any angular velocity at the end-effector, that is,

$$\omega_i = 0$$

From Eqs. (5.55) and (5.56), the $J_i(q)$ for a prismatic joint is, therefore

$$J_i(q) = \begin{bmatrix} J_{vi} \\ J_{\omega i} \end{bmatrix} = \begin{bmatrix} P_{i-1} \\ 0 \end{bmatrix} \tag{5.57}$$

Vector P_{i-1} in the above equation is computed using the coordinate transformations from base frame {0} to frame {i–1}. A unit vector in frame {i–1} along z_{i-1}-axis, with respect to frame {i–1} is given by $\hat{u} = [0 \quad 0 \quad 1]^T$.

The vector P_{i-1} is the transformation of this unit vector $\hat{u}$ with respect to base frame {0}. The homogeneous transformation matrix from frame {0} to frame {i–1} is

$${}^0T_1(q_1) \;...\; {}^{i-2}T_{i-1}(q_{i-1}) = {}^0T_{i-1} \tag{5.58}$$

The vector P_{i-1} is obtained as

$$P_{i-1} = {}^0R_{i-1}\, \hat{u} \tag{5.59}$$

where ${}^0R_{i-1}$ is the 3×3 orientation sub-matrix (the rotation matrix) of ${}^0T_{i-1}$. Note that P_{i-1} is the third column of the rotation matrix ${}^0R_{i-1}$.

5.5.3 The Rotary Joint Jacobian

Next, consider the joint i to be a rotary joint with angular velocity $\dot{\theta}_i$ about z_{i-1}-axis, as shown in Fig. 5.9. With all joints, except joint i, locked in position, the rotary joint rotates all the distal links from link i to link n at an angular velocity ω_i. For a rotary joint, the angular velocity will be given by

$$\omega_i = P_{i-1}\,\dot{\theta}_i \tag{5.60}$$

where P_{i-1} is the vector in frame {0} describing the axis of rotation of joint i, that is z_{i-1}-axis.

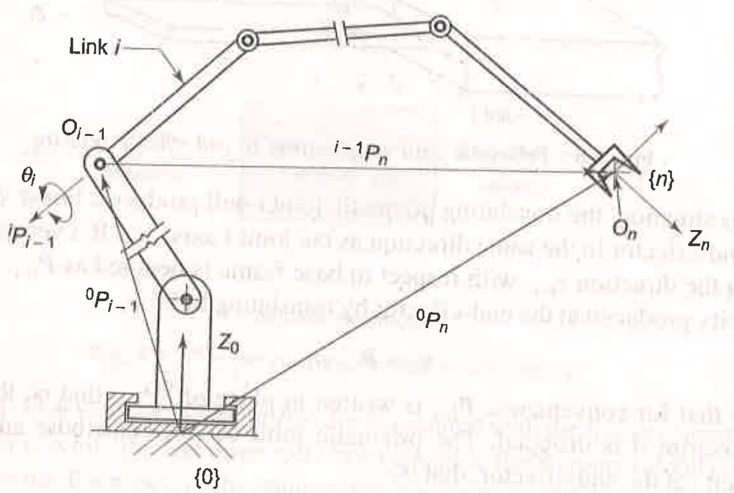


Fig. 5.9 *Motion of end-effecter generated by a rotary joint*

The angular velocity of link i also produces a linear velocity at the end-effector due to the rotation of all the distal links along with the end-effector about the origin O_{i-1} of frame $\{i-1\}$. From O_{i-1}, the end-effector frame, frame $\{n\}$ is defined by a position vector $^{i-1}P_n$, as shown in Fig. 5.9. From Eqs. (5.11) and (5.60), the linear velocity generated by ω_i, is

$$v_i = \omega_i \times {}^{i-1}P_n = (P_{i-1} \times {}^{i-1}P_n)\dot{\theta}_i \tag{5.61}$$

It follows from Eqs. (5.60) and (5.61), that for a rotary joint,

$$J_i = \begin{bmatrix} P_{i-1} \times {}^{i-1}P_n \\ P_{i-1} \end{bmatrix} \tag{5.62}$$

For computing J_i in Eq. (5.62), we need vectors P_{i-1} and $^{i-1}P_n$. The vector P_{i-1} is obtained from Eq. (5.59), while the vector $^{i-1}P_n$ is found from Fig. (5.9) as

$$^{i-1}P_n = {}^0P_n - {}^0P_{i-1}$$

The origin of frame $\{n\}$at the end-effector is $O_n = [\,0\ \ 0\ \ 0\ \ 1\,]^T$. This applies for origin of any frame i.e. for any value of n. The above vector equation can then be written as

or $\quad {}^{i-1}P_n = [{}^0T_1(q_1)\ \ldots {}^{n-1}T_n(q_n)\ O_n\,] - [{}^0T_1(q_1)\ \ldots {}^{i-2}T_{i-1}(q_{i-1})O_n]$

or $\quad {}^{i-1}P_n = {}^0T_nO_n - {}^0T_{i-1}\,O_n \qquad (5.63)$

The computation of right-hand side of Eq. (5.63) would yield the three elements of position vector and the fourth term will be unity, the scale factor of T matrix. The vector $^{i-1}P_i$ can be in homogeneous coordinates as 4×1 vector or in physical coordinates as 3×1 vector.

The manipulator Jacobian is obtained by determining all the elements using Eqs. (5.57), (5.59), (5.62) and (5.63). In summary, it is:

$$J_i(q) = \begin{bmatrix} J_{vi} \\ J_{\omega i} \end{bmatrix} = \begin{cases} \begin{bmatrix} \begin{bmatrix} P_{i-1} \\ 0 \end{bmatrix} \end{bmatrix} & \text{for a prismatic joint} \\ \begin{bmatrix} P_{i-1} \times {}^{i-1}P_n \\ P_{i-1} \end{bmatrix} & \text{for a revolute joint} \end{cases} \tag{5.64}$$

Equation (5.64) is a simple, systematic method of computing Jacobian of a manipulator. If, for any joint, the displacement axis is not z-axis, the above equations have to be appropriately modified. Note that the Jacobian matrix depends on the frame in which end-effector velocity is expressed.

5.6 JACOBIAN INVERSE

In practice, to make the manipulator's end-effector track a specified trajectory with a given velocity profile, it is required to coordinate individual joint motions. In other words, for a given end-effector velocity V_e, the corresponding joint velocities $(\dot{q})$ must be found that will cause the end-effector to move at the desired velocity.

In Section 5.5, it has been shown that the Jacobian of a manipulator defines a linear mapping between the vector $\dot{q}$ of the joint velocities and end-effector velocity V_e, from joint space to Cartesian space [Eq. (5.52)] as

$$V_e = J(q)\,\dot{q} \tag{5.65}$$

From this, the reverse mapping from Cartesian space to joint space is

$$\dot{q} = J^{-1}(q)\,V_e \tag{5.66}$$

It is known that for inverse of a matrix to exist, it must be a square matrix. The Jacobian, in general, for an n-DOF manipulator is a $6 \times n$ matrix. Therefore, only for a 6-DOF manipulator, the Jacobian J is a 6×6 square matrix. The inverse matrix J^{-1} exists, if J is nonsingular at the current configuration, in which case Eq. (5.66) determines the individual joint velocities required to obtain desired end-effector velocity.

Consequently at a configuration, where the Jacobian inverse exists, it is possible to move the end-effector in an arbitrary direction with an arbitrary velocity. For, a 6-DOF manipulator it means that all the six components of end-effector linear and angular velocities can be controlled by varying the six joint rates.

In earlier chapters, it has been shown that a manipulator must have at least 6-DOF to locate the end-effector in an arbitrary location. Similarly, 6-DOF are necessary for moving the end-effector in arbitrary direction with an arbitrary speed.

The above discussion is restricted to manipulator with exactly 6-DOF. Nevertheless, there are manipulators with less than 6-DOF for many applications. For example, to manipulate objects lying within a cylindrical workspace, the 4-DOF SCARA robot described in Examples 3.6 can be used. It can be shown that only four end-effector velocities, v_{x4}, v_{y4}, v_{z4}, and ω_{z4} can be controlled by varying the four joint velocities at any given configuration. Hence, the Cartesian velocity vector is $\boldsymbol{V}_4 = [\, v_{x4}\ v_{y4}\ v_{z4}\ \omega_{z4}\,]^T$ and Jacobian $\boldsymbol{J}(\boldsymbol{q})$ is a 4×4 matrix with $\boldsymbol{q} = [\theta_1\ \theta_2\ d_3\ \theta_4]^T$ and $\dot{\boldsymbol{q}} = [\dot{\theta}_1\ \ \dot{\theta}_2\ \ \dot{d}_3\ \ \dot{\theta}_4]^T$.

If a manipulator has less 6-DOF ($n < 6$), it has n controllable joint rates. As a result, there will be n controllable Cartesian velocity components. If this is acceptable for the jobs to be performed by the manipulator, its Jacobian is $n \times n$ square matrix, provided it is not in a singularity configuration. Jacobian singularities are discussed in the next section.

5.7 JACOBIAN SINGULARITIES

The manipulator Jacobian $\boldsymbol{J}$, a function of the configuration $\boldsymbol{q}$, may become rank-deficient or singular at certain configuration in Cartesian space. In such cases, the inverse Jacobian does not exist and Eq. (5.66) is not valid. Those manipulator configurations at which $\boldsymbol{J}$ becomes noninvertible are termed as Jacobian singularities and the configuration is itself called singular.

At a singular configuration, the Jacobian matrix $\boldsymbol{J}$ is not full rank; hence, its column vectors are linearly dependent. This means that there exists at least one direction in which the end-effector cannot be moved irrespective of values chosen for joint velocities $\dot{q}_1$ to $\dot{q}_n$.

The study of manipulator singularities is of great significance for the following reasons:

(a) It is not possible to give an arbitrary motion to end-effector; that is, singularities represent configurations at which structural mobility of the manipulator is reduced.
(b) At a singularity, no solution may exist for the inverse Jacobian problem.
(c) In the neighborhood of a singularity, small velocities in the Cartesian space require very high velocities in the joint space. This causes problems when the manipulator is required to track a trajectory that passes close to the singularity.

At a singular configuration, the manipulator loses one or more degrees of freedom. The singular configurations are classified into two categories based on the location of end-effector in the workspace.

(i) *Boundary singularities* occur when the end-effector is on the boundary of the workspace, that is, the manipulator is either fully stretched out or fully retracted. For example, consider the case of a two-link, 2-DOF-planar arm fully stretched out, as shown in Fig. 5.10. In this configuration, two links are in a straight line and the end-effector can be moved only in a direction perpendicular to the two links because it cannot move out of the workspace. Thus, the manipulator loses one degree of freedom. A similar situation will occur with θ_2 equal to 180°. Boundary singularities can be avoided by ensuring that the manipulator is not driven to boundaries of the reachable workspace during its work cycle.

(ii) *Interior singularities* occur when the end-effector is located inside the reachable workspace of the manipulator. These are caused when two or more joint axes become collinear or at specific end-effector configurations. For example, many spherical wrists cause interior singularities due to the lining up of two or more joint axes. These singularities cause serious problems such as path planning and control. They can occur anywhere in the reachable workspace. For certain manipulator configurations, the interior singularity points form a volume within the workspace called *void*.

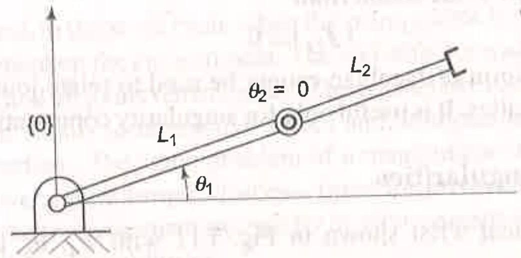


Fig. 5.10 *Two-link, 2-DOF planar manipulator fully stretched out*

In all the situations, it is essential that singularities are avoided. Therefore, one important criterion for a good design of manipulator configuration is to minimize the singularities.

5.7.1 Computation of Singularities

The computation of internal singularities can be carried out by analyzing the rank of the Jacobian matrix. The Jacobian matrix loses its rank and becomes illconditioned at values of joint variables $\boldsymbol{q}$ at which its determinant vanishes, that is,

$$|\,\boldsymbol{J}\,| = 0 \tag{5.67}$$

The solutions of Eq. (5.67) give the singular configurations of a manipulator. For manipulators that have 3-DOF arm and 3-DOF spherical wrists, it is possible to simplify the problem of singularity computation by dividing the problem into two separate problems:

- Computation of singularities resulting from motion of first three joints called *arm singularities*.
- Computation of singularities resulting from the motion of wrist joints called *wrist singularities*.

This is achieved by partitioning the Jacobian matrix into four 3 × 3 sub-matrices as

$$J = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} \tag{5.68}$$

As the singularities are typical of a configuration and are not dependent on frames chosen for kinematic analysis, the origin of the end-effector frame can be chosen at the end-of-arm point. This will make

$$J_{12} = 0 \tag{5.69}$$

In such a situation computation of determinant is greatly simplified, as,

$$|J| = |J_{11}|\,|J_{22}| \tag{5.70}$$

Hence, for a manipulator with a spherical wrist, the arm singularities are found from

$$|J_{11}| = 0 \tag{5.71}$$

and wrist singularities are found from

$$|J_{22}| = 0 \tag{5.72}$$

Note that this form of Jacobian cannot be used to relate joint velocities and end-effector velocities. It is useful only for singularity computations.

5.7.2 Wrist Singularities

Consider a spherical wrist shown in Fig. 5.11 with θ_4, θ_5, and θ_6 as joint variables.

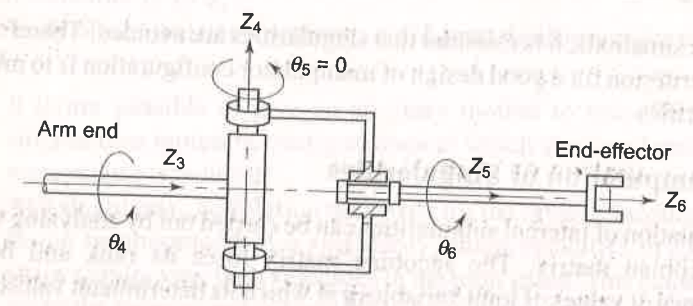


Fig. 5.11 *Spherical wrist at a singularity configuration*

It is known that a singularity occurs whenever two joint axes are aligned. The kinematic structure of the wrist reveals that only axis z_3 and axis z_5 can be aligned. For the configuration in Fig. 5.11 this occurs whenever

$$\theta_5 = 0 \text{ or } \theta_5 = \pi \tag{5.73}$$

The loss of mobility is caused by the fact that rotations of equal magnitude but in opposite direction about z_3- and z_5-axes do not produce any end-effector rotation. If axes z_3 and z_5 are aligned, it is not possible to rotate the wrist about the axis orthogonal to axes z_3 and z_4. This singularity can occur anywhere within the reachable workspace of the manipulator and consequently requires special care to be taken in programming wrist motions.

5.7.3 Arm Singularities

Arm singularities are characteristics of the manipulator's arm configuration and can be analyzed by considering the determinant of the upper left-hand corner 3 × 3 sub-matrix of Jacobian J.

If arm singularities are identified in the workspace, the mechanical designs can be modified in such a manner that arm singularities are minimized. Alternately, they can be suitably avoided in the end-effector path planning. The specific steps for computation of arm singularities are illustrated through the examples at the end of this chapter.

5.8 STATIC ANALYSIS

There are instances in the work cycle when the manipulator is required to exert a force and/or moment on the environment. The end-effector makes a contact with the environment and all joints remain static. The contact between the end-effector and environment results in interactive forces and moments at the end-effector and environment interface. The static problem of a manipulator is to determine the relationship between joint torques/forces—force for prismatic joint, torque for rotary joint and the force/moment exerted by its environment on the end-effector under static equilibrium conditions.

The force/torque at each joint of the manipulator are transmitted through the arm linkages to the end-effector, where the resultant force and moment acts on the environment. A manipulator carrying an object at its end point is equivalent to a force applied to end-effector by the environmental contact as discussed above. In both situations, the end point of the manipulator deflects by a magnitude determined by the stiffness of the manipulator. The end-point stiffness is an important characteristic of a manipulator that determines the accuracy of the manipulator and plays an important role in the control of mechanical interaction with the environment.

5.8.1 Force and Moment Balance

Consider an n-DOF manipulator with force and torque acting at each joint, which determine the torque/force to be exerted by the joint actuators, respectively. To find the force and torque acting on each joint requires the force and moment (torque) balance for the individual links. Figure 5.12 shows the free body diagram of link i with joint i and joint $(i+1)$ connecting link $(i-1)$ and link $(i+1)$, respectively. The corresponding coordinate frames at the two ends are frame $\{i-1\}$ and frame $\{i\}$ as shown.

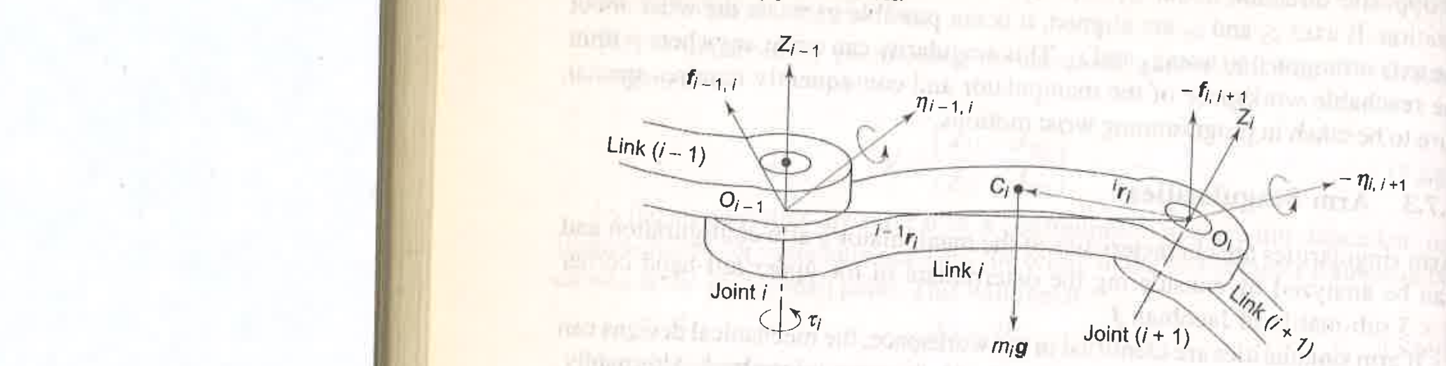


Fig. 5.12 *Forces and moments acting on link i in static equilibrium*

Let the vector $f_{i-1,i}$ denote the force acting on link i due to link $(i-1)$, where the first subscript denotes the link exerting the force, which acts on the link of second subscript. The vector $f_{i,i+1}$, therefore, represents the force applied by link i on link $(i+1)$. The force exerted by link $(i+1)$ on link i will be $-f_{i,i+1}$ because it is equal and opposite to force applied by link i on link $(i+1)$.

If m_i denotes the mass of link i and g is the 3×1 vector representing acceleration due to gravity, the gravity force acting at the centroid C_i of link i, is denoted by $m_i g$. Because the manipulator is at rest, each link is in static equilibrium, the force equilibrium gives

$$f_{i-1,i} - f_{i,i+1} + m_i g = 0 \tag{5.74}$$

All vectors in the above equation are with reference to the base frame $\{0\}$.

Also, for a link in static equilibrium, the vector sum of all moments acting on it about any arbitrary point is zero. Let $\eta_{i-1,i}$ represents the moment applied to link i by link $(i-1)$ with respect to frame $\{0\}$. Therefore, $-\eta_{i,i+1}$ is the moment applied to link i by link $(i+1)$. The moment balance about the centroid C_i of link i, see Fig. 5.12, gives

$$\eta_{i-1,i} + {}^0\boldsymbol{R}_{i-1}({}^{i-1}r_i + {}^i r_i) \times f_{i-1,i} - \eta_{i,i+1} + {}^0\boldsymbol{R}_i(-{}^i r_i) \times (-f_{i,i+1}) = 0$$

$$\text{for } i = 1, 2, \ldots, n \tag{5.75}$$

where $^{i-1}r_i$ is the position vector from origin O_{i-1} to origin O_i, $^i r_i$ is the position vector from origin O_i to centroid C_i and rotation matrix $\boldsymbol{R}$ is used to express all moment vectors in terms of frame $\{0\}$.

Equations (5.74) and (5.75) for each link give a set of n equations. Thus, there are $2n$ simultaneous equations involving $2(n+1)$ forces and moments. To solve these equations one force and one moment must be known. The force and moment applied by the manipulator on the environment can be specified to solve these equations. For $i = n$, $f_{n,n+1}$ and $\eta_{n,n+1}$ are the independent endpoint force and moment, respectively exerted by the manipulator on the environment.

The reaction force and moment from the environment to the manipulator, at its end point are $-f_{n,n+1}$ and $\eta_{n,n+1}$, respectively. The endpoint force and moment are combined into a 6-D vector, called *end-point force vector*, $\mathcal{F}$ as

$$\mathcal{F} = \begin{bmatrix} f_{n,n+1} \\ \eta_{n,n+1} \end{bmatrix} \tag{5.76}$$

Note that the negative sign has been dropped and will be appropriately taken care when used in a force/moment balance equations. Thus, for any n-DOF manipulator, the interactive forces and moments between links, $f_{i,i+1}$ and $\eta_{i,i+1}$ can be obtained from Eqs. (5.74) and (5.75) by performing inward iterations from $i = n$ down to 1.

Next, to determine the joint torques/forces to be exerted by the actuators to keep the manipulator in static equilibrium, let τ denote the $n \times 1$ vector of generalized joint torques

$$\tau = [\tau_1 \ \tau_2 \ \ldots \ \tau_n]^T \tag{5.77}$$

where τ_i is the generalized drive torque applied by the actuator i driving joint i and is defined as

$$\tau_i = \begin{cases} \text{Actuator force,} & \text{if joint } i \text{ is prismatic} \\ \text{Actuator torque,} & \text{if joint } i \text{ is revolute} \end{cases} \tag{5.78}$$

For a revolute joint i, the actuator at that joint will bear only that component of the moment $\eta_{i-1,i}$ which is in the direction of joint axis z_i, while the other components of the moment are borne by the joint structure. This gives, for a revolute joint, with $^{i-1}\hat{z}_{i-1}$ as the unit vector in the direction of the joint axis

$$\tau_i = [{}^0\boldsymbol{R}_{i-1} \ {}^{i-1}\hat{z}_{i-1}]^T \eta_{i-1,i} \tag{5.79}$$

Similarly, if joint i is prismatic, the actuator will bear that component of $f_{i-1,i}$ which is in the direction of joint axis, that is

$$\tau_i = [{}^0\boldsymbol{R}_{i-1} \ {}^{i-1}\hat{z}_{i-1}]^T f_{i-1,i} \tag{5.80}$$

5.8.2 The Jacobian in Statics

The relationship between the joint torques and the endpoint force/torque vector is derived using the *principle of virtual work*. This is used to determine the joint torques necessary to exert a given end-effector force and moment.

Consider an infinitesimal displacement of δq_i due to torque τ_i at joint i. Hence, δq is the $n \times 1$ vector of infinitesimal joint displacements. These displacements, also called virtual displacements, of a mechanical system need only satisfy the kinematic constraints. The end-point force and moment also act on the mechanical system producing infinitesimal endpoint displacement dx_e and rotation $\delta\phi_e$, respectively. Thus, the virtual work δW done by the forces and moments is given by

$$\delta W = \tau_1 \delta q_1 + \ldots + \tau_n \delta q_n - (f_{n,n+1})^T \delta x_e - (\eta_{n,n+1})^T \delta\phi_e$$

or
$$\delta W = \tau^T \delta q - \mathcal{F}^T \delta p \tag{5.81}$$
where δp is the $n \times 1$ vector of infinitesimal end-effector displacements (δx_e, $\delta \phi_e$) caused by the endpoint force $\mathcal{F}$. In the above, it is assumed that the joints of the mechanism are frictionless and the joint torques are the net torques that balance the endpoint force $\mathcal{F}$.

From Eq. (5.52), the Jacobian J relates infinitesimal joint displacement δq to infinitesimal end-effector displacement δp as:
$$\delta p = J(q)\ \delta q \tag{5.82}$$
Substituting Eq. (5.82) into Eq. (5.81) and rearranging gives
$$\delta W = (\tau - J(q)^T \mathcal{F})^T \delta q \tag{5.83}$$

[redacted]

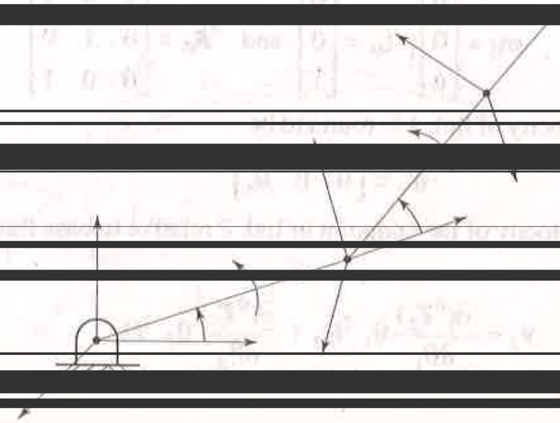
Comparing Eq. (5.87) with Eq. (5.27) gives
$$\omega = [0 \quad 0 \quad \dot{\alpha}]^T \tag{5.88}$$
as the angular velocity of the frame about z-axis.

Example 5.2 ***Velocity of RR-manipulator***

Calculate the velocity of the tip of the two-link, planar, RR-manipulator arm shown in Fig. 5.13.

Solution The frame assignments and joint-link parameters identification is carried out as discussed in Chapter 3 and are shown in Fig. 5.13 and Table 5.1.

[redacted]



[redacted]

$$\text{or} \quad {}^0\boldsymbol{T}_2 = \begin{bmatrix} C_{12} & -S_{12} & 0 & C_1 + C_{12} \\ S_{12} & C_{12} & 0 & S_1 + S_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.90}$$

Table 5.1 *Joint-link parameters for RR manipulator*

Link i	a_i	α_i	d_i	θ_i	$C\alpha_i$	$S\alpha_i$
1	1	0	0	θ_1	1	0
2	1	0	0	θ_2	1	0

Because both the joints of manipulator are rotary, according to Eq. (5.46) and Eq. (5.51), for endpoint of link 1, th it is origin of frame {1}, linear velocity is computed from

$$\boldsymbol{v}_1 = \frac{\partial({}^0\boldsymbol{T}_1)}{\partial\theta_1}\,\dot{\theta}_1\,{}^1\boldsymbol{D}_1$$

with

$$\frac{\partial({}^0\boldsymbol{T}_1)}{\partial\theta_1} = \begin{bmatrix} -S_1 & -C_1 & 0 & -S_1 \\ C_1 & -S_1 & 0 & C_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; \text{ and } {}^1\boldsymbol{D}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Note that the origin of frame {1}is ${}^1\boldsymbol{D}_1$. The linear velocity of link 1 is, thus

$$\boldsymbol{v}_1 = [-S_1 \quad -C_1 \quad 0 \quad 0]^T\,\dot{\theta} \tag{5.91}$$

The angular velocity is computed from

$$\omega_1 = \omega_0 + {}^0\boldsymbol{R}_0\,\hat{z}_0\,\dot{\theta}_1$$

with

$$\omega_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}; \ \hat{z}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ and } {}^0\boldsymbol{R}_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The angular velocity of link 1 is found to be

$$\omega_1 = \begin{bmatrix} 0 & 0 & \dot{\theta}_1 \end{bmatrix}^T \tag{5.92}$$

The linear velocity of the endpoint of link 2 relative to base frame is obtained as:

$$\boldsymbol{v}_2 = \frac{\partial({}^0\boldsymbol{T}_2)}{\partial\theta_1}\dot{\theta}_1\,{}^2\boldsymbol{D}_2 + \frac{\partial({}^0\boldsymbol{T}_2)}{\partial\theta_2}\dot{\theta}_2\,{}^2\boldsymbol{D}_2 \tag{5.93}$$

From Eq. (5.90)

$$\frac{\partial({}^0\boldsymbol{T}_2)}{\partial\theta_1} = \begin{bmatrix} -S_1C_2 - C_1S_2 & S_1S_2 - C_1C_2 & 0 & -S_1 - S_1C_2 - C_1S_2 \\ -S_1S_2 + C_1C_2 & -S_1C_2 - C_1S_2 & 0 & C_1 - S_1S_2 + C_1S_2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{5.94}$$

$$\text{and} \quad \frac{\partial({}^0\boldsymbol{T}_2)}{\partial\theta_2} = \begin{bmatrix} -C_1S_2 - S_1C_2 & -C_1C_2 + S_1S_2 & 0 & -C_1S_2 - S_1C_2 \\ C_1C_2 - S_1S_2 & -C_1S_2 - S_1C_2 & 0 & C_1C_2 - S_1S_2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{5.95}$$

The position vector of endpoint of link 2 is origin of frame {2}, that is

$${}^2\boldsymbol{D}_2 = [0 \quad 0 \quad 0 \quad 1]^T$$

Substituting these gives,

$$\boldsymbol{v}_2 = \begin{bmatrix} -S_1\dot{\theta}_1 - S_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ C_1\dot{\theta}_1 + C_{12}(\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix} \tag{5.96}$$

And for the angular velocity of the endpoint,

$$\omega_2 = \omega_1 + {}^0\boldsymbol{T}_1\,z_1\,\dot{\theta}_2 \tag{5.97}$$

From Eq. (5.89) and Eq. (5.92) this simplifies to

$$\omega_2 = \begin{bmatrix} 0 & 0 & (\dot{\theta}_1 + \dot{\theta}_2) \end{bmatrix}^T \tag{5.98}$$

Example 5.3 ***Jacobian of articulated arm***

Determine the manipulator Jacobian matrix for the 3-DOF articulated arm discussed in Example 3.3 and shown in Fig. 5.14.

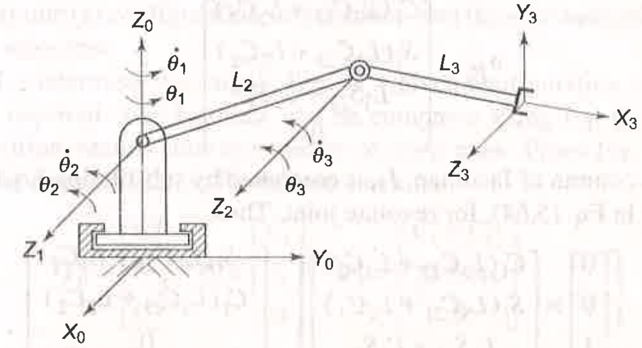


Fig. 5.14 *3-DOF articulated manipulator arm*

Solution Each column of Jacobian matrix is computed separately and all the columns are combined to form the total Jacobian matrix. The joint displacements

θ_1, θ_2, and θ_3 and joint velocities $\dot{\theta}_1$ $\dot{\theta}_2$, and $\dot{\theta}_3$ are shown in the figure and the transformation matrices are given in Eqs. (3.19)–(3.22).

The Jacobian matrix column J_1 for joint 1, which is a rotary joint, is determined as follows:

From Eq. (5.59), the joint axis vector P_0 (P_{i-1} for $i = 1$) is

$$\boldsymbol{P}_0 = {}^0\boldsymbol{R}_0\, \hat{\boldsymbol{u}} \tag{5.99}$$

The transformation matrix ${}^0T_0$ and rotation matrix ${}^0\boldsymbol{R}_0$ are the identity matrices. Thus,

$$\boldsymbol{P}_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.100}$$

The end-effector position vector (for $i = 1$ and $n = 3$) is determined from Eq. (5.63).

$${}^0\boldsymbol{P}_3 = {}^0\boldsymbol{T}_n\boldsymbol{O}_n - {}^0\boldsymbol{T}_0\boldsymbol{O}_n$$

or

$${}^0\boldsymbol{P}_3 = {}^0\boldsymbol{T}_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} - {}^0\boldsymbol{T}_0 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Substituting ${}^0\boldsymbol{T}_3$ from Eq. (3.22), in above equation gives

$${}^0\boldsymbol{P}_3 = \begin{bmatrix} C_1C_{23} & -C_1S_{23} & S_1 & C_1(L_3C_{23}+L_2C_2) \\ S_1C_{23} & -S_1S_{23} & -C_1 & S_1(L_3C_{23}+L_2C_2) \\ S_{23} & C_{23} & 0 & L_3S_{23}+L_2S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

or

$${}^0\boldsymbol{P}_3 = \begin{bmatrix} C_1(L_3C_{23}+L_2C_2) \\ S_1(L_3C_{23}+L_2C_2) \\ L_3S_{23}+L_2S_2 \\ 0 \end{bmatrix} \tag{5.101}$$

The first column of Jacobian, J_1, is computed by substituting Eq. (5.100) and Eq. (5.101) in Eq. (5.64), for revolute joint. Thus,

$$J_1 = \begin{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} C_1(L_3C_{23}+L_2C_2) \\ S_1(L_3C_{23}+L_2C_2) \\ L_3S_{23}+L_2S_2 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} -S_1(L_3C_{23}+L_2C_2) \\ C_1(L_3C_{23}+L_2C_2) \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.102}$$

By following the similar steps for joint 2 and joint 3, J_2 and J_3 are obtained as

$$J_2 = \begin{bmatrix} -C_1(L_3S_{23}+L_2S_2) \\ -S_1(L_3S_{23}+L_2S_2) \\ L_3C_{23}+L_2C_2 \\ S_1 \\ -C_1 \\ 0 \end{bmatrix} \tag{5.103}$$

and

$$J_3 = \begin{bmatrix} -L_3C_1S_{23} \\ -L_3S_1S_{23} \\ L_3C_{23} \\ S_1 \\ -C_1 \\ 0 \end{bmatrix} \tag{5.104}$$

Combining the three columns, the total Jacobian matrix for the articulated arm is

$$J = \begin{bmatrix} -S_1(L_3C_{23}+L_2C_2) & -C_1(L_3S_{23}+L_2S_2) & -L_3C_1S_{23} \\ C_1(L_3C_{23}+L_2C_2) & -S_1(L_3S_{23}+L_2S_2) & -L_3S_1S_{23} \\ 0 & L_3C_{23}+L_2C_2 & L_3C_{23} \\ 0 & S_1 & S_1 \\ 0 & -C_1 & -C_1 \\ 1 & 0 & 0 \end{bmatrix} \tag{5.105}$$

Example 5.4 *Singularities and joint velocities of a 2-DOF arm*

For the 2-DOF-planar arm shown in Fig. 5.13, assuming both the links of unit length, determine (a) configuration singularities, and (b) joint velocities in terms of endpoint velocities.

Solution To determine the singularities of a given configuration manipulator Jacobian is required. The Jacobian can be computed using Eq. (5.57), which requires position vectors and unit vector of joint axes. From Eq. (5.89) and Eq. (5.90) the position vectors for various links are

$${}^0\boldsymbol{P}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix};\quad {}^0\boldsymbol{P}_1 = \begin{bmatrix} C_1 \\ S_1 \\ 0 \end{bmatrix};\quad {}^0\boldsymbol{P}_2 = \begin{bmatrix} C_1+C_{12} \\ S_1+S_{12} \\ 0 \end{bmatrix} \tag{5.106}$$

The unit vectors along the revolute joint axes are

$$\hat{\boldsymbol{u}}_0 = \hat{\boldsymbol{u}}_1 = [0 \quad 0 \quad 1]^T \tag{5.107}$$

The Jacobian is given by (from Eq. 5.64)

$$J = \begin{bmatrix} \hat{u}_0 \times ({}^0P_2 - {}^0P_0) & \hat{u}_1 \times ({}^0P_2 - {}^0P_1) \\ \hat{u}_0 & \hat{u}_1 \end{bmatrix} \tag{5.108}$$

Substituting from Eqs. (5.106) and (5.107)

$$J = \begin{bmatrix} -S_1 - S_{12} & -S_{12} \\ C_1 + C_{12} & C_{12} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \tag{5.109}$$

(a) Configuration singularities In the Jacobian, Eq. (5.109) three rows are null and one row is unity. The arm has only 2-DOF, that is, it requires only two variables to be controlled. Hence, the upper 2×2 block of the Jacobian is only significant and sufficient for computing positions of two links. The significant Jacobian, to describe the relationship between the joint velocities and the arm-endpoint linear velocity, is therefore

$$J' = \begin{bmatrix} -S_1 - S_{12} & -S_{12} \\ C_1 + C_{12} & C_{12} \end{bmatrix} \tag{5.110}$$

It is instructive to note that Eq. (5.91) and Eq. (5.96) of Example 5.2, when written in the form

$$V_e = \begin{bmatrix} v \\ \omega \end{bmatrix} = J' \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \tag{5.111}$$

lead to the same result for J as in Eq. (5.109) from which J' of Eq. (5.110) follows.

Singularities of the configuration occur when determinant of J' vanishes. From Eq. (5.110), the determinant of J' is

$$|J'| = \begin{vmatrix} -S_1 - S_{12} & -S_{12} \\ C_1 + C_{12} & C_{12} \end{vmatrix} = S_2 \tag{5.112}$$

Note that the Jacobian determinant is independent of first joint variable θ_1. The determinant vanishes for non-zero link lengths whenever

$$\theta_2 = 0 \text{ or } \theta_2 = \pi$$

Hence, singular configurations occur when arm-endpoint is located either on the outer boundary ($\theta_2 = 0$) or on the inner boundary ($\theta_2 = \pi$) of the reachable workspace, that is, fully extended or fully retracted. At these values of θ_2, the two columns of J' become linearly dependent implying (i) the Jacobian loses rank (and the rank becomes one); (ii) the tip velocity components v_x and v_y are not independent, and (iii) the two links are collinear, as in Fig. 5.10.

(b) Joint velocities The joint velocities in terms of endpoint velocities v_x and v_y are obtained from Eq. (5.54). That is

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = J' \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

or

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = [J']^{-1} \begin{bmatrix} v_x \\ v_y \end{bmatrix} \tag{5.113}$$

The inverse of J' is

$$[J']^{-1} = \frac{1}{S_2} \begin{bmatrix} C_{12} & S_{12} \\ -C_1 - C_{12} & S_1 + S_{12} \end{bmatrix} \tag{5.114}$$

Thus, the joint velocities are

$$\dot{\theta}_1 = \frac{v_x C_{12} + v_y S_{12}}{S_2} \tag{5.115}$$

$$\dot{\theta}_2 = \frac{-v_x (C_1 + C_{12}) + v_y (S_1 + S_{12})}{S_2} \tag{5.116}$$

At regions near the singular points, the denominator in Eqs. (5.115) and (5.116) is almost zero, that is, to move the endpoint in the vicinity of the singularities excessively large joint velocities are required.

The joint velocity profiles for tracking a specified trajectory at specified speed (v_x, v_y) can be obtained by finding the angles that correspond to the endpoint position on the trajectory, that is, $\theta_1(t)$ and $\theta_2(t)$ substituting these into Eqs. (5.115) and (5.116) and plotting the joint velocities with respect to time.

Example 5.5 *Singularities of articulated arm configuration*

Determine the singularities of the 3-DOF articulated arm configuration discussed in Example 5.3.

Solution A close examination of the Jacobian of the articulated arm, Eq. (5.105) indicates that only three of the six rows of the Jacobian are linearly independent. It is also known that the arm has only three degrees of freedom, that is, it requires only three variables to be controlled. Hence, the upper 3×3 block of Jacobian is only significant and worth considering. The significant Jacobian is

$$J' = \begin{bmatrix} -S_1(L_3C_{23} + L_2C_2) & -C_1(L_3S_{23} + L_2S_2) & -L_3C_1S_{23} \\ C_1(L_3C_{23} + L_2C_2) & -S_1(L_3S_{23} + L_2S_2) & -L_3S_1S_{23} \\ 0 & L_3C_{23} + L_2C_2 & L_3C_{23} \end{bmatrix} \tag{5.117}$$

It is worth noting that for this 3-DOF arm endpoint angular velocity ω_x and ω_y are not independent, since $S_1\omega_y = -C_1\omega_x$ and this structure does not allow arbitrary endpoint angular velocity ω.

For determination of singularities, consider the Jacobian J', in Eq. (5.117). Its determinant is

$$|J'| = -L_2L_3S_3(L_2C_2 + L_3C_{23}) \tag{5.118}$$

It is seen from the determinant that it is independent of first joint variable. The determinant vanishes for nonzero link lengths when $S_3=0$ and/or $L_2C_2 + L_3C_3 = 0$.

Case 1 $S_3 = 0$ is possible for $\theta_3 = 0$ or $\theta_3 = \pi$.

At either of the values of θ_3, two links of arm are aligned, fully outstretched or retracted. This is similar to singularity condition of the RR planar manipulator discussed in Example 5.4.

Case 2 $L_2C_2 + L_3C_{23} = 0$.

This occurs when the endpoint lies on the axis of base frame, z_0. That is

$$p_x = p_y = 0$$

Example 5.6 *Singularities of RPY wrist*

Determine the singularities of the RPY wrist discussed in Example 3.4.

Solution To determine the singularities of the wrist, its Jacobian is determined first. For the 3-DOF RPY wrist, the link transformation matrices were obtained in Example 3.4, Eqs. (3.24) – (3.26), as

$$^0T_1 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.119}$$

$$^1T_2 = \begin{bmatrix} -S_2 & 0 & C_2 & 0 \\ C_2 & 0 & S_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.120}$$

$$^2T_3 = \begin{bmatrix} C_3 & -S_3 & 0 & 0 \\ S_3 & C_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.121}$$

The rotational transformation matrices for the three links of the wrist are given by the upper-right-3 × 3 submatrix of the respective transformation matrices in Eqs. (5.119), (5.120) and (5.121), as

$$^0R_1 = \begin{bmatrix} C_1 & 0 & S_1 \\ S_1 & 0 & -C_1 \\ 0 & 1 & 0 \end{bmatrix} \tag{5.122}$$

$$^1R_2 = \begin{bmatrix} -S_2 & 0 & C_2 \\ C_2 & 0 & S_2 \\ 0 & 1 & 0 \end{bmatrix} \tag{5.123}$$

$$^2R_3 = \begin{bmatrix} C_3 & -S_3 & 0 \\ S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{5.124}$$

The Jacobian for the 3-DOF RPY wrists is computed using Eq. (5.62) as all the three joints are revolute joints. That is

$$J_i = \begin{bmatrix} P_{i-1} \times {}^{i-1}P_n \\ P_{i-1} \end{bmatrix} \tag{5.125}$$

with P_{i-1} and $^{i-1}P_n$ given by Eq. (5.59) and Eq. (5.63), respectively.

For $i = 1$ and $n = 3$, the first coloumn of Jacobian J_1 is given by

$$J_1 = \begin{bmatrix} P_0 \times {}^0P_3 \\ P_0 \end{bmatrix} \tag{5.126}$$

where from Eq. (5.59)

$$P_0 = {}^0R_0\,\hat{u} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.127}$$

and from Eq. (5.63) and Eqs. (5.119), (5.120) and (5.121)

$$^0P_3 = {}^0T_3\,O_3 - {}^0T_0O_3$$

or

$$^0P_3 = \begin{bmatrix} -C_1S_2C_3 + S_1S_3 & C_1S_2S_3 + S_1C_3 & C_1C_2 & 0 \\ -S_1S_2C_3 - C_1S_3 & S_1S_2S_3 - C_1C_3 & S_1C_2 & 0 \\ C_2C_3 & -C_2S_3 & S_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{5.128}$$

Thus,

$$J_1 = \begin{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.129}$$

Similarly, for $i = 2$,

$$P_1 = {}^0R_1\,\hat{u} = \begin{bmatrix} C_1 & 0 & S_1 \\ S_1 & 0 & -C_1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} S_1 \\ -C_1 \\ 0 \end{bmatrix} \tag{5.130}$$

$$ {}^1P_3 = {}^0T_3\,O_3 - {}^0T_1\,O_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{5.131}$$

and

$$J_2 = \begin{bmatrix} P_1 \times {}^1P_3 \\ P_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ S_1 \\ -C_1 \\ 0 \end{bmatrix} \tag{5.132}$$

and for $i = 3$,

$$P_2 = {}^0R_2\,\hat{u} = \begin{bmatrix} -C_1S_2 & S_1 & C_1C_2 \\ -S_1S_2 & -C_1 & S_1C_2 \\ C_2 & 0 & S_2 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} C_1C_2 \\ S_1C_2 \\ S_2 \end{bmatrix} \tag{5.133}$$

$$ {}^2P_3 = {}^0T_3\,O_3 - {}^0T_2\,O_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{5.134}$$

(*Note:* ${}^0R_2$ and ${}^0T_2$ are obtained from Eqs. (5.119), (5.120) and Eqs. (5.122), (5.123), respectively). Thus,

$$J_3 = \begin{bmatrix} P_2 \times {}^2P_3 \\ P_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ C_1C_2 \\ S_1C_2 \\ S_2 \end{bmatrix} \tag{5.135}$$

From Eqs. (5.129), (5.132) and (5.135) the Jacobian for the RPY wrist is

$$J = [J_1 \quad J_2 \quad J_3] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & S_1 & C_1C_2 \\ 0 & -C_1 & S_1C_2 \\ 1 & 0 & S_2 \end{bmatrix} \tag{5.136}$$

Note that the first three rows of this Jacobian for the wrist are zero. This means that the wrist links have no linear velocities. This is expected because a wrist is used for orientation only.

The singularities of the RPY wrist are determined using the last three rows of the Jacobian in Eq. (5.136). The pseudo-Jacobian is

$$J' = \begin{bmatrix} 0 & S_1 & C_1C_2 \\ 0 & -C_1 & S_1C_2 \\ 1 & 0 & S_2 \end{bmatrix} \tag{5.137}$$

The wrist singularities will occur when inverse of J' cannot be found, or in other words J' becomes ill conditioned. This occurs whenever determinant of J' vanishes. Thus, the conditions of singularity are found from

$$|J'| = \begin{vmatrix} 0 & S_1 & C_1C_2 \\ 0 & -C_1 & S_1C_2 \\ 1 & 0 & S_2 \end{vmatrix} = 0 \tag{5.138}$$

or $$S_1(S_1C_2 - 0) + C_1C_2(0 + C_1) = 0$$

or $$C_2 = 0 \tag{5.139}$$

Hence, singularities of the RPY wrist are those positions of joints 2 where

$$\theta_2 = 90° \quad \text{or} \quad 270° \tag{5.140}$$

Example 5.7 ***RPR arm Jacobian***

For the manipulator shown in Figure 5.15 below, obtain the Jacobian to express the Cartesian velocities in terms of the joint velocities. Obtain the singularities of the manipulator.

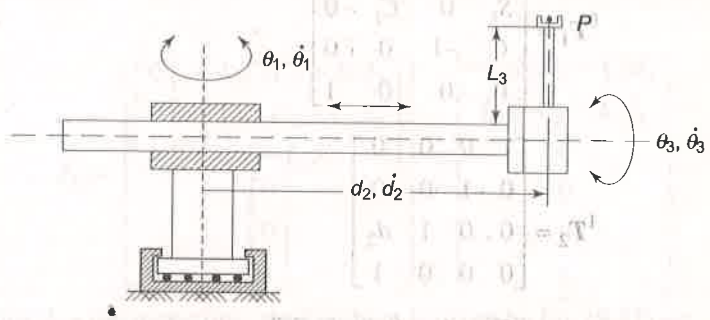


Fig. 5.15 *A 3-DOF RPR arm of a manipulator*

Solution In order to determine the Jacobian of the arm, first its forward kinematics model in terms of transformation matrices is required.

To determine the transformation matrices, frames are assigned as per Algorithm 3.1 and these are shown in Fig. 5.16. The joint-link parameters

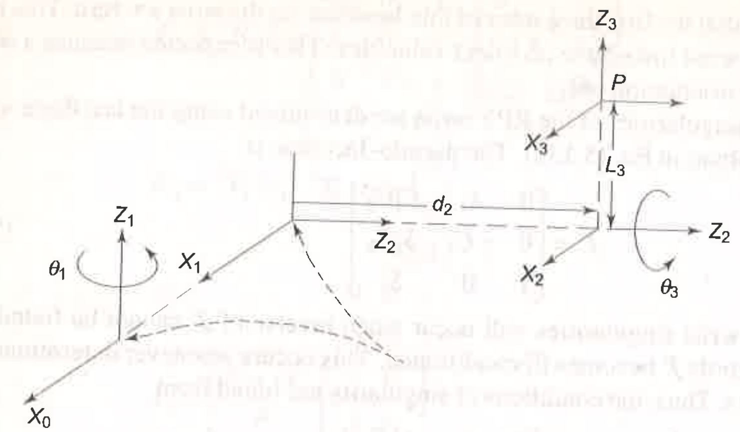


Fig. 5.16 *Frame assignment for RPR arm*

obtained there from are tabulated in Table 5.2 and the three transformation matrices obtained from these are given in Eqs. (5.141), (5.142) and (5.143).

Table 5.2 *Joint-link parameters for RPR arm*

i	θ_i	α_i	a_i	d_i	$C\theta$	$S\theta$	$C\alpha$	$S\alpha$
1	θ_1	–90°	0	0	C_1	S_1	0	–1
2	0	0	0	d_2	1	0	1	0
3	θ_3	90°	L_3	0	C_3	S_3	0	1

$$ {}^0T_1 = \begin{bmatrix} C_1 & 0 & -S_1 & 0 \\ S_1 & 0 & C_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.141} $$

$$ {}^1T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.142} $$

$$ {}^2T_3 = \begin{bmatrix} C_3 & 0 & S_3 & L_3C_3 \\ S_3 & 0 & -C_3 & L_3S_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.143} $$

From the individual transformation matrices following products of transformation matrices are computed:

$$ {}^0T_2 = {}^0T_1\,{}^1T_2 = \begin{bmatrix} C_1 & 0 & -S_1 & -S_1d_2 \\ S_1 & 0 & C_1 & C_1d_2 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.144} $$

$$ {}^0T_3 = {}^0T_2\,{}^2T_3 = \begin{bmatrix} C_1C_3 & -S_1 & C_1S_3 & L_3C_1C_3 - S_1d_2 \\ S_1C_3 & C_1 & S_1S_3 & L_3S_1C_3 + C_1d_2 \\ -S_3 & 0 & C_3 & -L_3S_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.145} $$

The link 1 is a revolute link and Jacobian for link 1 is computed using Eq. (5.62) as:

$$ \boldsymbol{J}_1 = \begin{bmatrix} \boldsymbol{P}_0 \times {}^0\boldsymbol{P}_3 \\ \boldsymbol{P}_0 \end{bmatrix} \tag{5.146} $$

where from Eq. (5.59)

$$ \boldsymbol{P}_0 = {}^0\boldsymbol{R}_0\,\hat{\boldsymbol{u}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.147} $$

and from Eq. (5.63)

$$ {}^0\boldsymbol{P}_3 = {}^0T_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} - {}^0T_0 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} L_3C_1C_3 - S_1d_2 \\ L_3S_1C_3 + C_1d_2 \\ -L_3S_3 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} L_3C_1C_3 - S_1d_2 \\ L_3S_1C_3 + C_1d_2 \\ -L_3S_3 \\ 0 \end{bmatrix} \tag{5.148} $$

Thus,

$$ \boldsymbol{J}_1 = \begin{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} L_3C_1C_3 - S_1d_2 \\ L_3S_1C_3 + C_1d_2 \\ -L_3S_3 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} -L_3S_1C_3 - C_1d_2 \\ L_3C_1C_3 - S_1d_2 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \tag{5.149} $$

For link 2, a prismatic link, the Jacobian is given by Eq. (5.57) as:

$$ \boldsymbol{J}_2 = \begin{bmatrix} \boldsymbol{P}_1 \\ 0 \end{bmatrix} \tag{5.150} $$

where

$$ \boldsymbol{P}_1 = {}^0\boldsymbol{R}_1\,\hat{\boldsymbol{u}} = \begin{bmatrix} C_1 & 0 & -S_1 \\ S_1 & 1 & C_1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -S_1 \\ C_1 \\ 0 \end{bmatrix} \tag{5.151} $$

Thus,

$$J_2 = \begin{bmatrix} -S_1 \\ C_1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{5.152}$$

For the revolute joint 3, the Jacobian is computed from

$$J_3 = \begin{bmatrix} P_2 \times {}^2P_3 \\ P_2 \end{bmatrix} \tag{5.153}$$

where

$$P_2 = {}^0R_2\, \hat{u} = \begin{bmatrix} C_1 & 0 & -S_1 \\ S_1 & 0 & C_1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -S_1 \\ C_1 \\ 0 \end{bmatrix} \tag{5.154}$$

and

$${}^2P_3 = {}^0T_3\, O_n - {}^0T_2\, O_n$$

or
$${}^2P_3 = \begin{bmatrix} L_3C_1C_3 - S_1d_2 \\ L_3S_1C_3 + C_1d_2 \\ -L_3S_3 \\ 1 \end{bmatrix} - \begin{bmatrix} C_1 & 0 & -S_1 & -S_1d_2 \\ S_1 & 0 & C_1 & C_1d_2 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} L_3C_1C_3 \\ L_3S_1C_3 \\ -L_3S_3 \\ 0 \end{bmatrix} \tag{5.155}$$

Substituting Eqs. (5.154) and (5.155) in Eq. (5.153) gives

$$J_3 = \begin{bmatrix} \begin{bmatrix} -S_1 \\ C_1 \\ 0 \end{bmatrix} \times \begin{bmatrix} L_3C_1C_3 \\ L_3S_1C_3 \\ -L_3S_3 \end{bmatrix} \\ \begin{bmatrix} -S_1 \\ C_1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} -L_3C_1S_3 \\ -L_3S_1S_3 \\ -L_3C_3 \\ -S_1 \\ C_1 \\ 0 \end{bmatrix} \tag{5.156}$$

The Jacobian for the manipulator is, therefore, obtained by combining Eqs. (5.149), (5.152) and (5.156).

$$J = \begin{bmatrix} -L_3S_1C_3 - C_1d_2 & -S_1 & -L_3C_1S_3 \\ L_3C_1C_3 - S_2d_2 & C_1 & -L_3S_1S_3 \\ 0 & 0 & -L_3C_3 \\ 0 & 0 & -S_1 \\ 0 & 0 & C_1 \\ 1 & 0 & 0 \end{bmatrix} \tag{5.157}$$

For determination of singularities, consider first three rows of J in Eq. (5.157) as the pseudo Jacobian J':

$$J' = \begin{bmatrix} -L_3S_1C_3 - d_2C_1 & -S_1 & -L_3C_1S_3 \\ L_3C_1C_3 - S_1d_2 & C_1 & L_3S_1S_3 \\ 0 & 0 & -L_3C_3 \end{bmatrix} \tag{5.158}$$

The determinant of J' is

$$|J'| = L_3d_2S_3\left(C_1^2 + S_1^2\right) = L_3d_2S_3 \tag{5.159}$$

The singularity condition is

$$|J'| = 0$$

or

$$L_3\, d_2\, C_3 = 0 \tag{5.160}$$

Since L_3 and d_2 cannot be zero, singularities occurs at $C_3 = 0$

or

$$\theta_3 = 90° \quad \text{or} \quad 270° \tag{5.161}$$

Example 5.8 ***Static forces in articulated arm***

The 3-DOF articulated arm of Example 5.3 applies a force $\mathcal{F}$ with its end-effector. Determine the torques required at the joints to maintain static equilibrium.

Solution Assume that the force $\mathcal{F}$ is acting at the origin of frame {3}(see Fig. 3.16) and is given by

$$\mathcal{F} = \begin{bmatrix} f_x & f_y & f_z & 0 & 0 & 0 \end{bmatrix}^T \tag{5.162}$$

Note that $\mathcal{F}$ is 6 × 1 Cartesian force-moment vector.

The joint torques are a function of the configuration and applied force and are given by Eq. (5.85) as

$$\tau = J(q)^T\mathcal{F} \tag{5.163}$$

The Jacobian of the articulated arm was obtained in Example 5.3, Eq. (5.105) as

$$J = \begin{bmatrix} -S_1(L_3C_{23} + L_2C_2) & -C_1(L_3S_{23} + L_2S_2) & -L_3C_1S_{23} \\ C_1(L_3C_{23} + L_2C_2) & -S_1(L_3S_{23} + L_2S_2) & -L_3S_1S_{23} \\ 0 & L_3C_{23} + L_2C_2 & L_3C_{23} \\ 0 & S_1 & S_1 \\ 0 & -C_1 & -C_1 \\ 1 & 0 & 0 \end{bmatrix} \tag{5.164}$$

The transpose of J is

$$J^T = \begin{bmatrix} -S_1(L_3C_{23} + L_2C_2) & C_1(L_3C_{23} + L_2C_2) & 0 & 0 & 0 & 1 \\ -C_1(L_3S_{23} + L_2S_2) & -S_1(L_3S_{23} + L_2S_2) & L_3C_{23} + L_2C_2 & S_1 & -C_1 & 0 \\ -L_3C_1S_{23} & -L_3S_1S_{23} & L_3C_{23} & S_1 & -C_1 & 0 \end{bmatrix} \tag{5.165}$$

Substituting Eqs. (5.162) and (5.165) in Eq. (5.163) gives

$\tau =$

$$\begin{bmatrix} -S_1(L_3C_{23}+L_2C_2) & C_1(L_3C_{23}+L_2C_2) & 0 & 0 & 0 & 1 \\ -C_1(L_3S_{23}+L_2S_2) & -S_1(L_3S_{23}+L_2S_2) & L_3C_{23}+L_2C_2 & S_1 & -C_1 & 0 \\ -L_3C_1S_{23} & -L_3S_1S_{23} & L_3C_{23} & S_1 & -C_1 & 0 \end{bmatrix} \begin{bmatrix} f_x \\ f_y \\ f_z \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{5.166}$$

or

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \begin{bmatrix} -S_1(L_3C_{23}+L_2C_2)f_x + C_1(L_3C_{23}+L_2C_2)f_y \\ -C_1(L_3S_{23}+L_2S_2)f_x - S_1(L_3S_{23}+L_2S_2)f_y + (L_3C_{23}+L_2C_2)f_z \\ -L_3C_1S_{23}f_x - L_3S_1S_{23}f_y + L_3C_{23}f_z \end{bmatrix} \tag{5.167}$$

EXERCISES

5.1 Show that the overall differential transformation due to three differential rotations of δ_x, δ_y, and δ_z about x-, y-, and x-axes, respectively, is independent of the order in which rotations are made. In the case of differential change $\sin\theta \to \delta$ and $\cos\theta \to 1$.

5.2 Show that the three differential rotations of δ_x, δ_y, and δ_z made in any order about the x-, y-, and z-axes, respectively are equivalent to a differential rotation of $d\theta$ about axis K.

5.3 Given a coordinate frame $\{A\}$

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5.168}$$

Find the differential transformation of A corresponding to a differential rotation $\Delta\theta = -0.1\hat{i} + 0\hat{j} + 0\hat{k}$ followed by a differential translation of $\Delta d = 0\hat{i} + 0.5\hat{j} + 1\hat{k}$ with respect to base frame.

5.4 A planar manipulator arm with one rotary and one prismatic joint is shown in Fig. E5.4.

(a) Compute the Jacobian matrix for the arm.

(b) What will be the joint velocities if the endpoint is required to move at a constant velocity along a flat surface?

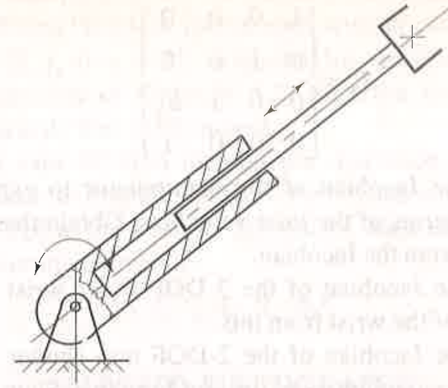


Fig. E5.4 *A 2-DOF planar arm with a rotary and a prismatic joint*

5.5 In Example 5.4, the Jacobian of a 2-DOF manipulator with $L_1 = L_2 = 1$ was obtained. Determine the Jacobian of the 2-DOF RR manipulator with $L_1 \neq L_2$ and show that the significant Jacobian is

$$J = \begin{bmatrix} -L_1S_1 - L_2S_{12} & -L_2S_{12} \\ L_1C_1 + L_2C_{12} & L_2C_{12} \end{bmatrix} \tag{5.169}$$

Determine the singularities and joint velocities of this manipulator.

5.6 Determine (a) Jacobian (b) Singularities and (c) Joint velocities for a 3-DOF planar arm with the revolute joints.

5.7 Compute the Jacobian for the spherical configuration manipulator in Fig. 4.7, Example 4.4.

5.8 Compute the Jacobian for the SCARA manipulator of Fig. 3.22, Example 3.6. Find the singularities for the manipulator.

5.9 Determine the singularities of the spherical configuration manipulator of Fig. 4.7, Example 4.4.

5.10 Find the singularities of the 3R, roll-pitch-yaw wrist in Fig. 3.18.

5.11 For the 6-DOF Stanford manipulator in Fig. 3.26 obtain the Jacobian.

5.12 Determine the singularities of the six degrees of freedom manipulator in Example 3.8.

5.13 For the 5-DOF manipulator shown in Fig. E3.14 compute the Jacobian and determine the singularities, if any.

5.14 Show that the spherical configuration manipulator in Exercise 5.7 has singularities at $q = [\, q_1\ 0\ q_3 \,]^T$. Which axes are collinear?

5.15 For a 3-DOF manipulator arm, the link transformation matrices are:

$${}^0T_1 = \begin{bmatrix} C_1 & -S_1 & 0 & C_1 \\ S_1 & C_1 & 0 & S_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix};\quad {}^1T_2 = \begin{bmatrix} C_2 & 0 & -S_2 & C_2 \\ S_2 & 0 & C_2 & S_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix};$$

$$^2T_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5.170)$$

Determine the Jacobian of the manipulator to express the Cartesian velocities in terms of the joint velocities. Obtain the singularities of the manipulator from the Jacobian.

5.16 Determine the Jacobian of the 3-DOF Euler wrist and determine the singularities of the wrist from this.

5.17 Determine the Jacobian of the 2-DOF arm shown in Fig. E5.17, and determine the singularities of the arm from this. State assumptions made, if any.

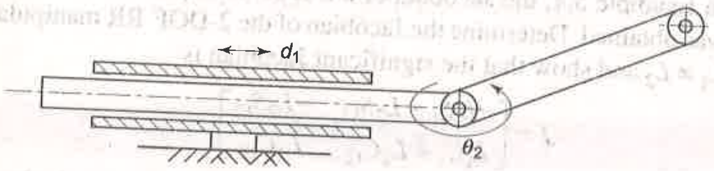


Fig. E5.17 *A 2-DOF planar arm with a rotary and a prismatic joint*

5.18 For the robotic arm shown in Fig. E5.18 obtain the Jacobian to express the Cartesian velocities in terms of joint velocities.

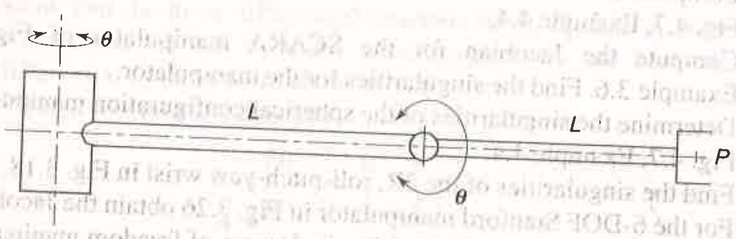


Fig. E5.18 *Two degree of freedom RR manipulator*

5.19 What are the singularities of a manipulator? How are they classified and determined? Explain briefly.

5.20 Jacobian of an n-DOF manipulator is a $6 \times n$ matrix. How can its inverse be determined to compute the kinematic singularities of the manipulator?

5.21 What is a skew symmetric matrix? How this matrix is related to the angular velocity of a link of an n-DOF manipulator?

5.22 How can a skew symmetric matrix be used to determine the velocity of a link if the position of the link, its angular velocity and rotational transformation matrix are known?

5.23 Determine the total derivative of the homogeneous transformation matrix of a link ^iT_j.

5.24 For a given manipulator, are the velocity Jacobian and the static force Jacobian different? Explain your answer with an example.

5.25 A force $\mathcal{F} = [f_x\ f_y\ 0\ 0\ 0\ 0]^T$ is applied by end-effector of the two link planar manipulator of Example 5.2. Find the required joint torques assuming a gravity free environment.

5.26 For the four axes SCARA manipulator discussed in Example 3.6 and whose Jacobian is computed in Exercise 5.8 above, determine the force/torques required at the joints to maintain a static equilibrium when end effector is exerting a force of

$$\mathcal{F} = [f_x\ f_y\ f_x\ \eta_x\ \eta_y\ \eta_z]^T \quad (5.171)$$

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6

Dynamic Modeling

During the work cycle a manipulator must accelerate, move at constant speed, and decelerate. This time-varying position and orientation of the manipulator is termed as its dynamic behaviour. Time-varying torques are applied at the joints (by the joint actuators) to balance out the internal and external forces. The internal forces are caused by motion (velocity and acceleration) of links. Inertial, Coriolis, and frictional forces are some of the internal forces. The external forces are the forces exerted by the environment. These include the "load" and gravitational forces. As a result, links and joints have to withstand stresses caused by force/torque balance across these.

In this chapter, the mathematical model for the dynamic behaviour of the manipulator is developed. The mathematical equations, often referred as *manipulator dynamics*, are a set of *equations of motion* (EOM) that describe the dynamic response of the manipulator to input actuator torques.

The dynamic model of a manipulator is useful for computation of torque and forces required for execution of a typical work cycle, which is vital information for the design of links, joints, drives, and actuators. The dynamic behaviour of the manipulator provides relationship between joint actuator torques and motion of links for simulation and design of control algorithms. The manipulator control maintains the dynamic response of the manipulator to obtain the desired performance, which directly depends on the accuracy of the dynamic model and efficiency of the control algorithms. The control problem requires specifying the control strategies to achieve the desired response and performance. Simulations of manipulator motion permits testing of control strategies, motion planning, and performance studies without a physical prototype of the manipulator.

The serial link manipulator represents a complex dynamic system, which can be modeled by systematically using known physical laws of Lagrangian mechanics or Newtonian mechanics. Approaches such as *Lagrange-Euler* (LE), which is "energy-based", and *Newton-Euler* (NE), based on "force-balance", can be systematically applied to develop the manipulator EOM. Assuming rigid body motion, no backlash, no friction and neglecting effects of control component dynamics, the resulting EOM are a set of second order, coupled, nonlinear differential equations, consisting of inertia loading and coupling reaction forces between joints. Another method, the generalized d'Alembert principle, provides an "equivalent" dynamic model.

The Newton-Euler and Lagrange-Euler formulations of the dynamic model provide a closed-form solution, which are computationally intensive making real-time control based on such a dynamic model, inefficient, if not impossible. To improve the computational speed, recursive methods and approximate models based on simplifying assumptions have been developed. These approximate models, however, result in suboptimal dynamic performance and restrict arm movement to low speeds. The LE and NE models, presented in this chapter, provide a symbolic solution to manipulator dynamics and give insight into the control problem.

6.1 LAGRANGIAN MECHANICS

A scalar function called *Lagrange function* or *Lagrangian* $\mathcal{L}$ is defined as the difference between the total kinetic energy $\mathcal{K}$ and the total potential energy $\mathcal{P}$ of a mechanical system.

$$\mathcal{L} = \mathcal{K} - \mathcal{P} \tag{6.1}$$

The Lagrange-Euler dynamic formulation is based on a set of generalized coordinates to describe the system variables. In the generalized coordinates, generalized displacement '$\boldsymbol{q}$' is used as a joint variable, which describes a linear displacement $\boldsymbol{d}$ for a prismatic joint and angular displacement θ for a rotary joint, and $\dot{q}$ describes linear velocity $\dot{\boldsymbol{d}}(=v)$ and angular velocity $\dot{\theta}(=\omega)$ for prismatic and rotary joints, respectively, as discussed in earlier chapters. Similarly, generalized torque 'τ' required at the joint to produce desired dynamics represents the force f for a prismatic joint and torque τ for a revolute joint, as seen in Eq. (5.78). Because kinetic and potential energies are function of $\boldsymbol{q}_i$ and $\dot{\boldsymbol{q}}_i$ ($i = 1, \ldots \boldsymbol{n}$), so is the Lagrangian $\mathcal{L}$.

The dynamic model based on Lagrange-Euler formulation is obtained from the Lagrangian, as a set of equations,

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i \quad \text{for } i = 1, 2, \ldots, n \tag{6.2}$$

The left-hand side of dynamic equations can be interpreted as sum of the torques/forces due to kinetic and potential energy present in the system. The right-hand side τ_i is the joint torque for joint i that is provided by the actuator i. If $\tau_i = 0$, it means that joint i does not move and if $\tau_i \neq 0$, the manipulator movement is modified by the actuator at joint i.

6.2 TWO DEGREE OF FREEDOM MANIPULATOR—DYNAMIC MODEL

The dynamics of a simple manipulator is worked out to illustrate the Lagrange-Euler formulation and to clarify the problems involved in the dynamic modeling. A planar, 2-DOF manipulator with both rotary joints, as shown in Fig. 6.1, is considered and its dynamic model is obtained using direct geometric approach before discussing the general formulation. For the manipulator, coordinate frames {0} and {1}, joint variables θ_1 and θ_2, link lengths L_1 and L_2, and mass of links m_1 and m_2, respectively, are shown in the figure. The mass of each link is assumed to be a point mass located at the center of mass of each link and links are assumed to be slender members. The linear and angular velocities are v_1, v_2, $\dot{\theta}_1$ and $\dot{\theta}_2$, respectively.

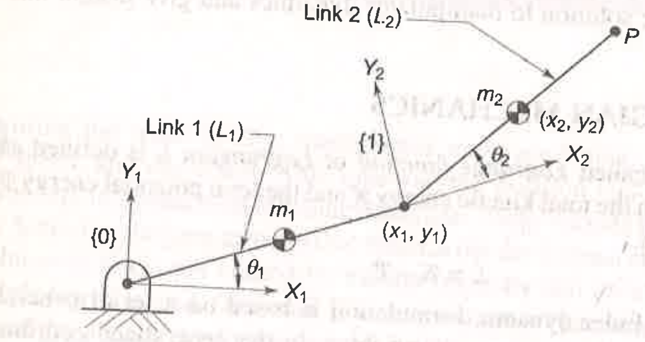

Fig. 6.1 *A 2-DOF planar articulated (RR) arm*

The Lagrangian requires kinetic and potential energies of the manipulator. The kinetic energy of a rigid body (a link), can be expressed as:

$$\mathcal{K} = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 \tag{6.3}$$

where v is the linear velocity, ω is the angular velocity, m is the mass, and I is the moment of inertia of the rigid body at its center of mass.

Thus, the kinetic energy for the link 1 with the linear velocity $v_1 = \frac{1}{2} L_1 \dot{\theta}_1$, angular velocity $\omega_1 = \dot{\theta}_1$, moment of inertia $I_1 = \frac{1}{12} m_1 L_1^2$, and mass m_1 is

$$\mathcal{K}_1 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} I_1 \omega_1^2 = \frac{1}{8} m_1 L_1^2 \dot{\theta}_1^2 + \frac{1}{24} m_1 L_1^2 \dot{\theta}_1^2 = \frac{1}{6} m_1 L_1^2 \dot{\theta}_1^2 \tag{6.4}$$

and its potential energy is

$$\mathcal{P}_1 = \frac{1}{2} m_1 g L_1 \sin \theta_1 \tag{6.5}$$

where g is the magnitude of acceleration due to gravity in the negative y-axis direction.

For the second link, link 2, the Cartesian position coordinates (x_2, y_2) of the center of mass of link are:

$$\begin{aligned} x_2 &= L_1 \cos \theta_1 + \frac{1}{2} L_2 \cos (\theta_1 + \theta_2) \\ y_2 &= L_1 \sin \theta_1 + \frac{1}{2} L_2 \sin (\theta_1 + \theta_2) \end{aligned} \tag{6.6}$$

Differentiating Eq. (6.6) gives the components of velocity of link 2 as

$$\begin{aligned} \dot{x}_2 &= - L_1 \sin \theta_1 \dot{\theta}_1 - \frac{1}{2} L_2 \sin (\theta_1 + \theta_2)(\dot{\theta}_1 + \dot{\theta}_2) \\ \dot{y}_2 &= L_1 \cos \theta_1 \dot{\theta}_1 + \frac{1}{2} L_2 \cos (\theta_1 + \theta_2)(\dot{\theta}_1 + \dot{\theta}_2) \end{aligned} \tag{6.7}$$

From these components, the square of the magnitude of velocity of the end of link 2 is

$$v_2^2 = \dot{x}_2^2 + \dot{y}_2^2$$

or

$$\begin{aligned} v_2^2 = {} & L_1^2 S_1^2 \dot{\theta}_1^2 + \frac{1}{4} L_2^2 S_{12}^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 + L_1 L_2 S_1 S_{12} (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2) \\ & + L_1^2 C_1^2 \dot{\theta}_1^2 + \frac{1}{4} L_2^2 C_{12}^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 + L_1 L_2 C_1 C_{12} (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2) \end{aligned}$$

Simplifying

$$v_2^2 = L_1^2 \dot{\theta}_1^2 + \frac{1}{4} L_2^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 + L_1 L_2 C_2 (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2) \tag{6.8}$$

where $C_i = \cos \theta_i$, $S_i = \sin \theta_i$, $C_{12} = \cos (\theta_1 + \theta_2)$ and $S_{12} = \sin (\theta_1 + \theta_2)$.

Thus, the kinetic energy of link 2 with $\omega_2 = \dot{\theta}_1 + \dot{\theta}_2$ and $I_2 = \frac{1}{12} m_2 L_2^2$ is

$$\begin{aligned} \mathcal{K}_2 &= \frac{1}{2} m_2 v_2^2 + \frac{1}{2} I_2 \omega_2^2 \\ &= \frac{1}{2} m_2 [L_1^2 \dot{\theta}_1^2 + \frac{1}{4} L_2^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 + L_1 L_2 C_2 (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2)] \\ &\quad + \frac{1}{24} m_2 L_2^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \\ &= \frac{1}{2} m_2 L_1^2 \dot{\theta}_1^2 + \frac{1}{6} m_2 L_2^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\dot{\theta}_1 \dot{\theta}_2) + \frac{1}{2} m_2 L_1 L_2 C_2 (\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2) \end{aligned} \tag{6.9}$$

The potential energy of link 2, from Eq. (6.6), is

$$\mathcal{P}_2 = m_2 g L_1 S_1 + \frac{1}{2} m_2 g L_2 S_{12} \tag{6.10}$$

The Lagrangian $\mathcal{L} = \mathcal{K} - \mathcal{P} = \mathcal{K}_1 + \mathcal{K}_2 - \mathcal{P}_1 - \mathcal{P}_2$ is obtained from Eqs. (6.4), (6.5), (6.9), and (6.10). Rearranging and simplifying, the Lagrangian is

$$\mathcal{L} = \frac{1}{2}\left(\frac{1}{3}m_1 + m_2\right)L_1^2\dot{\theta}_1^2 + \frac{1}{6}m_2L_2^2(\dot{\theta}_1{}^2 + \dot{\theta}_2{}^2 + 2\dot{\theta}_1\dot{\theta}_2) + \frac{1}{2}m_2L_1L_2C_2(\dot{\theta}_1^2 + \dot{\theta}_1\dot{\theta}_2) - \left(\frac{1}{2}m_1 + m_2\right)gL_1S_1 - \frac{1}{2}m_2gL_2S_{12} \quad (6.11)$$

The Lagrange-Euler formulation for link 1, Eq. (6.2), gives the torque τ_1 at joint 1 as

$$\tau_1 = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1}\right) - \frac{\partial \mathcal{L}}{\partial \theta_1} \quad (6.12)$$

The Lagrangian in Eq. (6.11) is differentiated wrt θ_1 and $\dot{\theta}_1$ to give

$$\frac{\partial \mathcal{L}}{\partial \theta_1} = -\left(\frac{1}{2}m_1 + m_2\right)g\,L_1C_1 - \frac{1}{2}m_2gL_2C_{12} \quad (6.13)$$

and

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} = \left(\frac{1}{3}m_1 + m_2\right)L_1^2\dot{\theta}_1 + \frac{1}{3}m_2L_2^2(\dot{\theta}_1 + \dot{\theta}_2) + \frac{1}{3}m_2L_1L_2C_2(2\dot{\theta}_1 + \dot{\theta}_2) \quad (6.14)$$

Differentiating Eq. (6.14) wrt time

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1}\right) = \left[\left(\frac{1}{3}m_1 + m_2\right)L_1^2 + \frac{1}{3}m_2L_2^2 + m_2L_1L_2C_2\right]\ddot{\theta}_1 + m_2\left[\frac{1}{3}L_2^2 + \frac{1}{2}L_1L_2C_2\right]\ddot{\theta}_2 - m_2L_1L_2S_2\dot{\theta}_1\dot{\theta}_2 - \frac{1}{2}m_2L_1L_2S_2\dot{\theta}_2^2 \quad (6.15)$$

Substituting the results of Eqs. (6.13) and (6.15) into Eq. (6.12), the torque at joint 1 is obtained as

$$\tau_1 = \left[\left(\frac{1}{3}m_1 + m_2\right)L_1^2 + \frac{1}{3}m_2L_2^2 + m_2L_1L_2C_2\right]\ddot{\theta}_1 + m_2\left[\frac{1}{3}L_2^2 + \frac{1}{2}L_1L_2C_2\right]\ddot{\theta}_2 - m_2L_1L_2S_2\dot{\theta}_1\dot{\theta}_2 - \frac{1}{2}m_2L_1L_2S_2\dot{\theta}_2^2 + \left(\frac{1}{2}m_1 + m_2\right)g\,L_1C_1 + \frac{1}{2}m_2g\,L_2C_{12} \quad (6.16)$$

Similarly, the derivatives of Lagrangian, Eq. (6.11), for joint 2 are

$$\frac{\partial \mathcal{L}}{\partial \theta_2} = -\frac{1}{2}m_2L_1L_2S_2(\dot{\theta}_1^2 + \dot{\theta}_1\dot{\theta}_2) - \frac{1}{2}m_2gL_2C_{12} \quad (6.17)$$

and

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} = \frac{1}{3}m_2L_2^2(\dot{\theta}_1 + \dot{\theta}_2) + \frac{1}{2}m_2L_1L_2C_2\dot{\theta}_1 \quad (6.18)$$

and

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_2}\right) = \left[\frac{1}{3}m_2L_2^2 + \frac{1}{2}m_2L_1L_2C_2\right]\ddot{\theta}_1 + \frac{1}{3}m_2L_2^2\ddot{\theta}_2 - \frac{1}{2}m_2L_1L_2S_2\dot{\theta}_1\dot{\theta}_2 \quad (6.19)$$

Again from Eq (6.12),

$$\tau_2 = m_2\left[\frac{1}{3}L_2^2 + \frac{1}{2}L_1L_2C_2\right]\ddot{\theta}_1 + \frac{1}{3}m_2L_2^2\ddot{\theta}_2 + \frac{1}{2}m_2L_1L_2S_2\dot{\theta}_1^2 + \frac{1}{2}m_2gL_2C_{12} \quad (6.20)$$

Equations (6.16) and (6.20) are the EOM (dynamic model) of the 2-link planar manipulator. Because both the joints are revolute, the generalized torques τ_1 and τ_2 represent the actual joint torques.

Torque equation, Eqs. (6.16) and (6.20) can be written in the generalized form, which will be described in Section 6.4, as

$$\begin{aligned}\tau_1 &= M_{11}\ddot{\theta}_1 + M_{12}\ddot{\theta}_2 + H_1 + G_1\\ \tau_2 &= M_{21}\ddot{\theta}_1 + M_{22}\ddot{\theta}_2 + H_2 + G_2\end{aligned} \quad (6.21)$$

where

$$M_{11} = \left[\left(\frac{1}{3}m_1 + m_2\right)L_1^2 + \frac{1}{3}m_2L_2^2 + m_2L_1L_2C_2\right]$$

$$M_{12} = M_{21} = m_2\left[\frac{1}{3}L_2^2 + \frac{1}{2}L_1L_2C_2\right]$$

$$M_{22} = \frac{1}{3}m_2L_2^2$$

$$H_1 = -m_2L_1L_2S_2\dot{\theta}_1\dot{\theta}_2 - \frac{1}{2}m_2L_1L_2S_2\dot{\theta}_2^2$$

$$H_2 = \frac{1}{2}m_2L_1L_2S_2\dot{\theta}_1^2$$

$$G_1 = \left[\left(\frac{1}{2}m_1 + m_2\right)L_1C_1 + \frac{1}{2}m_2L_2C_{12}\right]g$$

$$G_2 = \frac{1}{2}m_2L_2C_{12}g$$

These coefficients are defined as

M_{ii} = effective inertia,
M_{ij} = effective coupling inertia,
H_i = centrifugal and Coriolis acceleration forces

This direct formulation approach becomes quite cumbersome when a manipulator with more than 2-DOF is analyzed. In the following sections, the derivation of EOM for an n-DOF manipulator, based on homogeneous coordinate transformation matrices is presented.

6.3 LAGRANGE-EULER FORMULATION

The Lagrange-Euler formulation is a systematic procedure for obtaining the dynamic model of an n-DOF manipulator. The n-DOF open kinematic chain serial link manipulator has n joint position or displacement variables, $q = [q_1, \ldots, q_n]^T$. The LE formulation [Eq. (6.2)] establishes the relation between the joint positions, velocities, accelerations, and the generalized torques applied to the manipulator. The generalized torques are the nonconservative torques contributed by joint actuators, joint friction forces, and induced joint torques. The induced joint torques are the torques at the joints due to contact or interaction of the end-effector with the environment. In the present discussion only joint actuator torques are considered.

The derivation of EOM using LE formulation is carried out in the following subsections. It makes use of link transformation matrices T, which are obtained from the kinematic modeling discussed in Chapter 3. First, the link velocity is computed and next the link inertia tensor is obtained. These are used to compute kinetic energy. Then potential energy is calculated and next, the Lagrangian is formed, which is substituted in Eq. (6.2) to get the dynamic model.

6.3.1 VELOCITY OF A POINT ON THE MANIPULATOR

For computing the kinetic energy of a link of an n-DOF manipulator, link velocity is required. The concepts of Chapter 5 are extended to compute the velocity of each link.

Consider a point P on link i of an n-link (n-DOF) manipulator, as shown in Fig. 6.2. The coordinate frames, frame {0}, frame {i–1}, and frame {i} are chosen as per convention. The position vector ${}^i r_i$ describes the point P on the link with respect to frame {i}. In homogeneous coordinate notation,

$$ {}^i r_i = [x_i \quad y_i \quad z_i \quad 1]^T \tag{6.22}$$

The position of point P with respect to base coordinate is

$$ {}^0 r_i = {}^0T_i \, {}^i r_i = \left({}^0T_1 \, {}^1T_2 \cdots {}^{i-1}T_i\right) {}^i r_i \tag{6.23}$$

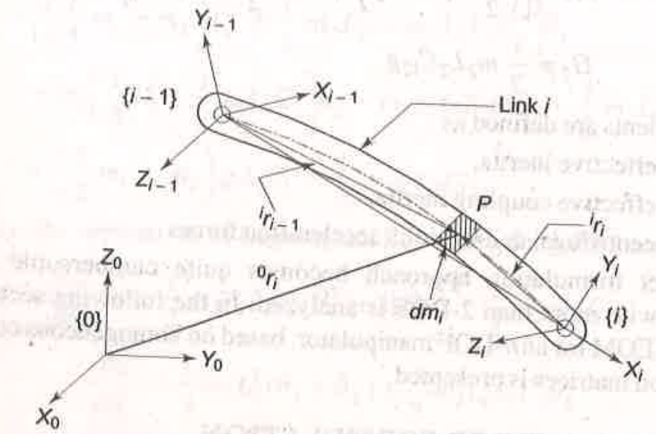

Fig. 6.2 *Velocity of a point on the link*

where ${}^{i-1}T_i$ is given by Eq. (3.3) with $q_i = \theta_i$ for a rotary joint or $q_i = d_i$ for a prismatic joint. The velocity of point P with respect to base coordinates, frame {0}, is obtained from Eq. (5.46), as

$$ {}^0 v_i \equiv v_i = {}^0\dot{r}_i = \left[\sum_{j=1}^{i} \frac{\partial {}^0T_i}{\partial q_j}\dot{q}_j\right] {}^i r_i \tag{6.24}$$

where the fact that ${}^i\dot{r}_i = 0$ is used.

The transformation matrix ${}^0T_i$ involves complex trigonometric terms and its partial derivative with respect to q_j, required in Eq. (6.24), involves complex computation. The following steps simplify the computation of partial derivative of the homogeneous transformation matrix.

Consider the transformation matrix ${}^{j-1}T_j$ for link j given by Eq. (3.4) for a rotary joint, that is,

$$ {}^{j-1}T_j = \begin{bmatrix} C\theta_j & -S\theta_j C\alpha_j & S\theta_j S\alpha_j & a_j C\theta_j \\ S\theta_j & C\theta_j C\alpha_j & -C\theta_j S\alpha_j & a_j S\theta_j \\ 0 & S\alpha_j & C\alpha_j & d_j \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.25}$$

where a_j, d_j, α_j, and θ_j have the usual meaning. The partial derivative of Eq. (6.25) with respect to θ_j is

$$ \frac{\partial {}^{j-1}T_j}{\partial \theta_j} = \begin{bmatrix} -S\theta_j & -C\theta_j C\alpha_j & C\theta_j S\alpha_j & -a_j S\theta_j \\ C\theta_j & -S\theta_j C\alpha_j & S\theta_j S\alpha_j & a_j C\theta_j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.26}$$

Comparison of Eqs. (6.25) and (6.26) gives a pattern. It is observed that Eq. (6.26) can be obtained from Eq. (6.25) by

- interchanging row 1 with row 2,
- changing the sign of row 1, and
- making row 3 and row 4 zero.

Hence, partial derivative of ${}^{j-1}T_j$ with respect to θ_j can be obtained using the above steps and without actually differentiating the terms. The same result can be obtained using matrix operations, which is more convenient in performing the operations using a computer. Mathematically these steps can be carried out using a 4×4 matrix Q_j defined as

$$ Q_j = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{(for revolute joint)} \tag{6.27}$$

and premultiplying ${}^{j-1}T_j$ with Q_j, that is,

$$Q_j\,{}^{j-1}T_j = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} C\theta_j & -S\theta_j C\alpha_j & S\theta_j S\alpha_j & a_j C\theta_j \\ S\theta_j & C\theta_j C\alpha_j & -C\theta_j S\alpha_j & a_j S\theta_j \\ 0 & S\alpha_j & C\alpha_j & d_j \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{or} \quad Q_j\,{}^{j-1}T_j = \begin{bmatrix} -S\theta_j & -C\theta_j C\alpha_j & C\theta_j S\alpha_j & -a_j S\theta_j \\ C\theta_j & -S\theta_j C\alpha_j & S\theta_j S\alpha_j & a_j C\theta_j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.28}$$

The result is same as Eq. (6.26). Hence,

$$\frac{\partial\,{}^{j-1}T_j}{\partial \theta_j} = Q_j\,{}^{j-1}T_j \tag{6.29}$$

The partial derivative equation, Eq. (6.29) also applies for a prismatic joint, with Q_j defined as

$$Q_j = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{(for prismatic joint)} \tag{6.30}$$

The reader may verify Eq. (6.29) with Q_j in Eq. (6.30) for a prismatic joint. Since ${}^0T_i = {}^0T_1\,{}^1T_2 \cdots {}^{i-1}T_i$, its partial derivative ${}^0T_i$ with respect to q_j is

$$\frac{\partial({}^0T_i)}{\partial q_j} = {}^0T_1\,{}^1T_2 \cdots {}^{j-2}T_{j-1}\,\frac{\partial({}^{j-1}T_j)}{\partial q_j}\,{}^jT_{j+1} \cdots {}^{i-1}T_i$$

Using Eq. (6.29), this simplifies to

$$\frac{\partial({}^0T_i)}{\partial q_j} = {}^0T_1\,{}^1T_2 \cdots {}^{j-2}T_{j-1}\,(Q_j\,{}^{j-1}T_j)\,{}^jT_{j+1} \cdots {}^{i-1}T_i = {}^0T_{j-1}\,Q_j\,{}^{j-1}T_i \tag{6.31}$$

This result is valid only for $j \le i$. Hence, for $i = 1, 2, \ldots, n$,

$$\frac{\partial({}^0T_i)}{\partial q_j} = \begin{cases} {}^0T_{j-1}\,Q_j\,{}^{j-1}T_i & \text{for } j \le i \\ 0 & \text{for } j > i \end{cases} \tag{6.32}$$

It is worth noting that the partial derivative of homogeneous transformation matrix ${}^{i-1}T_i$ with respect to q_j represents the effect of motion of joint j on link i. The link velocity v_i in Eq. (6.24) is, thus, simplified using Eq. (6.32) as

$$v_i = \sum_{j=1}^{i} {}^0T_{j-1}\,Q_j\,{}^{j-1}T_i\,\dot{q}_j\,{}^ir_i \tag{6.33}$$

6.3.2 The Inertia Tensor

The mass of the link contributes inertia forces during motion of the link. The mass properties, which reflect all the inertial loads with respect to rotations about the origin of frame of interest, are represented by a *moment of inertia tensor*. It is a 4×4 symmetric matrix, which characterizes the distribution of mass of a rigid body (the link i). The moment of inertia tensor is defined as

$$I_i = \begin{bmatrix} \int x_i^2\,dm_i & \int x_i y_i\,dm_i & \int x_i z_i\,dm_i & \int x_i\,dm_i \\ \int x_i y_i\,dm_i & \int y_i^2\,dm_i & \int y_i z_i\,dm_i & \int y_i\,dm_i \\ \int x_i z_i\,dm_i & \int y_i z_i\,dm_i & \int z_i^2\,dm_i & \int z_i\,dm_i \\ \int x_i\,dm_i & \int y_i\,dm_i & \int z_i\,dm_i & \int dm_i \end{bmatrix} \tag{6.34}$$

where dm_i is the mass of the element on link i located at ${}^ir_i = [x_i \;\; y_i \;\; z_i \;\; 1]^T$, as shown in Fig. 6.2. Using the moment of inertia, I_{kk}, cross product of inertia, $I_{kl}\,(k \neq l)$ and first moments of body, the inertia tensor in Eq. (6.34) I_i is expressed as (see Appendix C for details).

$$I_i = \begin{bmatrix} \frac{1}{2}(-I_{xx} + I_{yy} + I_{zz}) & I_{xy} & I_{xz} & m_i\bar{x}_i \\ I_{xy} & \frac{1}{2}(I_{xx} - I_{yy} + I_{zz}) & I_{yz} & m_i\bar{y}_i \\ I_{xz} & I_{yz} & \frac{1}{2}(I_{xx} + I_{yy} - I_{zz}) & m_i\bar{z}_i \\ m_i\bar{x}_i & m_i\bar{y}_i & m_i\bar{z}_i & m_i \end{bmatrix} \tag{6.35}$$

where m_i is the mass of the link i and ${}^i\bar{r}_i = [\bar{x}_i \;\; \bar{y}_i \;\; \bar{z}_i \;\; 1]^T$ is its center of mass. The moment of inertia tensor I_i for link i depends on the mass distribution of the link and not on its position or rate of change of position.

6.3.3 The Kinetic Energy

The kinetic energy of the differential mass dm_i on link i, for $i = 1, 2, \ldots, n$, located at ${}^0r_i$ and moving with velocity ${}^0v_i\,(= v_i)$ with respect to the base frame {0} is

$$d\mathcal{K}_i = \frac{1}{2}\,dm_i (v_i)^2 \tag{6.36}$$

The trace operator $\left(\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{ii}\right)$ is used to obtain $(v_i)^2$ as,

$$v_i^2 = v_i \cdot v_i = {}^0\dot{r}_i \cdot {}^0\dot{r}_i = \mathrm{Tr}({}^0\dot{r}_i\,{}^0\dot{r}_i^T) = \mathrm{Tr}\,(v_i v_i^T) \tag{6.37}$$

Substituting v_i from Eq. (6.33) in Eq. (6.37) and the result in Eq. (6.36), the kinetic energy of the differential mass is obtained as

$$d\mathcal{K}_i = \frac{1}{2}\,\mathrm{Tr}\left[\left(\sum_{j=1}^{i} {}^0T_{j-1}\,Q_j\,{}^{j-1}T_i\,\dot{q}_j\,{}^ir_i\right)\left(\sum_{k=1}^{i} {}^0T_{k-1}\,Q_k\,{}^{k-1}T_i\,\dot{q}_k\,{}^ir_i\right)^T\right] dm_i$$

or $$d\mathcal{K}_i = \frac{1}{2}\mathrm{Tr}\left[\sum_{j=1}^{i}\sum_{k=1}^{i}\left({}^0T_{j-1}Q_j\,{}^{j-1}T_i\right){}^i\bar{r}_i\,{}^i\bar{r}_i^T dm_i\left({}^0T_{k-1}Q_k\,{}^{k-1}T_i\right)^T\dot{q}_j\dot{q}_k\right] \tag{6.38}$$

The total kinetic energy of link i is then

$$\mathcal{K}_i = \int d\mathcal{K}_i$$

$$\mathcal{K}_i = \frac{1}{2}\mathrm{Tr}\left[\sum_{j=1}^{i}\sum_{k=1}^{i}\left({}^0T_{j-1}Q_j\,{}^{j-1}T_i\right)\int{}^i\bar{r}_i\,{}^i\bar{r}_i^T dm_i\left({}^0T_{k-1}Q_k\,{}^{k-1}T_i\right)^T\dot{q}_j\dot{q}_k\right] \tag{6.39}$$

The integral term $\int {}^i\bar{r}_i\,{}^i\bar{r}_i^T dm_i$ in Eq. (6.39) is the moment of inertia tensor I_i given by Eq. (6.35). Therefore, $\mathcal{K}_i$ is

$$\mathcal{K}_i = \frac{1}{2}\mathrm{Tr}\left[\sum_{j=1}^{i}\sum_{k=1}^{i}\left({}^0T_{j-1}Q_j\,{}^{j-1}T_i\right)I_i\left({}^0T_{k-1}Q_k\,{}^{k-1}T_i\right)^T\dot{q}_j\dot{q}_k\right] \tag{6.40}$$

Thus, for n-DOF manipulator, total kinetic energy of manipulator is

$$\mathcal{K} = \sum_{i=1}^{n}\mathcal{K}_i = \frac{1}{2}\sum_{i=1}^{n}\mathrm{Tr}\left[\sum_{j=1}^{i}\sum_{k=1}^{i}\left({}^0T_{j-1}Q_j\,{}^{j-1}T_i\right)I_i\left({}^0T_{k-1}Q_k\,{}^{k-1}T_i\right)^T\dot{q}_j\dot{q}_k\right]$$

Exchanging the trace and sum operations

$$\mathcal{K} = \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{i}\mathrm{Tr}\left[\left({}^0T_{j-1}Q_j\,{}^{j-1}T_i\right)I_i\left({}^0T_{k-1}Q_k\,{}^{k-1}T_i\right)^T\right]\dot{q}_j\dot{q}_k \tag{6.41}$$

Note that the kinetic energy $\mathcal{K}$ of the manipulator is a scalar and is a function of joint position and velocity $(\boldsymbol{q}, \dot{\boldsymbol{q}})$.

6.3.4 The Potential Energy

The potential energy $\mathcal{P}_i$ of link i in a gravity field g is

$$\mathcal{P}_i = -m_i\,g({}^0\bar{r}_i) = -m_i\,g\,{}^0T_i\,{}^i\bar{r}_i \tag{6.42}$$

where vectors ${}^0\bar{r}_i$ and ${}^i\bar{r}_i$ represent the center of mass of link i with respect to frame $\{0\}$ and with respect to frame $\{i\}$, respectively, and the acceleration due to gravity $g = \begin{bmatrix} g_x & g_y & g_z & 0\end{bmatrix}^T$ is the 4×1 gravity vector with respect to base frame $\{0\}$. The negative sign indicates that work is done on the system to raise link i against gravity. The total potential energy of the manipulator is sum of the potential energy of the links, that is,

$$\mathcal{P} = \sum_{i=1}^{n}\mathcal{P}_i = -\sum_{i=1}^{n}m_i\,g\,{}^0T_i\,{}^i\bar{r}_i \tag{6.43}$$

Because ${}^0\bar{r}_i$ is a function of q_i, the potential energy of a manipulator is described by a scalar formula as a function of joint displacements $\boldsymbol{q}$.

6.3.5 Equations of Motion

The Lagrangian, $\mathcal{L} = \mathcal{K} - \mathcal{P}$, from Eqs. (6.41) and (6.43), is given by

$$\mathcal{L} = \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{i}\mathrm{Tr}\left[\left({}^0T_{j-1}Q_j\,{}^{j-1}T_i\right)I_i\left({}^0T_{k-1}Q_k\,{}^{k-1}T_i\right)^T\right]\dot{q}_j\dot{q}_k + \sum_{i=1}^{n}m_i\,g\,{}^0T_i\,{}^i\bar{r}_i \tag{6.44}$$

According to the Lagrange-Euler dynamic formulation, the generalized torque τ_i of the actuator at joint i, to drive link i of the manipulator, is given by Eq. (6.2)

$$\tau_i = \frac{d}{dt}\left(\frac{\delta\mathcal{L}}{\delta\dot{q}_i}\right) - \frac{\delta\mathcal{L}}{\delta q_i}$$

By substituting $\mathcal{L}$ and carrying out the differentiation, the generalized torque τ_i applied to link i of n-DOF manipulator is obtained. The detailed derivations are carried out in Appendix D. The final EOM (dynamic model) is

$$\tau_i = \sum_{j=1}^{n}M_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n}\sum_{k=1}^{n}h_{ijk}\,\dot{q}_j\dot{q}_k + G_i \quad \text{for } i = 1, 2, \ldots, n \tag{6.45}$$

where

$$M_{ij} = \sum_{p=\max(i,j)}^{n}\mathrm{Tr}\left[d_{pj}I_p d_{pi}^T\right] \tag{6.46}$$

$$h_{ijk} = \sum_{p=\max(i,j,k)}^{n}\mathrm{Tr}\left[\frac{\partial(d_{pk})}{\partial q_p}I_p d_{pi}^T\right] \tag{6.47}$$

$$G_i = -\sum_{p=i}^{n}m_p\,g\,d_{pi}\,{}^p\bar{r}_p \tag{6.48}$$

and $$d_{ij} = \begin{cases} {}^0T_{j-1}Q_j\,{}^{j-1}T_i & \text{for } j \le i \\ 0 & \text{for } j > i \end{cases} \tag{6.49}$$

and $$\frac{\partial d_{ij}}{\partial q_k} = \begin{cases} {}^0T_{j-1}Q_j\,{}^{j-1}T_{k-1}Q_k\,{}^{k-1}T_i & \text{for } i \ge k \ge j \\ {}^0T_{k-1}Q_k\,{}^{k-1}T_{j-1}Q_j\,{}^{j-1}T_i & \text{for } i \ge j \ge k \\ 0 & \text{for } i < j \text{ or } i < k \end{cases} \tag{6.50}$$

Equation (6.45) is the dynamic model of the manipulator and gives a set of n nonlinear, coupled, second order ordinary differential equations for n-links of the n-DOF manipulator. These equations are the equations of motion or the dynamic equations of motion for the manipulator. Note that the mass of the payload would also contribute to the inertia and its position is continuously changing during the work cycle. This inertia is not included in the above EOM.

The physical meaning of various terms in Eqs. (6.45) to (6.49) is described as follows.

1. The coefficients of the $\ddot{q}_j$ terms in these equations represent inertia. It is known as effective inertia when acceleration of joint i causes a torque at joint i, and coupling inertia when acceleration at joint j causes a torque at joint i. In other words, the coefficient M_{ij} is related to acceleration of the joint and represent inertia loading of the actuator. In summary
   - M_{ii} = effective inertia at joint i where the driving torque τ_i acts.
   - M_{ij} = coupling inertia between joint i and joint j. It is the reaction torque $M_{ij}\ddot{q}_j$ at joint i induced by acceleration at joint j. Reverse applies equally with torque $M_{ji}\ddot{q}_j$ as torque at joint j due to acceleration of joint i.
   - Since $\text{Tr}(A) = \text{Tr}(A^T)$, it can be shown that $M_{ij} = M_{ji}$.
2. The coefficient h_{ijk} represents the velocity induced reaction torque at joint i, the first index. The indices j and k are related to velocities of joints j and joint k, whose dynamic interplay induces a reaction torque at joint i. In particular, a term of the form $h_{ijj}\dot{q}_j^2$ is the centrifugal force acting at joint i due to velocity at joint j and a term of the form $h_{ijk}\,\dot{q}_j\,\dot{q}_k$ is known as the Coriolis force acting at joint i due to velocities at joint j and k. In particular,
   - h_{ijk} = Coriolis force coefficient generated by the velocities of joint j and joint k and "felt" at joint i. Coriolis force acting at joint i due to velocities at joint j and joint k is a combination term of $h_{ijk}\,\dot{q}_j\,\dot{q}_k$ + $h_{ijk}\,\dot{q}_k\,\dot{q}_j$.
   - h_{ijj} = Centrifugal force coefficient at joint i generated due to angular velocity at joint j. The centrifugal force acting at joint i due to velocity at joint j is given by $h_{ijj}\dot{q}_j^2$.
   - $h_{ijk} = h_{ikj}$ because the Coriolis force acts at joint i due to velocities of joints j and k and suffix order does not matter.
   - $h_{iii} = 0$, Coriolis force at joint i is not due to joint velocity itself
3. The terms involving gravity g represent the gravity generated moment at joint i. The coefficient G_i is the gravity loading force at joint i due to the links i to n. The gravity term is a function of the current position.
4. τ_i = generalized force applied at joint i due to motion of links.
5. q_i = joint displacement for joint i
6. $\dot{q}_i$ = velocity of joint i
7. $\ddot{q}_i$ = acceleration of joint i
8. The terms d_{ij} determine the rate of change of points ${}^i r_i$ on link i, that is, the position and orientation of frame $\{i\}$ relative to the base coordinate frame as q_j changes.

In dynamic model equations [Eq. (6.45)], inertial and gravity terms are significant in manipulator control as they affect positioning accuracy and servo

stability, which in turn determine the repeatability of the manipulator. The Coriolis and centrifugal forces are significant for high-speed motion of the manipulator.

The Lagrange-Euler formulation discussed above has the following characteristics:

- It is systematic and describes motion in real physical terms.
- The equations of motion obtained are analytical and compact.
- The matrix-vector form of equations (inertia matrix, centrifugal, and Coriolis force matrix, and gravitational force vector) is appealing for calculations and control system design.
- The control problem can be simplified by designing the structure of the manipulator with minimal joint couplings, that is, coefficients M_{ij} and h_{ijk} may be reduced or eliminated.
- The model is computationally intensive and is not amenable to online control.

An algorithm to get the dynamic model of any n-DOF manipulator using LE formulation is described in the next section.

6.3.6 The LE Dynamic Model Algorithm

The Lagrange-Euler formulation to derive the closed-form equations of motion (dynamic model) of a manipulator can be summarized in the form of an algorithm.

Algorithm 6.1 ➢ LE Formulation of Dynamic Equations

This algorithm carries out the complete dynamic formulation of an n-DOF manipulator that satisfies the condition for existence of closed-form geometric solutions. The various steps are

Step 1 Assign frames $\{0\}, \ldots, \{n\}$ using DH notation (Algorithm 3.1) such that frame $\{i\}$ is oriented (aligned) with principal axis of link i.

Step 2 Obtain the link transformation matrix ${}^{i-1}T_i$ for each link and from these compute product matrices ${}^0T_2$, ${}^0T_3$ and so on, which are required for computing the coefficients d_{ij} and its derivatives, using

$$ {}^jT_k = \prod_{p=j+1}^{k} {}^{p-1}T_p \quad \text{for } j = 0, 1, 2, \ldots, n-2;\; k = j+2, \ldots, n \tag{6.51} $$

Step 3 Define Q_i for each link using Eq. (6.27) or Eq. (6.30), depending on whether the joint is revolute or prismatic.

Step 4 For each link i determine the inertia tensor I_i with respect to the frame $\{i\}$ (see Appendix C).

Step 5 Compute d_{ij} for $i, j = 1, 2, \ldots, n$ using Eq. (6.49).

Step 6 Compute the inertia coefficients M_{ij} for $i, j = 1, 2, \ldots n$ using Eq. (6.46).

Step 7 Compute the velocity coupling coefficients h_{ijk} for $i, j, k = 1, 2, \ldots, n$ using Eqs. (6.47) and (6.50).

Step 8 Compute gravity loading terms G_i for each link, $i = 1, 2, \ldots, n$ using Eq. (6.48).

Step 9 Substitute all the coefficients in Eq. (6.45) to formulate the i^{th} equation for torque τ_i.

An example is worked out using Algorithm 6.1 to get the dynamic model of a two-link planar manipulator.

Example 6.1 ***EOM for 2-DOF planar manipulator using LE formulation***

Determine the equations of motion (dynamic model) for the 2-DOF RR-planar manipulator arm using the Lagrange-Euler formulation. Assume both links have equal length ($L_1 = L_2 = L$) and have equal mass ($m_1 = m_2 = m$). Assume further that the links are slender members with a uniform mass distribution, that is, the center of mass of each link is located at the mid point of the link.

Solution The dynamic model of the two link, 2-DOF planar manipulator was developed in Section 6.3 using a direct approach. The manipulator is shown in Fig. 6.1.

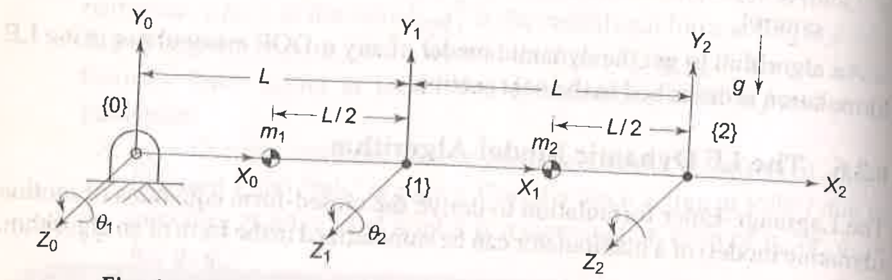

Fig. 6.3 *Frame assignments for two-link, planar RR manipulator*

The LE formulation begins with the determination of kinematic model. Applying step 1 of Algorithm 6.1, the frame assignment is carried out as shown in Fig. 6.3 and the joint-link parameters are tabulated in Table 6.1.

Table 6.1 *Joint-link parameters for RR planar manipulator*

Link i	a_i	α_i	d_i	θ_i	$C\alpha_i$	$S\alpha_i$
1	L	0	0	θ_1	1	0
2	L	0	0	θ_2	1	0

The link-transformation matrices and the required products of transformation matrices are obtained as per step 2. These are given by Eqs. (6.52) and (6.53).

$$ {}^0T_1 = \begin{bmatrix} C_1 & -S_1 & 0 & LC_1 \\ S_1 & C_1 & 0 & LS_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1T_2 = \begin{bmatrix} C_2 & -S_2 & 0 & LC_2 \\ S_2 & C_2 & 0 & LS_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.52}$$

$$ {}^0T_2 = \begin{bmatrix} C_{12} & -S_{12} & 0 & L(C_{12}+C_1) \\ S_{12} & C_{12} & 0 & L(S_{12}+S_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.53}$$

where $C_{12} = \cos(\theta_1 + \theta_2)$ and $S_{12} = \sin(\theta_1 + \theta_2)$. Applying step 3, Q_i matrices for rotary joints 1 and 2, from Eq. (6.27), are

$$ Q_1 = Q_2 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.54}$$

Next step (step 4) requires computation of inertia tensors. The inertia tensors I_1 and I_2 for two slender links of length with mass $m_1 = m_2 = m$ at the centroid of the link and $L_1 = L_2 = L$, with respect to frame $\{i\}$, $i = 1, 2$, are computed using Table C.1 in Appendix C as:

$$ I_1 = I_2 = \begin{bmatrix} \frac{1}{3}mL^2 & 0 & 0 & -\frac{1}{2}mL \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{2}mL & 0 & 0 & m \end{bmatrix} \tag{6.55}$$

Step 5 of Algorithm 6.1 requires the computation of matrices d_{ij}, which are required to compute all other coefficients in Eqs (6.46) – (6.48). According to Eq. (6.49)

$$ d_{ij} = \begin{cases} {}^0T_{j-1}\, Q_j\, {}^{j-1}T_i & \text{for } j \le i \\ 0 & \text{for } j > i \end{cases}$$

For $i, j = 1, 2$, the four d_{ij} matries are: d_{11}, d_{12}, d_{21}, and d_{22}. These are computed now. For $i = j = 1$, d_{11} is computed as:

$$ d_{11} = {}^0T_0\, Q_1\, {}^0T_1 = Q_1\, {}^0T_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} C_1 & -S_1 & 0 & LC_1 \\ S_1 & C_1 & 0 & LS_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note that ${}^0T_0$ is the identity matrix.

or

$$ d_{11} = \begin{bmatrix} -S_1 & -C_1 & 0 & -LS_1 \\ C_1 & -S_1 & 0 & LC_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.56}$$

Similarly, $\quad d_{12} = 0$ (because $j > i$) $\quad$ (6.57)

$$d_{21} = {}^0T_0\, Q_1\, {}^0T_2 = \begin{bmatrix} -S_{12} & -C_{12} & 0 & -L(S_{12}+S_1) \\ C_{12} & -S_{12} & 0 & L(C_{12}+C_1) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.58}$$

Finally,

$$d_{22} = {}^0T_1\, Q_2\, {}^1T_2 = \begin{bmatrix} -S_{12} & -C_{12} & 0 & -LS_{12} \\ C_{12} & -S_{12} & 0 & LC_{12} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.59}$$

Step 6 is applied to compute the elements of inertia matrix $M(M_{ij})$ using Eq. (6.46), that is

$$M_{ij} = \sum_{p=\max(i,j)}^{n} \mathrm{Tr}\left[d_{pj} I_p d_{pi}^T \right]$$

Using the d_{ij} and I_i from Eqs (6.55) to (6.59), the effective inertia coefficients M_{11} and M_{12} are computed, first, as:

$$M_{11} = \mathrm{Tr}(d_{11}\, I_1\, d_{11}^T) + \mathrm{Tr}(d_{21}\, I_2\, d_{21}^T) \tag{6.60}$$

Now,

$$d_{11} I_1 d_{11}^T = \begin{bmatrix} -S_1 & -C_1 & 0 & -LS_1 \\ C_1 & -S_1 & 0 & LC_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{3}mL^2 & 0 & 0 & -\frac{1}{2}mL \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{2}mL & 0 & 0 & m \end{bmatrix} \begin{bmatrix} -S_1 & C_1 & 0 & 0 \\ -C_1 & -S_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -LS_1 & LC_1 & 0 & 0 \end{bmatrix}$$

or

$$d_{11}\, I_1\, d_{11}^T = \begin{bmatrix} \frac{1}{3}mL^2S_1^2 & -\frac{1}{3}mL^2S_1C_1 & 0 & 0 \\ -\frac{1}{3}mL^2S_1C_1 & \frac{1}{3}mL^2C_1^2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.61}$$

Thus,

$$\mathrm{Tr}\,(d_{11}\, I_1\, d_{11}^T) = \frac{1}{3}mL^2(S_1^2 + C_1^2) = \frac{1}{3}mL^2 \tag{6.62}$$

and

$$d_{21}\, I_2\, d_{21}^T = \begin{bmatrix} -S_{12} & -C_{12} & 0 & -L(S_{12}+S_1) \\ C_{12} & -S_{12} & 0 & L(C_{12}+C_1) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{3}mL^2 & 0 & 0 & -\frac{1}{2}mL \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{2}mL & 0 & 0 & m \end{bmatrix} d_{21}^T$$

$$= \begin{bmatrix} mL^2(\frac{1}{6}S_{12} + \frac{1}{2}S_1) & 0 & 0 & -mL(\frac{1}{2}S_{12} + S_1) \\ -mL^2(\frac{1}{6}C_{12} + \frac{1}{6}C_1) & 0 & 0 & mL(\frac{1}{2}C_{12} + C_1) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -S_{12} & C_{12} & 0 & 0 \\ -C_{12} & -S_{12} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -L(S_{12}+S_1) & L(C_{12}+C_1) & 0 & 0 \end{bmatrix}$$

$$= mL^2 \begin{bmatrix} \frac{1}{3}S_{12}^2 + S_1S_{12} + S_1^2 & \begin{matrix} -(\frac{1}{2}S_{12}C_{12} + \frac{1}{2}C_1S_{12} \\ + \frac{1}{2}S_1C_{12} + S_1C_1) \end{matrix} & 0 & 0 \\ \begin{matrix} -(\frac{1}{3}S_{12}C_{12} + \frac{1}{2}C_1S_{12} \\ + \frac{1}{2}S_1C_{12} + S_1C_1) \end{matrix} & \frac{1}{3}C_{12}^2 + C_1C_{12} + C_1^2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.63}$$

Thus,

$$\mathrm{Tr}(d_{21}\, I_2\, d_{21}^T) = mL^2\left(\frac{4}{3} + C_2\right) \tag{6.64}$$

Substituting Eq. (6.62) and Eq. (6.64) in Eq. (6.60) and simplifying gives

$$M_{11} = \frac{5}{3}mL^2 + mL^2C_2 \tag{6.65}$$

Similarly,

$$M_{22} = \mathrm{Tr}(d_{22}\, I_2\, d_{22}^T) = \frac{1}{3}mL^2 \tag{6.66}$$

The coupling inertia coefficients are computed as

$$M_{12} = M_{21} = \mathrm{Tr}(d_{22}\, I_2\, d_{21}^T) = \frac{1}{3}mL^2 + \frac{1}{2}mL^2C_2 \tag{6.67}$$

From the inertia coefficients obtained above, the inertia matrix for the manipulator is

$$M(\theta) = \begin{bmatrix} mL^2(\frac{5}{3} + C_2) & mL^2(\frac{1}{3} + \frac{1}{2}C_2) \\ mL^2(\frac{1}{3} + \frac{1}{2}C_2) & \frac{1}{3}mL^2 \end{bmatrix} \tag{6.68}$$

In step 7, the Coriolis and centrifugal force coefficients (or velocity coupling coefficients), h_{ijk} for $i, j, k = 1, 2$ are obtained using Eq. (6.47), that is

$$h_{ijk} = \sum_{p=\max(i,j,k)}^{n} \mathrm{Tr}\left[\frac{\partial(d_{pk})}{\partial q_p} I_p d_{pi}^T \right]$$

From Eqs. (6.50) and (6.55), the centrifugal acceleration coefficients are

$$h_{111} = 0$$

$$h_{122} = \mathrm{Tr}\left[\frac{\partial(d_{22})}{\partial q_2} I_2 d_{21}^T \right]$$

$$= \mathrm{Tr}\,({}^0T_1\, Q_2\, {}^1T_1\, Q_2\, {}^1T_2\, I_2\, d_{21}^T) = -\frac{1}{2}mL^2S_2 \tag{6.69}$$

$$h_{211} = \mathrm{Tr}\,({}^0T_0 Q_1\, {}^0T_0 Q_1\, {}^0T_2\, I_2\, d_{22}^T) = \frac{1}{2}mL^2S_2$$

$$h_{222} = 0$$

and the Coriolis acceleration coefficients are

$$h_{112} = h_{121} = \mathrm{Tr}\,({}^0T_0\,Q_1\,{}^0T_1\,Q_2\,{}^1T_2\,I_2\,d_{21}^T) = -\frac{1}{2}mL^2S_2$$

$$h_{212} = h_{221} = \mathrm{Tr}\,({}^0T_0\,Q_1\,{}^0T_1\,Q_2\,{}^1T_2\,I_2\,d_{22}^T) = 0 \tag{6.70}$$

The Coriolis and centrifugal coefficient terms are computed using the series summation.

For $i = 1$, $H_1 = \sum_{j=1}^{2}\sum_{k=1}^{2} h_{1jk}\dot\theta_j\dot\theta_k = h_{111}\dot\theta_1^2 + h_{112}\dot\theta_1\dot\theta_2 + h_{121}\dot\theta_1\dot\theta_2 + h_{122}\dot\theta_2^2$

$$= -mL^2S_2\dot\theta_1\dot\theta_2 - \frac{1}{2}mL^2S_2\dot\theta_2^2 \tag{6.71}$$

and for $i = 2$, $H_2 = \sum_{j=1}^{2}\sum_{k=1}^{2} h_{2jk}\dot\theta_j\dot\theta_k = h_{211}\dot\theta_1^2 + h_{212}\dot\theta_1\dot\theta_2 + h_{221}\dot\theta_1\dot\theta_2 + h_{222}\dot\theta_2^2$

$$= \frac{1}{2}mL^2S_2\dot\theta_1^2 \tag{6.72}$$

Thus, the Coriolis and centrifugal coefficient matrix $\boldsymbol{H}$ is, therefore,

$$\boldsymbol{H}(\theta,\dot\theta) = \begin{bmatrix} H_1 \\ H_2 \end{bmatrix} = \begin{bmatrix} -mL^2S_2\dot\theta_1\dot\theta_2 - \frac{1}{2}mL^2S_2\dot\theta_2^2 \\ \frac{1}{2}mL^2S_2\dot\theta_1^2 \end{bmatrix} \tag{6.73}$$

Step 8 is applied to compute the gravity loading at the two joints. From Eqs. (6.48) and (6.56) – (6.59),

$$G_1 = -(mg\,d_{11}\,{}^1\bar r_1 + mg\,d_{21}\,{}^2\bar r_2) \tag{6.74}$$

The mass of the links has been assumed to be concentrated at the centroid of the links. The centroid is located from the origins of frame {1} and frame {2}

$${}^1\bar r_1 = {}^2\bar r_2 = \begin{bmatrix} -\frac{1}{2}L & 0 & 0 & 1 \end{bmatrix}^T \tag{6.75}$$

and the gravity is in the negative y-direction giving $\boldsymbol{g} = [0 \quad -g \quad 0 \quad 0]$ Hence,

$$mg\boldsymbol{d}_{11}\,{}^1\bar r_1 = m[0 \quad -g \quad 0 \quad 0]\begin{bmatrix} -S_1 & -C_1 & 0 & -LS_1 \\ C_1 & -S_1 & 0 & LC_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} -\frac{1}{2}L \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{2}mgLC_1 \tag{6.76}$$

$$mgd_{12}\,{}^2\bar r_2 = m[0 \quad -g \quad 0 \quad 0]\begin{bmatrix} -S_{12} & -C_{12} & 0 & -L(S_{12}+S_1) \\ C_{12} & -S_{12} & 0 & L(C_{12}+C_1) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} -\frac{1}{2}L \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= -mgL\left(\frac{1}{2}C_{12} + C_1\right) \tag{6.77}$$

Finally,
$$G_1 = \frac{1}{2}mgLC_{12} + \frac{3}{2}mgLC_1 \tag{6.78}$$

Similarly,
$$G_2 = -mg\,d_{22}\,{}^2\bar r_2 = \frac{1}{2}mgLC_{12} \tag{6.79}$$

The complete dynamic model is obtained in the final step, step 9, by substituting the above results, Eqs (6.68), (6.73), (6.78), and (6.79) into Eq. (6.45). The equations of motion in the vector-matrix form are

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} mL^2(\frac{5}{3}+C_2) & mL^2(\frac{1}{3}+\frac{1}{2}C_2) \\ mL^2(\frac{1}{3}+\frac{1}{2}C_2) & \frac{1}{3}mL^2 \end{bmatrix}\begin{bmatrix} \ddot\theta_1 \\ \ddot\theta_2 \end{bmatrix} + \begin{bmatrix} -mL^2S_2\dot\theta_1\dot\theta_2 - \frac{1}{2}mL^2S_2\dot\theta_2^2 \\ \frac{1}{2}mL^2S_2\dot\theta_1^2 \end{bmatrix} + \begin{bmatrix} \frac{1}{2}mgL\,C_{12} + \frac{3}{2}mgLC_1 \\ \frac{1}{2}mgL\,C_{12} \end{bmatrix} \tag{6.80}$$

The reader should verify that the dynamic model in Eq. (6.21) is identical to dynamic model in Eq. (6.80) for $m_1 = m_2 = m$ and $L_1 = L_2 = L$. Equations (6.80) are rather complex for a simple manipulator. Obviously, the dynamic equations for a 6-DOF manipulator would be quite complex.

6.4 NEWTON-EULER FORMULATION

The second approach to dynamic modeling of robotic manipulators is discussed now. The Newton-Euler (NE) formulation is based on the Newton's second law and d'Alembert principle. The balance of all forces acting on a link of the manipulator leads to a set of equations, whose structure allows a recursive solution. A forward recursion, which describes the kinematic relations of a moving coordinate frame, is performed for propagating velocities and accelerations, followed by a backward recursion for propagating forces and moments. Initially, it is assumed that the position, velocity, and acceleration of each joint, $(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})$ are known. The joint torques required to cause these time-dependent motions to realize a trajectory are computed using the recursive NE dynamic equations of motion. To understand the Newton-Euler formulation, some basic concepts of kinetics are reviewed first.

6.4.1 Newton's Equation, Euler's Equation

The robotic manipulator is an open kinematic chain of rigid links connected with 1-DOF joints. The mass distribution of a link is completely characterized by the location of the centre of mass and the inertia tensor of the link. The forces or torques required for moving the links, and accelerating or decelerating them, are a function of the mass distribution and inertia tensor of the links.

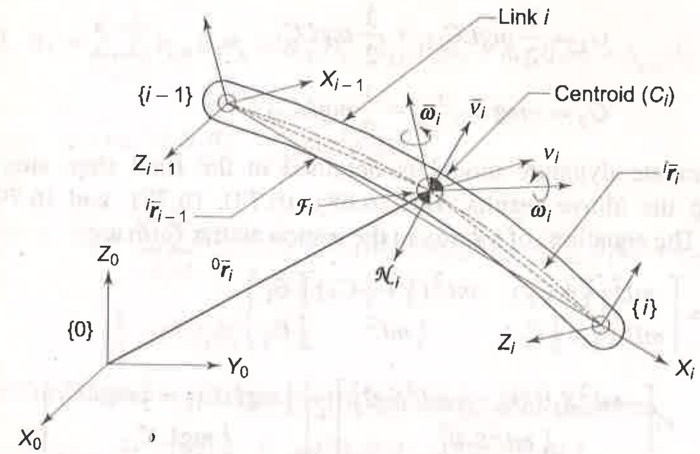

Fig. 6.4 *The geometry and kinematics of link i for Newton-Euler formulation*

Consider the rigid link i of the manipulator kinematic chain connected between joints i and $(i + 1)$. The frames at the two ends are frame $\{i–1\}$ and frame $\{i\}$ as shown in Fig. 6.4, with reference to inertial frame $\{0\}$. The centre of mass of the link is C_i and the parameters listed below characterize the geometry and kinematics:

$^{i-1}\bar{r}_i$ = position vector of C_i from frame $\{i–1\}$,

$^{i}\bar{r}_i$ = position vector of C_i from frame $\{i\}$,

m_i = mass of link,

I_i = inertia tensor of link with respect to a frame $\{C_i\}$, whose origin is located at the center of mass of the link C_i and orientation of frame $\{C_i\}$ is same as the orientation of the base frame $\{0\}$,

$\bar{v}_i$ = linear velocity of center of mass of link,

$\dot{\bar{v}}_i$ = linear acceleration of center of mass,

ω_i = angular velocity of link,

$\dot{\omega}_i$ = angular acceleration of link,

$\mathcal{F}_i$ = total external force acting at the centre of mass of link,

$\mathcal{N}_i$ = total external moment acting on link at the centre of mass of link.

The translational motion of the link in terms of the balance of forces is described by the *Newton's equation*. The force $\mathcal{F}_i$, acting at the centre of mass of the link is given by

$$\mathcal{F}_i = m_i \dot{\bar{v}}_i \tag{6.81}$$

where $\dot{\bar{v}}_i$ is the linear acceleration of the link.

The Euler equation for the rotational motion of the link describes the moment balance about the centre of mass of the link. The angular velocity of the link ω_i and the moment of inertia tensor I_i relate to the total moment $\mathcal{N}_i$ acting on link as

$$\mathcal{N}_i = \frac{d}{dt}(I_i\omega_i) = I_i\dot{\omega}_i + \omega_i \times (I_i\omega_i) \tag{6.82}$$

where the second term is the gyroscopic torque induced by the dependence of I_i on link's orientation with respect to base frame.

Equations (6.81) and (6.82) are the Newton-Euler equations that are recursively applied to compute the inertia force and torque acting at the center of mass of each link of the manipulator.

6.4.2 Kinematics of Links

To compute the inertial forces and torques acting on the links it is necessary to compute the linear and rotational velocity and acceleration of the centre of mass of each link of the manipulator at any given instant. The kinematic relationship of the moving-rotating links with respect to the base coordinate system is first described as a set of mathematical equations.

Two adjacent links i and $(i–1)$, forming the joint i, are shown in Fig. 6.5. The orthogonal coordinate frames are established as described in Section 3.4 with

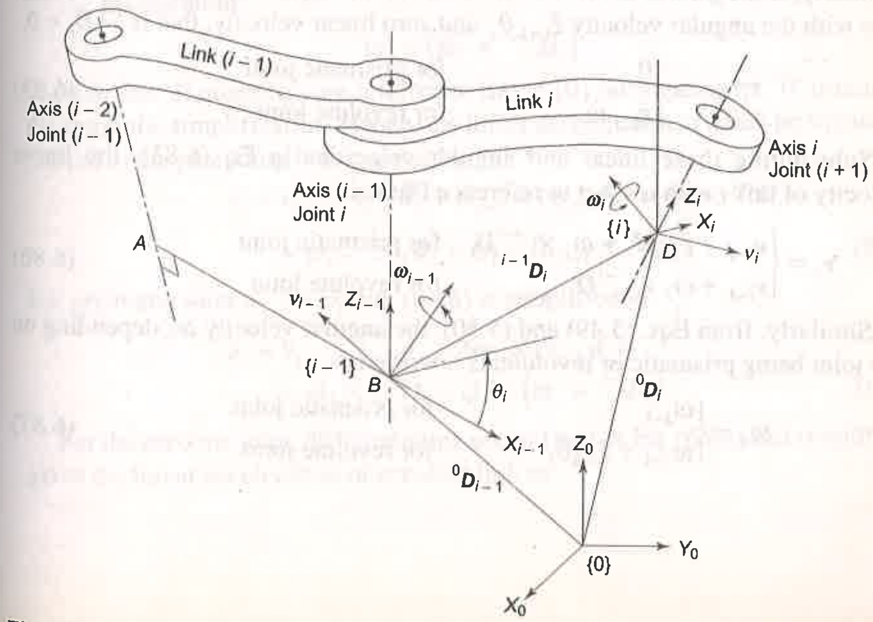

Fig. 6.5 *Characterization of two adjacent links forming the joint i for NE formulation*

frame{0} as the base coordinate frame; frame $\{i-1\}$ at joint i attached to link $(i-1)$ with point B as the origin, and frame $\{i\}$ at joint $(i+1)$ attached to link i with origin D. The origin D and origin B are located by position vectors ${}^0\boldsymbol{D}_i$ and ${}^0\boldsymbol{D}_{i-1}$, with respect to the base frame {0}, respectively, and the position vector ${}^{i-1}\boldsymbol{D}_i$ locates the origin D from the origin B with respect to the base coordinate system.

Let the linear and angular velocities of frame $\{i-1\}$, with respect to the base frame {0}, be $\boldsymbol{v}_{i-1}$ and ω_{i-1}, respectively, and ω_i be the angular velocity of frame $\{i\}$ with respect to the base frame {0}. Let ${}^{i-1}\omega_i$ be the relative angular velocity of frame $\{i\}$ with respect to frame $\{i-1\}$ referred to base frame {0}. *Note that superscript '0' is omitted for quantities expressed in the base frame* {0}.

The linear and angular velocities of frame $\{i\}$ (or origin D), as a function of translational and rotational velocities of link $\{i-1\}$, with respect to base frame {0} were determined in Chapter 5. [Eqs. (5.14) and (5.51). The linear velocity $\boldsymbol{v}_i$ and angular velocity ω_i of the frame $\{i\}$ with respect to base frame {0} are, respectively,

$$\boldsymbol{v}_i = \boldsymbol{v}_{i-1} + {}^{i-1}\dot{\boldsymbol{D}}_i + \omega_{i-1} \times {}^{i-1}\boldsymbol{D}_i \tag{6.83}$$

$$\omega_i = \omega_{i-1} + {}^{i-1}\omega_i \tag{6.84}$$

where ${}^{i-1}\dot{\boldsymbol{D}}_i$ denotes the velocity of frame $\{i\}$ with respect to origin of frame $\{i-1\}$ expressed in base frame {0}.

Recalling the behavior of prismatic and revolute joints discussed in Chapter 5, the linear and angular velocities are computed. If joint i is prismatic, link i travels in the direction of axes z_{i-1} with a linear joint velocity $\dot{d}_i$ relative to link $(i-1)$, that is, ${}^{i-1}\dot{\boldsymbol{D}}_i = \hat{z}_{i-1}\dot{d}_i$, with $\hat{z}_{i-1}$ as the unit vector along z_{i-1}-axis. The angular velocity of prismatic link i is same as that of link $(i-1)$, that is, $\omega_{i-1} = \omega_i$. Similarly, if the joint is revolute, the link i has an angular motion about the z_{i-1} axis with the angular velocity $\hat{z}_{i-1}\dot{\theta}_i$ and zero linear velocity, that is ${}^{i-1}\dot{\boldsymbol{D}}_i = 0$.

$$ {}^{i-1}\omega_i = \begin{cases} 0 & \text{for prismatic joint} \\ \hat{z}_{i-1}\dot{\omega}_i & \text{for revolute joint} \end{cases} \tag{6.85}$$

Substituting these linear and angular velocities in Eq. (6.83), the linear velocity of link i with respect to reference frame is

$$\boldsymbol{v}_i = \begin{cases} \boldsymbol{v}_{i-1} + \hat{z}_{i-1}\dot{d}_i + \omega_i \times {}^{i-1}\boldsymbol{D}_i & \text{for prismatic joint} \\ \boldsymbol{v}_{i-1} + \omega_i \times {}^{i-1}\boldsymbol{D}_i & \text{for revolute joint} \end{cases} \tag{6.86}$$

Similarly, from Eqs. (5.49) and (5.50), the angular velocity ω_i, depending on the joint being prismatic or revolute, is rewritten as

$$\omega_i = \begin{cases} \omega_{i-1} & \text{for prismatic joint} \\ \omega_{i-1} + \hat{z}_{i-1}\dot{\theta}_i & \text{for revolute joint} \end{cases} \tag{6.87}$$

Remember that the linear velocity is associated with a point and angular velocity is associated with a body. Hence, $\boldsymbol{v}_i$, the linear velocity of link i is the velocity of the origin of the frame $\{i\}$, and the angular velocity of the frame $\{i\}$, ω_i is the angular velocity of the whole link i.

6.4.3 Link Acceleration

The linear acceleration of link i (i.e. frame $\{i\}$) is obtained by differentiating Eq. (6.86) with respect to time. The required complex derivational details are not given here. Only qualitative arrangements are advanced. For a prismatic joint the following terms contribute to the linear acceleration of link i

1. $\dot{\boldsymbol{v}}_{i-1}$
2. $\hat{z}_{i-1}\ddot{d}_i$
3. Angular velocity ω_{i-1} interacting with linear velocity $\hat{z}_{i-1}\dot{d}_i$ contributes $\omega_{i-1} \times (\dot{z}_{i-1}\dot{d}_i)$.
4. Angular acceleration $\dot{\omega}_i$ interacting with distance ${}^{i-1}\boldsymbol{D}_i$ contributes $\dot{\omega}_i \times {}^{i-1}\boldsymbol{D}_i$.
5. Angular velocity $\dot{\omega}_i$ interacting with linear velocity $\hat{z}_{i-1}\dot{d}_i$ contributes $\dot{\omega}_i \times (\hat{z}_{i-1}\dot{d}_i)$.
6. Angular velocity ω_{i-1} interact with ${}^{i-1}\boldsymbol{D}_i$ to cause linear velocity $(\dot{\omega}_i \times {}^{i-1}\boldsymbol{D}_i)$ which interacting with angular velocity ω_i cause the linear acceleration

$$\omega_i \times (\dot{\omega}_i \times {}^{i-1}\boldsymbol{D}_i)$$

Note that all quantities are referred to frame {0} but superscript '0' is omitted for symbolic simplification. Hence, the linear acceleration of a link prismatically jointed to the preceding link is given by

$$\begin{aligned}\dot{\boldsymbol{v}}_i &= \dot{\boldsymbol{v}}_{i-1} + \hat{z}_{i-1}\ddot{d}_i + \omega_{i-1} \times (\hat{z}_{i-1}\dot{d}_i) + \dot{\omega}_i \times {}^{i-1}\boldsymbol{D}_i \\ &+ \omega_i \times \hat{z}_{i-1}\dot{d}_i + \omega_i \times (\omega_{i-1} \times {}^{i-1}\boldsymbol{D}_i)\end{aligned} \tag{6.88}$$

For prismatic joint $\omega_{i-1} = \omega_i$, Eq. (6.88) is simplified to

$$\begin{aligned}\dot{\boldsymbol{v}}_i &= \dot{\boldsymbol{v}}_{i-1} + \hat{z}_{i-1}\ddot{d}_i + 2\omega_i \times (\hat{z}_{i-1}\dot{d}_i) \\ &+ \dot{\omega}_i \times {}^{i-1}\boldsymbol{D}_i + \omega_i \times (\omega_i \times {}^{i-1}\boldsymbol{D}_i)\end{aligned} \tag{6.89}$$

For the revolute joint, differentiating second part of Eq. (6.86) and simplifying gives the linear acceleration of revolute link as

$$\dot{v}_i = \dot{v}_{i-1} + \dot{\omega}_i \times {}^{i-1}\boldsymbol{D}_i + \omega_i \times \left(\omega_i \times {}^{i-1}\boldsymbol{D}_i\right) \tag{6.90}$$

Combining Eqs. (6.89) and (6.90), the linear acceleration of link i is given by

$$\dot{v}_i = \begin{cases} \dot{v}_{i-1} + \hat{z}_{i-1}\ddot{d}_i + 2\omega_i \times (\hat{z}_{i-1}\dot{d}_i) + \\ \dot{\omega}_i \times {}^{i-1}\boldsymbol{D}_i + \omega_i \times \left(\omega_i \times {}^{i-1}\boldsymbol{D}_i\right) & \text{for prismatic joint} \\ \dot{v}_{i-1} + \omega_i \times {}^{i-1}\boldsymbol{D}_i + \omega_i \times \left(\omega_i \times {}^{i-1}\boldsymbol{D}_i\right) & \end{cases} \tag{6.91}$$

Similarly, Eq.(6.87) is differentiated wit[illegible] accelerations for prismatic and revolute links [illegible] are:

$$\dot{\omega}_i = \begin{cases} \dot{\omega}_{i-1} & \text{for pris[illegible]} \\ \dot{\omega}_{i-1} + \hat{z}_{i-1}\ddot{\theta}_i + \omega_{i-1} \times \hat{z}_{i-1}\dot{\theta}_i & \text{for revolute joint} \end{cases} \tag{6.92}$$

The linear velocity and acceleration of the center of mass of link i are, respectively,

$$\bar{v}_i = v_i + \omega_i \times {}^i\bar{r}_i \tag{6.93}$$

and

$$\dot{\bar{v}}_i = \dot{v}_i + \dot{\omega}_i \times {}^i\bar{r}_i + \omega_i \times \left(\omega_i \times {}^i\bar{r}_i\right) \tag{6.94}$$

where ${}^i\bar{r}_i$ is the position vector of centre of mass of link i from frame $\{i\}$ with respect to base frame $\{0\}$.

6.4.4 Recursive Newton-Euler Formulation

The recursive formulation of dynamic equations based on NE equations is now carried out from the above kinematic information of each link. In the recursive formulation the serial open kinematic chain structure of a manipulator is exploited. The NE formulation requires two passes over the links of the manipulator, one for computing the velocities and accelerations of the links and, second, to compute joint forces and torques, as shown in Fig. 6.6.

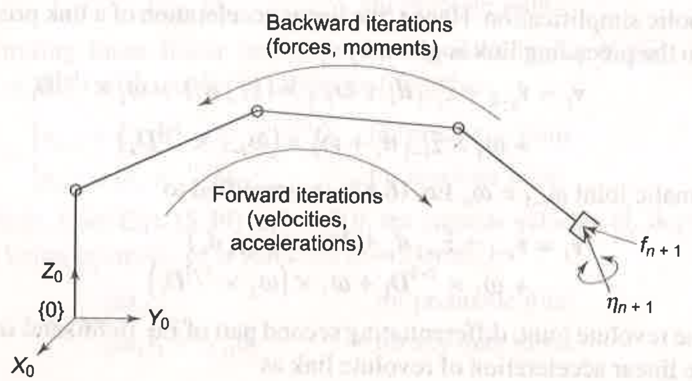

Fig. 6.6 *Two-pass recursive Newton-Euler formulation of dynamic equations*

The *forward iteration* or *outward iteration* is carried out to compute the velocities and accelerations of each link recursively, starting at the base and propagating forward to the end-effector. The boundary conditions are base velocity, linear and angular, which are zero, and the boundary acceleration is the gravitational acceleration.

In the *backward* or *inward iteration*, the forces and moments acting on each link are computed using the Newton's and Euler's equations.

First, the force and moment that must be applied to the centre of mass of each link are computed, and from these, the force and moment that must be applied at each joint are computed starting from the end-effector and moving inward (working backward) to the base. The end-effector force f_{n+1} and moment η_{n+1} exerted by the environment or the grasped object provide the necessary boundary conditions.

6.4.5 Forward Iteration

In the equations for velocity and acceleration, Eqs. (6.86), (6.87), (6.91), and (6.92), all the physical parameters are referenced to the base frame $\{0\}$. Because the parameters referred to the base frame change, as the manipulator is moving, the computations are complex. The computations are much simplified by referring all velocities, accelerations, inertia tensors, and location of centre of mass of each link to their own link-coordinate frames. The reference to link frame $\{i\}$ results in constant inertia tensor $\boldsymbol{I}_i$ and constant vectors appear in the equations, simplifying the numerical computations. The change of frames is accomplished by using the 3×3 rotational transformation matrices, which give the kinematic relationship between the links.

The 3×3 rotation matrix ${}^{i-1}\boldsymbol{R}_i$ transforms rotation of any vector with reference to frame $\{i\}$ to the frame $\{i-1\}$. The rotation matrix ${}^{i-1}\boldsymbol{R}_i$ is the upper left 3×3 sub-matrix of ${}^{i-1}\boldsymbol{T}_i$. That is,

$${}^{i-1}\boldsymbol{T}_i = \left[\begin{array}{ccc:c} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right] = \left[\begin{array}{c:c} {}^{i-1}\boldsymbol{R}_i & {}^{i-1}\boldsymbol{D}_i \\ \hdashline 0\ 0\ 0 & 1 \end{array}\right] \tag{6.95}$$

with

$${}^{i-1}\boldsymbol{R}_i = \begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i \\ 0 & S\alpha_i & C\alpha_i \end{bmatrix}; \text{ and } {}^{i-1}\boldsymbol{D}_i = \begin{bmatrix} a_i C\theta_i \\ a_i S\theta_i \\ d_i \end{bmatrix} \tag{6.96}$$

It has been shown earlier, [Eq. (2.15)] that

$$({}^{i-1}\boldsymbol{R}_i)^{-1} = {}^i\boldsymbol{R}_{i-1} = ({}^{i-1}\boldsymbol{R}_i)^T \tag{6.97}$$

Thus, using the rotational transformation matrices, it is possible to express the vectors related to link i with respect to link frame $\{i\}$ instead of base frame $\{0\}$*. This gives constant vectors instead of variable vectors, simplifying the numerical computations. Applying this the velocity and acceleration relationships of Eqs. (6.87), (6.91), and (6.94) are modified for the outward iteration as below. All the variables in these equations are now referred to their own frame.

$$ {}^{i}\omega_i = \begin{cases} {}^{i}\boldsymbol{R}_{i-1}\,{}^{i-1}\omega_{i-1} & \text{for prismatic joint} \\ {}^{i}\boldsymbol{R}_{i-1}[{}^{i-1}\omega_{i-1} + \hat{z}_0\dot{\theta}_i] & \text{for revolute joint} \end{cases} \tag{6.98} $$

$$ {}^{i}\dot{\omega}_i = \begin{cases} {}^{i}\boldsymbol{R}_{i-1}\,{}^{i-1}\dot{\omega}_{i-1} & \text{for prismatic joint} \\ {}^{i}\boldsymbol{R}_{i-1}[{}^{i-1}\dot{\omega}_{i-1} + \hat{z}_0\ddot{\theta}_i + {}^{i-1}\omega_{i-1} \times \hat{z}_0\dot{\theta}_i] & \text{for revolute joint} \end{cases} \tag{6.99} $$

---

* *Transforming vectors to link's own frame*

This is indicated here for some typical terms in the expressions of Eqs. (6.98), (6.99) and (6.100).

(1) ω_i, ω_{i-1} terms:

$$ \omega_i = {}^{i}\omega_i,\quad {}^{i}R_0\,\omega_{i-1} = {}^{i}R_{i-1}\,{}^{i-1}R_0\omega_{i-1} = {}^{i}R_{i-1}\,{}^{i-1}\omega_{i-1} $$

These results apply to linear velocities $\boldsymbol{v}_i$ and $\boldsymbol{v}_{i-1}$ and also to angular and linear acceleration $\dot{\omega}_i$, $\dot{\omega}_{i-1}$; $\dot{\boldsymbol{v}}_i$ and $\dot{\boldsymbol{v}}_{i-1}$.

(2) $\hat{z}_{i-1}\dot{d}_i$, $\hat{z}_{i-1}\ddot{d}_i$, $\hat{z}_{i-1}\dot{\theta}_i$, $\hat{z}_{i-1}\ddot{\theta}_i$ terms:

These terms are of one kind, therefore

$$ {}^{i}\boldsymbol{R}_0\hat{z}_{i-1}\dot{d}_i = {}^{i}\boldsymbol{R}_{i-1}\,{}^{i-1}\boldsymbol{R}_0\hat{z}_{i-1}\dot{d}_i = {}^{i}\boldsymbol{R}_{i-1}\hat{z}_0\dot{d}_i $$

where $\hat{z}_0$ is $\hat{z}_{i-1}$-axis referred to its own frame. These apply for all i and $\hat{z}_0 = [0 \quad 0 \quad 1]^T$ always.

(3) (i) $\omega_i \times (\hat{z}_{i-1}\dot{d}_i)$ term:

$$ {}^{i}\boldsymbol{R}_0\,\omega_i \times ({}^{i}\boldsymbol{R}_0\hat{z}_{i-1}\dot{d}_i) = {}^{i}\omega_i \times ({}^{i}\boldsymbol{R}_{i-1}\hat{z}_0 d_i) $$

(ii) $\omega_i \times {}^{i-1}\boldsymbol{D}_i$ term:

$$ {}^{i}\boldsymbol{R}_0\,\omega_i \times ({}^{i}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i) = {}^{i}\omega_i \times ({}^{i}\boldsymbol{R}_{i-1}\,{}^{i-1}\boldsymbol{D}_i) $$

(iii) $\omega_i \times (\omega_i \times {}^{i-1}\boldsymbol{D}_i)$:

$$ {}^{i}\boldsymbol{R}_0\,\omega_i \times \left[{}^{i}\boldsymbol{R}_0\,\omega_i \times ({}^{i}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i)\right] = {}^{i}\omega_i \times {}^{i}\omega_i \times ({}^{i-1}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i) $$

$$ {}^{i}\dot{\boldsymbol{v}}_i = \begin{cases} {}^{i}\boldsymbol{R}_{i-1}[{}^{i-1}\dot{\boldsymbol{v}}_{i-1} + \hat{z}_0\ddot{d}_i] + 2\,{}^{i}\omega_i \times ({}^{i}\boldsymbol{R}_{i-1}\hat{z}_0\dot{d}_i) \\ \quad + {}^{i}\omega_i \times ({}^{i}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i) + {}^{i}\omega_i \times \left[{}^{i}\omega_i \times ({}^{i}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i)\right] & \text{for prismatic joint} \\ {}^{i}\boldsymbol{R}_{i-1}\,{}^{i-1}\dot{\boldsymbol{v}}_{i-1} + {}^{i}\dot{\omega}_i \times ({}^{i}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i) + \\ {}^{i}\omega_i \times \left[{}^{i}\omega_i \times ({}^{i}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i)\right] & \text{for revolute joint} \end{cases} \tag{6.100} $$

The linear acceleration of the center of mass is given by

$$ {}^{i}\dot{\bar{\boldsymbol{v}}}_i = {}^{i}\dot{\boldsymbol{v}}_i + {}^{i}\omega_i \times {}^{i}\bar{\boldsymbol{r}}_i + {}^{i}\omega_i \times \left({}^{i}\omega_i \times {}^{i}\bar{\boldsymbol{r}}_i\right) \tag{6.101} $$

where $\hat{z}_0 = [0 \quad 0 \quad 1]^T$ is the unit vector in the direction of z_0-axis and the matrix products $({}^{i}\boldsymbol{R}_0\,{}^{i-1}\boldsymbol{D}_i)$ in Eq. (6.100) are simplified, using Eq. (6.96) and (6.97), to yield

$$ {}^{i}\boldsymbol{R}_{i-1}\,{}^{i-1}\boldsymbol{D}_i = \begin{bmatrix} C_i & S_i & 0 \\ -S_iC\alpha_i & C_iC\alpha_i & S\alpha_i \\ S_iS\alpha_i & -C_iC\alpha_i & C\alpha_i \end{bmatrix} \begin{bmatrix} a_iC_i \\ a_iS_i \\ d_i \end{bmatrix} = \begin{bmatrix} a_i \\ d_iS\alpha_i \\ d_iC\alpha_i \end{bmatrix} \tag{6.102} $$

with $C_i = C\theta_i = \cos\theta_i$ and $S_i = S\theta_i = \sin\theta_i$.

Equations (6.98) to (6.101) give forward Newton-Euler equations. The forward iteration starts at the base, that is $i = 0$. The initial conditions for forward NE recursion for a fixed (inertial) base of manipulator are

$$ \begin{aligned} {}^{0}\boldsymbol{v}_0 &= 0 \\ {}^{0}\dot{\boldsymbol{v}}_0 &= 0 \\ {}^{0}\omega_0 &= 0 \\ {}^{0}\dot{\omega}_0 &= 0 \end{aligned} \tag{6.103} $$

The effect of gravity can be included by considering the linear accelerations of base frame as

$$ {}^{0}\dot{\boldsymbol{v}}_0 = \boldsymbol{g} = \left[g_x \quad g_y \quad g_z\right]^T \tag{6.104} $$

The gravity loading on each link is then automatically propagated through the links by the forward recursion.

6.4.6 Backward Iteration

Once the velocities and accelerations of each link are known, the forces and moments acting on each link are computed by starting at the end-effector and working backwards.

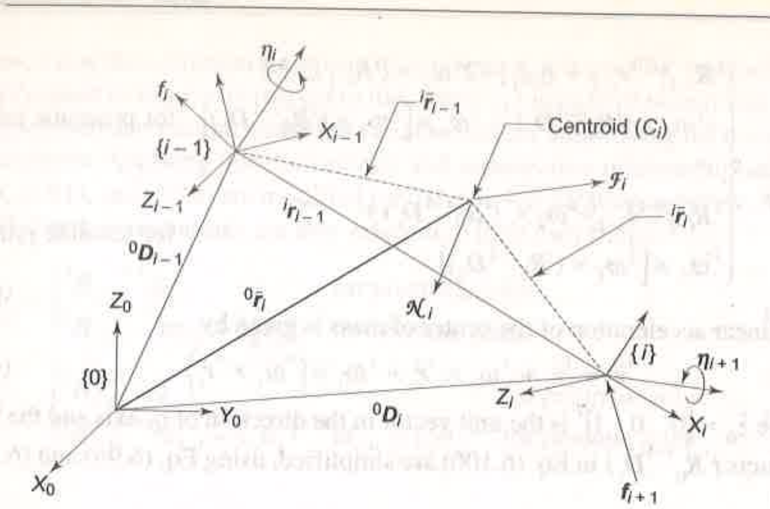

Fig. 6.7 *Forces and moments acting on a link and its centroid*

Let f_i and η_i be the force and moment exerted by link $(i-1)$ on link i at the origin of frame $\{i-1\}$ with respect to base frame $\{0\}$ as shown in Fig. 6.7. Further, link i exerts force f_{i+1} and moment η_{i+1} on link $(i+1)$ at the origin of frame $\{i\}$. The reaction force f_{i+1} and moment η_{i+1} act on link i in the opposite direction. This force and moment would, therefore, have a negative sign attached to them. The total external force $\mathcal{F}_i$ and $\mathcal{N}_i$ acting on link i are given by Eqs. (6.81) and (6.82) with these observations, the force and moment balance equations for link i from Fig. 6.7 are

$$\mathcal{F}_i = f_i - f_{i+1} \tag{6.105}$$

$$\mathcal{N}_i = \eta_i - \eta_{i+1} + ({}^0\boldsymbol{D}_{i-1} - {}^0\bar{\boldsymbol{r}}_i) \times f_i - ({}^0\boldsymbol{D}_i - {}^0\bar{\boldsymbol{r}}_i) \times f_{i+1} \tag{6.106}$$

Substituting f_i from Eq. (6.105) in Eq. (6.106)

$$\mathcal{N}_i = \eta_i - \eta_{i+1} + ({}^0\boldsymbol{D}_{i-1} - {}^0\bar{\boldsymbol{r}}_i) \times (\mathcal{F}_i + f_{i+1}) - ({}^0\boldsymbol{D}_i - {}^0\bar{\boldsymbol{r}}_i) \times f_{i+1} \tag{6.107}$$

or $$\mathcal{N}_i = \eta_i - \eta_{i+1} + ({}^0\boldsymbol{D}_{i-1} - {}^0\boldsymbol{D}_i) \times f_{i+1} + ({}^0\boldsymbol{D}_{i-1} - {}^0\bar{\boldsymbol{r}}_i) \times \mathcal{F}_i \tag{6.108}$$

From Fig. 6.7, the position vectors are related as

$$({}^0\boldsymbol{D}_{i-1} - {}^0\bar{\boldsymbol{r}}_i) = -({}^{i-1}\boldsymbol{D}_i + {}^i\bar{\boldsymbol{r}}_i) \text{ and } ({}^0\boldsymbol{D}_{i-1} - {}^0\boldsymbol{D}_i) = -{}^{i-1}\boldsymbol{D}_i \tag{6.109}$$

Therefore,

$$\mathcal{N}_i = \eta_i - \eta_{i+1} - \left({}^{i-1}\boldsymbol{D}_i + {}^i\bar{\boldsymbol{r}}_i\right) \times \mathcal{F}_i - {}^{i-1}\boldsymbol{D}_i \times f_{i+1} \tag{6.110}$$

These equations can be rewritten as recursive backward iteration equations from Eqs. (6.105) and (6.110) as

$$f_i = \mathcal{F}_i + f_{i+1} \tag{6.111}$$

$$\eta_i = \eta_{i+1} + {}^{i-1}\boldsymbol{D}_i \times f_{i+1} + \left({}^{i-1}\boldsymbol{D}_i + {}^i\bar{\boldsymbol{r}}_i\right) \times \mathcal{F}_i + \mathcal{N}_i \tag{6.112}$$

The joint torque τ_i at joint i is the sum of the projections of η_i onto the z_{i-1}-axis, the axis about which a revolute joint moves. However, if the joint is prismatic, it translates along z_{i-1}-axis, and the joint torque τ_i is the sum of the projections of f_i onto the z_{i-1}-axis. Hence, the generalized torque for joint i in base frame $\{0\}$ is given by

$$\tau_i = \begin{cases} f_{i-1}^T \ \hat{z}_{i-1} & \text{for prismatic joint} \\ \eta_{i-1}^T \ \hat{z}_{i-1} & \text{for revolute joint} \end{cases} \tag{6.113}$$

These equations [Eqs. (6.81), (6.82), (6.111), (6.112), and (6.113)] are transformed to joint-link coordinates using the rotation transformation matrix $\boldsymbol{R}$ to give final backward recursive equations as

$${}^i\mathcal{F}_i = m_i \, {}^i\dot{\bar{v}}_i \tag{6.114}$$

$${}^i\mathcal{N}_i = \boldsymbol{I}_i \, {}^i\dot{\omega}_i + {}^i\omega_i \times \left(\boldsymbol{I}_i \, {}^i\omega_i\right) \tag{6.115}$$

$${}^if_i = {}^i\mathcal{F}_i + {}^i\boldsymbol{R}_{i+1} \, {}^{i+1}f_{i+1} \tag{6.116}$$

$$\begin{aligned} {}^i\eta_i = {}&{}^i\boldsymbol{R}_{i+1} \, {}^{i+1}\eta_{i+1} + ({}^i\boldsymbol{R}_0 \, {}^{i-1}\boldsymbol{D}_i) \times {}^i\boldsymbol{R}_{i+1} \, {}^{i+1}f_{i+1} \\ &+ ({}^i\boldsymbol{R}_0 \, {}^{i-1}\boldsymbol{D}_i + {}^i\boldsymbol{R}_0 \, {}^i\bar{\boldsymbol{r}}_i) \times {}^i\mathcal{F}_i + {}^i\mathcal{N}_i \end{aligned} \tag{6.117}$$

and $$\tau_i = \begin{cases} {}^if_i^T \, {}^i\boldsymbol{R}_{i-1} \, \hat{z}_0 & \text{for prismatic joint} \\ {}^i\eta_i^T \, {}^i\boldsymbol{R}_{i-1} \, \hat{z}_0 & \text{for revolute joint} \end{cases} \tag{6.118}$$

The above five equations, Eq.(6.114) to (6.118), are recursive and can be used to compute the forces and moments (f_i, η_i) at the joints for $i = 1, 2, \ldots, n$ for a n-DOF manipulator when the force and moment exerted by the end-effector on the external object or environment f_{n+1} and η_{n+1} are known.

In summary, recursive Newton-Euler equation of motion are a set of forward and backward recursive equations with the kinematics and dynamics of each link referred to link coordinate system. These equations are systematically reproduced in Table 6.2.

The recursive equations are amenable to an algorithmic approach for obtaining the dynamic model of the manipulator. The algorithm for applying NE recursive approach to obtain dynamic model follows.

Algorithm 6.2 ➢ Newton-Euler formulation

This algorithm generates the joint torque equations for all the joint actuators of an n-DOF manipulator. The steps of algorithm are:

Step 0. Define the variables

n = number of degrees of freedom = number of links = number of joints

Joint variables: q_i, $\dot{q}_i$, $\ddot{q}_i$, for $i = 1, 2, \ldots, n$

Link variables: $\mathcal{F}_i, f_i, \mathcal{N}_i, \eta_i, \tau_i$

Table 6.2 *Recursive Newton-Euler equations of motion*

Initial Conditions:

$$ {}^0\omega_0 = {}^0\dot{\omega}_0 = {}^0v_0 = 0 $$

$$ {}^0\dot{v}_0 = g = \begin{bmatrix} g_x & g_y & g_z \end{bmatrix}^T $$

set f_{n+1}, η_{n+1} to given end-effector force and moment.

Forward Iteration: for $i = 1, 2, \ldots, n$

For revolute joint $\quad {}^i\omega_i = {}^iR_{i-1}\left[{}^{i-1}\omega_{i-1} + \hat{z}_0\dot{\theta}_i\right]$

$$ {}^i\dot{\omega}_i = {}^iR_{i-1}\left[{}^{i-1}\dot{\omega}_{i-1} + \hat{z}_0\ddot{\theta}_i + {}^{i-1}\omega_{i-1} \times \hat{z}_0\dot{\theta}_i\right] $$

$$ {}^i\dot{v}_i = {}^iR_{i-1}\,{}^{i-1}\dot{v}_{i-1} + {}^i\dot{\omega}_i \times ({}^iR_0\,{}^{i-1}D_i) + {}^i\omega_i \times \left[{}^i\omega_i \times ({}^iR_0\,{}^{i-1}D_i)\right] $$

For prismatic joint $\quad {}^i\omega_i = {}^iR_{i-1}\,{}^{i-1}\omega_{i-1}$

$$ {}^i\omega_i = {}^iR_{i-1}\,{}^{i-1}\omega_{i-1} $$

$$ {}^i\dot{v}_i = {}^iR_{i-1}\left[{}^{i-1}\dot{v}_{i-1} + \hat{z}_0\ddot{d}_i\right] + 2\,{}^i\omega_i \times ({}^iR_{i-1}\hat{z}_0\dot{d}_i) + {}^i\omega_i \times \left({}^iR_0\,{}^{i-1}D_i\right) + {}^i\omega_i \times \left[{}^i\omega_i \times ({}^iR_0\,{}^{i-1}D_i)\right] $$

$$ {}^i\dot{\bar{v}}_i = {}^i\dot{v}_i + {}^i\omega_i \times {}^i\bar{r}_i + {}^i\omega_i \times \left[{}^i\omega_i \times {}^i\bar{r}_i\right] $$

where $\hat{z}_0 = [0 \quad 0 \quad 1]^T$,

${}^{i-1}D_i$ is referred to frame $\{0\}$, when referred to frame $\{i\}$, it is

$$ ({}^iR_0\,{}^{i-1}D_i) = \begin{bmatrix} a_i \\ d_iS\alpha_i \\ d_iC\alpha_i \end{bmatrix} $$

${}^i\bar{r}_i$ is the center of mass of link i referred to link frame $\{i\}$.

Backward Iteration: for $i = 1, 2, \ldots, n$

$$ {}^i\mathcal{F}_i = m_i\,{}^i\dot{\bar{v}}_i $$

$$ {}^i\mathcal{N}_i = I_i\,{}^i\dot{\omega}_i + {}^i\omega_i \times \left(I_i\,{}^i\omega_i\right) $$

$$ {}^if_i = {}^i\mathcal{F}_i + {}^iR_{i+1}\,{}^{i+1}f_{i+1} $$

$$ {}^i\eta_i = {}^iR_{i+1}\,{}^{i+1}\eta_{i+1} + ({}^iR_0\,{}^{i-1}D_i) \times {}^iR_{i+1}\,{}^{i+1}f_{i+1} + ({}^iR_0\,{}^{i-1}D_i + {}^iR_0\,{}^i\bar{r}_i) \times {}^iF_i + {}^i\mathcal{N}_i $$

where I_i is the top left 3×3 submatrix of inertia tensor of link i about its center of mass with respect to frame $\{i\}$.

The joint torque: i = $1, 2, 3 \cdots, n$

$$ \tau_i = \begin{cases} {}^if_i^T\,{}^iR_{i-1}\,\hat{z}_0 & \text{for prismatic joint} \\ {}^i\eta_i^T\,{}^iR_{i-1}\,\hat{z}_0 & \text{for revolute joint} \end{cases} $$

Step 1. Set the initial conditions

$$ {}^0\omega_0 = {}^0\dot{\omega}_0 = {}^0v_0 = 0 $$

$$ {}^0\dot{v}_0 = g = \begin{bmatrix} g_x & g_y & g_z \end{bmatrix}^T $$

Step 2. Assign frames $\{0\}, \ldots, \{n\}$using DH notation (Algorithm 3.1) such that frame $\{i\}$ is oriented with principle axes of link i. Obtain rotational transformation matrices ${}^{i-1}R_i$, their products and inverses ${}^iR_{i-1}$, centre of mass of links, inertia tensors I_i of links at centre of mass with respect to frame $\{i\}$.

Forward Iteration

Step 3. Set $i = 1$

Step 4. Compute: ${}^i\omega_i$, ${}^i\dot{\omega}_i$, ${}^i\dot{v}_i$ and ${}^i\dot{\bar{v}}_i$ using equations in Table 6.2.

Step 5. If $i = n$, then go to step 6; otherwise set $i = i+1$ and go to step 4.

Backward Iteration

Step 6. Set f_{n+1} = required end-effector force and η_{n+1} = required end-effector moment. Note that if end-effector is free to move in space, it has no load and required force and moment are zero.

Step 7. Compute $\mathcal{F}_i$, $\mathcal{N}_i$, f_i, η_i and τ_i using equations in Table 6.2.

Step 8. If $i = 1$, then stop; otherwise set $i = i-1$ and go to step 7.

Example 6.2 *NE formulation of EOM for 2-DOF planar manipulator*

The NE formulation to obtain the EOM is illustrated by considering the 2-DOF planar manipulator discussed in Example 6.1, Fig. 6.3.

Solution The Algorithm 6.2 is applied, step-by-step, to obtain the EOM using the recursive Newton-Euler formulation. The base of the robot is fixed and the gravity is assumed to be acting in negative y_0-direction. Thus, the initial conditions are set as per step 1 for the 2-DOF manipulator as

$$ {}^0v_0 = 0; \quad {}^0\omega_0 = 0; \quad {}^0\dot{\omega}_0 = 0; \quad {}^0\dot{v}_0 = [0 \quad -g \quad 0]^T \tag{6.119} $$

Next step (step 2) requires assignment of frames and based on it determination of geometrical parameters, rotational transform matrices and inertia tensors for the two links. Because, the manipulator is the same as in Example 6.1, the rotational transformation matrices from Eqs. (6.52) and (6.53) are:

$$ {}^{i-1}R_i = \begin{bmatrix} C_i & -S_i & 0 \\ S_i & C_i & 0 \\ 0 & 0 & 1 \end{bmatrix}, \; i = 1, 2; \text{ and } {}^0R_2 = \begin{bmatrix} C_{12} & -S_{12} & 0 \\ S_{12} & C_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{6.120} $$

and their inverses are

$$ {}^iR_{i-1} = \begin{bmatrix} C_i & S_i & 0 \\ -S_i & C_i & 0 \\ 0 & 0 & 1 \end{bmatrix}, \; i = 1, 2; \text{ and } {}^2R_0 = \begin{bmatrix} C_{12} & S_{12} & 0 \\ -S_{12} & C_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{6.121} $$

The location of center of mass for each length i of length L with respect to its frame $\{i\}$ is given by

$$^{i}\bar{r}_i = \begin{bmatrix} -\frac{1}{2}L \\ 0 \\ 0 \end{bmatrix}, \quad i = 1, 2 \tag{6.122}$$

and from Eq. (6.52), the relative positions of origin of frames are

$$^{0}\boldsymbol{D}_1 = \begin{bmatrix} LC_1 \\ LS_1 \\ 0 \end{bmatrix}; \quad ^{1}\boldsymbol{D}_2 = \begin{bmatrix} LC_2 \\ LS_2 \\ 0 \end{bmatrix} \tag{6.123}$$

and from Eq. (6.102)

$$^{1}\boldsymbol{R}_0\,^{0}\boldsymbol{D}_1 = \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix}; \quad ^{2}\boldsymbol{R}_0\,^{1}\boldsymbol{D}_2 = \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix} \tag{6.124}$$

The inertia tensors of each link with mass m at the center of mass with respect to the link frame from Eq. (6.55) is

$$\boldsymbol{I}_1 = \boldsymbol{I}_2 = \begin{bmatrix} \frac{1}{3}mL^2 & 0 & 0 & -\frac{1}{2}mL \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{2}mL & 0 & 0 & m \end{bmatrix} \tag{6.125}$$

Now, forward iterations are carried out as per step 3–5 for $i = 1, 2$.

For $i = 1$

The angular velocity of revolute joint is calculated from Eq. (6.98) with ω_0 and $^{1}\boldsymbol{R}_0$ from Eqs. (6.119) and (6.121), respectively

$$^{1}\omega_1 = {}^{1}\boldsymbol{R}_0\,[{}^{0}\omega_0 + \hat{z}_0\dot{\theta}_1] = \begin{bmatrix} C_1 & S_1 & 0 \\ -S_1 & C_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 \tag{6.126}$$

The angular acceleration is computed using Eq. (6.99) for revolute joint with $^{0}\omega_0 = {}^{0}\dot{\omega}_0 = 0$ as:

$$^{1}\dot{\omega}_1 = {}^{1}\boldsymbol{R}_0\left[{}^{0}\dot{\omega}_0 + \hat{z}_0\ddot{\theta}_1 + {}^{0}\omega_0 \times \hat{z}_0\dot{\theta}_1\right]$$

or

$$^{1}\dot{\omega}_1 = {}^{1}\boldsymbol{R}_0\hat{z}_0\ddot{\theta}_1 = \begin{bmatrix} C_1 & S_1 & 0 \\ -S_1 & C_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\ddot{\theta}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\ddot{\theta}_1 \tag{6.127}$$

The linear acceleration for revolute joint is computed from Eq. (6.100), as

$$^{1}\dot{v}_1 = {}^{1}\boldsymbol{R}_0\,{}^{0}\dot{v}_0 + {}^{1}\dot{\omega}_1 \times {}^{1}\boldsymbol{R}_0\,{}^{0}\boldsymbol{D}_1 + {}^{1}\omega_1 \times [{}^{1}\omega_1 \times ({}^{1}\boldsymbol{R}_0\,{}^{0}\boldsymbol{D}_1)]$$

Substituting values from Eqs. (6.119), (6.121), (6.124), (6.126) and (6.127)

$$^{1}\dot{v}_1 = \begin{bmatrix} C_1 & S_1 & 0 \\ -S_1 & C_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\ddot{\theta}_1 \times \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 \times \left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 \times \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix}\right\}$$

$$= \begin{bmatrix} -gS_1 - L\dot{\theta}_1^2 \\ -gC_1 + L\ddot{\theta}_1 \\ 0 \end{bmatrix} \tag{6.128}$$

and from Eq. (6.101), the linear acceleration of the centre of mass of link 1 is

$$^{1}\dot{\bar{v}}_1 = {}^{1}\dot{v}_1 + {}^{1}\dot{\omega}_1 \times {}^{1}\bar{r}_1 + {}^{1}\omega_1 \times \left({}^{1}\omega_1 \times {}^{1}\bar{r}_1\right)$$

Substituting $^{1}\dot{v}_1$, $^{1}\omega_1$, $^{1}\dot{\omega}_1$ and $^{1}\bar{r}_1$ from Eqs. (6.122), (6.126), (6.127) and (6.128), gives:

$$^{1}\dot{\bar{v}}_1 = \begin{bmatrix} -gS_1 - L\dot{}_1^{2} \\ -gC_1 + L\ddot{}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\ddot{\theta}_1 \times \begin{bmatrix} -\frac{1}{2}L \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 \times \left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 \times \begin{bmatrix} -\frac{1}{2}L \\ 0 \\ 0 \end{bmatrix}\right\}$$

$$= \begin{bmatrix} -gS_1 - \frac{1}{2}L\dot{\theta}_1^2 \\ -gC_1 + \frac{1}{2}L\ddot{\theta}_1 \\ 0 \end{bmatrix} \tag{6.129}$$

For $i = 2$

The angular velocity is

$$^{2}\omega_2 = {}^{2}\boldsymbol{R}_1\left[{}^{1}\omega_1 + \hat{z}_0\dot{\theta}_2\right] = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_2\right\}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}(\dot{\theta}_1 + \dot{\theta}_2) \tag{6.130}$$

and angular acceleration is

$$^{2}\dot{\omega}_2 = {}^{2}\boldsymbol{R}_1\left[{}^{1}\dot{\omega}_1 + \hat{z}_0\ddot{\theta}_2 + {}^{1}\omega_1 \times \hat{z}_0\dot{\theta}_2\right]$$

or

$$^{2}\dot{\omega}_2 = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\ddot{\theta}_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\ddot{\theta}_2 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_1 \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\dot{\theta}_2\right\}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}(\ddot{\theta}_1 + \ddot{\theta}_2) \tag{6.131}$$

The linear velocity of link 2 is

$$ {}^2\boldsymbol{v}_2 = {}^2\boldsymbol{R}_1\,{}^1\boldsymbol{v}_1 + {}^2\dot{\omega}_2 \times ({}^2\boldsymbol{R}_0\,{}^1\boldsymbol{D}_2) + {}^2\omega_2 \times \left[{}^2\omega_2 \times ({}^2\boldsymbol{R}_0\,{}^1\boldsymbol{D}_2)\right] $$

Substituting values from Eqs. (6.121), (6.124), (6.128), (6.130), and (6.131),

$$ {}^2\boldsymbol{v}_2 = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -gS_1 - L\dot{\theta}_1^2 \\ -gC_1 + L\ddot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (\ddot{\theta}_1 + \ddot{\theta}_2) \times \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix} $$

$$ + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (\dot{\theta}_1 + \dot{\theta}_2) \times \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (\dot{\theta}_1 + \dot{\theta}_2) \times \begin{bmatrix} L \\ 0 \\ 0 \end{bmatrix} \right\} $$

or

$$ {}^2\boldsymbol{v}_2 = \begin{bmatrix} L\ddot{\theta}_1 S_2 - L\dot{\theta}_1^2 C_2 - L\dot{\theta}_1^2 - L\dot{\theta}_2^2 - 2L\dot{\theta}_1\dot{\theta}_2 - gS_{12} \\ L\ddot{\theta}_1 C_2 + L\ddot{\theta}_1 + L\ddot{\theta}_2 + L\dot{\theta}_1^2 S_2 - gC_{12} \\ 0 \end{bmatrix} \tag{6.132} $$

and the linear acceleration of link 2 is

$$ {}^2\dot{\bar{v}}_2 = {}^2\dot{v}_2 + {}^2\dot{\omega}_2 \times {}^2\bar{r}_2 + {}^2\omega_2 \times \left[{}^2\omega_2 \times {}^2\bar{r}_2\right] $$

Substituting values obtained above

$$ {}^2\dot{v}_2 = \begin{bmatrix} L\ddot{\theta}_1 S_2 - L\dot{\theta}_1^2 C_2 - L\dot{\theta}_1^2 - L\dot{\theta}_2^2 - 2L\dot{\theta}_1\dot{\theta}_2 - gS_{12} \\ L\ddot{\theta}_1 C_2 + L\ddot{\theta}_1 + L\ddot{\theta}_2 + L\dot{\theta}_1^2 S_2 - gC_{12} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (\ddot{\theta}_1 + \ddot{\theta}_2) $$

$$ \times \begin{bmatrix} -\frac{1}{2}L \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (\dot{\theta}_1 + \dot{\theta}_2) \times \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (\dot{\theta}_1 + \dot{\theta}_2) \times \begin{bmatrix} -\frac{1}{2}L \\ 0 \\ 0 \end{bmatrix} \right\} $$

or

$$ {}^2\dot{v}_2 = \begin{bmatrix} L\ddot{\theta}_1 S_2 - L\dot{\theta}_1^2 C_2 - \frac{1}{2}L\dot{\theta}_1^2 - \frac{1}{2}L\dot{\theta}_2^2 - L\dot{\theta}_1\dot{\theta}_2 - gS_{12} \\ L\ddot{\theta}_1 C_2 + \frac{1}{2}L\ddot{\theta}_1 + \frac{1}{2}L\ddot{\theta}_2 + L\dot{\theta}_1^2 S_2 - gC_{12} \\ 0 \end{bmatrix} \tag{6.133} $$

Backward iterations

Backward iterations are now carried out for $i = 2, 1$, according to step 6–8. The end of link 2 is assumed to move freely, that is, no-load condition, giving

$$ {}^3f_3 = {}^3\eta_3 = 0 \tag{6.134} $$

For $i = 2$

From Eqs. (6.114) to (6.117)

$$ {}^2\mathcal{F}_2 = m\,{}^2\dot{\bar{v}}_2 $$

or

$$ {}^2\mathcal{F}_2 = \begin{bmatrix} mL\ddot{\theta}_1 S_2 - mL\dot{\theta}_2^2 C_2 - \frac{1}{2}mL\dot{\theta}_1^2 - \frac{1}{2}mL\dot{\theta}_2^2 - mL\dot{\theta}_1\dot{\theta}_2 - mgS_{12} \\ mL\ddot{\theta}_1 C_2 + \frac{1}{2}mL\ddot{\theta}_1 + \frac{1}{2}mL\ddot{\theta}_2 + mL_1\dot{\theta}_1^2 S_2 - mgC_{12} \\ 0 \end{bmatrix} \tag{6.135} $$

$$ {}^2\mathcal{N}_2 = \boldsymbol{I}_2\,{}^2\dot{\omega}_2 + {}^2\omega_2 \times \left(\boldsymbol{I}_2\,{}^2\omega_2\right) = [0\ 0\ 0]^T $$

$$ {}^2f_2 = {}^2\mathcal{F}_2 + {}^2\boldsymbol{R}_3\,{}^3f_3 = {}^2\mathcal{F}_2 $$

$$ {}^2\eta_2 = {}^2\boldsymbol{R}_3\,{}^3\eta_3 + ({}^2\boldsymbol{R}_0\,{}^1\boldsymbol{D}_2) \times {}^2\boldsymbol{R}_3\,{}^3f_3 + ({}^2\boldsymbol{R}_0\,{}^1\boldsymbol{D}_2 + {}^2\boldsymbol{R}_0\,{}^2\bar{r}_2) \times {}^2\mathcal{F}_2 + {}^2\mathcal{N}_2 $$

or

$$ {}^2\eta_2 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2}mL^2C_2\ddot{\theta}_1 + \frac{1}{3}mL^2\ddot{\theta}_1 + \frac{1}{3}mL^2\ddot{\theta}_2 + \frac{1}{2}mL^2S_2\dot{\theta}_2^2 - \frac{1}{2}mLgC_{12} \end{bmatrix} \tag{6.136} $$

For $i = 1$

$$ {}^1\mathcal{F}_1 = m\,{}^1\dot{\bar{v}}_1 = \begin{bmatrix} -mgS_1 - \frac{1}{2}mL\dot{\theta}_1^2 \\ -mgC_1 + \frac{1}{2}mL\ddot{\theta}_1 \\ 0 \end{bmatrix} \tag{6.137} $$

$$ {}^1\mathcal{N}_1 = \boldsymbol{I}_1\,{}^1\dot{\omega}_1 + {}^1\omega_1 \times (\boldsymbol{I}_1\,{}^1\omega_1) = [0\ 0\ 0]^T $$

$$ {}^1f_1 = {}^1\mathcal{F}_1 + {}^1\boldsymbol{R}_2\,{}^2f_2 $$

or

$$ {}^1f_1 = \begin{bmatrix} -mgS_1 - \frac{1}{2}mL\dot{\theta}_1^2 \\ -mgC_1 + \frac{1}{2}mL\ddot{\theta}_1 \\ 0 \end{bmatrix} + $$

$$ \begin{bmatrix} C_2 & -S_2 & 0 \\ S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} mL\ddot{\theta}_1 S_2 - mL\dot{\theta}_2^2 C_2 - \frac{1}{2}mL\dot{\theta}_1^2 - \frac{1}{2}mL\dot{\theta}_2^2 - mL\dot{\theta}_1\dot{\theta}_2 - mgS_{12} \\ mL\ddot{\theta}_1 C_2 + \frac{1}{2}mL\ddot{\theta}_1 + \frac{1}{2}mL\ddot{\theta}_2 + mL_1\dot{\theta}_1^2 S_2 - mgC_{12} \\ 0 \end{bmatrix} $$

Simplifying,

$$ {}^1f_1 = \begin{bmatrix} -\frac{1}{2}mLS_2\ddot{\theta}_1 - \frac{1}{2}mLS_2\ddot{\theta}_2 - \frac{3}{2}mL\dot{\theta}_1^2 - \frac{1}{2}mLC_2\dot{\theta}_1^2 - \frac{1}{2}mLC_2\dot{\theta}_2^2 - mLC_2\dot{\theta}_1\dot{\theta}_2 - 2mgS_1 \\ \frac{3}{2}mL\ddot{\theta}_1 + \frac{1}{2}mLC_2\ddot{\theta}_1 + \frac{1}{2}mLC_2\ddot{\theta}_2 - \frac{1}{2}mLS_2\dot{\theta}_1^2 - \frac{1}{2}mLS_2\dot{\theta}_2^2 - mLS_2\dot{\theta}_1\dot{\theta}_2 - 2mgC_1 \\ 0 \end{bmatrix} \tag{6.138} $$

Similarly,

$$ {}^1\eta_1 = {}^1\boldsymbol{R}_2\,{}^2\eta_2 + ({}^1\boldsymbol{R}_0\,{}^0\boldsymbol{D}_1) \times {}^1\boldsymbol{R}_2\,{}^2f_2 + ({}^1\boldsymbol{R}_0\,{}^0\boldsymbol{D}_1 + {}^1\boldsymbol{R}_0\,{}^1\bar{r}_1) \times {}^1\boldsymbol{F}_1 + {}^1\mathcal{N}_1 $$

$$
{}^1\eta_1 = \begin{bmatrix} 0 \\ 0 \\ mL^2C_2\ddot{\theta}_1 - mL^2S_2\dot{\theta}_2^2 + mLgC_{12} + mL^2(\ddot{\theta}_1 + \ddot{\theta}_2) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ m_1L^2\ddot{\theta}_1 + m_1LgC_1 \end{bmatrix}
$$

$$
+ \begin{bmatrix} 0 \\ 0 \\ mL^2\ddot{\theta}_1 - mL^2S_2(\dot{\theta}_1 + \dot{\theta}_2)^2 + mLgS_2S_{12} + mL^2C_2(\ddot{\theta}_1 + \ddot{\theta}_2) + mLgC_2C_{12} \end{bmatrix} \tag{6.139}
$$

The joint torques for two joints are finally, obtained using Eq. (6.118).

$$\tau_1 = {}^1\eta_1^T \, {}^1\boldsymbol{R}_0 \, \hat{z}_0$$

or
$$\tau_1 = \frac{5}{3}mL^2\ddot{\theta}_1 + \frac{1}{3}mL^2\ddot{\theta}_2 + mL^2C_2\ddot{\theta}_1 + \frac{1}{2}mL^2C_2\ddot{\theta}_2$$

$$- \frac{1}{2}mL^2S_2\dot{\theta}_2^2 - mL^2S_2\dot{\theta}_1\dot{\theta}_2 + \frac{1}{2}mLgC_{12} + \frac{3}{2}mLgC_1 \tag{6.140}$$

and
$$\tau_2 = \frac{1}{2}mL^2C_2\ddot{\theta}_1 + \frac{1}{3}mL^2\ddot{\theta}_1 + \frac{1}{3}mL^2\ddot{\theta}_2 + \frac{1}{2}mL^2S_2\dot{\theta}_1^2$$

$$+ \frac{1}{2}mLgC_{12} \tag{6.141}$$

The above equations, Eqs. (6.140) and (6.141), are written in matrix form as:

$$
\begin{bmatrix}\tau_1 \\ \tau_2\end{bmatrix} = \begin{bmatrix} mL^2(\frac{5}{3} + C_2) & mL^2(\frac{1}{3} + \frac{1}{2}C_2) \\ mL^2(\frac{1}{3} + \frac{1}{2}C_2) & \frac{1}{3}mL^2 \end{bmatrix}\begin{bmatrix}\ddot{\theta}_1 \\ \ddot{\theta}_2\end{bmatrix}
$$

$$
+ \begin{bmatrix} -mL^2S_2\dot{\theta}_1\dot{\theta}_2 - \frac{1}{2}mL^2S_2\dot{\theta}_2^2 \\ \frac{1}{2}mL^2S_2\dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} \frac{1}{2}mgLC_{12} + \frac{3}{2}mgLC_1 \\ \frac{1}{2}mgLC_{12} \end{bmatrix} \tag{6.142}
$$

This is same as equation (6.80), obtained from the Lagrange-Euler formulation in Example 6.1.

6.5 COMPARISON OF LAGRANGE-EULER AND NEWTON-EULER FORMULATIONS

Two approaches to dynamic modeling of robotic manipulators have been discussed above. Both Lagrange-Euler and Newton-Euler formulations give the relationship between the joint and end-effector forces/torques and the manipulator joint-link position, velocities, and accelerations. These formulations can be compared on the basis of their computational requirements.

The mathematical operations required to compute τ for an n-DOF manipulator in Lagrange-Euler formulation is estimated to be (see Appendix D)

$$
\begin{aligned}
&\text{Multiplications:} \quad \frac{128}{3}n^4 + \frac{512}{3}n^3 + \frac{844}{3}n^2 + \frac{76}{3}n \\
&\text{Additions:} \quad \frac{98}{3}n^4 + \frac{786}{3}n^3 + \frac{673}{3}n^2 + \frac{107}{3}n
\end{aligned} \tag{6.143}
$$

In the Newton-Euler formulation, the computations are dependent on the configuration, in addition to the number of degrees of freedom n. For a typical manipulator the numbers of computations for NE formulation are estimated to be

$$
\begin{aligned}
&\text{Multiplications} : 117n - 24 \\
&\text{Additions} \quad : 103n - 21
\end{aligned} \tag{6.144}
$$

The recursive Newton-Euler formulation has a computational complexity of order $O(n)$, that is, mathematical operations of multiplication and addition are proportional to n, the number of degrees of freedom of the manipulator. This is markedly less compared to the Lagrange-Euler formulation that has a complexity of order $O(n^4)$. The drawback of recursive formulation is that it is not as amenable to simple physical interpretation because the closed-form Lagrange-Euler formulation is. However, recursive formulation is much more efficient in terms of computational effort, particularly as the number of degrees of freedom n increases, they are ideally suited for computer implementation.

Example 6.3 ***Application of dynamic model to study the torque variations***

Determine the variation of joint torques from the dynamic model of 2-DOF manipulator discussed in Example 6.1. Take the following data for the manipulator:

$$m_1 = m_2 = 6 \text{ kg} \quad \text{and} \quad L_1 = L_2 = 1 \text{ m}$$

The specified motion cycle is: Both joints move from rest with equal velocity, attain maximum velocity then decelerate and come to rest, traversing $\pi/2$ rad in 1 s.

Solution The given values of m and L are substituted in Eq. (6.142). The resulting joint torque equations for the 2-DOF manipulator are:

$$
\begin{bmatrix}\tau_1 \\ \tau_2\end{bmatrix} = \begin{bmatrix} 10 + 6C_2 & 2 + 2C_2 \\ 2 + 2C_2 & 2 \end{bmatrix}\begin{bmatrix}\ddot{\theta}_1 \\ \ddot{\theta}_2\end{bmatrix} + \begin{bmatrix} -6S_2\dot{\theta}_1\dot{\theta}_2 - 3S_2\dot{\theta}_2^2 \\ 3S_2\dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} 2gC_{12} + 9gC_1 \\ 3gC_{12} \end{bmatrix} \tag{6.145}
$$

For the specified motion the velocity profile for both joints is a triangular velocity profile, as shown in Fig. 6.8(a). The variation of position of each link is obtained by integrating the velocity profile and values of all other terms in Eq. (6.145) are computed as a function of time. The time history of position, acceleration, components of torque (M_{ij}), and net torque for this velocity profile for each joint is shown in Fig. 6.8(b) to Fig. 6.8(h).

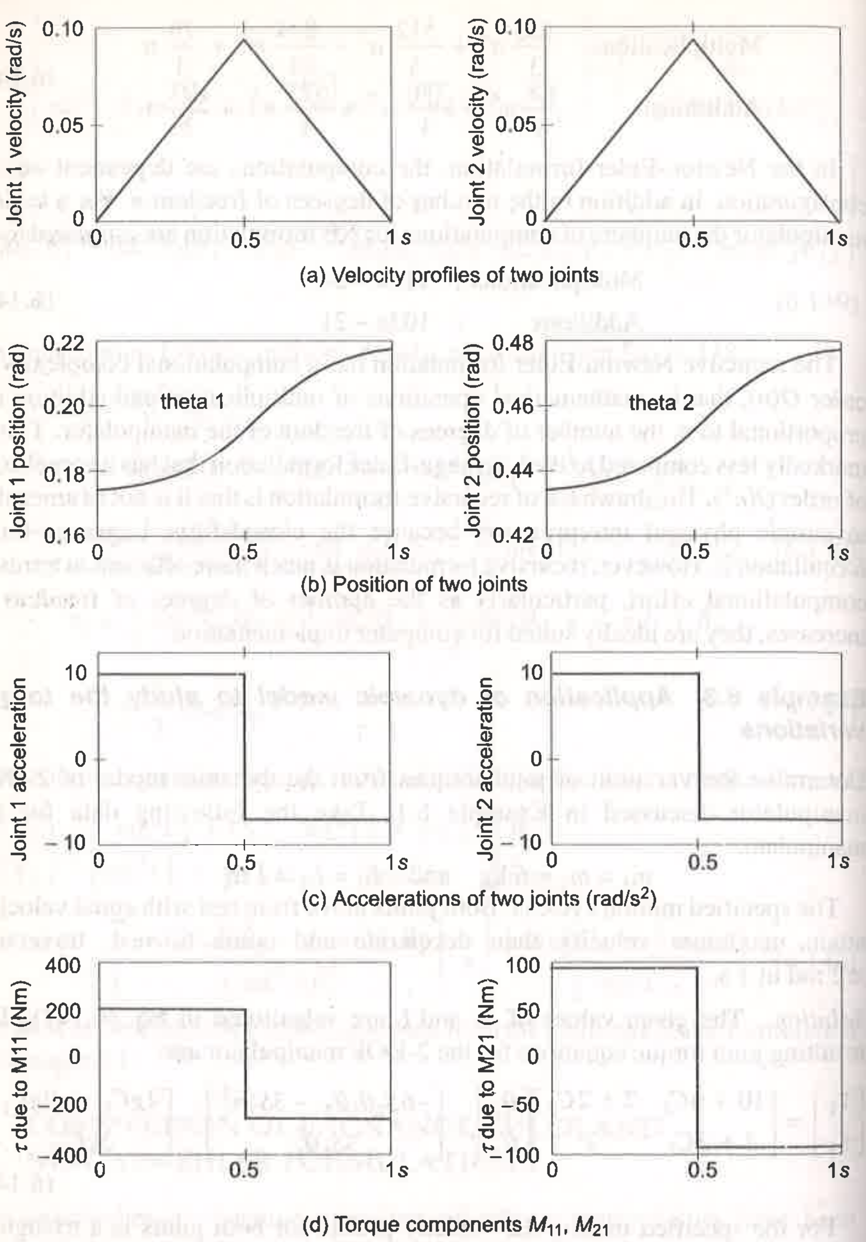

Fig. 6.8 *Time histories of position, velocity, acceleration and torques for the two joints of the 2-DOF manipulator (Contd….)*

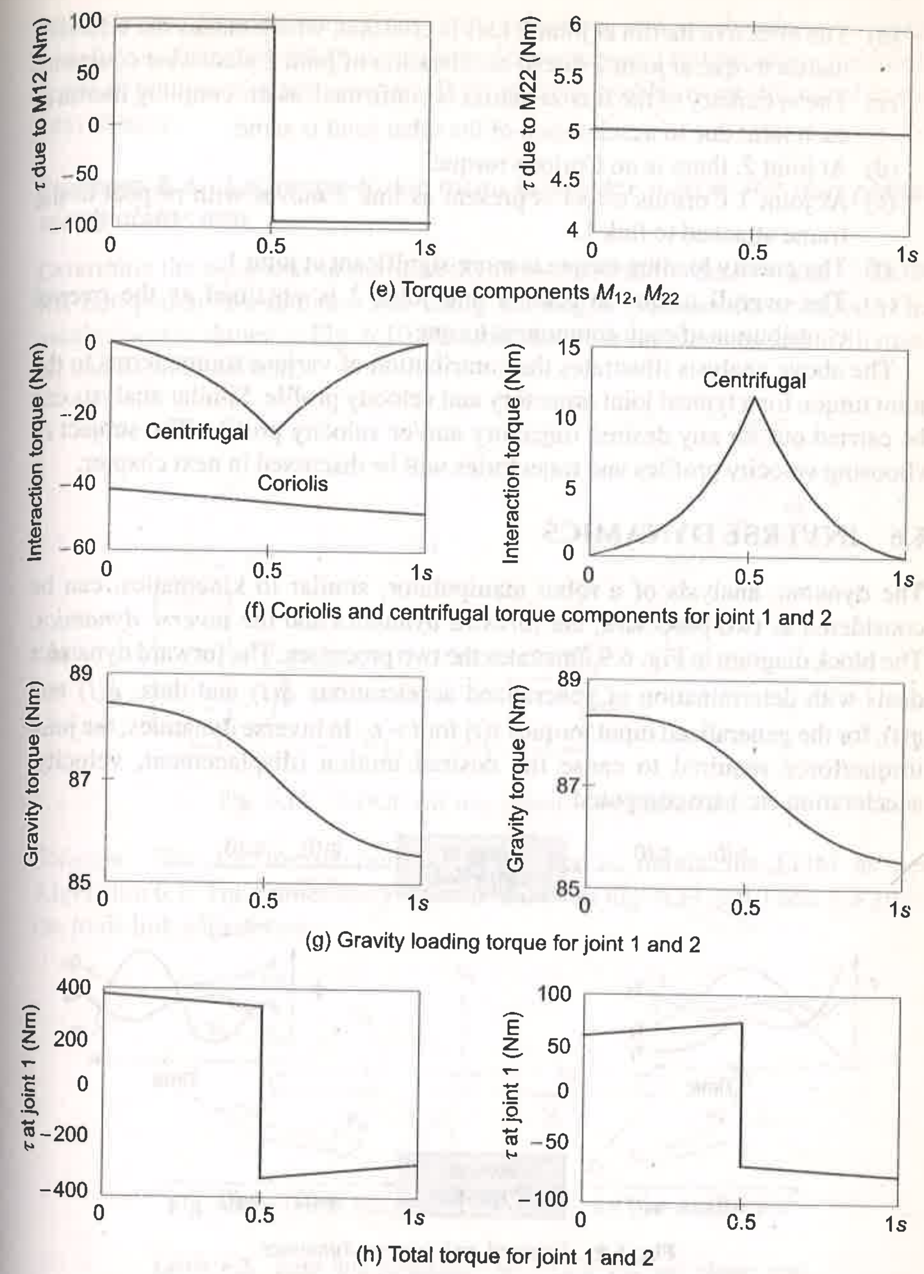

Fig. 6.8 *Time histories of position, velocity, acceleration and torques for the two joints of the 2-DOF manipulator.*

From Eq. (6.145) and the time history of various torque components, following observations are made:

(a) The effective inertia at joint 1 due to acceleration of joint 1 follows joint acceleration (see Fig. 6.8 (c) and (d)).

(b) The effective inertia at joint 2 axis is constant, which makes the effective inertia torque at joint 2 due to acceleration of joint 2 piecewise constant.
(c) The symmetry of the inertia matrix is confirmed, as the coupling inertia at each joint due to acceleration of the other joint is same.
(d) At joint 2, there is no Coriolis torque.
(e) At joint 1 Coriolis effect is present as link 2 moves with respect to the frame attached to link 1.
(f) The gravity loading torque is more significant at joint 1.
(g) The overall torque at joint 1 and joint 2 is obtained as the overall contribution of each component torque.

The above analysis illustrates the contribution of various torque terms to the joint torque for a typical joint trajectory and velocity profile. Similar analysis can be carried out for any desired trajectory and/or velocity profile. The subject of choosing velocity profiles and trajectories will be discussed in next chapter.

6.6 INVERSE DYNAMICS

The dynamic analysis of a robot manipulator, similar to kinematics, can be considered as two processes, the *forward dynamics* and the *inverse dynamics*. The block diagram in Fig. 6.9 illustrates the two processes. The forward dynamics deals with determination of generalized accelerations $\ddot{q}(t)$ and thus, $\dot{q}(t)$ and $q(t)$, for the generalized input torques $\tau(t)$ for $t > t_0$. In inverse dynamics, the joint torque/force required to cause the desired motion (displacement, velocity, acceleration etc.) are computed.

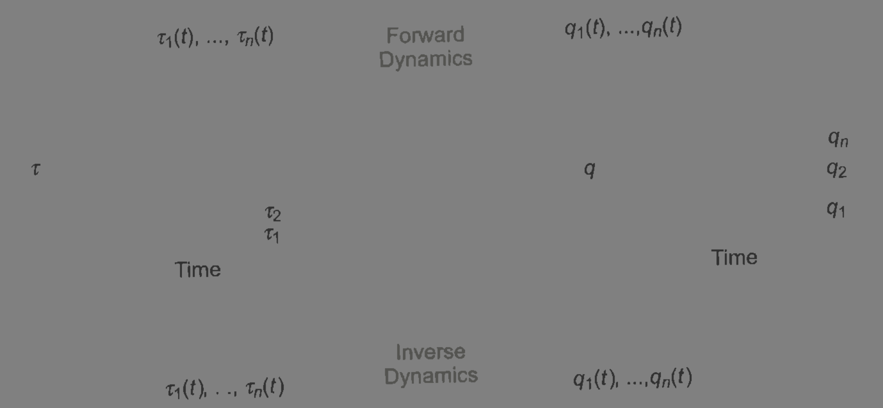

Fig. 6.9 *Forward and inverse dynamics*

which is specified in terms of positions, velocities, and accelerations, is useful both for verifying feasibility of the desired trajectory and compensating nonlinear terms in the linear control models. Both of these problems are discussed in the next chapter.

Example 6.4 *Lagrange-Euler formulation for 2-DOF RR non-planar manipulator arm*

Determine the equations of motions (dynamic model) for the two-link, 2-DOF, RR non-planar manipulator arm using the Lagrange-Euler formulation. The manipulator is shown in Fig. 6.10. Assume that both links are slender with mass m_1 and m_2 at distal end.

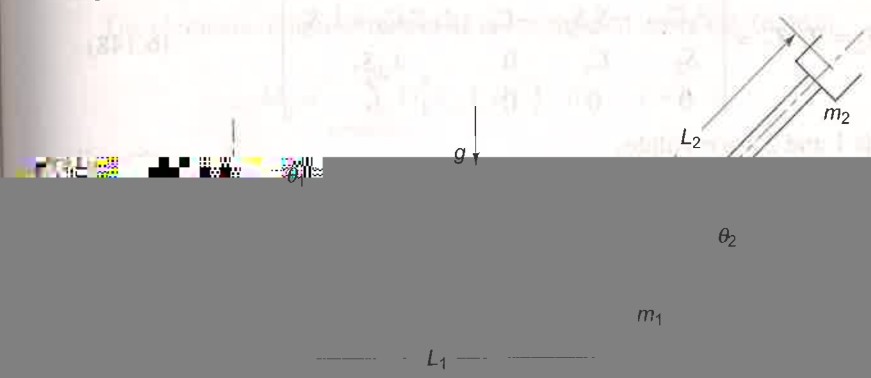

Fig. 6.10 *2-DOF RR non-planar manipulator arm*

Solution The LE formulation is carried out to obtain the EOM as per Algorithm 6.1. The frames assignment is shown in Fig. 6.11 and Table 6.3 gives the joint-link parameters.

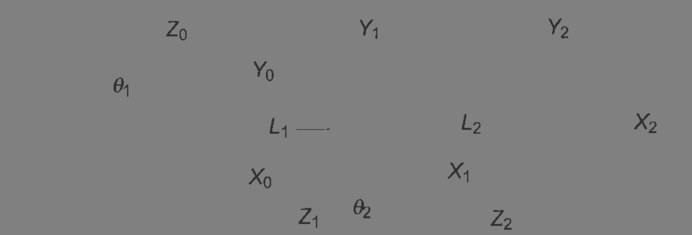

Fig. 6.11 *Frame assignment for 2-DOF RR non-planar arm*

Table 6.3 *Joint link parameters for 2-DOF RR non-planar arm*

i	a_i	α_i	d_i	θ_i	$C\theta_i$	$S\theta_i$	$C\alpha_i$	$S\alpha_i$
1	L_1	90°	0	θ_1	C_1	S_1	0	1
2	L_2	0	0	θ_2	C_2	S_2	1	0

The link transformation matrices and the overall transformation matrix are:

$$
{}^0\boldsymbol{T}_1 = \begin{bmatrix} C_1 & 0 & S_1 & L_1C_1 \\ S_1 & 0 & -C_1 & L_1S_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.146}
$$

$$
{}^1\boldsymbol{T}_2 = \begin{bmatrix} C_2 & -S_2 & 0 & L_2C_2 \\ S_2 & C_2 & 0 & L_2S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.147}
$$

$$
{}^0\boldsymbol{T}_2 = {}^0\boldsymbol{T}_1{}^1\boldsymbol{T}_2 = \begin{bmatrix} C_1C_2 & -C_1S_2 & S_1 & L_2C_1C_2 + L_1C_1 \\ S_1C_2 & -S_1S_2 & -C_1 & L_2S_1C_2 + L_1S_1 \\ S_2 & C_2 & 0 & L_2S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.148}
$$

Since joints 1 and 2 are revolute,

$$
\boldsymbol{Q}_1 = \boldsymbol{Q}_2 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.149}
$$

The inertia tensors $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$ for two slender links, with point mass m_1 and m_2 at the distal ends of the links are, respectively:

$$
\boldsymbol{I}_1 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & m_1 \end{bmatrix} \tag{6.150}
$$

$$
\boldsymbol{I}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & m_2 \end{bmatrix} \tag{6.151}
$$

The coefficient $\boldsymbol{d}_{ij}$ for $i, j = 1, 2$ are first computed using Eq. (6.49), that is

$$
\boldsymbol{d}_{ij} = \begin{cases} \boldsymbol{T}_{j-1}\,\boldsymbol{Q}_j\,{}^{j-1}\boldsymbol{T}_i & \text{for } j \le i \\ 0 & \text{for } j > i \end{cases} \tag{6.152}
$$

which gives $\boldsymbol{d}_{11}, \boldsymbol{d}_{12}, \boldsymbol{d}_{21}$ and $\boldsymbol{d}_{22}$ as:

$$
\boldsymbol{d}_{11} = {}^0\boldsymbol{T}_0\boldsymbol{Q}_1{}^0\boldsymbol{T}_1 = \begin{bmatrix} -S_1 & 0 & C_1 & -L_1S_1 \\ C_1 & 0 & S_1 & L_1C_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}
$$

$\boldsymbol{d}_{12} = 0$ (because $j > i$)

$$
\boldsymbol{d}_{21} = {}^0\boldsymbol{T}_1\boldsymbol{Q}_2{}^1\boldsymbol{T}_2 = \begin{bmatrix} -S_1C_2 & S_1S_2 & C_1 & -L_2S_1C_2 - L_1S_1 \\ C_1C_2 & -C_1S_2 & S_1 & L_2C_1C_2 + L_1C_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.153}
$$

$$
\boldsymbol{d}_{22} = {}^0\boldsymbol{T}_1\boldsymbol{Q}_2{}^1\boldsymbol{T}_2 = \begin{bmatrix} -C_1S_2 & -C_1C_2 & 0 & -L_2C_1S_2 \\ -S_1S_2 & -S_1C_2 & 0 & -L_2S_1S_2 \\ C_2 & -S_2 & 0 & L_2C_2 \\ 0 & 0 & 0 & 0 \end{bmatrix}
$$

The elements of inertia matrix $\boldsymbol{M}$ are computed using Eq. (6.46).

$$
M_{ij} = \sum_{p=\max(i,j)}^{n} \mathrm{Tr}\left[\boldsymbol{d}_{pj}\boldsymbol{I}_p\boldsymbol{d}_{pi}^T\right] \quad \text{for } i, j = 1, 2 \tag{6.154}
$$

This gives,

$$
M_{11} = \mathrm{Tr}(\boldsymbol{d}_{11}\boldsymbol{I}_1\boldsymbol{d}_{11}^T) + \mathrm{Tr}(\boldsymbol{d}_{21}\boldsymbol{I}_2\boldsymbol{d}_{21}^T) \tag{6.155}
$$

or $\quad M_{11} = m_1L_1^2 + (L_2C_2 + L_1)^2 m_2$

and $\quad M_{22} = \mathrm{Tr}(\boldsymbol{d}_{22}\boldsymbol{I}_2\boldsymbol{d}_{22}^T) = m_2L_2^2$

$$
M_{12} = M_{21} = \mathrm{Tr}(\boldsymbol{d}_{22}\boldsymbol{I}_2\boldsymbol{d}_{21}^T) = 0
$$

The Coriolis and centrifugal force coefficients (or velocity coupling coefficients), h_{ijk} for $i, j, k = 1, 2$ are computed using Eq. (6.47).

$$
h_{ijk} = \sum_{\mathrm{p}=\max(i,j,k)}^{n} \left[\frac{\partial(\boldsymbol{d}_{pk})}{\partial q_p}\boldsymbol{I}_p\boldsymbol{d}_{pi}^T\right] \tag{6.156}
$$

The partial derivatives in Eq. (6.156) are obtained from Eq. (6.50). Thus,

$$
\begin{aligned}
h_{111} &= 0 \\
h_{222} &= 0 \\
h_{122} &= \mathrm{Tr}\left[{}^0\boldsymbol{T}_1\,\boldsymbol{Q}_2\,{}^1\boldsymbol{T}_1\,\boldsymbol{Q}_2\,{}^1\boldsymbol{T}_2\,\boldsymbol{I}_2\,\boldsymbol{d}_{21}^T\right] \\
&= -L_2C_1C_2m_2(-L_2S_1C_2 - L_1S_1) - L_2S_1C_2m_2(L_2C_1C_2 + L_1C_1) = 0 \\
h_{211} &= \mathrm{Tr}\left[{}^0\boldsymbol{T}_0\,\boldsymbol{Q}_1\,{}^0\boldsymbol{T}_0\,\boldsymbol{Q}_1\,{}^0\boldsymbol{T}_2\,\boldsymbol{I}_2\,\boldsymbol{d}_{22}^T\right] \\
&= -L_2C_1S_2m_2(-L_2C_1C_2 - L_1C_1) - L_2S_1S_2m_2(-L_2S_1C_2 - L_1S_1) \qquad (6.157) \\
&= m_2S_2(L_2^2C_2 + L_1L_2) \\
h_{121} &= h_{112} = \mathrm{Tr}\left[{}^0\boldsymbol{T}_0\,\boldsymbol{Q}_1\,{}^0\boldsymbol{T}_1\,\boldsymbol{Q}_2\,{}^1\boldsymbol{T}_2\,\boldsymbol{I}_2\,\boldsymbol{d}_{21}^T\right] \\
&= L_2S_1S_2m_2(-L_2S_1C_2 - L_1S_1) - L_2C_1S_2m_2(L_2C_1C_2 + L_1C_1) \\
&= -m_2S_2(L_2^2C_2 + L_1L_2) \\
h_{221} &= h_{212} = \mathrm{Tr}\left[{}^0\boldsymbol{T}_0\,\boldsymbol{Q}_1\,{}^0\boldsymbol{T}_1\,\boldsymbol{Q}_2\,{}^1\boldsymbol{T}_2\,\boldsymbol{I}_2\,\boldsymbol{d}_{22}^T\right] = 0
\end{aligned}
$$

The Coriolis and centrifugal torque terms are computed using the series summation:

$$H_i = \sum_{j=1}^{n}\sum_{k=1}^{n} h_{ijk}\,\dot{\theta}_j\dot{\theta}_k \tag{6.158}$$

$$H_1 = \sum_{j=1}^{2}\sum_{k=1}^{2} h_{ijk}\dot{\theta}_j\dot{\theta}_k = h_{111}\dot{\theta}_1^2 + h_{112}\dot{\theta}_1\dot{\theta}_2 + h_{121}\dot{\theta}_1\dot{\theta}_2 + h_{122}\dot{\theta}_2^2 \tag{6.159}$$

Substituting values of h_{ijk} from Eq. (6.157) and simplifying gives

$$H_1 = -2m_2S_2(L_2^2C_2 + L_1L_2)\dot{\theta}_1\dot{\theta}_2 \tag{6.160}$$

Similarly,

$$H_2 = h_{211}\dot{\theta}_1^2 + h_{212}\dot{\theta}_1\dot{\theta}_2 + h_{221}\dot{\theta}_1\dot{\theta}_2 + h_{222}\dot{\theta}_2^2$$

or

$$H_2 = m_2S_2(L_2^2C_2 + L_1L_2)\dot{\theta}_1^2 \tag{6.161}$$

Thus,

$$\boldsymbol{H}(\theta,\dot{\theta}) = \begin{bmatrix} H_1 \\ H_2 \end{bmatrix} = \begin{bmatrix} -2m_2S_2(L_2^2C_2 + L_1L_2)\dot{\theta}_1\dot{\theta}_2 \\ m_2S_2(L_2^2C_2 + L_1L_2)\dot{\theta}_1^2 \end{bmatrix} \tag{6.162}$$

The mass of the links is at the distal end of the links, that is, at origin of frame {1} and frame {2}. Thus,

$$^1\bar{r}_1 = {}^2\bar{r}_2 = [0 \quad 0 \quad 0 \quad 1]^T \tag{6.163}$$

and the gravity is in the –ve direction of z-axis of frame {0}, that is,

$$\boldsymbol{g} = [0 \quad 0 \quad -g] \tag{6.164}$$

where $g = 9.8062$ m/s^2. Therefore, the gravity loading at the joint 1 is

$$G_1 = (m_1\boldsymbol{g}\boldsymbol{d}_{11}\,{}^1\bar{r}_1 + m_2\boldsymbol{g}\boldsymbol{d}_{21}\,{}^2\bar{r}_2) = 0 \tag{6.165}$$

Similarly, at joint 2

$$G_2 = -m_2\boldsymbol{g}\boldsymbol{d}_{22}\,{}^2\bar{r}_2 = m_2gL_2C_2 \tag{6.166}$$

Hence, equations of motion in the vector matrix form are obtained from Eqs. (6.155), (6.162) (6.165) and (6.166) as

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} m_1L_1^2 + (L_2C_2 + L_1)^2m_2 & 0 \\ 0 & m_2L_2^2 \end{bmatrix}\begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \begin{bmatrix} -2m_2S_2(L_2^2C_2 + L_1L_2)\dot{\theta}_1\dot{\theta}_2 \\ m_2S_2(L_2^2C_2 + L_1L_2)\dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} 0 \\ m_2gL_2C_2 \end{bmatrix} \tag{6.167}$$

Example 6.5 ***NE formulation of EOM for two link, 2-DOF nonplanar manipulator***

As the last example, consider the NE formulation to obtain the EOM for the two-link, 2-DOF nonplanar manipulator arm discussed in Example 6.4 and shown in Fig. 6.10. The assumption about the link masses is same as in Example 6.4, that is, the link masses m_1 and m_2 are at the distal end of the respective links.

Solution The recursive NE formulation is carried out using equations in Table 6.2 (Algorithm 6.2) to obtain the EOM. The base of the arm is fixed and the gravity is assumed to be acting in negative z_0-direction. Thus, the initial conditions are set as:

$$^0\boldsymbol{v}_0 = {}^0\omega_0 = {}^0\dot{\omega}_0 = 0; \quad {}^0\dot{v}_0 = [0 \quad 0 \quad -g]^T \tag{6.168}$$

where $g = 9.8062$ m/s^2.

The assignment of frames was carried out in Example 6.4, Fig. 6.11 and transformation matrices were determined in, Eqs. (6.146)-(6.148). The rotational transformation matrices from these equations are:

$$^0\boldsymbol{R}_1 = \begin{bmatrix} C_1 & 0 & S_1 \\ S_1 & 0 & -C_1 \\ 0 & 1 & 0 \end{bmatrix},\ {}^1\boldsymbol{R}_2 = \begin{bmatrix} C_2 & -S_2 & 0 \\ S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix},\ {}^0\boldsymbol{R}_2 = \begin{bmatrix} C_1C_2 & -C_1S_2 & S_1 \\ S_1C_2 & -S_1S_2 & -C_1 \\ S_2 & C_2 & 1 \end{bmatrix} \tag{6.169}$$

and their inverses are

$$^0\boldsymbol{R}_1 = \begin{bmatrix} C_1 & -S_1 & 0 \\ 0 & 0 & 1 \\ S_1 & -C_1 & 0 \end{bmatrix},\ {}^2\boldsymbol{R}_1 = \begin{bmatrix} C_2 & -S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix},\ {}^2\boldsymbol{R}_0 = \begin{bmatrix} C_{12} & S_{12} & S_2 \\ -S_{12} & C_{12} & C_2 \\ -S_1 & -C_1 & 1 \end{bmatrix} \tag{6.170}$$

The location of center of mass for each link with respect to its frame $\{i\}$ is given by

$$^1\bar{r}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},\quad {}^2\bar{r}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{6.171}$$

and from Eq. (6.52), the relative positions of origin of frames are

$$^0\boldsymbol{D}_1 = \begin{bmatrix} L_1C_1 \\ L_1S_1 \\ 0 \end{bmatrix};\quad {}^1\boldsymbol{D}_2 = \begin{bmatrix} L_2C_2 \\ L_2S_2 \\ 0 \end{bmatrix} \tag{6.172}$$

and from Eq. (6.102)

$$^1\boldsymbol{R}_0\,{}^0\boldsymbol{D}_1 = \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix};\quad {}^2\boldsymbol{R}_0\,{}^1\boldsymbol{D}_2 = \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix} \tag{6.173}$$

The inertia tensors of the two links with mass m_1 and m_2 at the distal end with respect to the link frame are as given by upper 3×3 submatrix of Eqs. (6.150) and (6.151) of Example 6.4.

Now, forward iterations are carried out for $i = 1, 2$.

For i = 1

The angular velocity of first joint is calculated from Eq. (6.98) as:

$$ {}^1\omega_1 = {}^1\boldsymbol{R}_0[{}^0\omega_0 + \hat{z}_0\dot{\theta}_1] = \begin{bmatrix} C_1 & S_1 & 0 \\ 0 & 0 & 1 \\ S_1 & -C_1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \dot{\theta}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dot{\theta}_1 \tag{6.174} $$

The angular acceleration is computed using Eq. (6.99) for revolute joint with as:

$$ {}^1\dot{\omega}_1 = {}^1\boldsymbol{R}_0\left[{}^0\dot{\omega}_0 + \hat{z}_0\ddot{\theta}_1 + {}^0\omega_0 \times \hat{z}_0\dot{\theta}_1\right] $$

or
$$ {}^1\dot{\omega}_1 = {}^1\boldsymbol{R}_0\hat{z}_0\ddot{\theta}_1 = \begin{bmatrix} C_1 & S_1 & 0 \\ 0 & 0 & 1 \\ S_1 & -C_1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \ddot{\theta}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \ddot{\theta}_1 \tag{6.175} $$

The linear acceleration for revolute joint is computed from Eq. (6.100), as

$$ {}^1\dot{\boldsymbol{v}}_1 = {}^1\boldsymbol{R}_0\,{}^0\dot{\boldsymbol{v}}_0 + {}^1\dot{\omega}_1 \times {}^1\boldsymbol{R}_0\,{}^0\boldsymbol{D}_1 + {}^1\omega_1 \times [{}^1\omega_1 \times ({}^1\boldsymbol{R}_0\,{}^0\boldsymbol{D}_1)] $$

Substituting values from Eqs. (6.119), (6.121), (6.124), (6.126) and (6.127)

$$ {}^1\dot{\boldsymbol{v}}_1 = \begin{bmatrix} C_1 & S_1 & 0 \\ 0 & 0 & 1 \\ S_1 & -C_1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \ddot{\theta}_1 \times \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dot{\theta}_1 \times \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dot{\theta}_1 \times \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix} \right\} $$

$$ = \begin{bmatrix} -L_1\dot{\theta}_1^2 \\ g \\ -L_1\ddot{\theta}_1 \end{bmatrix} \tag{6.176} $$

and from Eq. (6.101), the linear acceleration of the centre of mass of link 1 is

$$ {}^1\dot{\bar{\boldsymbol{v}}}_1 = {}^1\dot{\boldsymbol{v}}_1 + {}^1\dot{\omega}_1 \times {}^1\bar{\boldsymbol{r}}_1 + {}^1\omega_1 \times \left({}^1\omega_1 \times {}^1\bar{\boldsymbol{r}}_1\right) $$

Substituting ${}^1\dot{\boldsymbol{v}}_1$, ${}^1\omega_1$, ${}^1\dot{\omega}_1$ and ${}^1\bar{\boldsymbol{r}}_1$ from Eqs. (6.122), (6.126), (6.127) and (6.128), gives:

$$ {}^1\dot{\bar{\boldsymbol{v}}}_1 = \begin{bmatrix} -L\dot{\theta}_1^2 \\ g \\ L_1\ddot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \ddot{\theta}_1 \times \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dot{\theta}_1 \times \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dot{\theta}_1 \times \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} = \begin{bmatrix} -L_1\dot{\theta}_1^2 \\ g \\ -L_1\ddot{\theta}_1 \end{bmatrix} \tag{6.177} $$

For i = 2

The angular velocity is

$$ {}^2\omega_2 = {}^2\boldsymbol{R}_1\left[{}^1\omega_1 + \hat{z}_0\dot{\theta}_2\right] = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dot{\theta}_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \dot{\theta}_2 \right\} = \begin{bmatrix} S_2\dot{\theta}_1 \\ C_2\dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \tag{6.178} $$

and angular acceleration is

$$ {}^2\dot{\omega}_2 = {}^2\boldsymbol{R}_1\left[{}^1\dot{\omega}_1 + \hat{z}_0\ddot{\theta}_2 + {}^1\omega_1 \times \hat{z}_0\dot{\theta}_2\right] $$

or
$$ {}^2\dot{\omega}_2 = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \ddot{\theta}_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \ddot{\theta}_2 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \dot{\theta}_1 \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \dot{\theta}_2 \right\} = \begin{bmatrix} C_2\dot{\theta}_1\dot{\theta}_2 + S_2\ddot{\theta}_1 \\ -S_2\dot{\theta}_1\dot{\theta}_2 + C_2\ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \tag{6.179} $$

The linear velocity of link 2 is

$$ {}^2\dot{\boldsymbol{v}}_2 = {}^2\boldsymbol{R}_1\,{}^1\dot{\boldsymbol{v}}_1 + {}^2\dot{\omega}_2 \times ({}^2\boldsymbol{R}_0\,{}^1\boldsymbol{D}_2) + {}^2\omega_2 \times \left[{}^2\omega_2 \times ({}^2\boldsymbol{R}_0\,{}^1\boldsymbol{D}_2)\right] $$

Substituting values from Eqs. (6.121), (6.124), (6.128), (6.130) and (6.131),

$$ {}^2\dot{\boldsymbol{v}}_2 = \begin{bmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -L_1\dot{\theta}_1^2 \\ g \\ -L_1\ddot{\theta}_1 \end{bmatrix} + \begin{bmatrix} C_2\dot{\theta}_1\dot{\theta}_2 + S_2\ddot{\theta}_1 \\ -S_2\dot{\theta}_1\dot{\theta}_2 + C_2\ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \times \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} S_2\dot{\theta}_1 \\ C_2\dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \times \left\{ \begin{bmatrix} S_2\dot{\theta}_1 \\ C_2\dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \times \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix} \right\} $$

or
$$ {}^2\dot{\boldsymbol{v}}_2 = \begin{bmatrix} -L_1\dot{\theta}_1^2 C_2 - C_2^2 L_2\dot{\theta}_1^2 - L_2\dot{\theta}_2^2 + gS_2 \\ L_2\ddot{\theta}_2 + L_1\dot{\theta}_1^2 S_2 + S_2C_2L_2\dot{\theta}_1^2 + gC_2 \\ -L_1\ddot{\theta}_1 - L_2C_2\ddot{\theta}_1 + 2S_2L_2\dot{\theta}_1\dot{\theta}_2 \end{bmatrix} \tag{6.180} $$

and the linear acceleration of link 2 is

$$ {}^2\dot{\bar{\boldsymbol{v}}}_2 = {}^2\dot{\boldsymbol{v}}_2 + {}^2\dot{\omega}_2 \times {}^2\bar{\boldsymbol{r}}_2 + {}^2\omega_2 \times \left[{}^2\omega_2 \times {}^2\bar{\boldsymbol{r}}_2 1\right] $$

Substituting values obtained above

$$ {}^2\dot{\bar{\boldsymbol{v}}}_2 = \begin{bmatrix} -L_1\dot{\theta}_1^2 C_2 - C_2^2 L_2\dot{\theta}_1^2 - L_2\dot{\theta}_2^2 + gS_2 \\ L_2\ddot{\theta}_2 + L_1\dot{\theta}_1^2 S_2 + S_2C_2L_2\dot{\theta}_1^2 + gC_2 \\ -L_1\ddot{\theta}_1 - L_2C_2\ddot{\theta}_1 + 2S_2L_2\dot{\theta}_1\dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} C_2\dot{\theta}_1\dot{\theta}_2 + S_2\ddot{\theta}_1 \\ -S_2\dot{\theta}_1\dot{\theta}_2 + C_2\ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} S_2\dot{\theta}_1 \\ C_2\dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \times \left\{ \begin{bmatrix} S_2\dot{\theta}_1 \\ C_2\dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \times \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix} \right\} $$

or
$$ {}^2\dot{\bar{\boldsymbol{v}}}_2 = \begin{bmatrix} -L_1\dot{\theta}_1^2 C_2 - C_2^2 L_2\dot{\theta}_1^2 - L_2\dot{\theta}_2^2 + gS_2 \\ L_2\ddot{\theta}_2 + L_1\dot{\theta}_1^2 S_2 + S_2C_2L_2\dot{\theta}_1^2 + gC_2 \\ -L_1\ddot{\theta}_1 - L_2C_2\ddot{\theta}_1 + 2S_2L_2\dot{\theta}_1\dot{\theta}_2 \end{bmatrix} \tag{6.181} $$

Backward iterations

Backward iterations are now carried out for $i = 2, 1$. The end of link 2 is assumed to move freely, that is, no-load condition, giving

$$ {}^3f_3 = {}^3\eta_3 = 0 \tag{6.182} $$

For i = 2

From Eqs. (6.114) to (6.117)

$$ {}^2\mathcal{F}_2 = \boldsymbol{m}_2\,{}^2\dot{\bar{\boldsymbol{v}}}_2 $$

or
$$ {}^2\mathcal{F}_2 = m_2 \begin{bmatrix} -L_1\dot{\theta}_1^2 C_2 - C_2^2 L_2\dot{\theta}_1^2 - L_2\dot{\theta}_2^2 + gS_2 \\ L_2\ddot{\theta}_2 + L_1\dot{\theta}_1^2 S_2 + S_2C_2L_2\dot{\theta}_1^2 + gC_2 \\ -L_1\ddot{\theta}_1 - L_2C_2\ddot{\theta}_1 + 2S_2L_2\dot{\theta}_1\dot{\theta}_2 \end{bmatrix} \tag{6.183} $$

$$^2\mathcal{N}_2 = \boldsymbol{I}_2\ ^2\dot{\omega}_2 + {}^2\omega_2 \times \left(\boldsymbol{I}_2\ ^2\omega_2\right) = \begin{bmatrix}0 & 0 & 0\end{bmatrix}^T$$

$$^2\boldsymbol{f}_2 = {}^2\mathcal{F}_2 + {}^2\boldsymbol{R}_3\ ^3\boldsymbol{f}_3 = {}^2\mathcal{F}_2 \tag{6.184}$$

$$^2\boldsymbol{\eta}_2 = {}^2\boldsymbol{R}_3\ ^3\boldsymbol{\eta}_3 + ({}^2\boldsymbol{R}_0\ ^1\boldsymbol{D}_2) \times {}^2\boldsymbol{R}_3\ ^3\boldsymbol{f}_3 + ({}^2\boldsymbol{R}_0\ ^1\boldsymbol{D}_2 + {}^2\boldsymbol{R}_0\ ^2\bar{\boldsymbol{r}}_2) \times {}^2\mathcal{F}_2 + {}^2\mathcal{N}_2$$

or
$$^2\boldsymbol{\eta}_2 = m_2 L_2 \begin{bmatrix} 0 \\ L_1\ddot{\theta}_1 + C_2 L_2\ddot{\theta}_1 - 2S_2 L_2\dot{\theta}_1\dot{\theta}_2 \\ L_2\ddot{\theta}_2 + L_1 S_2\dot{\theta}_1^2 + S_2 C_2 L_2\dot{\theta}_1^2 + gC_2 \end{bmatrix} \tag{6.185}$$

For $i = 1$

$$^1\mathcal{F}_1 = m_1\ ^1\dot{\bar{v}}_1 = m_1 \begin{bmatrix} -L_1\dot{\theta}_1^2 \\ g \\ -L_1\ddot{\theta}_1 \end{bmatrix} \tag{6.186}$$

$$^1\mathcal{N}_1 = \boldsymbol{I}_1\ ^1\dot{\omega}_1 + {}^1\omega_1 \times (\boldsymbol{I}_1\ ^1\omega_1) = \begin{bmatrix}0 & 0 & 0\end{bmatrix}^T$$

$$^1\boldsymbol{f}_1 = {}^1\mathcal{F}_1 + {}^1\boldsymbol{R}_2\ ^2\boldsymbol{f}_2$$

Substituting from Eqs (6. 169), (6.184) and (6.186) gives

$$^1\boldsymbol{f}_1 = \begin{bmatrix} -m_1 L\dot{\theta}_1^2 \\ m_1 g \\ -m_1 L\ddot{\theta}_1 \end{bmatrix} + \begin{bmatrix} C_2 & -S_2 & 0 \\ S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} m_2 \begin{bmatrix} -L_1\dot{\theta}_1^2 C_2 - L_2 C_2^2\dot{\theta}_1^2 - L_2\dot{\theta}_2^2 + gS_2 \\ L_2\ddot{\theta}_2 + L_1\dot{\theta}_1^2 S_2 + S_2 C_2 L_2\dot{\theta}_1^2 + gC_2 \\ -L_1\ddot{\theta}_1 - L_2 C_2\ddot{\theta}_1 + 2S_2 L_2\dot{\theta}_1\dot{\theta}_2 \end{bmatrix}$$

Simplifying,

$$^1\boldsymbol{f}_1 = \begin{bmatrix} -m_2 S_2 L_2\ddot{\theta}_2 - m_1 L_1\dot{\theta}_1^2 - m_2(L_1 + L_2 C_2)\dot{\theta}_1^2 - m_2 C_2 L_2\dot{\theta}_2^2 \\ (m_1 + m_2)g + m_2 L_2\dot{\theta}_2^2 + m_2 C_2 L_2\ddot{\theta}_2 \\ -m_1 L_1\ddot{\theta}_1 - m_2 L_1\ddot{\theta}_1 - m_2 L_2 C_2\ddot{\theta}_1 + 2S_2 L_2\dot{\theta}_1\dot{\theta}_2 \end{bmatrix} \tag{6.187}$$

Similarly,

$$^1\boldsymbol{\eta}_1 = {}^1\boldsymbol{R}_2\ ^2\boldsymbol{\eta}_2 + ({}^1\boldsymbol{R}_0\ ^0\boldsymbol{D}_1) \times {}^1\boldsymbol{R}_2\ ^2\boldsymbol{f}_2 + ({}^1\boldsymbol{R}_0\ ^0\boldsymbol{D}_1 + {}^1\boldsymbol{R}_0\ ^1\bar{\boldsymbol{r}}_1) \times {}^1\mathcal{F}_1 + {}^1\mathcal{N}_1$$

$$^1\boldsymbol{\eta}_1 = \begin{bmatrix} -m_2 S_2 L_2(L_1 + L_2 C_2)\ddot{\theta}_1 + 2m_2 S_2 L_2\dot{\theta}_1\dot{\theta}_2 \\ m_2 C_2 L_2(L_1 + L_2 C_2)\ddot{\theta}_1 + 2m_2 C_2 L_2\dot{\theta}_1\dot{\theta}_2 \\ m_2 S_2 L_2(L_1 + L_2 C_2)\dot{\theta}_1^2 + m_2 L_2^2\ddot{\theta}_2 + m_2 L_2 C_2 g \end{bmatrix} + \begin{bmatrix} 0 \\ m_1 L_1^2\ddot{\theta}_1 \\ -m_1 L_1 g \end{bmatrix}$$
$$+ \begin{bmatrix} 0 \\ m_2 L_1(L_1 + L_2 C_2)\ddot{\theta}_1 - 2m_2 S_2 L_1 L_2\dot{\theta}_1\dot{\theta}_2 \\ m_2 L_1 g - m_2 S_2 L_1 L_2\dot{\theta}_1^2 + m_2 C_2 L_1 L_2\ddot{\theta}_2 \end{bmatrix} \tag{6.188}$$

Joint torques for the two joints are, finally, obtained using Eq. (6.118) as

$$\tau_1 = {}^1\boldsymbol{\eta}_1^T\ ^1\boldsymbol{R}_0\ \hat{z}_0$$

or
$$\tau_1 = m_1 L_1^2\ddot{\theta}_1 + m_2(L_1^2 + L_2^2 C_2^2 + 2L_1 L_2 C_2)\ddot{\theta}_1 - m_2 S_2(L_2^2 C_2 + L_1 L_2)\dot{\theta}_1\dot{\theta}_2 \tag{6.189}$$

and
$$\tau_2 = L_2 m_2(L_2\ddot{\theta}_2 + L_1 S_2\dot{\theta}_1^2 + L_2 S_2 C_2\dot{\theta}_1^2 + gC_2 \tag{6.190}$$

The above equations, Eqs. (6.189) and (6.190), are written in matrix form as:

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} m_1 L_1^2 + (L_2 C_2 + L_1)^2 m_2 & 0 \\ 0 & m_2 L_2^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix}$$
$$+ \begin{bmatrix} -2m_2 S_2(L_2^2 C_2 + L_1 L_2)\dot{\theta}_1\dot{\theta}_2 \\ m_2 S_2(L_2^2 C_2 + L_1 C_2)\dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} 0 \\ m_2 g L_2 C_2 \end{bmatrix} \tag{6.191}$$

This EOM is same as equation (6.167), obtained from the Lagrange-Euler formulation in Example 6.4.

EXERCISES

6.1 Show that the velocity coupling vector for a one-axis robot is always zero, that is

$$\boldsymbol{h}(\dot{q}, q) = 0 \tag{6.192}$$

6.2 Prove that $\dfrac{\partial({}^0\boldsymbol{T}_i)}{\partial q_j} = 0$ for $j > i$ for all $i, j = 1, 2, \ldots, \boldsymbol{n}$.

6.3 Determine the partial derivative of d_{ij} with respect to $\boldsymbol{q}_k$ where $\boldsymbol{d}_{ij}$ is

$$\boldsymbol{d}_{ij} = \begin{cases} {}^0\boldsymbol{T}_{j-1}\,\boldsymbol{Q}_j\ {}^{j-1}\boldsymbol{T}_i & \text{for } j \le i \\ 0 & \text{for } j > i \end{cases} \tag{6.193}$$

and show that

$$\frac{\partial \boldsymbol{d}_{ij}}{\partial \boldsymbol{q}_k} = \begin{cases} {}^0\boldsymbol{T}_{j-1}\,\boldsymbol{Q}_j\ {}^{j-1}\boldsymbol{T}_{k-1}\,\boldsymbol{Q}_k\ {}^{k-1}\boldsymbol{T}_i & \text{for } i \ge k \ge j \\ {}^0\boldsymbol{T}_{k-1}\,\boldsymbol{Q}_k\ {}^{k-1}\boldsymbol{T}_{j-1}\,\boldsymbol{Q}_j\ {}^{j-1}\boldsymbol{T}_i & \text{for } i \ge j \ge k \\ 0 & \text{for } i < j \text{ or } i < k \end{cases} \tag{6.194}$$

6.4 Determine the dynamic model of a 1-DOF, 1-axis planar manipulator with one rotary joint (the inverted pendulum). Assume the link to be a thin cylinder (slender member) with length L and mass m acting at the centroid of the link. Obtain direct solution and solution using Lagrange-Euler formulation and compare the two.

6.5 Find the dynamic model of a 2-DOF Cartesian manipulator shown in Fig. E6.5 using Newton-Euler formulation. Assume that the mass of the links and the actuators to drive them are m_1, and m_2, respectively, and are concentrated at the centroid of each link. Take link lengths to be L_1 and L_2.

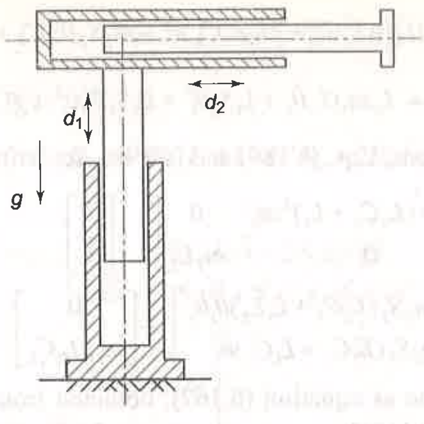

Fig. E6.5 *A 2-DOF Cartesian manipulator*

6.6 If the axis of second joint makes an angle of 60° with the axis of first joint for the two-link Cartesian manipulator instead of 90° in Fig. 6.12, find the dynamic model. Compare the resulting model with one obtained in Exercise 6.5.

6.7 Derive the equation of motion for the 1-DOF manipulator in Exercise 6.4 using recursive Newton-Euler formulation. Verify that the result is equivalent to that of Exercise 6.4.

6.8 Obtain the dynamic model of the two-link RP manipulator shown in Fig. E6.8 by determining total kinetic and potential energy of the manipulator. Assume that the links are thin homogeneous cylinders with radii b, mass m_1, and m_2, and lengths L_1 and L_2.

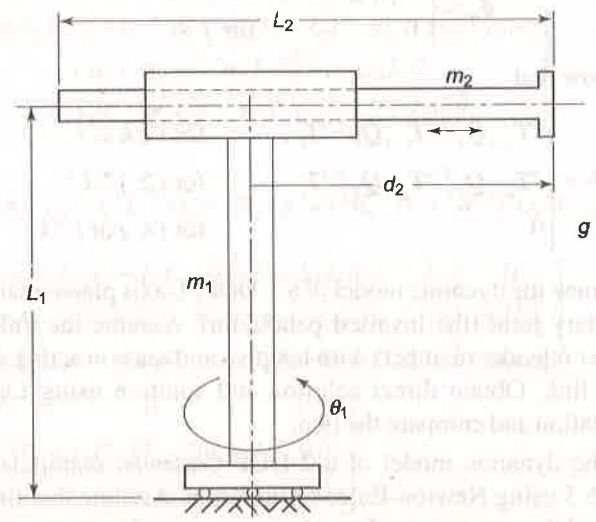

Fig. E6.8 *A two-link RP manipulator*

6.9 Determine the dynamic model of the manipulator in Exercise 6.8 using Newton-Euler formulation.

6.10 Find the dynamic equations of motion for the planar 2-DOF manipulator with one prismatic joint and one rotary joint, as shown in Fig. E6.10, using Lagrange-Euler formulation. The link parameters are shown in the figure.

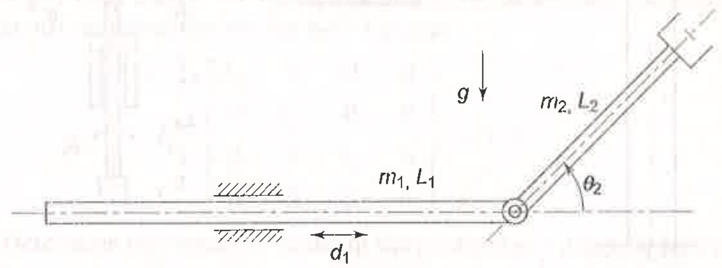

Fig. E6.10 *A 2-DOF planar manipulator with one prismatic and one rotary join*

6.11 Compute the joint torques for the two-link planar manipulator discussed in Example 6.1 for $\theta_1 = 45°$ and $\theta_2 = 60°$. Take other link parameters from Example 6.3.

6.12 Obtain the dynamic equations for a three-axes Cartesian arm configuration in Fig. 1.11 using the Newton-Euler and Lagrange-Euler formulations.

6.13 For the nonplanar 2-DOF manipulator with two rotary joints, shown in Fig. 6E.13, find the dynamic equations of motion using Lagrange-Euler formulation. Assume link masses, m_1 and m_2 to be unity and concentrated at the distal ends of the respective links.

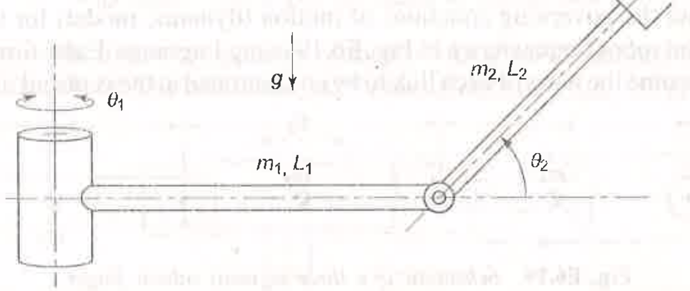

Fig. E6.13 *A 2-DOF non-planar manipulator with two rotary joins*

6.14 Determine the effective inertia, coupling inertia, and gravity terms for the 4-DOF SCARA manipulator shown in Fig. E6.14. Take the mass of four links as m_1, m_2, m_3, and m_4, and lengths as L_1, L_2, L_3, and L_4, respectively, acting at the centroid of each link. Assume each link to be a slender member. The SCARA manipulator shown in Fig. E6.14 is a simplified model of SCARA manipulator discussed in Example 3.5, Fig. 3.20, with $L_{11} = L_1$ and $L_{12} = 0$.

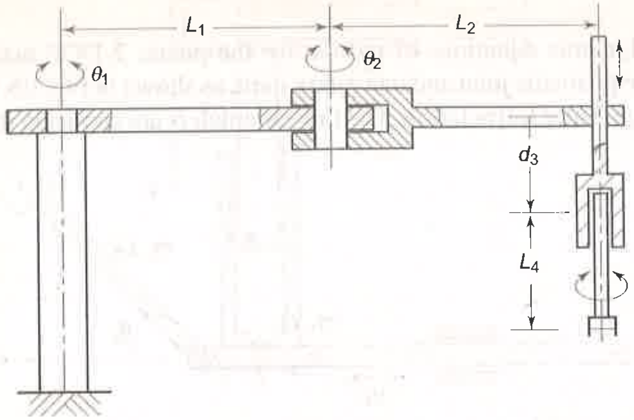

Fig. E6.14 *Schematic diagram of SCARA manipulator*

6.15 Obtain the equations to determine the joint torques (τ_1, τ_2, τ_3, and τ_4) using Lagrange-Euler formulation for the SCARA manipulator described in Exercise 6.14.

6.16 Formulate the dynamic model of the 2-link manipulator in Fig. E6.10, Exercise 6.10 using the Newton-Euler formulation. Identify the centrifugal force, Coriolis force, and gravity loading for each joint.

6.17 Derive the dynamic model for the 3-DOF cylindrical arm (see Fig. 1.12) using Lagrange-Euler formulation.

6.18 For the SCARA manipulator in Exercise 6.14, plot the time history of position and torque for each joint assuming length of each link as 1 unit and mass of each link as 1 unit.

6.19 Find the governing equations of motion (dynamic model) for the three-joint robot finger shown in Fig. E6.19 using Lagrange-Euler formulation. Assume the mass of each link to be concentrated at the centroid of the link.

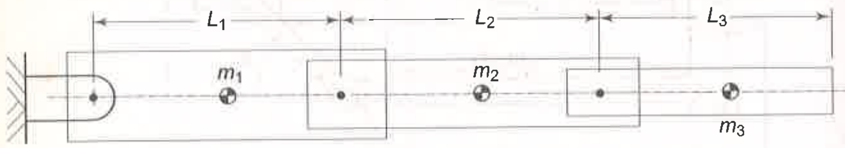

Fig. E6.19 *Schematic of a three-segment robotic finger*

6.20 For the 2-DOF manipulator in Example 6.1 assume that the links have a length of 2 units and a mass of 1 unit, each. Calculate the linear velocity and acceleration and angular velocity and acceleration of the centroids of the two links as a function of time, if the joint angle of both joints varies with the time function:

$$\theta_i(t) = 2t + t^2 \tag{6.195}$$

6.21 A n-DOF manipulator is required to work at very low velocities only. What simplifications will result from this condition in the Lagrange-Euler formulation? Explain.

6.22 If a manipulator is to be used in a gravity-free environment, say space, what will be the effect of this on the dynamic model. Which forces will be significant under these conditions?

6.23 A two link, 2-DOF RP manipulator is shown in Fig. E6.23. Assuming the centroid inertia tensors for both links as

$$\boldsymbol{I}_i = \begin{bmatrix} I_{x_i} & 0 & 0 & 0 \\ 0 & I_{y_i} & 0 & 0 \\ 0 & 0 & I_{z_i} & 0 \\ 0 & 0 & 0 & m_i \end{bmatrix}; \; i = 1,2 \tag{6.196}$$

Determine the dynamic model of the manipulator using (i) Newton-Euler formulation (ii) Lagrange-Euler formulation.

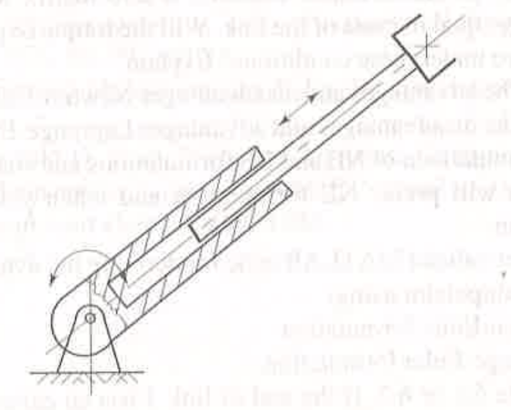

Fig. E6.23 *Two degree of freedom RP manipulator*

6.24 Derive the dynamic equations of for the 3-DOF RPR manipulator shown in Fig. E6.24. The centroidal inertia tensors of link 1 and 3 is given by

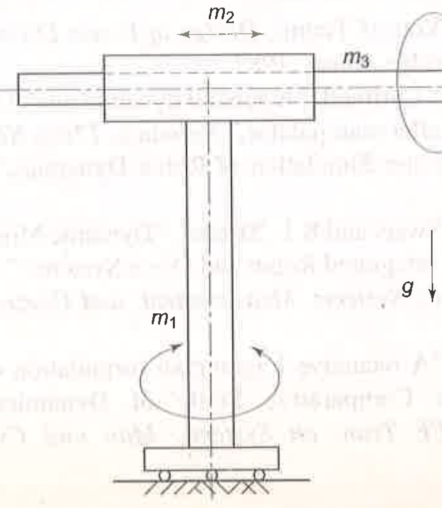

Fig. E6.24 *A 3-DOF RPR arm*

$$I_i = \begin{bmatrix} I_{x_i} & 0 & 0 & 0 \\ 0 & I_{y_i} & 0 & 0 \\ 0 & 0 & I_{z_i} & 0 \\ 0 & 0 & 0 & m_i \end{bmatrix}; i = 1,3 \tag{6.197}$$

The link 2 has a point mass m_2 located at the origin of its frame. Use, both, Newton-Euler and Lagrange-Euler formulations and compare the results.

6.25 Write a generalized MATLAB code for deriving the inertia tensor of a link. It should take input about the link geometry, the reference frame about which inertia tensor is required and assumptions about location of center of mass, if any.

6.26 Under the assumption that the mass of a slender link is concentrated at its distal end, the inertia tensor becomes a zero matrix with only element (4,4) being equal to mass of the link. Will the torque required to move the link be zero under these conditions? Explain.

6.27 Describe the advantages and disadvantages Newton-Euler formulation.

6.28 Describe the disadvantages and advantages Lagrange-Euler formulation.

6.29 Make a comparison of NE and LE formulations and state the situation(s) when you will prefer NE formulation and when you will prefer LE formulation.

6.30 Write a generalized MATLAB code for deriving the dynamic model of an *n*-DOF manipulator using:
(a) Newton-Euler formulation
(b) Lagrange-Euler formulation.

6.31 In Example 6.1 or 6.2, if the end of link 2 has an external load of 1 kg, what will be the change in the dynamic model obtained?

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7

Trajectory Planning

The end-effector of the manipulator is required to move in a particular fashion to accomplish a specified task. The previous chapters discussed the mathematical models for geometric, kinematic, static, and dynamic behaviour of mechanical manipulators, which form the background for this chapter. The execution of the specific task requires the manipulator to follow a preplanned path, which is the larger problem of motion or trajectory planning and motion control for the manipulator. The goal of trajectory planning is to describe the requisite motion of the manipulator as a time sequence of joint/link/end-effector locations and derivatives of locations, which are generated by "interpolating" or "approximating" the desired path by a polynomial function. These time base sequence locations, also called time law or time history, obtained from the trajectory planning serve as reference input or "control set points" to the manipulator's control system. The control system, in turn, assures that the manipulator executes the planned trajectories. The motion control is discussed in the next chapter.

This chapter focuses on various trajectory-planning issues. The chapter begins with describing the terminology involved in trajectory planning. Some techniques for trajectory generation for two types of motion, point-to-point motion, and continuous-path motion, are presented. First, the problem of trajectory planning in joint space is considered and then Cartesian space planning is discussed. The chapter concludes with examples illustrating various trajectory-planning techniques.

The user usually defines the desired trajectory by a number of parameters. A polynomial function is identified to "approximate" the trajectory. The trajectory planning consists of generating a time sequence of values of different parameters attained by the polynomial interpolating the trajectory.

Every manipulator has at least capability to move from an initial location (position and orientation) to a final desired location. The kinematics and dynamics of the manipulator govern the transition of posture with actuators exerting the

desired torques. As no limits must be violated and no unmodelled behaviour should be executed, it is necessary to devise planning algorithms that generate smooth trajectories.

One of the main objectives of any trajectory-planning algorithm is to achieve a smooth motion of the manipulator. A smooth function is one that is continuous and has a continuous first derivative. The smooth motion has the important advantage of reducing the vibrations and wear of the mechanical system.

7.1 DEFINITIONS AND PLANNING TASKS

To begin with, a trajectory planner may be thought of as a black box to which the user inputs a list of parameters and constraints describing the desired trajectory. The trajectory planner generates a time sequence of intermediate configurations expressed in either joint or Cartesian coordinate frames. The block diagram in Fig. 7.1 shows the parameters involved in the trajectory-planning problem. To solve the trajectory-planning problem the terminology used in trajectory planning is discussed first.

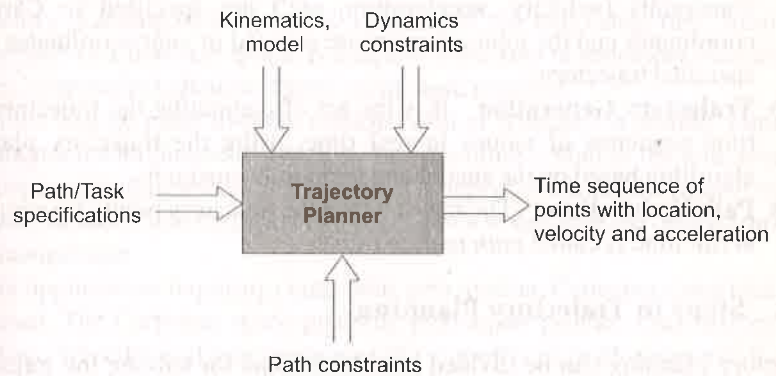


Fig. 7.1 *The trajectory planning problem*

7.1.1 Terminology

Trajectory planning involves certain terms and it is important to understand their meaning without ambiguity, to avoid confusion between terms. In this section, some of the frequently used terms in trajectory planning are defined.

(i) **Path** A path is the locus of points (either in the joint space or in the Cartesian space) to be traversed by the manipulator to execute the specified task. A path is a purely geometric (spatial) description of the motion.

(ii) **Trajectory** A trajectory is a path with specified qualities of motion, that is, a path on which a time law is specified in terms of velocities and/or accelerations at each point. In other words, a trajectory is the time sequence (or time history) of position, velocity, and acceleration for each joint or end-effector of the manipulator. A trajectory is both a spatial and temporal representation of motion. It can be specified either in joint space or in Cartesian space.

(iii) **Knot Points or Via Points** Knot points or via points or interpolation points are the set of intermediate locations between the start and goal points on the trajectory through which the manipulator must pass enroute to the destination.

(iv) **Spline** It is the smooth function that passes through the set of via points.

(v) **Joint Space Trajectory Planning** In joint space trajectory planning each path point is specified in terms of a desired position and orientation of the end-effector frame relative to the base frame. Each of these points is converted into a set of desired joint positions by application of inverse kinematics. A smooth function is then found for each of the joints, which passes through these points.

(vi) **Cartesian Space Trajectory Planning** In Cartesian space trajectory planning, the path is explicitly specified in the Cartesian space. The path constraints (velocity, acceleration, etc.) are specified in Cartesian coordinates and the joint actuators are servoed in joint coordinates to the specified trajectory.

(vii) **Trajectory Generation** It is the act of computing the trajectory as a time sequence of values in real time, using the trajectory planning algorithm based on the spatial and temporal constraints.

(viii) **Path Update Rate** The rate at which the trajectory points are computed at run time is called *path update rate*.

7.1.2 Steps in Trajectory Planning

Trajectory planning can be divided into three steps for solving the trajectory-planning problem as outlined below.

(i) Task Description The first step in the motion-planning problem is to identify the kind of motion required. This specification of the requisite trajectory is the input to the trajectory-planning algorithm. The tasks can be grouped into three different categories.

In the case of applications such as *pick and place* operation, the task is specified as initial and final end-effector location. This is called *point-to-point* (PTP) motion. In a point-to-point motion, except for the constraint that the motion be smooth, no particular specification about the intermediate locations of the end-effector is given and the planner is free to formulate any convenient path. In such cases, the user specifies only the goal point (location) in Cartesian space for a known initial point.

If in addition to start and finish points, a specific path between them is required to be traced by the end-effector in Cartesian space, this is called *continuous path* (CP) motion and trajectory. Operations such as arc welding and plotting are examples of continuous path motion. In such cases, user specifies the type and parameters of the path to be tracked.

There is a third type of task description, where more than two points of the path are specified. This is done to ensure better monitoring of the executed trajectory in case of applications similar to those discussed for point-to-point motion. A typical example is a pick-n-place operation where obstacles are present in the path. Here, the first point on the path is called the "initial" point, while the last point is called the "goal" point. The intermediate locations are the via points. Together, these are referred to as *path points*.

Apart from the required task attributes, the user can also include temporal attributes such as the travel time in any of the task specifications discussed above. The task definition must not be misunderstood to mean that end-effector or manipulator is only required to traverse a path or trajectory. The task performed by the tool or gripper at a specific point in PTP motion or along the path in CP motion is not a part of trajectory planning. It is an additional task for the "tool" which may or may not be controlled by the robot controller.

(ii) Selecting and Employing a Trajectory Planning Technique The various trajectory-planning techniques fall into one of the two categories: *Joint space techniques* or *Cartesian space techniques*. Next step in trajectory planning is to select a particular technique based on the task required.

In the case of point-to-point motion (with or without via points), joint space techniques are employed in which motion planning is done at the joint level. The joint-space planning schemes generate time-dependent functions (time history) of all joint variables and their first two derivatives to describe the desired motion of the manipulator.

For applications requiring continuous path motion, Cartesian space techniques are used. The Cartesian space-planning techniques provide time history of the location, velocity, and acceleration of the end-effector with respect to the base. The corresponding joint variables and their derivatives are computed, using inverse kinematics.

(iii) Computing the Trajectory The final step is to compute the time sequence of values attained by the functions generated from the trajectory planning technique. These values are computed at a particular path update or sampling rate. The path update rate in real time, lies between 20 Hz and 200 Hz in typical industrial manipulator systems.

7.2 JOINT SPACE TECHNIQUES

To plan a trajectory, the values of joint variables have to be determined from the end-effector location specified by the user. It is useful to recall that joint-space techniques are used for point-to-point motion with or without via points and each path point is specified by the end-effector position and orientation with respect to the base in Cartesian space.

The first step in these techniques is to obtain the corresponding set of joint variable values for each specified path point, either by employing the inverse kinematics algorithm or the variable can be directly recorded if trajectory planning is performed by *teaching-by-showing technique*.

The next step is to find a smooth function $q(t)$ for each of the n joints of an n-DOF manipulator, which interpolates the joint-variable vector for each given path point, with respect to imposed constraints. In general, following features are required in a joint space trajectory-planning algorithm:

- An important temporal constraint is that the travelling time between any two path points is the same for each joint so as to make all the joints reach their corresponding path-point locations simultaneously. This guarantees the attainment of the desired location by tool with respect to the base at each path point.
- Joint locations and velocities be continuous functions of time (continuity of accelerations may be imposed, too) so as to ensure smooth motion.
- The interpolating function should not be computationally intensive.
- Nonsmooth trajectories and other similar undesirable effects be minimized.

For moving the end-effector from starting (initial) position q^s to a goal (target) position q^g in a certain amount of time, it is seen that there are innumerable functions that satisfy the aforesaid criteria as shown in Fig. 7.2. The selection of appropriate polynomial function plays a pivotal role in joint-space planning. For making a choice of polynomial, two of the common approaches are:

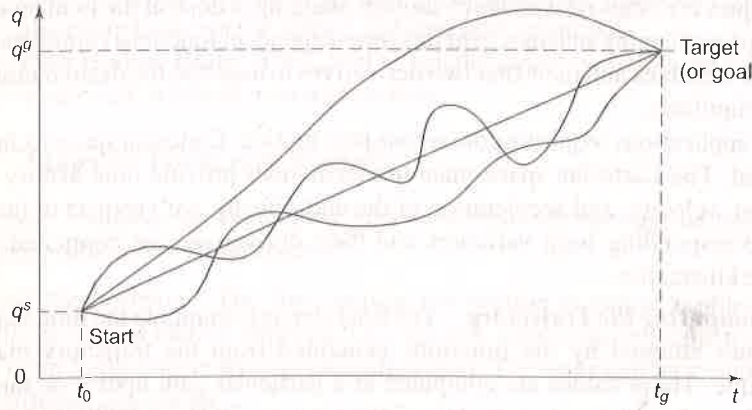


Fig. 7.2 *Smooth functions for motion to goal position*

(a) Choose the highest degree polynomial with at least as many coefficients as are the specified constraints.

(b) Split the trajectory into segments and use lower degree polynomial to interpolate each segment with smooth transition from one segment to next segment.

In the first approach, the higher degree polynomials tend to be more oscillatory and numerically less accurate. The second approach imposes additional constraints required for blending two segments for smooth motion.

Two polynomial functions are examined here, one in each of above approaches:

(i) a *cubic polynomial* in which cubics connect the path points of each joint in a smooth way, and

(ii) a *linear function with parabolic blends* in which each segment between two successive path points contains a linear function with parabolic blends near the path points.

Both these functions are widely used in joint-space techniques and are not very demanding from computational viewpoint. Before taking up these specific functions, general discussion on a p-degree polynomial function is carried out to illustrate the concepts of constraint identification and their use to compute polynomial coefficients.

7.2.1 Use of a p-Degree Polynomial as Interpolation Function

The selection of a single polynomial for the entire joint path depends on the number of constraints imposed and the type of motion desired. The minimum number of constraints for a smooth motion between two points are

(i) Initial position,
(ii) Initial velocity,
(iii) Final position, and
(iv) Final velocity.

With four constraints, a third-degree polynomial (cubic spline) with four coefficients can be used, that is,

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \tag{7.1}$$

If, in addition, the initial and final accelerations are also specified, then one may use a fifth-degree polynomial,

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \tag{7.2}$$

However, for practical purposes of simple pick-n-place task, just specifying a start and finish point is not enough for satisfactory performance. The reason is that while picking up an object the motion of the end-effector must be directed away from the supporting surface so that the gripper does not crash into the supporting surface. Thus, a "lift-off position" along the outward normal to the surface, from the initial position, is selected as additional specification. On the basis of similar considerations, a "set-down position" is selected to specify the correct approach direction. This gives, in all, four position constraints as

(i) Initial position,
(ii) Lift-off position,
(iii) Set-down position, and
(iv) Final position.

In a typical pick-n-place task, the manipulator begins moving at constant acceleration, away from the initial position in appropriate direction (for example, in vertically upward direction from a horizontal surface). After it has moved a safe distance, lift-off is complete. From then onwards, it moves at constant velocity, that is, with zero acceleration. The constant velocity motion continues until the set-down position from where it begins to decelerate ending at the final position with zero velocity.

A typical trajectory of this kind is shown in Fig. 7.3, which has six constraints, four position constraints given above and two acceleration constraints—constant acceleration in the initial phase and the constant deceleration in the final phase. The two velocity constraints are in fact subsumed by the two acceleration constraints.

These concepts for the end-effector motion can be extended to position, velocity, and acceleration of the joints. Further, additional constraints are required for continuity of each joint variable and its derivatives at every path point. In summary, the constraints that can be applied to a joint for joint interpolated trajectory planning can be grouped into four categories: initial point constraints, intermediate constraints, final point constraints, and time constraints. There are a number of possible constraints in each of these categories.

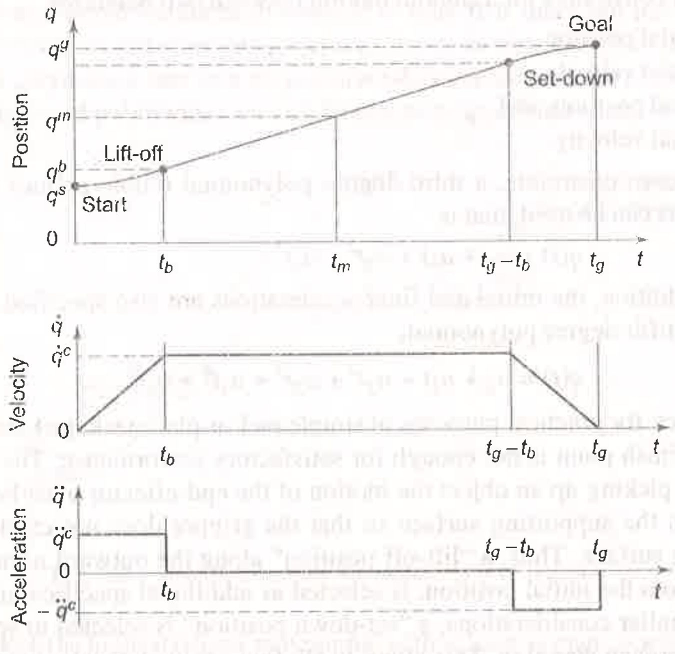


Fig. 7.3 *Trajectory of a joint with six constraints*

(a) *Initial point constraints* are the constraints on the initial or starting point.
   1. Position constraints,
   2. Velocity constraints, and
   3. Acceleration constraints.

(b) *Intermediate constraints* are the constraints on the intermediate points of the trajectory.

1. Lift-off position,
2. Continuity at lift-off position,
3. Continuity of velocity at lift-off point,
4. Continuity of acceleration at lift-off point,
5. Set-down position,
6. Continuity at set-down position,
7. Continuity of velocity at set-down point, and
8. Continuity of acceleration at set-down point.

Additional via points or knot points may be specified and at each of such point velocity/acceleration may be specified with additional constraint of continuity for smooth transition.

(c) *Final point constraints* are the constraints on the final or goal or target point of the trajectory.
   1. Position,
   2. Velocity, and
   3. Acceleration.

(d) *Time constraints are* the constraints imposed by the cycle time for the task.
   1. Total traversal time,
   2. Traversal time for each segment, and
   3. Duration of a specific polynomial interpolating a trajectory in case of blended trajectories.

A suitable polynomial function is selected such that the required constraints are satisfied and the trajectory is smooth. It is known that a p-degree polynomial has $(p+1)$ coefficients and can therefore satisfy $(p+1)$ constraints. Thus, for a trajectory with $(p+1)$ constraints, a p-degree polynomial can be the general choice for trajectory planning. However, if the degree of polynomial is high (say > 5), it becomes computationally intensive and it tends to cause extraneous motion.

The complexity can be reduced by splitting the trajectory into two or more segments so that in each segment polynomials of lower degree can be used to interpolate trajectory. For example, if a trajectory is split with two intermediate points, there will be three segments. For each of these three segments, lower degree polynomials can be used to interpolate the joint positions.

The type of polynomial used for interpolation of the trajectory also has options like

1. Linear position or constant velocity trajectory
2. Parabolic position or linear velocity or constant acceleration trajectory

Some other complex possibilities of polynomials are:

(a) *Linear trajectory with parabolic blends* — The first segment is interpolated with a quadratic polynomial (parabola) specifying trajectory from initial position to the "blend" point. The second segment is interpolated as linear or constant velocity followed by a quadratic polynomial in the last (third) segment, identical to first segment.

(b) *4-3-4 Trajectory* — A fourth-degree polynomial in the first segment followed by a cubic polynomial in the second segment followed by a fourth-degree polynomial in the last segment.

(c) *3-5-3 Trajectory* — A third-degree polynomial or cubic spline for the first segment, a fifth-degree polynomial for the second segment, and a third-degree polynomial in the last segment.

In segmented trajectories, the trajectories must blend smoothly at the intermediate points. These intermediate points may, sometimes, be only "virtual points" and the specification of the location of these points may not be required. Only the continuity constraints are imposed on them to solve the coefficients of the polynomials. An estimation of the computations involved with segmented trajectories is made as follows.

Suppose the trajectory is split into segments such that there are k path points. Thus, there will be $(k-2)$ intermediate or via points and $(k-1)$ polynomials will be required to connect the k path points. For each of the $(k-2)$ via points, there will be $(p+1)$ constraints for the p-degree polynomial and, say, there are six constraints on the initial and final points. Hence, in all there will be $(p+1)(k-2)+6$ constraints that must be solved to determine the trajectory consisting of $(k-1)$ p-degree polynomials.

A set of cubic polynomials ($p=3$) can be used for interpolation of trajectory, preserving the continuity in the first and second derivatives at the interpolation points. Such cubic polynomials are known as *cubic-spline functions*. The cubic-spline trajectories are most preferred among the interpolating functions as the cubic polynomial is the lowest degree polynomial for which both first and second derivatives exist, allowing for continuity in velocity and acceleration, and it is computationally more efficient. The number of constraint equations for a k path point trajectory interpolated with cubic polynomials will be $4k-2$. Cubic polynomial and other trajectories are discussed in the following sections. For example, a five-cubic polynomials interpolation will have five segments and six path or interpolation points and the number of constraint equations will be 22.

7.2.2 Cubic Polynomial Trajectories

The problem of obtaining cubic polynomial trajectories is considered first, for a point-to-point motion without via points and then it is extended to the case with via points.

Point-to-Point motion without via points In this scheme, the goal point and travel time are specified by the user. The set of joint-variable values for the given goal point are obtained from the inverse kinematics. It is assumed that the joint-variable values corresponding to the initial location of the end-effector are known.

A cubic trajectory is obtained for each joint variable, $q(t)$, of the n-DOF manipulator. The determination of the interpolation cubic polynomial is carried out for each of the n-joint variables.

Let q^s and q^g be the starting and goal-point values, respectively, of the joint variable q. Note that the generalized joint variable q_i denotes the joint variable (θ_i or d_i) for joint i of the n-DOF manipulator. Therefore, to denote time-dependent values of q_i, at different path points, a superscript is used, for example, q_i^j implies the value of $q_i(t)$ at path point j. The subscript i for joint i is dropped whenever only one joint variable is considered, for the sake of clarity. The manipulator motion begins at time $t = 0$ ($t_0 = 0$) and ends at $t = t_g$, that is, the travel time or time to reach goal is t_g.

For a smooth motion between the initial and goal points, the functions $q(t)$ and $\dot{q}(t)$ have to be smooth. This requirement imposes two constraints each on the joint position and velocity functions. Further, $q(t)$ is zero for $t < 0$ and $t > t_g$. This, along with the continuity requirements gives the four constraints as

$$\begin{aligned} q(0) &= q^s \\ q(t_g) &= q^g \\ \dot{q}(0) &= 0 \\ \dot{q}(t_g) &= 0 \end{aligned} \tag{7.3}$$

To satisfy these four constraints, the polynomial $q(t)$ must be of order three, that is, a cubic polynomial with four coefficients. This justifies our assumption that a cubic polynomial would give a smooth motion profile for joint variable q. Therefore, to describe the joint motion, assume that the cubic polynomial is

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \tag{7.4}$$

which gives a parabolic velocity profile

$$\dot{q}(t) = a_1 + 2a_2 t + 3a_3 t^2 \tag{7.5}$$

and a linear acceleration profile

$$\ddot{q}(t) = 2a_2 + 6a_3 t \tag{7.6}$$

Applying the constraints of Eq. (7.3) to Eqs. (7.4) and (7.5), gives the following set of four equations in four unknowns:

$$\begin{aligned} a_0 &= q^s \\ a_0 + a_1 t_g + a_2 t_g^2 + a_3 t_g^3 &= q^g \\ a_1 &= 0 \\ a_1 + 2a_2 t_g + 3a_3 t_g^2 &= 0 \end{aligned} \tag{7.7}$$

The solution of Eq. (7.7) gives the coefficients of cubic polynomial, Eq. (7.4), as

$$\begin{aligned} a_0 &= q^s \\ a_1 &= 0 \\ a_2 &= \frac{3}{t_g^2}(q^g - q^s) \\ a_3 &= -\frac{2}{t_g^3}(q^g - q^s) \end{aligned} \tag{7.8}$$

Thus, the cubic polynomial to interpolate the path connecting the initial joint position to final joint position is

$$q(t) = q^s + \frac{3}{t_g^2}(q^g - q^s)t^2 - \frac{2}{t_g^3}(q^g - q^s)t^3 \quad (7.9)$$

It may be noted that this solution is for the case when the velocity is zero both at the start and finish points.

The acceleration profile, though linear, has discontinuity at initial and final positions and the motion will be jerky at the start and at the goal point due to this. The variation of the joint variable, velocity, and acceleration are shown in Fig.7.4. The acceleration at the start and goal points is difficult to realize with physical actuators.

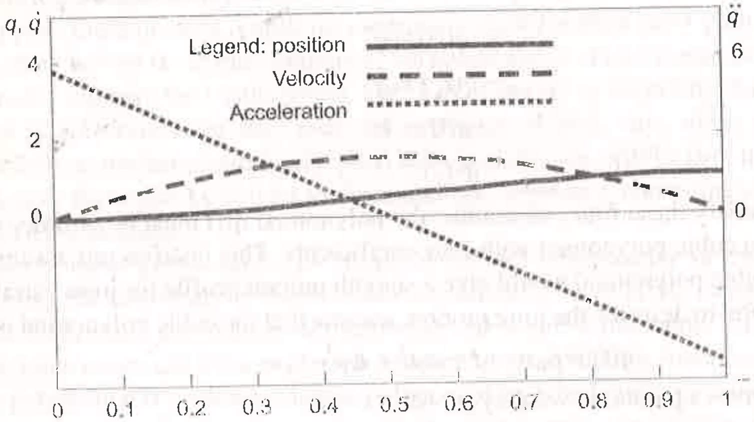


Fig. 7.4 *Cubic polynomials trajectory for PTP motion*

Point-to-point motion with via points Now consider the problem of determining the cubics that connect the given "k" path points smoothly. If the manipulator comes to rest at each via point, then the cubic solution of above section can be applied between each pair of successive via points. A different approach is needed when the manipulator is required to pass the via points without stopping or coming to rest at the via points. For a path with k path points, there are $(k-2)$ intermediate or via points, and $(k-1)$ polynomials are required to connect the k path points in a smooth way.

Usually, each path point is specified in terms of the desired end-effector position and orientation. Similar to the single goal point case, each of these path points is transformed into a set of joint-variable values by inverse kinematics. Assume that q_j denotes the value taken by joint variable q corresponding to path point j and the cubic polynomial connecting path points j and $(j+1)$ is $P_j(t)$. This cubic is defined in the time interval $t_{j-1} < t < t_j$, with $T_j = t_j - t_{j-1}$ denoting the travel time between path points j and $(j+1)$. These time intervals for k path point with $(k-2)$ via points are shown in Fig. 7.5. The total travel time for the path is $T(=t_g)$ with

$$T = \sum_{m=1}^{k-1} T_m \quad (7.10)$$

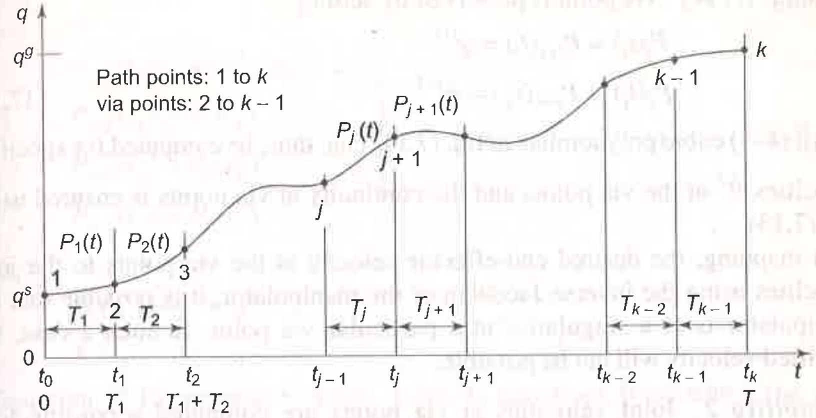


Fig. 7.5 *Time intervals for k path points and (k–2) via points*

Thus, the joint position function, $q(t)$ for the entire path over the time interval $0 < t < T$ is defined by $(k-1)$ cubic polynomials as

$$q(t) = \begin{cases} P_1(t) & 0 < t < T_1 \text{ or } t_0 < t < t_1 \\ P_2(t - T_1) & T_1 < t < T_1 + T_2 \\ \vdots & \\ P_j(t - \sum_{m=0}^{j-1} T_m) & \sum_{m=0}^{j-1} T_m < t < \sum_{m=0}^{j} T_m \text{ or } T_{j-1} < t < T_j \\ \vdots & \\ P_{k-1}(t - \sum_{m=0}^{k-2} T_m) & \sum_{m=0}^{k-2} T_m < t < \sum_{m=0}^{k-1} T_m \text{ or } T_{K-2} < t < T_{K-1} \end{cases} \quad (7.11)$$

For smooth trajectory at each via point, additional constraints are required. Once these constraints are specified, the $(k-1)$ cubic polynomials are determined as before. There are many possible ways of specifying the additional constraints at via points. Three possibilities are presented here.

Possibility 1 The desired velocity at each via point is specified in terms of Cartesian linear and angular velocity of the tool frame at that instant. These desired velocities are mapped to the joint velocities using the inverse Jacobian. The initial and goal point joint velocities are set to zero to ensure the continuity of $q(t)$ at the end points, as before. Thus, the set of constraints imposed on j^{th} cubic $P_j(t)$ between via points j and $(j+1)$ is given by

$$\begin{aligned} P_j(t_{j-1}) &= q^j \\ P_j(t_j) &= q^{j+1} \\ \dot{P}_j(t_{j-1}) &= \dot{q}^j \\ \dot{P}_j(t_j) &= \dot{q}^{j+1} \end{aligned} \quad (7.12)$$

The four constraints in Eq. (7.12) represent a set of four equations from which the four unknown coefficients of cubic $P_j(t)$ can be computed. The continuity of q^j and $\dot{q}^j(t)$ at j^{th} via point is preserved by setting

$$P_j(t_j) = P_{j+1}(t_j) = q^{j+1}$$

and
$$\dot{P}_j(t_j) = \dot{P}_{j+1}(t_j) = \dot{q}^{j+1} \tag{7.13}$$

All (k–1) cubic polynomials in Eq. (7.11) can, thus, be computed for specified velocities $\dot{q}^j$ at the via points and the continuity at via points is ensured using Eq. (7.13).

In mapping, the desired end-effector velocity at the via points to the joint velocities using the inverse Jacobian of the manipulator, it is possible that the manipulator is at a singularity at a particular via point. In such a case, the specified velocity will not be possible.

***Possibility* 2** Joint velocities at via points are computed according to a heuristic criterion. In this case, the constraints imposed on the cubics are same as those used in the previous case but the method of specifying via-point joint velocity is different.

The joint variables $q(t)$ at the path points along with the travel time T, are specified by the user. The joint velocities at via points are assigned using a criterion, which preserves the continuity of $q(t)$. One such criterion is to use the average velocity as

$$\begin{aligned}
&\dot{q}^1 = 0 \\
&\dot{q}^j = \begin{cases} 0 & \text{if } \operatorname{sign}(\dot{\bar{q}}^{j-1}) \neq \operatorname{sign}(\dot{\bar{q}}^j) \\ \frac{1}{2}(\dot{q}^{j-1} + \dot{q}^j) & \text{if } \operatorname{sign}(\dot{\bar{q}}^{j-1}) = \operatorname{sign}(\dot{\bar{q}}^j) \end{cases} \quad \text{for } j = 2, \cdots, (k-1) \\
&\dot{q}^k = 0
\end{aligned} \tag{7.14}$$

where $\dot{\bar{q}}^j = \dfrac{q^{j+1} - q^j}{T_j}$ is the average velocity for segment j.

These values of $\dot{q}^j$ given by Eq. (7.14) are used in Eq. (7.11) to obtain the unknown coefficients of the cubics.

***Possibility* 3** Another alternative to generate cubics, interpolating more than two path points, is to cause the joint accelerations to be continuous at the via points. This implies that the velocities at via points are automatically chosen by the system to satisfy acceleration continuity, which also ensures velocity continuity at via points.

Under these conditions, constraints at the end-points are same as the case of single goal point, that is, joint velocities are continuous at the end-points but accelerations are discontinuous. The velocity and acceleration continuity restrictions at the via points give the following constraints

$$\begin{aligned}
&P_1(t_0) = q^1 = q^s \\
&\dot{P}_1(t_0) = \dot{q}^1 = \dot{q}^s = 0 \\
&\cdots \\
&\left.\begin{aligned}
&P_{j-1}(t_j) = q^j P_{j-1}(t_j) = q^j \\
&P_{j-1}(t_j) = P_j(t_j) \\
&\dot{P}_{j-1}(t_j) = \dot{P}_j(t_j) \\
&\ddot{P}_{j-1}(t_j) = \ddot{P}_j(t_j)
\end{aligned}\right\} \quad \text{for } j = 2, \cdots, (k-1) \\
&\cdots \\
&P_k(t_k) = q^k = q^g \\
&\dot{P}_k(t_k) = \dot{q}^k = \dot{q}^g = 0
\end{aligned} \tag{7.15}$$

Equation (7.15) represents a set of 4(k–1) equations from which the four unknown coefficients for each of the (k–1) cubics can be obtained.

It is important to note that Eq. (7.15) guarantees that the path points are reached as per the temporal constraints specified as well as ensures the continuity of joint velocity and acceleration functions at the via points.

The three possibilities discussed above to connect the cubics smoothly at the via points require different amount of computations. The first requires the least computations, whereas the last one is most computationally intensive. The possibility 3 ensures the velocity and acceleration continuity at the via points giving a smooth motion throughout, while in possibility 2, jerks are possible at via points because acceleration continuity is not enforced. The heuristics used would also influence the trajectory. Possibility 1 is useful when it is desired to have specific velocities at the via points. This obviously requires the knowledge of desired velocities at via points, which may be a burden.

7.2.3 Linear Function with Parabolic Blends

Another possible trajectory is to use linear interpolation from start to goal point. This would cause the velocity to be discontinuous at the beginning and end of motion and would require infinite acceleration. Also, in the case of cubic trajectory for PTP motion without via points (section 7.2.2), the discontinuity of acceleration at the end-points is observed (see Fig.7.4) and the acceleration is not physically realizable. One method to avoid this difficulty is to use higher degree polynomial with more constraints, requiring more complex computations, as discussed in section 7.2.1. Alternately, it is possible to use a blended polynomial, which allows for verification whether the resulting velocities and accelerations can be supported by the physical mechanical manipulator.

A trapezoidal velocity profile, for example, imposes a constant acceleration in start phase, a cruise velocity, and a constant deceleration in the final phase, (see Fig. 7.3). The resulting trajectory has a linear segment with parabolic segments at both ends. In general, a blended trajectory is constructed using polynomial interpolation function with blending functions at both ends, which are splined together so that the entire path has continuity of position and velocity, including the smooth motion at resulting via points.

The trapezoidal velocity profile trajectory planning is discussed, first for point-to-point motion without via points and later, it is extended to the case of point-to-point motion with via points.

Blended trajectory for point-to-point motion without via points For the given start and target points, a smooth trajectory, which consists of a linear segment with parabolic blends near the end-points, is used. During the blend portion of the trajectory, constant acceleration is used to achieve a smooth velocity transition. The two parabolic blends are assumed to be identical, that is, they have the same constant acceleration. Therefore, parabolic blends near the path points are of same duration (t_b) and the whole trajectory is symmetric about the halfway point in time and position (t_m, q^m). Thus, the trajectory gives continuity of position, velocity, and acceleration throughout. The time variation of position for this trajectory is shown in Fig. 7.6.

For this type of trajectory planning, the user input is the target point position q^g, travel time t_g with known initial point q^s and the constant blend acceleration $\ddot{q}^c$ in two-end segments.

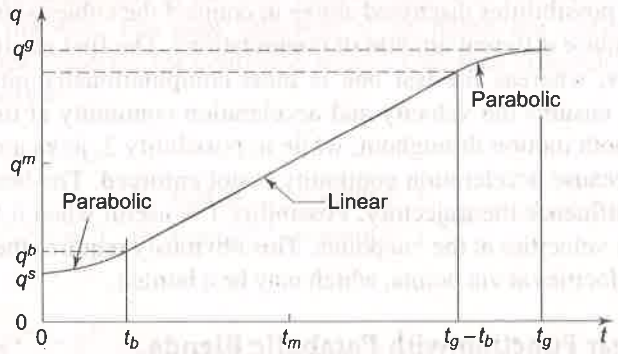


Fig. 7.6 *Time history of position for trajectory consisting of a linear segment with parabolic blends*

As can be seen from Fig. 7.6, the constraints are: both the initial and final velocities are zero and the joint position profile is symmetrical about the point (q^m, t_m) with

$$q^m = \frac{q^s + q^g}{2} \text{ and } t_m = \frac{t_g}{2} \tag{7.16}$$

The trajectory has to satisfy the additional constraint for smooth transition from parabolic blend to linear segment. For joint velocity to be continuous, the velocity at the end of the first blend (or beginning of second blend) must be equal to the velocity of the linear segment, that is, at the blend point (q^b, t_b)

$$\ddot{q}^c t_b = \frac{q^m - q^b}{t_m - t_b} \tag{7.17}$$

where q^b is the value of the joint variable at the end of the parabolic segment, at time t_b. For constant acceleration $\ddot{q}^c$, the joint position q^b, at the end of the first blend is given by

$$q^b = q^s + \frac{1}{2}\ddot{q}^c t_b^2 \tag{7.18}$$

Combining Eqs (7.16), (7.17), and (7.18) gives quadratic equation in t_b as

$$\ddot{q}^c t_b^2 - \ddot{q}^c t_g t_b + (q^g - q^s) = 0 \tag{7.19}$$

To predict the time history of q, $\dot{q}$, and $\ddot{q}$, with given q^s, q^g, and t_g, the only unknown parameter is t_b, the blend duration. Solving Eq. (7.19) for t_b, gives

$$t_b = \frac{1}{2}t_g - \frac{1}{2}\sqrt{\frac{t_g^2\ddot{q}^c - 4(q^g - q^s)}{\ddot{q}^c}} \tag{7.20}$$

Note that there are many solutions to Eq. (7.20), depending on the value of $\ddot{q}^c$ but the answer is always symmetric about the point (q^m, t_m). The constraint on the choice of acceleration during the blend, $\ddot{q}^c$, is

$$|\ddot{q}^c| \geq \frac{4|q^g - q^s|}{t_g^2}; \text{ and } \ddot{q}^c \neq 0 \tag{7.21}$$

Solution to Eq. (7.19) will not exist if the above condition is not satisfied. When equality occurs in Eq. (7.21), the two blends meet at t_m and there is no linear or constant velocity segment and only acceleration and deceleration segments are present, giving a triangular velocity profile.

On the other hand, as the acceleration becomes larger and larger, t_b becomes shorter and shorter. The trajectory tends to linear interpolation with acceleration tending to infinity or t_b tending to zero.

The trapezoidal velocity profile (linear interpolation with parabolic blends) trajectory using Eq. (7.20) generates the following sequence of polynomials for the time history of joint position, velocity and acceleration.

$$q(t) = \begin{cases} q^s + \frac{1}{2}\ddot{q}^c t^2 & 0 \leq t \leq t_b \\ q^s - \frac{1}{2}\ddot{q}^c t_b^2 + \ddot{q}^c t_b t & t_b < t \leq t_g - t_b \\ q^g - \frac{1}{2}\ddot{q}^c (t_g - t)^2 & t_g - t_b < t \leq t_g \end{cases} \tag{7.22}$$

$$\dot{q}(t) = \begin{cases} \ddot{q}^c t & 0 \leq t \leq t_b \\ \ddot{q}^c t_b & t_b < t \leq t_g - t_b \\ \ddot{q}^c (t_g - t) & t_g - t_b < t \leq t_g \end{cases} \tag{7.23}$$

$$\ddot{q}(t) = \begin{cases} \ddot{q}^c & 0 \leq t \leq t_b \\ 0 & t_b < t \leq t_g - t_b \\ -\ddot{q}^c & t_g - t_b < t \leq t_g \end{cases} \tag{7.24}$$

These polynomials confirm with Fig. 7.3 and Fig. 7.6. Alternative trajectory solutions are possible if cruise velocity $\dot{q}^c$ is specified instead of constant blend acceleration $\ddot{q}^c$.

Blended trajectory for point-to-point motion with via points Polynomial with linear-parabolic-blend splines can also be used to interpolate the given path points in a smooth way. As shown in Fig. 7.7, a smooth trajectory for a joint variable, q, is constructed with linear function connecting via points, and parabolic blends are added to the linear segments near the via points.

To maintain the continuity at via points, the parabolic blends of two segments must also have a smooth transition, This is achieved by choosing a continuous parabola for the two-blend segments at a via point. The result of this is that the polynomial $q(t)$ does not touch the via points and the via points become virtual or pseudo via points. This means that different methods to compute the blend duration and trajectory for end-points and for via points have to be employed.

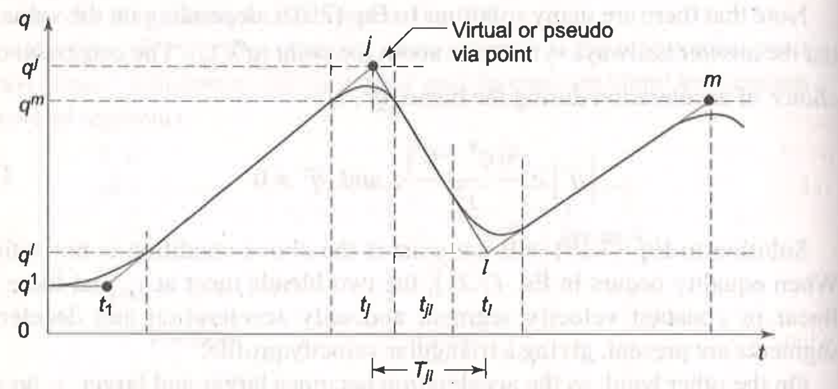


Fig. 7.7 *Multisegment linear-parabolic-blend splines*

The path points, travel time between successive path points and constant blend acceleration at path points constitute the input for the problem. The only missing parameter required to determine the time history of q, $\dot{q}$ and $\ddot{q}$ is the blend duration at each path point.

Assume that there are k path points and three consecutive path points are j, l, and m. Following notations are defined.

- q^j – value of joint variable q corresponding to path point j,
- $|\ddot{q}^j|$ – magnitude of constant blend acceleration at path point j,
- T_{jl} – total travel time between path points j and l,
- t_{jl} – travel time for the linear segment between path points j and l,
- t_j – duration of the blend around path point j,
- $\dot{q}^{jl}$ – constant joint velocity (or slope) of the linear segment connecting path points j and l.

For the via points, the blend duration t_{jl} near via point l is computed from specified acceleration at the via point $\ddot{q}^l$ and the constant linear velocities in two segments $\dot{q}^{jl}$ and $\dot{q}^{lm}$.

$$t_l = \frac{\dot{q}^{jl} - \dot{q}^{lm}}{\ddot{q}^l} \tag{7.25}$$

where $\ddot{q}^l = \text{sign}(\dot{q}^{jl} - \dot{q}^{lm})|\ddot{q}^l|$

The linear velocity in segment jl is given by

$$\dot{q}^{jl} = \frac{q^l - q^j}{T_{jl}} \tag{7.26}$$

The duration of the linear segment t_{jl}, assuming parabolic blend to be symmetric around the path points, is

$$t_{jl} = T_{jl} - \frac{1}{2}t_j - \frac{1}{2}t_l \tag{7.27}$$

Once again, there are many possible solutions, depending on the value of accelerations used in each blend. To compute the blend duration at the initial point $j = 1$, the fact that the joint velocity at the end of blend is the same as the velocity of the linear segment 12 (t_{12}), is used. This gives

$$\ddot{q}^1 t_1 = \frac{q^2 - q^1}{T_{12} - \frac{1}{2}t_1} \tag{7.28}$$

where $\ddot{q}^1 = \text{sign}(q^2 - q^1)|\ddot{q}^1|$

From Eq. (7.28) the blend time t_1 at the initial point and using t_1, the linear segment velocity and duration, $\dot{q}^{12}$ and t_{12} are computed as

$$t_1 = T_{12} - \sqrt{\frac{T_{12}^2\ddot{q}^1 - 2(q^2 - q^1)}{\ddot{q}^1}} \tag{7.29}$$

$$t_{12} = T_{12} - \frac{1}{2}t_2 - t_1 \tag{7.30}$$

and $$\dot{q}^{12} = \frac{q^2 - q^1}{T_{12} - \frac{1}{2}t_1} \tag{7.31}$$

Finally, the blend duration at the goal point is determined. Similar to the starting point, the velocity continuity constraint, in the middle segments connecting path points (k–1) and k, gives

$$\ddot{q}^k t_k = \frac{q^k - q^{k-1}}{T_{(k-1)k} - \frac{1}{2}t_k} \tag{7.32}$$

where $\ddot{q}^k = \text{sign}(\dot{q}^{k-1} - \dot{q}^k)|\ddot{q}^k|$

The solution for blend duration, linear velocity, and its duration are

$$t_k = T_{(k-1)k} - \sqrt{\frac{T^2_{(k-1)k}\ddot{q}^k - 2(q^k - q^{k-1})}{\ddot{q}^k}} \quad (7.33)$$

$$t_{(k-1)k} = T_{(k-1)k} - \frac{1}{2}t_{k-1} - t_k \quad (7.34)$$

$$\dot{q}^{(k-1)k} = \frac{q^k - q^{k-1}}{T_{(k-1)k} - \frac{1}{2}t_k} \quad (7.35)$$

Thus, all the requisite parameters to determine the time history of joint position, velocity, and acceleration are obtained using Eqs (7.25) through (7.35). The acceleration at each via point must be sufficiently large, as discussed in single segment case [see Eq. (7.21)]. To make computations simpler, same value of blend acceleration may be used for each via point.

7.3 CARTESIAN SPACE TECHNIQUES

In the previous section, some approaches to joint-space trajectory planning have been discussed. Another approach to trajectory planning is to use Cartesian coordinates and Cartesian space. In Cartesian space, the position and orientation of a rigid body can be clearly defined.

The problem of planning trajectories that enable the manipulator's end-effector to track a given path in Cartesian space is investigated in this section. The user specifies the desired end-effector path, the travelling time, and the tool orientations along the path.

To tackle the aforesaid problem, a parametric description for the desired end-effector path is required. This description specifies the spatial attributes of the requisite motion with respect to the base frame. First, the mathematics of parametric description of the path in Cartesian space is briefly reviewed. Then, some common techniques for Cartesian space trajectory planning are discussed.

7.3.1 Parametric Description of a Path

In parametric representation, the path (or curve) is represented as a function of single parameter, often this parameter is time. The arc length c of the curve is a function of time and can be used as parameter for the parametric representation of curves. Many other parameters can be used to describe a given path. However, in this discussion only the path coordinate c is used as the path parameter.

Figure 7.8 shows a Cartesian space path, $\mathcal{P}$ to be tracked by the end-effector of a given n-DOF manipulator, and the base frame {0}. If the path lies in plane it is a plane path, otherwise it is a 'twisted path'. Most tool-point trajectories are twisted paths.

Let P_s and P_g be the end-points of an open path $\mathcal{P}$ and $F(c)$ be a continuous vector function defined in the interval $[c_s, c_g]$. The path coordinate c of a generic point P on path $\mathcal{P}$ is the length of arc of $\mathcal{P}$ with extremes P and P_s. Usually, the starting point on the path P_s is the path coordinate origin and at origin of path, $c = 0$. This means that for any point P on the path, the path coordinate specifies the arc length of the path between P_s and P.

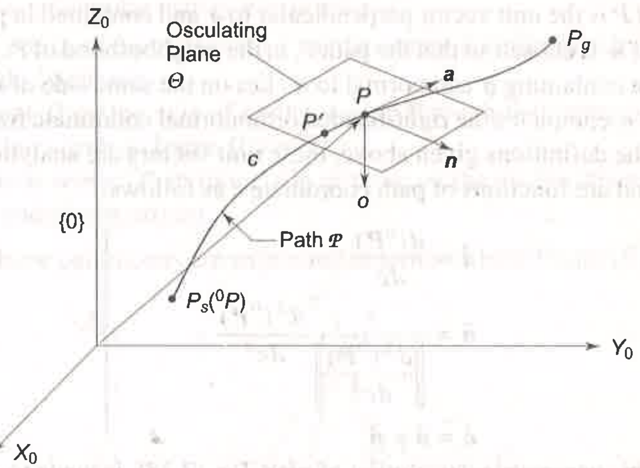


Fig. 7.8 *Path $\mathcal{P}$ relative to frame {0}*

It then follows that the path coordinate c can be used to express any point P on the path with respect to frame {0} as

$$^{n}\mathbf{P} = F(c) \quad (7.36)$$

Note that in Eq. (7.36), $^0\mathbf{P}$ is a 3×1 position vector of point P with respect to frame {0}. The vector function $F(c)$, thus, analytically expresses any point on the path $\mathcal{P}$ with respect to frame {0}. Hence, the path $\mathcal{P}$ can also be written as

$$^0\mathcal{P} = F(c) \quad (7.37)$$

Three unit vectors can be used to express point P, which form a right-handed orthonormal coordinate system with three orthonormal planes as shown in Fig. 7.9.

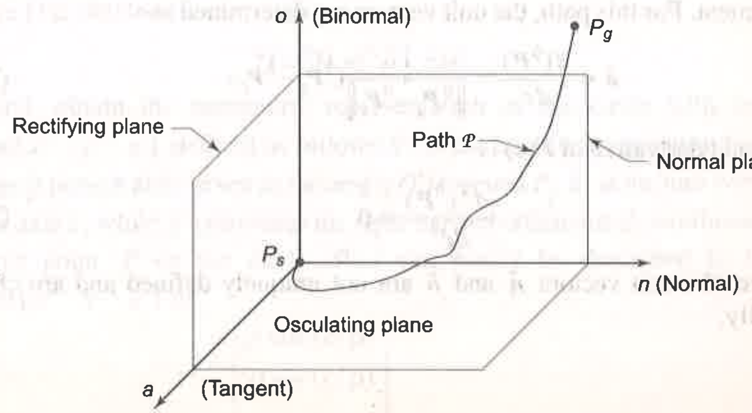


Fig. 7.9 *The orthonormal coordinate system for path $\mathcal{P}$*

First unit vector is the tangent unit vector $\hat{a}$, pointing in the increasing direction of c. The unit vector $\hat{a}$ lies in the osculating plane Θ at point P. This plane is the limit position of the plane containing the unit vector $\hat{a}$ and a point $P' \in \mathcal{P}$, when P' tends to P along the path, as shown in Fig. 7.8. The normal unit vector $\hat{n}$ at P is the unit vector perpendicular to $\hat{a}$ and contained in plane Θ. The direction of $\hat{n}$ is chosen so that the path $\mathcal{P}$, in the neighborhood of P, with respect to the plane containing $\hat{a}$ and normal to $\hat{n}$, lies on the same side of $\hat{n}$. Binormal unit vector $\hat{o}$ completes the right-handed orthonormal coordinate frame $\{n\ o\ a\}$.

As per the definitions given above, these unit vectors are analytically related to path $\mathcal{P}$ and are functions of path coordinate c as follows:

$$\left.\begin{aligned} \hat{a} &= \frac{d(^0\boldsymbol{P})}{dc} \\ \hat{n} &= \frac{1}{\left\| \dfrac{d^2(^0\boldsymbol{P})}{dc^2} \right\|} \frac{d^2(^0\boldsymbol{P})}{dc^2} \\ \hat{o} &= \hat{a} \times \hat{n} \end{aligned}\right\} \qquad (7.38)$$

Use of this parametric description of path, Eq. (7.38), is made to compute the

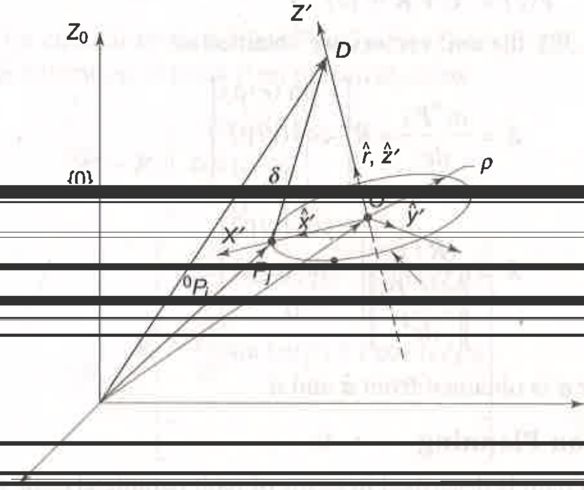
[redacted]

7.3.3 A Circular Path

The circular path $\mathcal{P}(c)$, shown in Fig. 7.10, is specified using the following parameters

(a) Radius of circular path is ρ in plane $x'y'$.
(b) Unit vector of the circle axis, $\hat{r}$, at the center of circle, directed according to right-hand rule,
(c) A point D on the axis of circle passing through the center of circle. Its position vector in frame $\{0\}$ is $^0\boldsymbol{D}$.
(d) Position vector $^0\boldsymbol{P}_j$ in frame $\{0\}$ of point on the circle (this serves as the path coordinate origin).

All the above parameters are expressed in terms of base frame $\{0\}$.



[redacted]

where $\rho = \|{}^0P_j - {}^0C\|$ gives the circle radius. Thus, the description of the circular path $\mathcal{P}'$ with respect to farme $\{x'\,y'\,z'\}$, is given by

$$\mathcal{P}'(c) = \begin{bmatrix} \rho\cos(c/\rho) \\ \rho\sin(c/\rho) \\ 0 \end{bmatrix} \tag{7.44}$$

Finally, the parametric description of circle with respect to frame {0} is obtained by mapping, $\mathcal{P}'$ to base frame as

$${}^0\mathcal{P}(c) = {}^0C + R\,\mathcal{P}'(c) \tag{7.45}$$

where R is the rotation matrix of frame $\{x'\,y'\,z'\}$ with respect to base frame {0}. Thus, the position vector of any point on the circle is given by

$${}^0P(c) = {}^0C + R\,\mathcal{P}'(c) \tag{7.46}$$

From Eq. (7.38), the unit vectors are obtained as

$$\hat{a} = \frac{d({}^0P)}{dc} = R\begin{bmatrix} -\sin(c/\rho) \\ \cos(c/\rho) \\ 0 \end{bmatrix} \tag{7.47}$$

$$\hat{n} = \frac{(R/\rho)}{\left\|\dfrac{d^2({}^0P)}{dc^2}\right\|}\begin{bmatrix} -\cos(c/\rho) \\ -\sin(c/\rho) \\ 0 \end{bmatrix} \tag{7.48}$$

The unit vector $\hat{o}$ is obtained from $\hat{a}$ and $\hat{n}$.

7.3.4 Position Planning

Once the given path is described in terms of path parameters, the next aim is to determine the time history of end-effector position, velocity, and acceleration consistent with the temporal constraints prescribed.

For a path ${}^0P$ to be traversed by a given n-DOF manipulator in a time t_g, first the extreme values of path coordinate c are determined. For the given path, assume $c = 0$ at time $t = 0$ is the initial position, and $c = c_g$ at $t = t_g$ is the goal position. Next, a function $c(t)$, which interpolates c between $c = 0$ and $c = c_g$ in a smooth way, is obtained. This smooth function $c(t)$ could either be a cubic polynomial or a linear segment with parabolic blends and is obtained as discussed in the case of joint-space schemes for point-to-point motion without via points.

Now that a time law is imposed on the path coordinate c, it is possible to obtain the time history of ${}^0P$ by substituting the values attained by $c(t)$, at each instant of time, in the Eq. (7.38).

The linear velocity of the end-effector is determined as

$${}^0\dot{P} = \dot{c}\frac{d({}^0P)}{dc} = \hat{a}\,\dot{c} \tag{7.49}$$

where $\hat{a} = \dfrac{d({}^0P)}{dc}$ as per Eq. (7.38).

Linear acceleration of the tool is obtained by differentiating Eq. (7.49) with respect to time,

$${}^0\ddot{P} = \dot{c}^2\frac{d^2({}^0P)}{dc^2} + \ddot{c}\frac{d({}^0P)}{dc} \tag{7.50}$$

Thus, for a Cartesian straight-line motion, ${}^0P$ is obtained from Eqs (7.40), (7.49) and (7.50) as

$${}^0\dot{P} = \frac{\dot{s}}{\|{}^0P_g - {}^0P_s\|}({}^0\dot{P}_g - {}^0\dot{P}_s) = \hat{a}\,\dot{c} \tag{7.51}$$

$${}^0\ddot{P} = \frac{\dot{s}}{\|{}^0P_g - {}^0P_s\|}({}^0\ddot{P}_g - {}^0\ddot{P}_s) = \hat{a}\,\ddot{c} \tag{7.52}$$

Similarly, for circular motion, from Eqs (7.47), (7.48), (7.49), and (7.50), the velocity and acceleration of point P on the circle are as

$${}^0\dot{P} = R\dot{c}\begin{bmatrix} -\sin(c/\rho) \\ \cos(c/\rho) \\ 0 \end{bmatrix} \tag{7.53}$$

$${}^0\ddot{P} = R\begin{bmatrix} -\dfrac{\dot{c}^2}{\rho}\cos(c/\rho) - \ddot{c}\sin(c/\rho) \\ -\dfrac{\dot{c}^2}{\rho}\sin(c/\rho) + \ddot{c}\cos(c/\rho) \\ 0 \end{bmatrix} \tag{7.54}$$

7.3.5 Orientation Planning

The next step in Cartesian space schemes is to obtain the time sequence of orientations attained by the end-effector. But, these orientations cannot be specified by rotation matrices because interpolation of two-rotation matrices does not always yield a valid rotation matrix (because orthonormality of columns is not maintained).

Hence, in planning orientations, either the Fixed or Euler angle representation outlined in Chapter 2, is used. Supposing, zyx-Euler angle representation is used to express the end-effector orientation with respect to the base, the angles ϕ, θ, and φ are defined as the Euler angles. It will be useful to define 3×1 orientation vector:

$$\alpha = [\phi \;\; \theta \;\; \varphi]^T \tag{7.55}$$

Because α is similar to ${}^0P$, discussed in position planning, any path on the end-effector's orientation along with temporal constraints can be imposed. The steps outlined as follows generate a smooth trajectory joining

$\alpha_s = [\phi_s \quad \theta_s \quad \varphi_s]^T$ to $\alpha_g = [\phi_g \quad \theta_g \quad \varphi_g]^T$ in time $t = t_g$.

Step 1. Obtain the parametric description of the given path joining α_s to α_g.

Step 2. Impose a time law on path coordinate, c, as done in position planning in Cartesian space.

Step 3. Obtain the time sequence of values attained by $\alpha(t)$ based on $c(t)$.

Step 4. As explained earlier for position planning, obtain the time history of $\dot{\alpha}(t)$ and $\ddot{\alpha}(t)$.

For example, a linear segment joint α_s and α_g has the following expressions for $\dot{\alpha}(t)$ and $\ddot{\alpha}(t)$.

$$\dot{\alpha}(t) = \frac{\dot{c}}{\|\alpha_g - \alpha_s\|}(\alpha_g - \alpha_s) \tag{7.56}$$

$$\ddot{\alpha}(t) = \frac{\ddot{c}}{\|\alpha_g - \alpha_s\|}(\alpha_g - \alpha_s) \tag{7.57}$$

7.4 JOINT-SPACE VERSUS CARTESIAN SPACE TRAJECTORY PLANNING

The joint-space trajectory planning is simple to implement and joint coordinates fully specify the position and orientation of end-effector. But when it comes to specifying a task or defining an obstacle in workspace, Cartesian coordinates offer a better choice. It is easy to specify the position and orientation of a rigid body in space in Cartesian coordinates. The obstacles and the path followed by the end-effector (or tool point) can be expressed in terms of simple transformations and easy to visualize in Cartesian space.

The main drawback with joint-space trajectory planning is that it does not account for the existence of obstacles in the workspace of a manipulator. The other drawback is that due to the nonlinear nature of the manipulator's kinematics model, it is difficult to predict the resulting end-effector motion that will be produced by a particular trajectory executed in the joint space. The above-mentioned drawbacks could be overcome by employing Cartesian space trajectory planning.

One of the main drawbacks with Cartesian space techniques is its computational complexity. Because the manipulator control system controls the joint variables, it is essential to resort to inverse kinematics/dynamics and determine the values of joint variables at every instant of time. This requires much more computations, which reduce the trajectory-sampling rate in real-time operation. As a result, the trajectory-tracking accuracy decreases. This problem is not there in joint- space schemes, where planning is done directly in joint space.

The other limitation with Cartesian space schemes is that the planned paths may have singularities or may pass close to manipulator singularities. In such a case, when singular configuration of manipulator is reached, one or more joint velocities may increase infinitely. One approach to avoid excessive joint rates is to limit joint rates within manageable limits, while computing the trajectory and to increase the travel time in such a region. However, doing so would violate the temporal attributes of the specified motion and this will make the overall motion slower.

It may also happen that the intermediate points of a planned path are unreachable although the start and goal points lie within the workspace. This happens primarily due to presence of voids within the workspace created by the limits over the movements of individual joints. The solution is to identify such voids and avoid them while planning Cartesian paths.

Cartesian space planning requires more complex algorithms and more computational ability, so that in runtime the joints are servoed to follow the desired Cartesian paths. To sum it up, Cartesian space planning offers a better and simpler representation of the task but at a higher computational cost.

SOLVED EXAMPLES

Example 7.1 ***Cubic spline trajectory***

The second joint of a SCARA manipulator is required to move from $\theta_2 = 30^\circ$ to 150° in 5 seconds. Find the cubic polynomial to generate the smooth trajectory for the joint. What is the maximum velocity and acceleration for this trajectory?

Solution The given parameters of the trajectory are:

$$\begin{aligned} \theta^s &= 30^\circ \qquad \theta^g = 105^\circ \\ v^s &= 0 \qquad v^g = 0 \\ &\qquad\quad t^g = 5\,s \end{aligned} \tag{7.58}$$

The four coefficients of the cubic polynomial are computed from the above using Eq. (7.8) as:

$$\begin{aligned} a_0 &= \theta^s = 30.0 \\ a_1 &= 0 \\ a_2 &= \frac{3}{t_g^2}(\theta^g - \theta^s) = 6.0 \\ a_3 &= -\frac{2}{t_g^3}(\theta^g - \theta^s) = -1.2 \end{aligned} \tag{7.59}$$

Hence the polynomials interpolating the position, velocity and acceleration of the joint are

$$\begin{aligned} \theta(t) &= 30.0 + 6.0t^2 - 1.2t^3 \\ \dot{\theta}(t) &= 12.0t - 3.6t^2 \\ \ddot{\theta}(t) &= 12.0 - 7.2t \end{aligned} \tag{7.60}$$

As expected, the velocity profile is parabolic and acceleration profile is linear. The time sequence of position, velocity, and acceleration can be easily generated at any specified sampling rate. The plot of these functions will be identical to

Fig. 7.4. The maximum values of velocity and acceleration are 7.5 deg/s and 12 deg/s^2, respectively.

Example 7.2 Point-to-point motion with via points with specified velocities

Determine the time-history of the position, velocity and acceleration of the end-effector of a 5-DOF manipulator, which moves from start to goal point via two intermediate points. The desired end-effector motion is specified in Table 7.1. Assume a cubic spline trajectory in each segment and the continuity of velocity at the via points is enforced.

Table 7.1 *Desired end-effector motion*

End-effector	*Path point (j)* 1	2	3	4
Position q^j (rad)	0	$3\pi/2$	$-\pi/2$	2π
Velocity $\dot{q}^j$(rad/s)	0	$-\pi/2$	π	0
Traversal time t_j (s)	0	3	5	2

Solution The given trajectory has two via points and, therefore, will have three segments and three cubic polynomials will be required to define the time-history of motion of end-effector. Let the three cubic polynomials be

$$\begin{aligned} P_1(t) &= a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3 \\ P_2(t) &= a_{20} + a_{21}t + a_{22}t^2 + a_{23}t^3 \\ P_3(t) &= a_{30} + a_{31}t + a_{32}t^2 + a_{33}t^3 \end{aligned} \tag{7.61}$$

where a_{ij} are the unknown coefficients of the three cubic polynomials.

The constraints imposed on these three polynomials are given by Eq. (7.12). Applying the constraints and substituting the boundary values from Table 7.1, for first segment cubic gives the following equations:

$$\begin{aligned} P_1(0) &= a_{10} = 0 \\ P_1(3) &= a_{10} + 3a_{11} + 9a_{12} + 27a_{13} = 3\pi/2 \\ \dot{P}_1(0) &= a_{11} = 0 \\ \dot{P}_1(3) &= a_{11} + 6a_{12} + 27a_{13} = \pi/2 \end{aligned} \tag{7.62}$$

The four coefficient for the first segment polynomial P_1 are computed from Eq. (7.62) as

$$\begin{aligned} a_{10} &= 0 \\ a_{11} &= 0 \\ a_{12} &= 2.09 \\ a_{13} &= -0.52 \end{aligned} \tag{7.63}$$

and the cubic for first segment is

$$P_1(t) = 2.09\, t^2 - 0.52\, t^3 \tag{7.64}$$

Similarly, for second cubic the constraints from Table 7.1 give

$$\begin{aligned} P_2(3) &= a_{20} + 3a_{21} + 9a_{22} + 27a_{23} = 3\pi/2 \\ P_2(8) &= a_{20} + 8a_{21} + 64a_{22} + 512a_{23} = -\pi/2 \\ \dot{P}_2(3) &= a_{21} + 6a_{22} + 27a_{23} = -\pi/2 \\ \dot{P}_2(8) &= a_{21} + 16a_{22} + 192a_{23} = \pi \end{aligned} \tag{7.65}$$

Equations (7.65) are solved to compute the from coefficients of polynomial P_2. The solution is

$$\begin{aligned} a_{20} &= -1.77 \\ a_{21} &= 7.36 \\ a_{22} &= -2.22 \\ a_{23} &= 0.16 \end{aligned} \tag{7.66}$$

and the cubic for the second segment is

$$P_2(t) = -1.77 + 7.36t - 2.22t^2 + 0.16t^3 \tag{7.67}$$

For the third cubic polynomial constraints from Table 7.1 give

$$\begin{aligned} q_3(8) &= a_{30} + 8a_{31} + 64a_{32} + 512a_{33} = -\pi/2 \\ q_3(10) &= a_{30} + 10a_{31} + 100a_{32} + 1000a_{33} = 2\pi \\ \dot{q}_3(8) &= a_{31} + 16a_{32} + 192a_{33} = \pi \\ \dot{q}_3(10) &= a_{31} + 20a_{32} + 300a_{33} = 0 \end{aligned} \tag{7.68}$$

The four constraints for the third cubic are computed as:

$$\begin{aligned} a_{30} &= 752.41 \\ a_{31} &= -267.04 \\ a_{32} &= 31.02 \\ a_{33} &= -1.18 \end{aligned} \tag{7.69}$$

and the cubic for the third and last segment is:

$$P_3(t) = 752.41 - 267.04t + 31.04t^2 - 1.18t^3 \tag{7.70}$$

The cubic polynomials P_1, P_2 and P_3 for the three segments of the point-to-point trajectory are, thus, give by Eqs. (7.64), (7.67) and (7.70). These cubic polynomials describing the position of end-effector are plotted in Fig. 7.11(a) and their first and second derivatives, which represent velocity and acceleration of end-effector, respectively, are plotted in Fig. 7.11(b) and Fig. 7.11(c).

From these figures, note that continuity of motion is maintained for position and velocity but there is discontinuity of acceleration at the via points.

Example 7.3 Point-to-point motion with via points with computed velocities

Consider the determination of point-to-point trajectory with two via points with the data in Example 7.2 where the velocities at via points are computed instead of being specified.

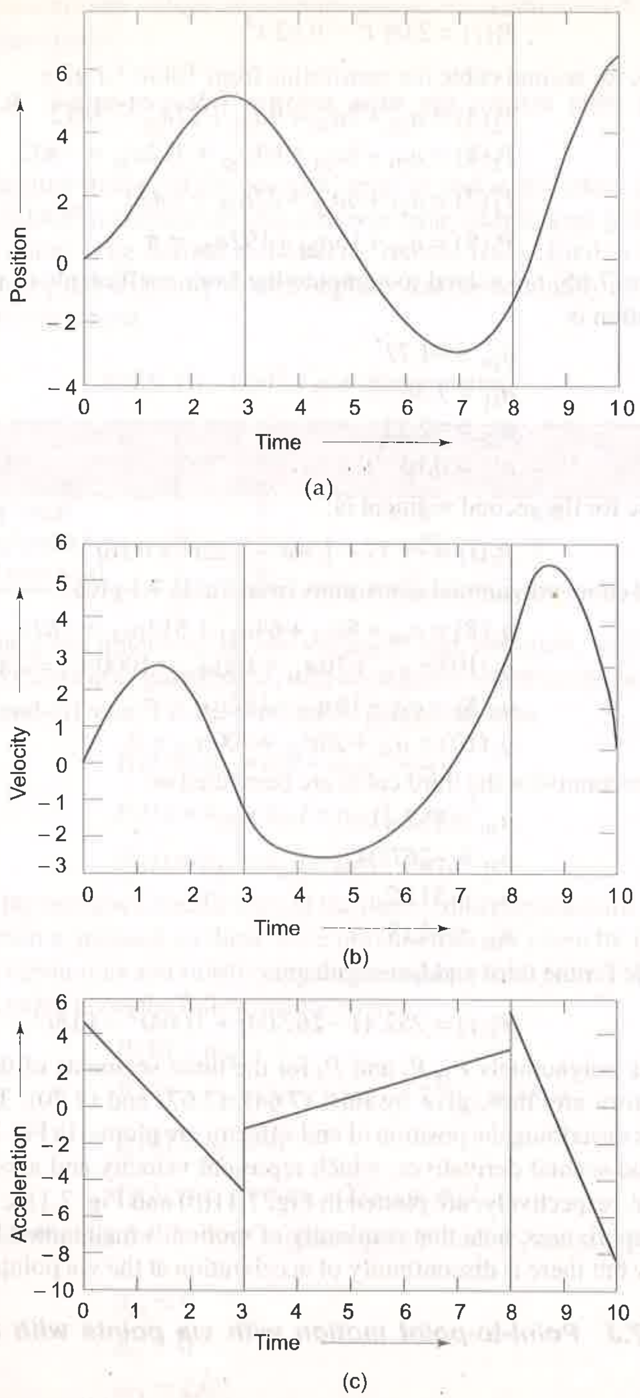


Fig. 7.11 *Time-history of position, velocity and acceleration for point-to-point motion with via points*

Solution As in Example 7.2, the starting and final velocities are zero, that is

$$\dot{q}^1 = \dot{q}^4 = 0 \tag{7.71}$$

The average velocities for the segments 1, 2 and 3 are computed using Eq. (7.14) and the data from Table 7.1. Thus

$$\dot{\bar{q}}^1 = \frac{q^2 - q^1}{T_1} = \frac{\frac{3\pi}{2} - 0}{3} = \frac{\pi}{2}$$

$$\dot{\bar{q}}^2 = \frac{q^3 - q^2}{T_2} = \frac{-\frac{\pi}{2} - \frac{3\pi}{2}}{2} = -\pi \tag{7.72}$$

$$\dot{\bar{q}}^3 = \frac{q^4 - q^3}{T_3} = \frac{2\pi + \frac{\pi}{2}}{3} = \frac{5\pi}{2}$$

According to Eq. (7.14), the via point velocities are computed from

$$\dot{q}^j = \begin{cases} 0 & \text{if } \operatorname{sign}(\dot{\bar{q}}^{j-1}) \neq \operatorname{sign}(\dot{\bar{q}}^j) \\ \frac{1}{2}(\dot{\bar{q}}^{j-1} + \dot{\bar{q}}^j) & \text{if } \operatorname{sign}(\dot{\bar{q}}^{j-1}) = \operatorname{sign}(\dot{\bar{q}}^j) \end{cases} \tag{7.73}$$

At via point 2

$$\dot{q}^2 = 0 \quad \text{because from Eq. (7.72) } \operatorname{sign}(\dot{\bar{q}}^1) \neq \operatorname{sign}(\dot{\bar{q}}^2) \tag{7.74}$$

and at via point 3

$$\dot{q}^3 = 0 \quad \text{since } \operatorname{sign}(\dot{\bar{q}}^2) \neq \operatorname{sign}(\dot{\bar{q}}^3) \tag{7.75}$$

Note that in this case the given path points positions result in zero velocity at both via points. The three cubic polynomials for the three segments of the trajectory are computed using the position and time constraints from Table 7.1 and velocity constraints from Eqs (7.71), (7.74) and (7.75). The three polynomials are as in Eq. (7.61).

Applying the given and computed constraints for the first polynomial, gives four equations as:

$$\begin{aligned} P_1(0) &= a_{10} = 0 \\ P_1(3) &= a_{10} + 3a_{11} + 9a_{12} + 27a_{13} = 3\pi/2 \\ \dot{P}_1(0) &= a_{11} = 0 \\ \dot{P}_1(3) &= a_{11} + 6a_{12} + 27a_{13} = 0 \end{aligned} \tag{7.76}$$

which give the four coefficients of $P_1(t)$ as:

$$\begin{aligned} a_{10} &= a_{11} = 0 \\ a_{12} &= 1.57 \\ a_{13} &= -0.35 \end{aligned} \tag{7.77}$$

Applying the constraints for the second joint yields the four equations as (see Eq. (7.65)):

$$\begin{aligned} P_2(3) &= a_{20} + 3a_{21} + 9a_{22} + 27a_{23} = 3\pi/2 \\ P_2(8) &= a_{20} + 8a_{21} + 64a_{22} + 512a_{23} = -\pi/2 \\ \dot{P}_2(3) &= a_{21} + 6a_{22} + 27a_{23} = 0 \\ \dot{P}_2(8) &= a_{21} + 16a_{22} + 192a_{23} = 0 \end{aligned} \tag{7.78}$$

which are solved to give four coefficients of the second segment cubic $P_2(t)$ as:

$$\begin{aligned} a_{20} &= -4.79 \\ a_{21} &= 7.24 \\ a_{22} &= -1.66 \\ a_{23} &= 0.10 \end{aligned} \tag{7.79}$$

Similarly, the solution for the four coefficients of the third polynomial $P_3(t)$ is

$$\begin{aligned} a_{30} &= 1.38 \times 10^3 \\ a_{31} &= -471.2 \\ a_{32} &= 53.01 \\ a_{33} &= -1.96 \end{aligned} \tag{7.80}$$

Substituting the computed values of the coefficients from Eqs (7.77), (7.79) and (7.80) into three cubic polynomials (Eq. (7.61)) gives the three cubics for the three segments as:

$$\begin{aligned} P_1(t) &= 1.57t^2 - 0.35t^3 \\ P_2(t) &= -4.79 + 7.24t - 1.66t^2 + 0.10t^3 \\ P_3(t) &= 1.38 \times 10^3 - 471.2t + 53.01t^2 - 1.96t^3 \end{aligned} \tag{7.81}$$

These cubic for the three segments are plotted and the time-history of position, velocity and acceleration are obtained as illustrated in Fig. 7.12. From these plots it is easy to observe that the continuity of position and velocity is maintained as in Example 7.2 and the acceleration is still discontinuous. Comparing these profiles with profiles in Fig. 7.11 reveals the effect of change in boundary conditions on the motion of the end-effector.

Example 7.4 ***Trajectory with a trapezoidal velocity profile***

The motion of a joint of 3-DOF arm is constrained by an actuator that can produce a maximum acceleration of 0.35 rad/s^2 and maximum velocity of 15 rad/s. If trapezoidal velocity profile is assumed, determine the trajectory if the joint moves by π rad in 10 s.

Solution A trapezoidal velocity profile implies a constant acceleration in the start phase, a cruise velocity and a constant deceleration in the arrival phase. The trajectory will be as shown in Fig. 7.3.

Assuming that the trajectory is symmetric and the traversal is with maximum possible acceleration, the parameters for the trajectory are:

$$\begin{aligned} q^s &= 0 & q^g &= \pi \\ \dot{q}^s &= 0 & \dot{q}^g &= 0 \\ t_s &= 0 & t_g &= 10 \text{ s} \\ v^c &= 15 \text{ rad/s} & \ddot{q}^c &= 0.35 \text{ rad/s}^2 \end{aligned} \tag{7.82}$$

From Eq. (7.16), at the mid-point of the trapezoidal trajectory

$$\begin{aligned} q^m &= \frac{q^s + q^g}{2} = \frac{\pi}{2} \\ t_m &= \frac{t^g}{2} = 5 \text{ s} \end{aligned} \tag{7.83}$$

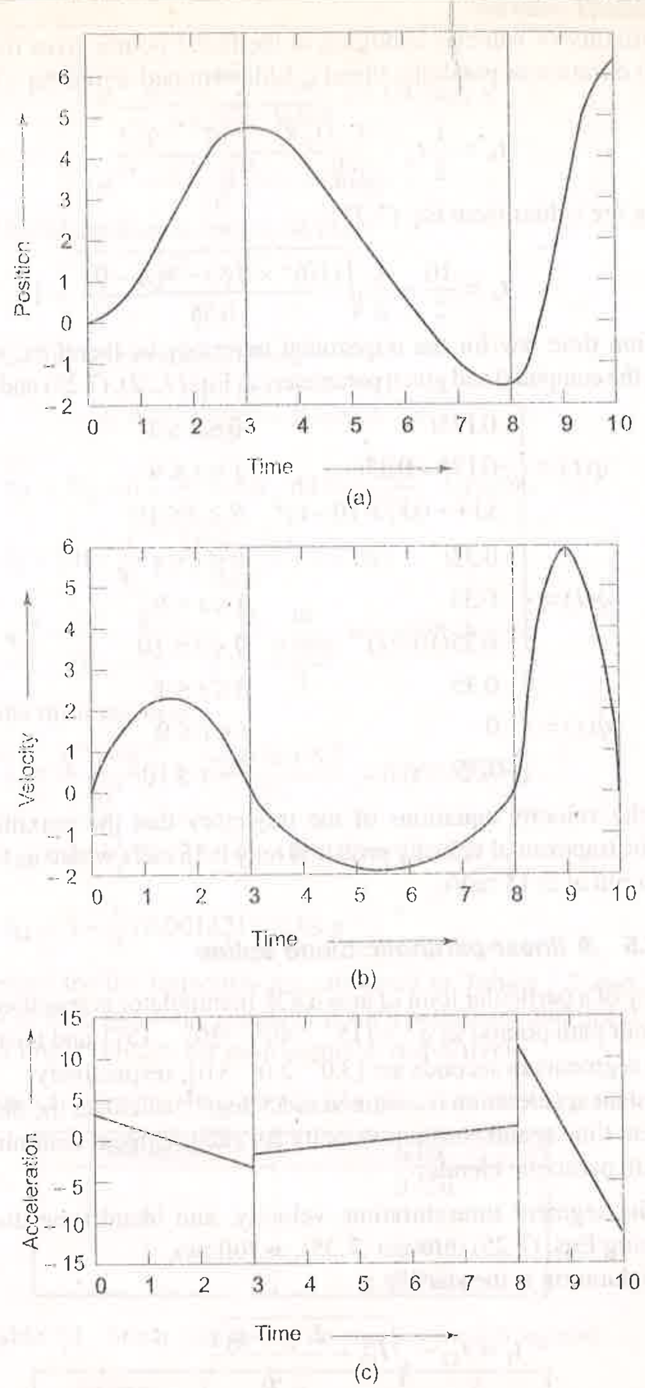


Fig. 7.12 *Time-history of position, velocity and accelerations for point-to-point motion with computed velocity at via points*

The continuity of velocity condition at the blend points gives the constraint from which duration of parabolic blend t_b is determined. From Eq. (7.20)

$$t_b = \frac{1}{2}t_g - \frac{1}{2}\sqrt{\frac{t_g^2\ddot{q}^c - 4(q^g - q^s)}{\ddot{q}^c}} \tag{7.84}$$

Substituting the values from Eq. (7.71)

$$t_b = \frac{10}{2} - \frac{1}{2}\sqrt{\frac{(10)^2 \times 0.35 - 4(\pi - 0)}{0.35}} = 1 \text{ s} \tag{7.85}$$

The motion time law for the trapezoidal trajectory is, therefore, obtained by substituting the computed and given parameters in Eqs (7.22), (7.23) and (7.24) as

$$q(t) = \begin{cases} 0.175t^2 & 0 \le t \le 1 \\ -0.175 + 0.35t & 1 < t \le 9 \\ 3.14 - 0.175(10-t)^2 & 9 < t \le 10 \end{cases}$$

$$\dot{q}(t) = \begin{cases} 0.35t & 0 \le t \le 1 \\ 0.35 & 1 < t \le 9 \\ 0.35(10-t) & 9 < t \le 10 \end{cases} \tag{7.86}$$

$$\ddot{q}(t) = \begin{cases} 0.35 & 0 \le t \le 1 \\ 0 & 1 < t \le 9 \\ -0.35 & 9 < t \le 10 \end{cases}$$

Note from the velocity equations of the trajectory that the maximum cruise velocity of the trapezoidal velocity profile is only 0.35 rad/s where as the actuator is capable to run at is 15 rad/s.

Example 7.5 ***A linear-parabolic-blend spline***

The trajectory of a particular joint of an n-DOF manipulator is specified with two via points (four path points) as $\boldsymbol{q}^j = [15^\circ \quad 40^\circ \quad 30^\circ \quad 15^\circ]$ and the travel time for the three segments in seconds are [3.0 2.0 3.0], respectively.

If the constant acceleration is assumed as 55 deg/s^2, calculate the blend time t_b linear segment time t_{jl} and constant velocity for each segment assuming a linear trajectory with parabolic blends.

Solution The segment time duration, velocity, and blend time duration are calculated using Eqs. (7.25) through (7.35), as follows.

The blend duration at the starting is

$$t_1 = T_{12} - \sqrt{T_{12}^2 - \frac{2(q_2 - q_1)}{\ddot{q}_1}}$$

or

$$t_1 = 3.0 - \sqrt{3.0^2 - \frac{2(40-15)}{55}} = 0.156 \text{ s} \tag{7.87}$$

The constant velocities for the linear segments are:

$$\dot{q}^{12} = \frac{40-15}{2 - 0.5 \times 0.156} = 13.0 \text{ deg / s}$$

$$\dot{q}^{23} = \frac{30-40}{2.0} = -5.0 \text{ deg / s} \tag{7.88}$$

The second blend duration is computed from:

$$t_2 = \frac{\dot{q}^{23} - \dot{q}^{12}}{\ddot{q}^2}$$

where $\ddot{q}^2 = -55$ (for deceleration). Thus,

$$t_2 = \frac{-5.0 - 13.0}{-55} = 0.33 \text{ s}$$

and

$$t_{12} = T_{12} - t_1 - \frac{t_2}{2} = 3.0 - 0.156 - \frac{0.33}{2} = 2.679 \text{ s} \tag{7.89}$$

Also

$$t_4 = 3.0 - \sqrt{9 + \frac{2(15-30)}{55.0}} = 0.092 \text{ s} \tag{7.90}$$

and

$$\dot{q}^{34} = \frac{q^4 - q^3}{T_{34} - \frac{t_4}{2}} = \frac{15-30}{3.0 - \frac{0.092}{2}} = -5.078 \text{ deg / s} \tag{7.91}$$

Similarly, for the third segment:

$$t_3 = \frac{\dot{q}^{34} - \dot{q}^{23}}{\ddot{q}^3} = \frac{-5.078 + 5.0}{55} = 0.00142 \text{ s}$$

$$t_{23} = 2 - \frac{1}{2}(0.33) - \frac{1}{2}(0.00142) = 1.834 \text{ s} \tag{7.92}$$

$$t_{34} = 3 - \frac{1}{2}(0.00142) = 2.99 \text{ s}$$

The parameters for the trajectory are tabulated in Tables 7.2 and 7.3 with blend duration of each parabolic segment at each path point and linear velocity and duration of linear velocity for each segment, respectively.

Table 7.2 *Blend duration for parabolic blends at each path point*

Path Point	t_b *(s)*
1	0.156
2	0.330
3	0.001
4	0.092

Table 7.3 *Linear velocity and its duration for each segment*

Segment	$\dot{q}$ *(deg/s)*	t *(s)*
1–2	13.00	2.68
2–3	–5.00	1.83
3–4	–5.08	2.99

Example 7.6 *Trajectory planning for a 3-DOF RRP arm*

For the 3-DOF manipulator arm discussed in Example 4.4, Fig. 4.7 design a linear trajectory with parabolic blends. The initial & final position of the end-effector is expressed by homogeneous transformation matrices T_s & T_g given below and time taken for traversal is 5 seconds.

$$T_s = \begin{bmatrix} 0.354 & 0.866 & -0.354 & -0.106 \\ -0.612 & 0.500 & 0.612 & 0.184 \\ 0.707 & 0 & 0.707 & 0.212 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{7.93}$$

$$T_g = \begin{bmatrix} 0.583 & -0.766 & 0.272 & 0.027 \\ 0.694 & 0.643 & 0.324 & 0.032 \\ -0.423 & 0 & 0.906 & 0.091 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{7.94}$$

Solution The forward kinematic model for the 3-DOF RRP arm was obtained in Example 4.4 and is given by Eq. (4.67) as:

$$^0T_3 = \begin{bmatrix} C_1C_2 & -S_1 & -C_1S_2 & -d_3C_1S_2 \\ S_1C_2 & C_1 & -S_1S_2 & -d_3S_1S_2 \\ S_2 & 0 & C_2 & d_3C_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ r_{21} & r_{22} & r_{23} & r_{24} \\ r_{31} & r_{32} & r_{33} & r_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{7.95}$$

The inverse kinematics of the arm was also carried out in Example 4.4 and solutions for the three joint variables were obtained as (see Eqs (4.71), (4.73) and (4.74)):

$$\begin{aligned} \theta_1 &= A\tan 2(-r_{24}, -r_{14}) \\ \theta_2 &= A\tan 2\left(\pm\sqrt{r_{14}^2 + r_{24}^2}, r_{34}\right) \\ d_3 &= +\sqrt{r_{14}^2 + r_{24}^2 + r_{34}^2} \end{aligned} \tag{7.96}$$

For the initial position of the end-effector specified by T_s, Eq. (7.93), the feasible solution obtained in Example 4.4 is (see Eq. (4.75)):

$$\begin{aligned} \theta_1^s &= -60^\circ \\ \theta_2^s &= 45^\circ \\ d_3^s &= 0.30 \text{ m} \end{aligned} \tag{7.97}$$

The joint variables corresponding to final position of the end-effector as specified by T_g are:

$$\begin{aligned} \theta_1^g &= 50^\circ \\ \theta_2^g &= -25^\circ \\ d_3^g &= 0.10 \text{ m} \end{aligned} \tag{7.98}$$

Now, to plan linear trajectory with parabolic blends for the three links, the default acceleration for each link should satisfy the constraint in Eq. (7.21).

$$|\ddot{q}^c| \geq \frac{4|q^g - q^s|}{t_g^2} \tag{7.99}$$

That is, for link 1

$$\ddot{q}_1^{\,c} \geq \frac{4 \times (50 - (-60))}{5^2} = 17.6 \quad \text{or choose } \ddot{q}_1^{\,c} = 20.0 \text{ deg/s}^2 \tag{7.100}$$

Similarly, for link 2 and link 3

$$\begin{aligned} \ddot{q}_2^{\,c} &\geq -11.2 \quad \text{or choose } \ddot{q}_2^{\,c} = -15.0 \text{ deg/s}^2 \\ \ddot{q}_3^{\,c} &\geq -0.032 \quad \text{or choose } \ddot{q}_3^{\,c} = -0.05 \text{ m/s}^2 \end{aligned} \tag{7.101}$$

The trajectory (time-history) for each link is computed using the Eqs (7.20), (7.22), (7.23) and (7.24).

For first link

$$t_{b1} = \frac{1}{2} \times 5 - \frac{1}{2}\sqrt{\frac{25 \times 20 - 4(50 - (-60))}{20}} = 1.63 \text{ s} \tag{7.102}$$

Therefore,

$$\begin{aligned} q_1(t) &= \begin{cases} -60 + 10t^2 & 0 \leq t \leq 1.63 \\ -86.57 + 32.6t & 1.63 < t \leq 3.37 \\ 50 - 10(5 - t)^2 & 3.37 < t \leq 5.0 \end{cases} \\ \dot{q}_1(t) &= \begin{cases} 20t & 0 \leq t \leq 1.63 \\ 32.6 & 1.63 < t \leq 3.37 \\ 20(5 - t) & 3.37 < t \leq 5.0 \end{cases} \\ \ddot{q}_1(t) &= \begin{cases} 20 & 0 \leq t \leq 1.63 \\ 0 & 1.63 < t \leq 3.37 \\ -20 & 3.37 < t \leq 5.0 \end{cases} \end{aligned} \tag{7.103}$$

For second link, with $t_g = 5$ s and $\ddot{q}_2^{\,c} = -15$ deg/s²

$$t_{b2} = 2.5 - \frac{1}{2}\sqrt{\frac{25 \times (-15) - 4(-25 - 45)}{-15}} = 1.24 \text{ s} \tag{7.104}$$

$$\begin{aligned} q_2(t) &= \begin{cases} 45 + 7.5t^2 & 0 \leq t \leq 1.24 \\ 56.53 - 18.6t & 1.24 < t \leq 3.76 \\ -25 + 7.5(5 - t)^2 & 3.76 < t \leq 5.0 \end{cases} \\ \dot{q}_2(t) &= \begin{cases} -15t & 0 \leq t \leq 1.24 \\ -18.6 & 1.24 < t \leq 3.76 \\ -15(5 - t) & 3.76 < t \leq 5.0 \end{cases} \\ \ddot{q}_2(t) &= \begin{cases} -15 & 0 \leq t \leq 1.24 \\ 0 & 1.24 < t \leq 3.76 \\ 15 & 3.76 < t \leq 5.0 \end{cases} \end{aligned} \tag{7.105}$$

For third link, with $t_g = 5$ s and $\ddot{q}_3^c = -0.05$ m/s

$$t_{b3} = 2.5 - \frac{1}{2}\sqrt{\frac{25\times(-0.05) - 4(0.10 - 0.30)}{-0.05}} = 1 \text{ s} \tag{7.106}$$

$$q_3(t) = \begin{cases} 0.3 - 0.025t^2 & 0 \le t \le 1 \\ 0.325 - 0.05t & 1 < t \le 4 \\ 0.10 + 0.025(5-t)^2 & 4 < t \le 5 \end{cases}$$

$$\dot{q}_3(t) = \begin{cases} -0.05t & 0 \le t \le 1 \\ 0.05 & 1 < t \le 4 \\ -0.05(5-t) & 4 < t \le 5 \end{cases} \tag{7.107}$$

$$\ddot{q}_3(t) = \begin{cases} -0.05 & 0 \le t \le 1 \\ 0 & 1 < t \le 4 \\ 0.05 & 4 < t \le 5 \end{cases}$$

EXERCISES

7.1. A one-degree of freedom manipulator with rotary joint is to move from 113° to 210° in 7 seconds. Find the coefficients of the cubic polynomial to interpolate a smooth trajectory. Plot the position, velocity and acceleration variation as a function of time.

7.2. A single cubic trajectory given by $q(t) = 30 + t^2 - 6t^3$ is used for a period of 3 seconds. Determine starting and goal position, velocity, and accelerations of the end-effector.

7.3. The joint in Example 7.1 has initial and final velocity of 1.0 deg/s and 1.2 deg/s at $\theta^s = 30°$ and $\theta^g = 105°$, respectively. Determine the smooth trajectory polynomial for the position of the joint.

7.4. The trajectory for a joint's motion between start and goal position in a pick-n-place operation is determined by dividing the motion in two segments. The interpolating polynomial for each segment is cubic and the acceleration is continuous at the via point. Determine the coefficients for the two cubics.

7.5. The motion of a joint from start to goal position is specified in terms of position, velocity and acceleration at the beginning and end of a path segment. Determine the coefficients of the fifth-degree (quintic) polynomial for interpolating the smooth trajectory in the segment.

7.6. The trajectory for point-to-point motion between two points is divided into three segments and a 4-3-4 trajectory plan is used. That is, the first segments is a fourth degree polynomial specifying the trajectory from start position to lift-off point; followed by a cubic polynomial in the mid trajectory segment up to the set-down point and the last segments trajectory is similar to the first segment, a fourth-degree polynomial, from set-down point to goal position. Determine the polynomial equations for the three segments. Assume appropriate constraints.

7.7. Determine the equations of polynomials for the three segments of a point-to-point trajectory between two points if a 3-5-3 trajectory plan is used.

7.8. The trajectory between two points is divided into five segments and five-cubic polynomial interpolation is to be used. The boundary conditions that the polynomials must satisfy are
(i) position constraints at start, lift-off, set-down and goal positions, and
(ii) continuity of velocity and acceleration at all the path points.
Determine the polynomial for each segment.

7.9. A rotary joint moves from –15° to +45° in 3 seconds. Determine the polynomial for a smooth trajectory, if the initial and final velocity and accelerations are zero.

7.10. A linear trajectory with parabolic blends is used instead of a cubic polynomial in Exercise 7.1. Determine the parameters for the trajectory. Assume constant acceleration of 30 deg/s^2

7.11. Calculate the parameters for a two-segment linear spline with parabolic blends with acceleration in blend region to be 30 deg/s^2 and duration of each segment as 3 seconds. The three path points for the joint are 15°, –5° and 30°. Plot the trajectory.

7.12. Determine the trajectory for a pick-n-place cycle, which has to pass through three via points using piecewise-linear interpolation with parabolic blends for each segment. The path points are $[0° \;\; 10° \;\; 45° \;\; 30° \;\; 5°]$ and the travel times for the segments are $[0.5 \;\; 1.5 \;\; 2.0 \;\; 1.0]$ seconds respectively. Assume that the constant acceleration is constant at 25 deg/s^2 in each segment.

7.13. Find the trajectory for a Cartesian space straight line path from $[1.0 \;\; 1.0 \;\; 1.0]^T$ to $[-1.0 \;\; 0 \;\; 1.0]^T$.

7.14. Compute the joint trajectory for $q^s = 0$ and $q^g = 4$ with null initial and final velocities and accelerations. Assume $t_g = 1$.

7.15. The velocity profile of a trajectory is trapezoidal with constant acceleration segment of 0.5 second duration and constant velocity of 10 deg/sec. Determine the parameters of the smooth trajectory for interpolating the time sequence of position with this type of trajectory.

7.16. The end-effector of two link planar manipulator with each link 1 m long, discussed in Section 6.2, is to move from initial position $(x_s, y_s) = (0.9, 1.6)$ to final position $(x_g, y_g) = (-1.6, 0.5)$. Determine the coefficients of cubic polynomials for each joint to generate a smooth trajectory.

7.17. If linear spline with parabolic blend is used instead of a cubic polynomial in Exercise 7.16, determine the parameters of the polynomials to interpolate both joint trajectories.

7.18. For a three-degree of freedom, cylindrical manipulator, design a linear trajectory with parabolic blends. The initial position of the end-effector is expressed by the homogeneous matrix $\boldsymbol{T}_s$ and the goal position by $\boldsymbol{T}_g$ as

$$T_s = \begin{bmatrix} 0 & 1 & 0 & 200 \\ 1 & 0 & 0 & 100 \\ 0 & 0 & -1 & -100 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7.108)$$

$$T_g = \begin{bmatrix} 1 & 0 & 0 & -50 \\ 0 & -1 & 0 & 200 \\ 0 & 0 & 0 & 400 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7.109)$$

7.19. For a two-segments continuous-acceleration trajectory, the path points are $\theta^s = 10°$, $\theta^g = 30°$ and $\theta^m = 22°$. The duration of two segments are 1.2 second and 1.0 second, respectively. Taking the velocity at via point as 15 deg/s, plot the position, velocity and accelera[illegible] of the interpolated trajectory as a function of time.

7.20. Third joint of a 5-DOF manipulator is $\theta_3 = 60°$ at start of work cycle. It is to move by angle of 45° in 3 seconds. Find the cubic trajectory for this joint to achieve the specified motion. The joint motion starts from rest and comes to rest at the destination point. Plot position, velocity and acceleration of the joint as a function of time.

7.21. Two joints of a SCARA manipulator are to move by 30° and 45° in 2 seconds and 3 seconds, respectively. Assuming the trajectory to be cubic for both the joints, determine the coefficients a_{ij} for the two cubic polynomials:

$$\theta_1(t) = a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3$$

$$\theta_2(t) = a_{20} + a_{21}t + a_{22}t^2 + a_{23}t^3$$

Will the trajectories for the two joints be different if the joints are considered at some location other than 0° at the start point?

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